



Spectrum Power 5
Information Model
Management - Graphics
Editor - v2.20

5.30

Reference Guide

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NOTE

For your own safety, observe the warnings and safety instructions contained in this document, if available.

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Preface

Notes on Safety

This manual is not a complete index of all safety measures required for operation of the equipment (module or device). However, it comprises important information that must be followed for personal safety, as well as to avoid material damage. Information is highlighted and illustrated as follows according to the degree of danger:



WARNING

WARNING means that death or severe injury **may** result if the measures specified are not taken.

- ✧ Comply with all instructions, in order to avoid death or severe injuries.



CAUTION

CAUTION means that medium-severe or slight injuries **can** occur if the specified measures are not taken.

- ✧ Comply with all instructions, in order to avoid moderate or minor injuries.



NOTE

Important information about the product, product handling or a certain section of the documentation which must be given particular attention.

Qualified Electrical Engineering Personnel

Only qualified electrical engineering personnel may commission and operate the equipment (module, device) described in this document. Qualified electrical engineering personnel in the sense of this manual are people who can demonstrate technical qualifications as electrical technicians. These persons may commission, isolate, ground and label devices, systems and circuits according to the standards of safety engineering. Only competent and authorized personnel should work with this product after becoming thoroughly familiar with all warnings, safety notices, operating instructions and maintenance procedures.

Proper Use

The product must not be used for any other purposes than that described in the technical documentation. If it is used together with third-party devices and components, these must be recommended or approved by Siemens AG. The successful and safe operation of this product is dependent on adequate transportation and proper handling, storage, installation, operation and maintenance.

Scope

This document provides specific and detailed information on how to use a particular product or product component.

**NOTE**

Note that the screenshots used in this document contain sample data which may not be available in some systems.

Typical Users

This document is designed for users that are already familiar with operational and technical aspects of power generation and power transmission and distribution as well as product concepts:

- **System Engineers**
The System Engineer is able to install and to customize the system. He needs deep knowledge about the internal structures and processes of the network control system.
- **Operators**
The Operator is controlling the network or the plants with the system therefore he needs to know how the network control system has to be operated.
- **Data Engineers**
The Data Engineer translates the real network of a utility or the generation plants to the data model of the network control system. He also maintains the data model.
- **Technical Specialists**
The Technical Specialist is responsible for a special technical area which is related to the network control system. This can be the planning department, which plans changes in the network and uses network analyses applications of the network control system or the RTU department, which has to connect the RTU to the network control system. The Technical Specialist has to know how specific functions of the network control system work in detail.

Structure of the Manual

- **Introduction:**
Basic information about the component
- **Workflow/Operations:**
Main use cases, workflows and operational steps for optimal use of the component
- **References:**
Reference and description of user interfaces and dialogs

Documentation Conventions

Example	Meaning
File > Open	Selecting the menu item Open on the File menu
F1	Pressing the F1 function key on the keyboard
Control+o	Holding down the Ctrl key and pressing the letter o on the keyboard
Control+Shift+o	Holding down the Ctrl key and the Shift key and pressing the letter o on the keyboard
Left-click	Clicking the left mouse button
Right-click	Clicking the right mouse button
Shift-click	Holding down the Shift key on the keyboard and clicking the left mouse button

Questions?

This document is part of a set of manuals that describes the complete product.

Various other product modules may be mentioned or discussed in this document. For more detailed information or if you have any questions about these products, please contact your Siemens representative.

Open Source Software

Among other things, this product contains Open Source Software that was developed by third parties. The Open Source Software included with this product and the relevant Open Source Software license agreement can be found in Readme_OSS. The Open Source Software program is protected by copyright. You are permitted to use the Open Source Software according to the Open Source Software license agreement. Should contradictions occur between the Open Source Software license conditions and the Siemens license conditions applicable to the product, the Open Source Software license agreement with respect to the Open Source Software shall prevail. The Open Source Software is provided free of charge. If stipulated by the Open Source Software license agreement, the source text of the software is available until the end of the third year after purchasing the product. Shipping charges will apply separately. We are liable for the product including the Open Source Software according to the license agreement for the product. Any liability arising from the use of the Open Source Software that exceeds the intended program flow of the product, as well as liability for defects that were caused by changes made to the Open Source Software, is excluded. If the product was modified, we cannot provide technical support.

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1 Graphics Editor in the IMM Environment

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1.1 Operational Steps

1.1.1 Opening the Graphics Editor

The goal of this task is to open the **Graphics Editor**.

The **Graphics Editor** is invoked within an **IMM Job** only. For additional information on how to start IMM and working with the Job management, refer to Reference Guide Information Model Management (IMM).

There are 2 modes available to open the **Graphics Editor**:

- Opening the **Graphics Editor** in the **Job Realtime (RT)**.
If RT job state is view-only, viewing of the current productive graphical displays is possible.
- Opening the **Graphics Editor** in any other **User-Job** allows editing of graphical displays.



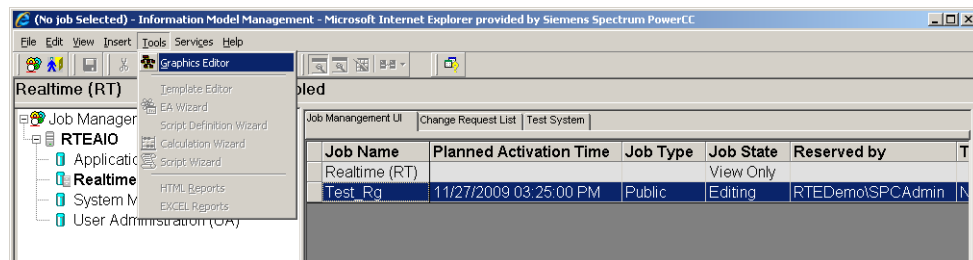
NOTE

To run the **Graphics Editor**, WinCC must be installed on the computer on which the **Graphics Editor** is used.

To open the **Graphics Editor**:

- Select a job in the RT dataset within the **Job Management User Interface (UI)**.
- Select **Tools > Graphics Editor** in the IMM menu bar.

The **Graphics Editor** is opened within the selected job.

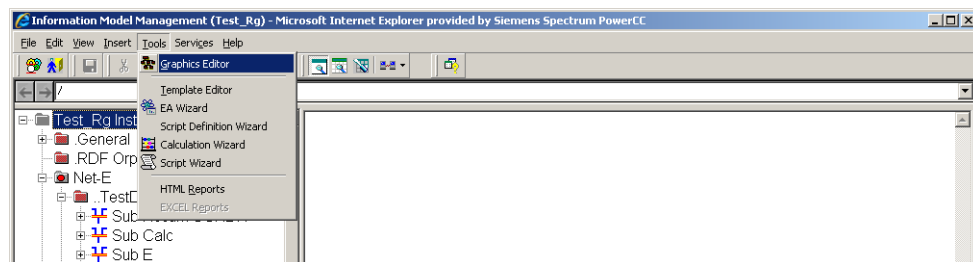


[immrg001-080210-01.tif, 1, en_US]

Figure 1-1 Opening Graphics Editor within the Selected Job

Alternative Operation

- Open a job in the RT dataset.
- Select **Tools > Graphics Editor** to open the **Graphics Editor** within the Job.



[immrg002-080210-01.tif, 1, en_US]

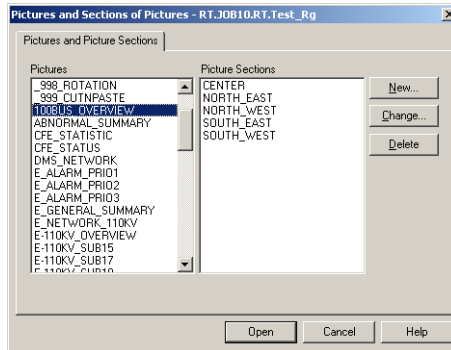
Figure 1-2 Opening Graphics Editor in a Job

1.1.2 Opening an Existing Worldmap or Worldmap View

The goal of this task is to open an existing worldmap or worldmap view in the **Graphics Editor**.

Worldmap

- Select **Open Worldmap** from the **Graphics Editor** toolbar.
The **Pictures and Sections of Pictures** dialog opens.



[immrg003-070409-01.tif, 1, en_US]

Figure 1-3 Worldmap

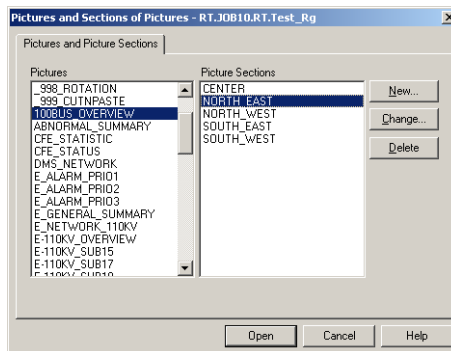
- Select the desired worldmap within the **Pictures** area.
- Click **Open**.
The selected worldmap opens on the drawing section of the **Graphics Editor**. The worldmap with the specified zoom factor is displayed.

Alternative Operation

- Double-click the desired worldmap within the **Pictures** area of the **Pictures and Sections of Pictures** dialog to open the worldmap.
- Alternately, use the keyboard to jump within the **Pictures** area to the desired worldmap, character by character.

Worldmap View

- Select **Open Worldmap** from the **Graphics Editor** toolbar.
The **Pictures and Sections of Pictures** dialog opens.



[immrg004-070409-01.tif, 1, en_US]

Figure 1-4 Worldmap View

- Select the desired view of the selected worldmap within the **Picture Sections** area.
- Click **Open**.
The selected worldmap view opens on the drawing section of the **Graphics Editor**. It displays the selected worldmap view completely. The view-specific coordinates and zoom factor are considered.



NOTE

By nature, the displayed content of the worldmap view is according to the presentation settings of the **Graphics Editor** (default presentation rule). It differs from the presentation at runtime.

Alternative Operation

- Double-click the desired worldmap view within the **Picture Sections** area of the **Pictures and Sections of Pictures** dialog.
This opens the worldmap view of the selected worldmap.
- Alternately, use the keyboard to jump within the **Picture Sections** area to the desired worldmap view, character by character.

1.1.3 Migrating Existing Worldmaps

Worldmaps created with the **Graphics Designer** of Spectrum PowerCC must be migrated for the **Graphics Editor**.

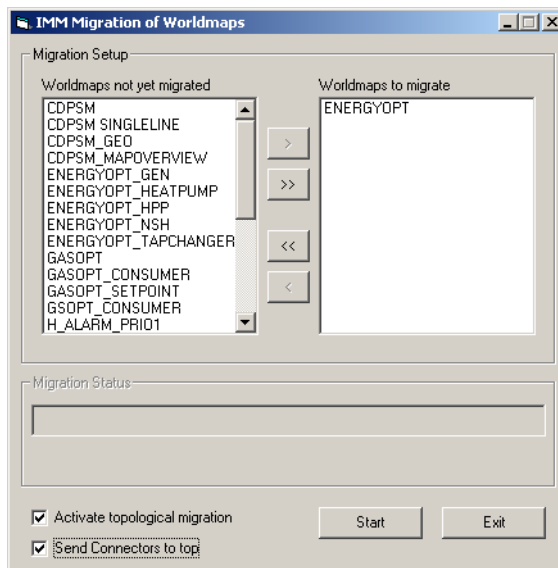
The goal of this task is to migrate existing worldmaps created with the old **Graphics Designer** of Spectrum PowerCC to the current **Graphics Editor** format.



NOTE

After updating your Spectrum PowerCC System to Spectrum Power 5.20 or higher, make sure that your worldmaps created with the **Graphics Designer** are stored within the legacy WinCC database.

- Select **Tools > Macros > AddIns > Migrate Worldmaps to IMM** from the **Graphics Editor** toolbar.
The **IMM Migration of Worldmaps** dialog opens.



[immrg003-080210-01.tif, 1, en_US]

Figure 1-5 Migration of Worldmaps

- Select the desired worldmap or multiple worldmaps to migrate within the **Migration Setup** area.
- Check **Activate topological migration** to consider the links from the graphical objects to the IMM domain data.

If unchecked, only the graphical data is migrated.

- Check **Send Connectors to top** to make sure that the connectors are created in front of all other objects within the layers of the worldmap. If unchecked, some connectors may be hidden by other graphical objects.
- Click **Start**.
The **Migration Status** bar shows the progress of the migration process.



NOTE

Undo action on migration of displays and worldmaps is not possible.
Migrating displays and worldmaps takes some time.

1.1.4 Importing / Exporting Library Objects

The goal of this task is to import / export library objects for use by other projects or reuse from other projects. Importing / exporting of library objects is supported with the IMM Import / Export. For additional information, refer to Reference Guide Information Model Management (IMM).

1.2 User Interface Description

1.2.1 Introduction

The **Graphics Editor** is an editor for creating displays and worldmaps and making them dynamic. It is based on the **WinCC Graphics Designer** and is adapted with features required for working within the IMM environment and the Spectrum Power Control Center domain.

Based on the Windows standard, it provides functions for the creation and modification of displays and world-maps used on the runtime user interface.

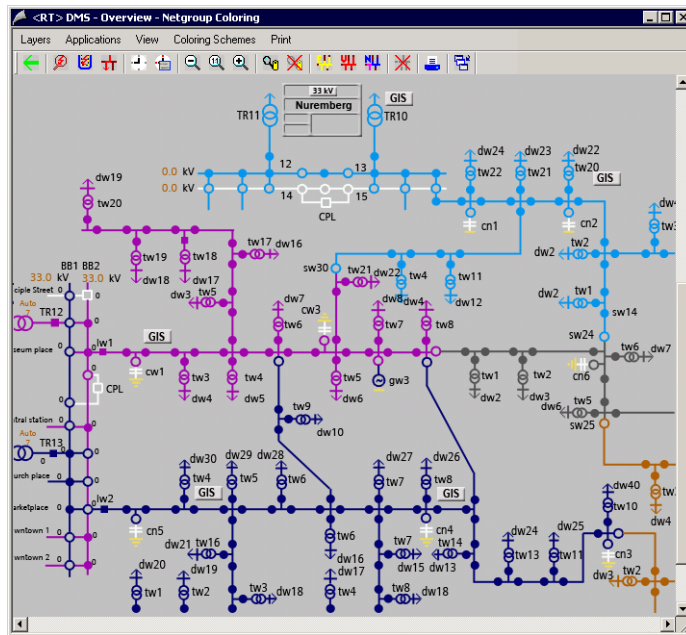


Figure 1-6 Example of a Network Diagram at Runtime

The familiar windows program interface makes it easy to begin using the complex program. Direct help provides fast answers to questions. You can set up a personalized working environment.

The **Graphics Editor** is fully integrated into the Spectrum Power **Information Model Management (IMM)**. The IMM job management controls the access or lock of network diagrams. For additional information on the IMM job management, refer to Reference Guide Information Model Management (IMM).

1.2.2 Display or Worldmap Migration

The **Graphics Editor** is the successor of the former used **WinCC Graphics Designer** for Spectrum PowerCC. Displays and worldmaps created with the **WinCC Graphics Designer** for Spectrum PowerCC must be converted to the current format of the **Graphics Editor**. The migration process adds the graphical data into the IMM database.

Migration is done with the **Graphics Editor** by the **Migrate Worldmaps to IMM** dialog. Worldmaps not yet migrated are displayed in the corresponding area within the dialog.



NOTE

After updating your Spectrum PowerCC System to Spectrum Power 5.20 or higher, make sure that your worldmaps created with the **Graphics Designer** are stored within the legacy WinCC database.

For sequential migration of worldmaps build with the **WinCC Graphics Designer** make sure that the names of these worldmaps differ from worldmap names that are already migrated and available in the IMM database. If not, your worldmap is not visible in the corresponding area within the **Migrate Worldmaps to IMM** dialog.

Selection is made whether only the graphical data gets migrated or the links to the IMM domain data is considered. By default **Activate topological migration** is selected to consider the IMM domain data relationships. By default also **Send Connectors to top** is selected to make sure that the connectors are created in front of all other objects within the layers of the worldmap.



NOTE

Undo action on migration of displays and worldmaps is not possible.
Migrating displays and worldmaps takes some time.

The main goal of the migration functionality is to get the layout and runtime functionality of the worldmaps similar to what was available in the previous releases of Spectrum PowerCC. The migrated worldmaps can not be used for technological editing in general. You have to adapt them case by case during further maintenance work.

1.2.3 Display or Worldmap Building

Displays and worldmaps are achieved by 2 methods:

- Designing graphical displays on the drawing section of the **Graphics Editor**.
- Import of graphic displays in **Enhanced Meta Files (EMF)**, **Windows Meta Files (WMF)**, and native **WinCC Graphic Files (PDL)** format.

EMF file format provides only the import of static information (for example, background information).
The same is true for WMF format.



NOTE

The IMM provides data exchange of all engineering data, including graphical data, also in **eXchange Data Format (XDF)** and **Common Information Model - Resource Description Format (CIM-RDF)**. The IMM supports complete, partial, and incremental data exchange.

For additional information on graphical data exchange in **CIM-RDF** or **XDF** format, refer to Reference Guide Information Model Management (IMM).

The preferred way of importing worldmaps within the Spectrum Power IMM is using **XDF** and **CIM-RDF** formats.



NOTE

Created displays are stored in the database. All Spectrum Power Control Center consoles, which are connected to this database have access to the displays. Access right is restricted to consoles or users according to Spectrum Power Control Center security strategy.

1.2.4 Technological Editing

Spectrum Power provides a domain topology model using terminal and connectivity node classes, which describe the technological object or instance data model. This domain topology is the base for all Spectrum Power Control Center applications.

One way of entering the domain topology is to derive it from the graphic model (single-line diagrams) while the graphic model is edited. This data modification is so-called technological editing.

The **Graphics Editor** provides the following use cases:

- Creation of new graphical objects and new domain data using templates
- Creation of new graphical objects and new domain data using copy and paste of existing graphical objects
- Creation of new graphical objects and linking them to existent domain data in one operation
- Connection or disconnection of graphic objects
- Creation and manipulation of the network topology with connecting graphical objects
- Deletion of domain data using deleting linked graphical objects
- Linking existent graphical objects to existent domain data.
- Creation of user-defined templates that contain domain and graphical data

1.2.5 Common Locking Strategy

1.2.5.1 Overview

Spectrum Power IMM provides a common locking strategy for editing data concurrently by different users. The main goal is to guarantee the integrity of the database and to prevent inconsistencies as far as possible. This strategy applies for graphical data and their links to domain data as well. For additional information on the common locking strategy, refer to Technical Specification Information Model Management (IMM).

IMM provides 2 different model maintenance concepts:

- Interlock-based job model
- Time-based job model

The job model selected for the IMM defines the locking strategy.

1.2.5.2 Object Lock Mechanism

The **Graphics Editor** provides an object lock mechanism, which supports the editing of multiple users inside the same worldmap. Specific worldmap areas are not locked, only single objects are locked.



NOTE

The object lock mechanism is used within the interlock-based job model mode.
For additional information, refer to Reference Guide Information Model Management (IMM).

Object Lock

Objects are locked if they are modified in another IMM job in the interlock-based job model mode. Locked objects cannot be modified or deleted. This keeps jobs independent from each other.

The objects modified by you are automatically locked for other jobs to inhibit concurrent changes of the same object in different jobs.

Display Locked Objects

All available locked objects of a worldmap are highlighted graphically within the worldmap.

The properties of a locked object are displayed by selecting the menu item **View > Graphical Properties**.



NOTE

Modification of the properties of locked objects is not possible.

1.2.5.3 Time-based Locking Mechanism

The **Graphics Editor** provides data integrity checks, which supports the editing of multiple users inside the same worldmap. Within the time-based job model, the same data instance can be multiple modified based on the specified planned activation date. If data changes of a job are in conflict with the changes of a job with later activation date, the user is notified of the conflict. As there is no dedicated object lock mechanism, highlighting of locked objects is not required.



NOTE

The time-based locking mechanism is only used within the time-based job model mode.
For additional information, refer to Reference Guide Information Model Management (IMM).

Data Integrity Checks

Objects are implicitly locked if they are in conflict with modifications in another job with later planned activation date in the time-based job model mode. Objects with conflicts cannot be modified or deleted to ensure data integrity.

The conflicts are presented in the **Errors** dialog when trying to perform the user operation.

1.2.6 Worldmap Concept

Spectrum Power has implemented a worldmap concept for creation, display, maintenance, and navigation of large network diagrams. Worldmaps are network displays, which consist of multiple stations in a single display.

This worldmap concept offers the necessary user interface features requested by large systems (for example, **DMS** systems). Both overview displays and fine granular detailed views are found and it is easy to navigate between them.

A worldmap is a 2-dimensional drawing plane, containing each presentation detail. It consists of layers and the presentation details are distributed to several layers. In a window on the console screen, only a small subset of the worldmap view is displayed. The worldmap concept provides for sophisticated methods to navigate the subset of details that are currently interesting for the operators work.



NOTE

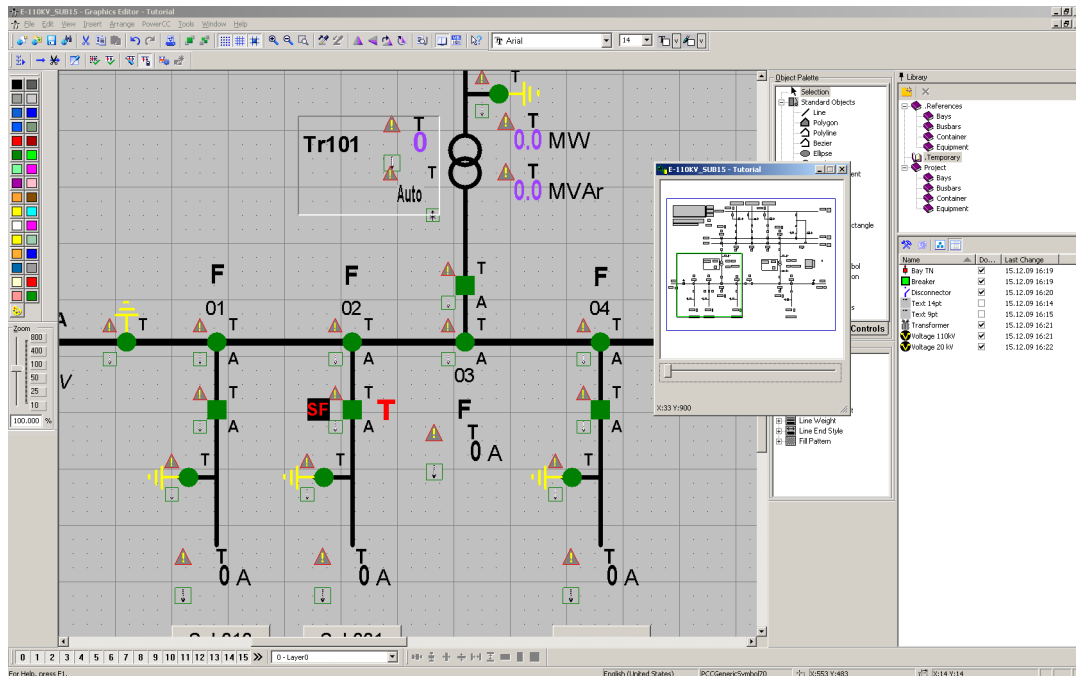
Layers are used to provide additional levels of details, which is switched on or off in the runtime system.

The following are the worldmap concepts:

- The drawing area consists of 32 layers.
Usually, all layers are visible together and a display consists of a set of all layers. The layer concept helps the display designer to organize the grouping of the network components. It is possible, however not recommended, to put all details within a single layer.
- Zooming is possible up to a zoom factor of 800 percent.
While zooming, the visibility of the displayed details (lines, character strings, symbols, etc.) is automatically kept within visible limits by decluttering.
- The decluttering function controls the visibility of the graphical objects.
There are 2 types of decluttering:
 - **Object-based** decluttering
An object is displayed when its size is greater than the configured lower limit and smaller than the configured upper limit.
 - **Layer-based** decluttering
Each object is assigned to a layer. Layers are defined to be visible only within a certain zoom range. Only those objects are displayed that belong to a currently visible layer.

Navigation Window

The **Navigation** window allows quick maneuver in large worldmap network diagrams. A subset of the worldmap along with the entire worldmap is viewed simultaneously.



[imgmge001-150313-01.tif, 1, en_US]

Figure 1-7 Navigation of a Worldmap

The presented worldmap shows how the subset of information in the worldmap window is located in the defined area. A green border in the navigation window identifies it.

1.2.7 Zooming and Panning

Zooming

You can zoom in to focus on details within the current network diagram or zoom out to view more operational network components and information.

The **Graphics Editor** supports 3 types of zooming methods:

- Zoom buttons
- Zoom slider
- Scroll wheel, if available (as used in MS Office)

Panning

The panning methods available in the **Graphics Editor** are MS Windows-based panning functions (scroll bars) as well as panning using the scroll wheel.

These methods allow to navigate from 1 side of a network diagram to another location. Additionally, the **Graphics Editor** provides the **Navigation** window that allows to navigate quickly in large network diagrams (worldmap).

There are also panning methods available during normal editing operations of a display. For example, object resizing, object definitions, object movement, rubberband selection, and area zooming. These operations cause an automatic panning as soon as the borders of the objects are touching the border of the drawing section of the display.

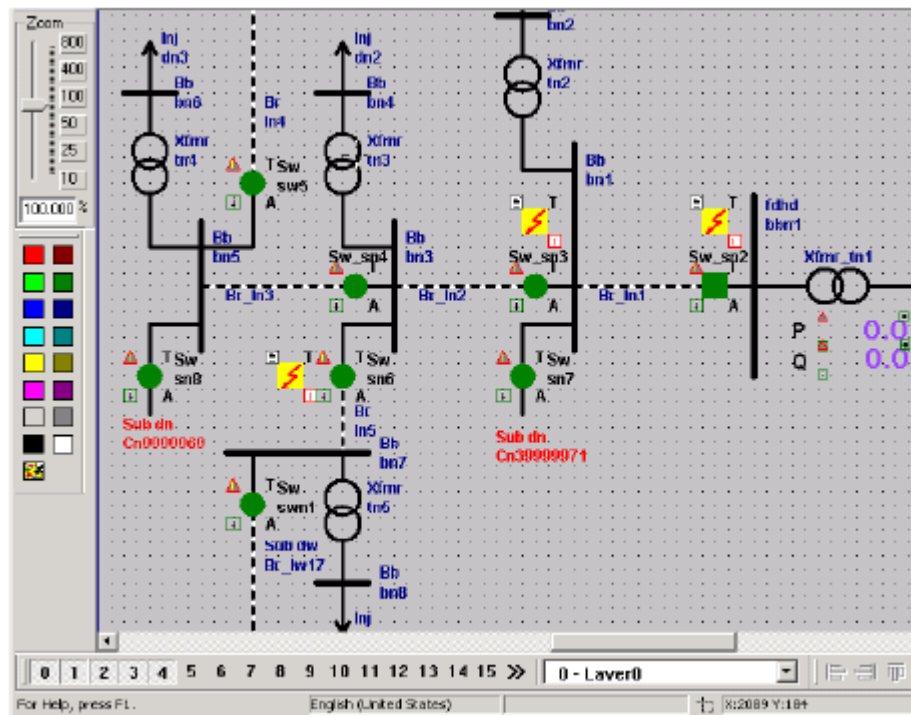
1.2.8 Layering and Decluttering

Layering

The graphical objects are distributed in different layers and therefore they do not have all details within one layer. Each layer is configured with a zoom ratio based on a specific number of pixels and optionally, alpha-numeric labels for the objects. Each layer is switched on (visible) and off (invisible) manually. If layer-based decluttering is activated, the visibility of layers is controlled using the current zoom factor.

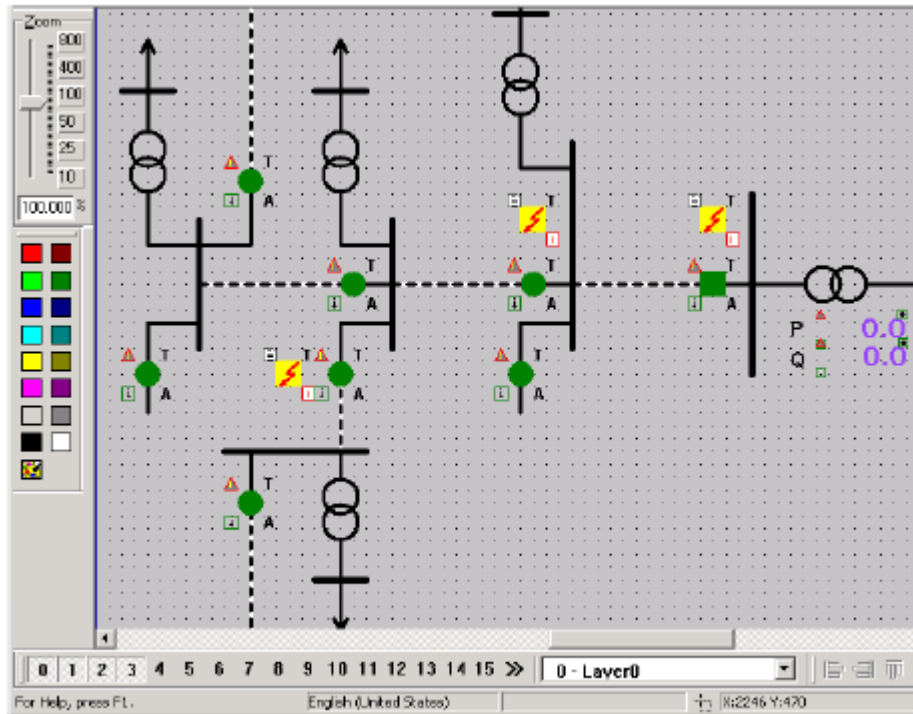
Layering is used for the following 2 purposes:

- To show or hide individual layers manually and to determine which details are displayed.
- To control the visibility of objects by defining a zoom range for each layer. It determines, which layers are visible at a certain zoom factor.



[immg003-210808-01.tif, 1, en_US]

Figure 1-8 Part of a Network Diagram that Displays 5 Active Layers (Layer 0 to Layer 4)



[immrg004-210808-01.tif, 1, en_US]

Figure 1-9 Network Diagram without the Texts (Only Layers 0 to 3 Are Visible, Layer 4 Containing the Texts Is Invisible)

The visible characteristics configured for each layer automatically display as each layer is invoked. Navigate between the layers by:

- Using the zoom functions to maneuver from one layer using zooming or decluttering to another.
- Manually selecting pre-configured layers.

Decluttering

The decluttering functionality is used to control the visibility of objects based on the specified lower and upper limits of the object sizes. It is also based on the visibility of the layers in which the object is located.



NOTE

For static texts, additional and independent presentation properties for zooming or layering and decluttering behavior are provided.

An example for the common use case is, keep the size of the static text while zooming in for better readability.

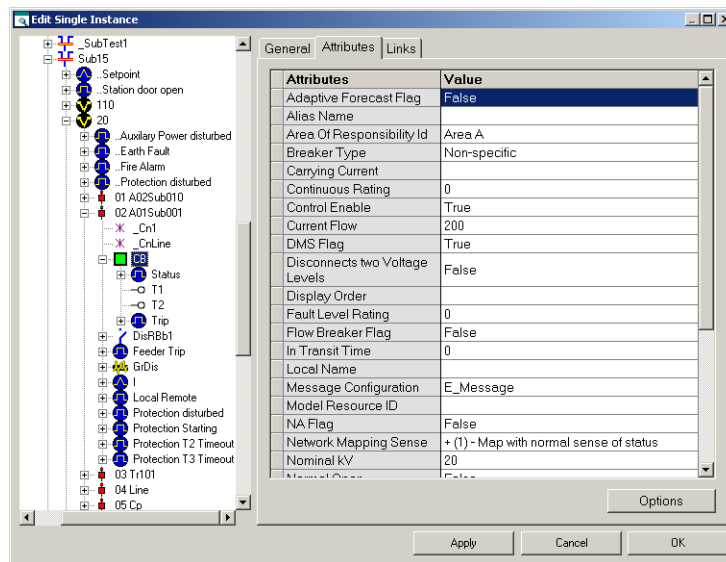
1.2.9 Related IMM Editors

1.2.9.1 Single Instance Editor

The **Graphics Editor** is fully integrated into the **Information Model Management (IMM)**. The **Single Instance Editor** modifies the instance data.

The **Single Instance Editor** is invoked by clicking the appropriate toolbar button of the **Graphics Editor**. If you select a graphic object on the drawing section, the corresponding instance level of the selected object opens in the **Single Instance Editor**. If no object is selected on the drawing section, the first level instances opens in the **Single Instance Editor**.

Right click the mouse to invoke the **Single Instance Editor** on an already linked object and select **Network Element Properties**.



[immrg057-090409-01.tif, 1, en_US]

Figure 1-10 Attributes Page of the Single Instance Editor

For additional information on the **Single Instance Editor**, refer to Reference Guide Information Model Management (IMM).



NOTE

The tree view on the right side of the single instance editor is used for navigation only.

1.2.9.2 Presentation Logic Editor

The presentation logics for a certain type of network element are defined in the **Presentation Logic Editor** of IMM. These presentation logics are assigned to an object in the **Graphics Editor**. For additional information on the **Presentation Logic Editor**, refer to Reference Guide Information Model Management (IMM).

1.2.10 Library Templates

The library of the **Graphics Editor** is a versatile tool for saving and managing graphic objects. It is used for creating displays and worldmaps together with domain data. The library provides for Spectrum Power a base set of predefined templates. Objects within the library can be imported and exported for intra-project reuse. The library instances are stored in the **Application Configuration (AC)** dataset. The associated path is: **IMM Configuration/Graphic Templates/...**

For additional information on import and export of IMM data, refer to Reference Guide Information Model Management (IMM).

1.2.11 Keyboard Shortcuts and Menu Bar

1.2.11.1 Keyboard Shortcuts



NOTE

Keys are all lower case, all upper case, or mixed lower case and upper case unless explicitly described.

To	Press
Open Graphics Editor from IMM Menu	Alt-tg
Close Graphics Editor (dependent of worldmap opened or not)	Alt-fe or Alt-fx, or Alt-F4



NOTE

The layout of the menu bar and the submenus is dependent if a worldmap is currently opened or not. This influences also the keyboard shortcut functions available.

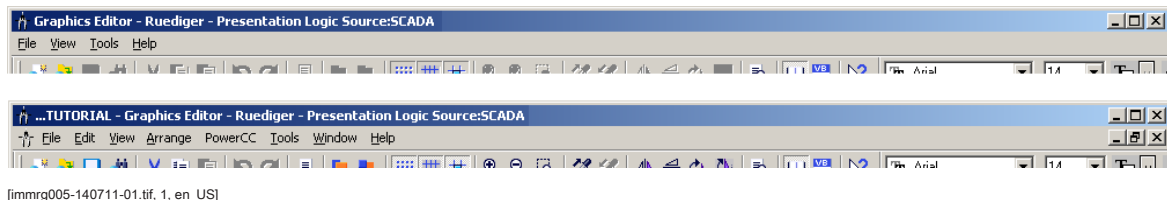


Figure 1-11 Dependent Graphics Editor Menu Bar

1.2.11.2 Menu Bar Shortcuts

Accessing Menus

To Display	Press
File menu	Alt-f
Edit menu	Alt-e
View menu	Alt-v
Arrange menu	Alt-a
PowerCC menu	Alt-p
Tools menu	Alt-t
Window menu	Alt-w
Help menu	Alt-h
Window System menu	Alt-SpaceBar

Accessing File Menu Functions

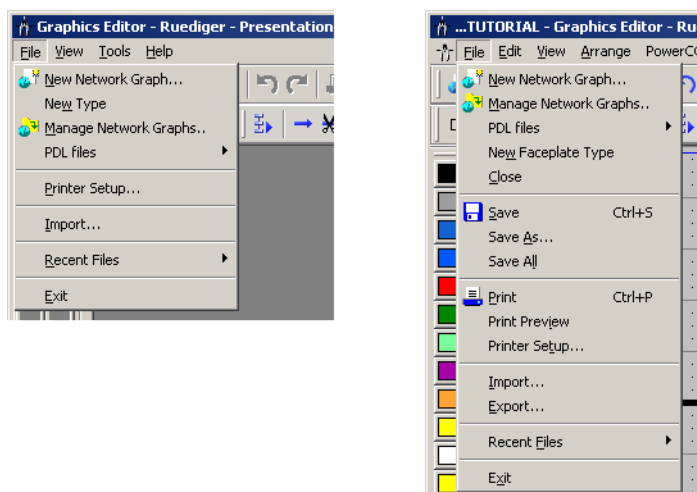
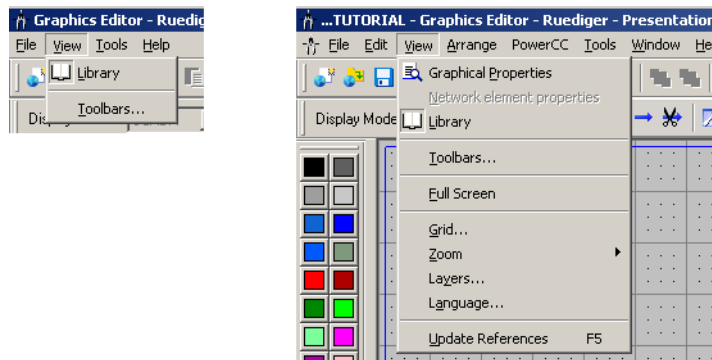


Figure 1-12 Dependent File Menu

To	Press
Create and open a new native WinCC picture	Ctrl-n

To	Press
Create and open a new Spectrum Power 5 worldmap	Alt-fn
Open an existing native WinCC picture	Ctrl-o
Manage Spectrum Power 5 worldmaps	Alt-fv
Create a new Faceplate Type. Not supported within Spectrum Power 5 and IMM	Alt-fw
Close a native WinCC picture or Spectrum Power 5 worldmap	Alt-fc or Ctrl-F4
Save a native WinCC picture or Spectrum Power 5 worldmap	Alt-fs or Ctrl-s
Copy a native WinCC picture or Spectrum Power 5 worldmap and save it under a different name	Alt-fa
Save all opened native WinCC pictures and Spectrum Power 5 worldmaps	Alt-fl
Print a native WinCC picture or Spectrum Power 5 worldmap	Alt-fp or Ctrl-p
Print preview of a native WinCC picture or Spectrum Power 5 worldmap	Alt-fi
Set up a printer	Alt-fp or Alt-ft
Import static graphics into a native WinCC picture or Spectrum Power 5 worldmap	Alt-fi
Export a native WinCC picture or Spectrum Power 5 worldmap in EMF or PDL format	Alt-fe
Get a list of all recently used native WinCC pictures or Spectrum Power 5 worldmaps to select one for opening	Alt-fr

Accessing View Menu Functions

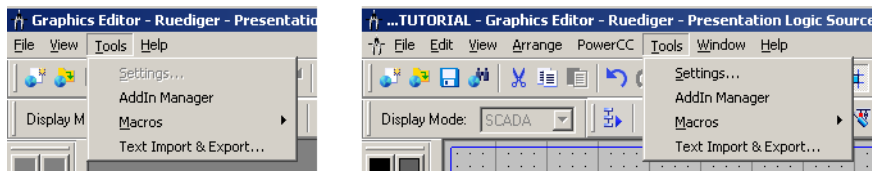


[image003-150313-01.tif, 1, en_US]

Figure 1-13 Dependent View Menu

To	Press
Show or hide the library	Alt-vl
Show or hide specific toolbars	Alt-vt
Open the Graphical Properties dialog	Alt-vp
Open the Single Instance Editor dialog	Alt-vn
Switch to full screen mode	Alt-vf
Open the Settings dialog for the grid	Alt-vg
Zoom into a native WinCC picture or Spectrum Power 5 worldmap	Alt-vzi
Zoom out a native WinCC picture or Spectrum Power 5 worldmap	Alt-vzo
Switch to zoom area selection mode	Alt-vzz
Open the Settings dialog for the layers	Alt-vy
Select the language	Alt-va
Update the references	Alt-vu or F5

Accessing Tools Menu Functions

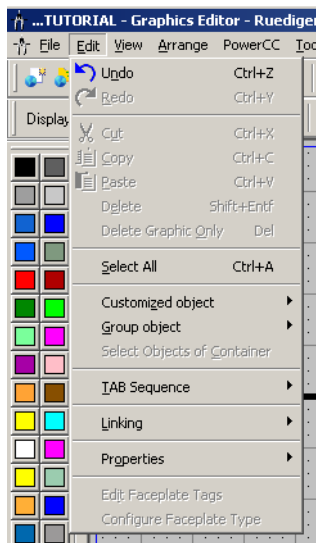


[image004-150313-01.tif, 1, en_US]

Figure 1-14 Dependent Tools Menu

To	Press
Open the Visual Basic Editor	Alt-tmv or Alt-F11
Migrate worldmaps to IMM	Alt-tmam
Open the Settings dialog	Alt-ts
Open the Add-In Manager dialog	Alt-ta

Accessing Edit Menu Functions



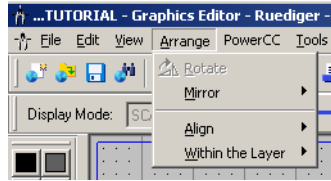
[immg009-140711-28.tif, 1, en_US]

Figure 1-15 Edit Menu with Submenus

To	Press
Undo non-technological editing operations	Alt-en or Ctrl-z, or Alt-Backspace
Redo non-technological editing operations	Alt-er or Ctrl-y
Cut selected objects and copy it into the clipboard	Alt-eu or Ctrl-x
Copy selected objects into the clipboard	Alt-ec or Ctrl-c, or Ctrl-Ins
Paste copied or cut objects	Alt-ep or Ctrl-v, or Shift-Ins
Delete objects including linked IMM domain data	Alt-ee or Shift-Del
Delete objects only graphically	Alt-eo or Del
Select all objects within the worldmap	Alt-es or Ctrl-a
Handle customized objects	Alt-ez
To group objects	Alt-egg
To ungroup objects	Alt-egu
Select all related graphics of a label object	Alt-ea
Define a TAB-Sequence	Alt-et
Link Tag Connections or Texts	Alt-el

To	Press
To copy properties of a selected object	Alt-eoc
To assign copied properties to objects	Alt-eoa

Accessing Arrange Menu Functions

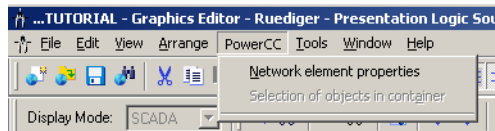


[immrg011-140711-01.tif, 1, en_US]

Figure 1-16 Arrange Menu with Submenus

To	Press
Rotate selected objects counter-clock wise	Alt-ar
Mirror selected objects horizontally	Alt-amh
Mirror selected objects vertically	Alt-amv
Align selected objects	Alt-aa
Bring the selected objects within the layer to front	Alt-awt or Ctrl-Plus
Send the selected objects within the layer to back	Alt-awb or Ctrl-Minus

Accessing Power 5 Menu Functions

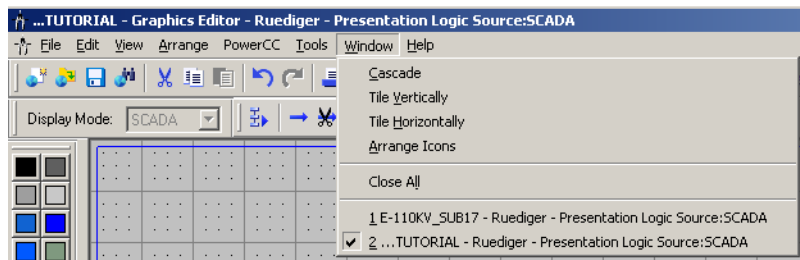


[immrg012-140711-01.tif, 1, en_US]

Figure 1-17 PowerCC Menu

To	Press
Open the Single Instance Editor dialog	Alt-pn
Select all related graphics of a container object	Alt-pz

Accessing Window Menu Functions



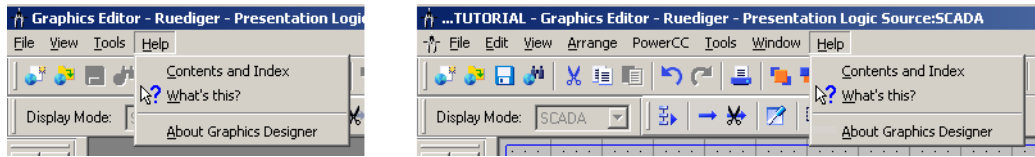
[immge005-150313-01.tif, 1, en_US]

Figure 1-18 Window Menu

To	Press
Cascade all opened native WinCC pictures or Spectrum Power 5 worldmaps	Alt-wc
Tiles all opened native WinCC pictures or Spectrum Power 5 worldmaps vertically	Alt-wv
Tiles all opened native WinCC pictures or Spectrum Power 5 worldmaps horizontally	Alt-wh

To	Press
Arranges Icons for all opened but minimized native WinCC pictures or Spectrum Power 5 worldmaps	Alt-wa
Close all opened native WinCC pictures or Spectrum Power 5 worldmaps	Alt-wl

Accessing Help Information



[image006-150313-01.tif, 1, en_US]

Figure 1-19 Dependent Help Menu

To View	Press
Native WinCC Graphics Designer Help information	Alt-hc or F1
Help information about work pane content (context-sensitive help)	Alt-hw or Shift-F1
Version information about the Graphics Editor	Alt-ha

1.2.11.3 Other Shortcuts

To	Press
Navigate between all opened native WinCC pictures and Spectrum Power 5 worldmaps	Ctrl-F6
Zoom with the mouse wheel	Ctrl-MouseWheel
Set default UI language	Ctrl-Shift-Alt-d
Set UI language to English	Ctrl-Shift-Alt-e
Set UI language to French	Ctrl-Shift-Alt-f
Cancel the operation	Esc
Select the object palette	Alt-o
Select the style palette	Alt-s
Select the zoom bar	Alt-z

Accessing Additional Editing Functions

To	Press
Move the selected object by 1 pixel	Shift-Arrow Key
Move the selected object by vertical grid points	Shift-PageUp or PageDown
Move the selected object without considering snap to grid	ObjectMove-Shift
Select or unselect within a multi selection	Shift-ObjectSelect
Duplicate objects	Ctrl-ObjectMoves
Create objects with same width and height (except line or polyline, or polygon)	Shift-ObjectCreation
Draw straight lines (line or polyline)	Shift-ObjectCreation or resizing, or pulling
Shift the reference point of a single object	Ctrl-Shift-LButtonDown
Shift the master reference point within a multi selection	Ctrl-Shift-LButtonDown
Highlight the anchor points of non-scalable static text	Ctrl-Spacebar
Shift the anchor points of non-scalable static text	Ctrl-Spacebar-LButtonDown
Shift connection points of selected objects	Ctrl-Shift-ConnectionPoint- pulling

To	Press
Create new corner point for single graphical object	Double-Click on existing corner point
Delete corner point for single graphical object	Ctrl-Double-Click on corner point to be deleted

2 Graphics Editor Basics

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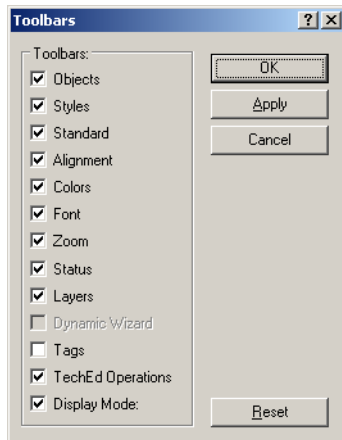
2.1 Operational Steps

2.1.1 Showing or Hiding the Toolbars and Palettes

The goal of this task is to change the appearance of the toolbars and palettes.

- Select **View > Toolbars...** in the menu bar.

The **Toolbars** dialog opens.



[rgimm002-140711-01.tif, 1, en_US]

Figure 2-1 Toolbars Dialog

If the check box is enabled, the corresponding toolbar or palette is displayed. If the check box is disabled, it is not displayed.

The **Apply** button shows the new settings and the **Redo** button restores the default settings.

- Click **OK** to save the new settings.

The dialog closes.



NOTE

After processing an OLE object, it sometimes occurs that the toolbars of the **Graphics Editor** are no longer displayed. You prevent this error after processing an OLE object by closing the server application first before exiting the **Graphics Editor**. For example, **MS Excel** or **MS Paint**.

To restore the display of toolbars, proceed as follows:

- Select all the objects available in the active worldmap.
- Open the **View** menu and select the **Toolbars...** option.
- Click **Redo** button in the **Toolbars** dialog.

Alternative Operation

- Open the **Toolbars** dialog with right-click on any free toolbar or palette space.

2.1.2 Arranging the Toolbars and Palettes

The goal of this task is to arrange the toolbars and palettes.

The toolbars and the palettes are originally arranged at the border of the screen. Move them to another position or drag them as windows on the desktop.

The changed view is saved when you exit from the **Graphics Editor**.

Special Features

- If a palette is located as a window on the desktop, its size is changeable.
- If a palette is moved from the workspace back to the border of the screen, its size is not adapted to the free space. Hence, set the palette size to the size of the free space before the move.



NOTE

If the individual palettes are enlarged and then moved to the border of the screen, the desktop is too small to drag the palette back onto it to reduce its size. Double-click the palette title, to detach the palette from the border of the screen. It is displayed as a window and you can reduce its size.

Moving a Palette as a Window on the Desktop

- Click the border or on the handle of the palette, with the left mouse button pressed.
- Drag the palette on the desktop.
The size of multiple palettes are changeable.

Moving a Palette to the Screen Border

- Click the title bar of the palette, with the left mouse button pressed.
- Drag the palette to the border of the screen.

Restoring the Original View

- Open the **View** menu and select the **Toolbars...** option.
- Click the **Redo** button.



NOTE

If you want to change the size of the palette, convert it into a window by double-clicking the palette title. Change the size, and move it back to the border by double-clicking the palette title again.

2.1.3 Showing or Hiding the Library

The goal of this task is to change the appearance of the library.

- Select **View > Library** in the menu bar or click .

The library is shown if not there or is hidden if already shown.

Changing the Appearance of the Library

The shown library changes its permanent appearance with the pin button. By activating the **Pin** button, the **Library** window is always shown. By deactivating the **Pin** button, the **Library** window is minimized to a library sidebar as soon as the mouse pointer leaves the library. The **Library** window is shown again as soon as the mouse pointer hovers the library sidebar.

- On the title bar of the **Library** window, click  to enable minimizing to a sidebar.
- On the title bar of the **Library** window, click  to disable minimizing to a sidebar.

2.1.4 Arranging the Library

The goal of this task is to arrange the library.

The library is originally arranged at the border of the screen. Move it to another position or drag it as a window on the desktop.

The changed view is saved when you exit from the **Graphics Editor**.

Special Features

- If the library is located as a window on the desktop or arranged at the border of the screen, its width is changeable.
- If the library is located as a window on the desktop, its height is changeable. Arranged at the border, the library always uses maximum height.
- The height of the library structure respective of the height of the library details is changed by moving the splitter between both.
- If the library is moved from the workspace back to the border of the screen, its height is adapted to the free space.

Moving the Library as a Window on the Desktop

- Click the title bar of the library, with the left mouse button pressed.
- Drag the library on the desktop.

Moving the Library to the Screen Border

- Click the title bar of the library, with the left mouse button pressed.
- Drag the library to the border of the screen.



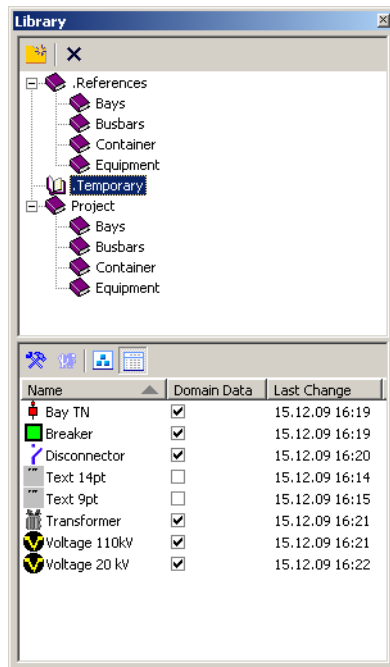
NOTE

Double-click the **Library** title bar also converts a library at the border of the screen to a window on the desktop and either ways.

2.1.5 Managing the Library

The goal of this task is to structure the library and the library objects.

The library is split between a library structure with folders. It contains library objects and a details area showing details for the library objects within a selected folder.



[immrg016-080210-01.tif, 1, en_US]

Figure 2-2 Layout of the Library

Adding Folders

- Select a folder in the library structure under which a new folder is to create.
- Select  or select **New Folder** from the context menu.
- Change the name as required and click **<RETURN>**.

Renaming Folders

- Select a folder in the library structure which is to rename.
- Change the name as required and click **<RETURN>**.

Deleting Folders

- Select a folder in the library structure which is to delete.
- Select  or select **Delete** from the context menu.
- Confirm the security inquiry with **OK**.
The selected folder with all associated library objects gets deleted.

Change Presentation of Library Objects

- Select  in the toolbar of the details area.
The library objects are shown as large symbols with the library objects name.
- Select  in the toolbar of the details area.
The library objects are shown with small symbols. Beside the symbols the name of the library object, if domain data is included and the date of the last modification is displayed in a tabular form. For example, a hook is displayed if domain data is attached, no hook means graphical library object only. Sorting is possible by clicking the desired headers of the 3 columns.

Renaming Library Objects

- Select a library object in the library details area which is to rename.
- Change the name as needed and click **<RETURN>**.

Changing Library Object Icons

- Select a library object in the library details area which gets a new icon.
- Select **Edit Icon...** from the context menu to open the **OPEN** menu to browse for a new icon.
- Select the desired icon and confirm with **Open**.
- To remove a user-defined icon select **Remove Icon** from the context menu.
The icon is replaced with the default icon.

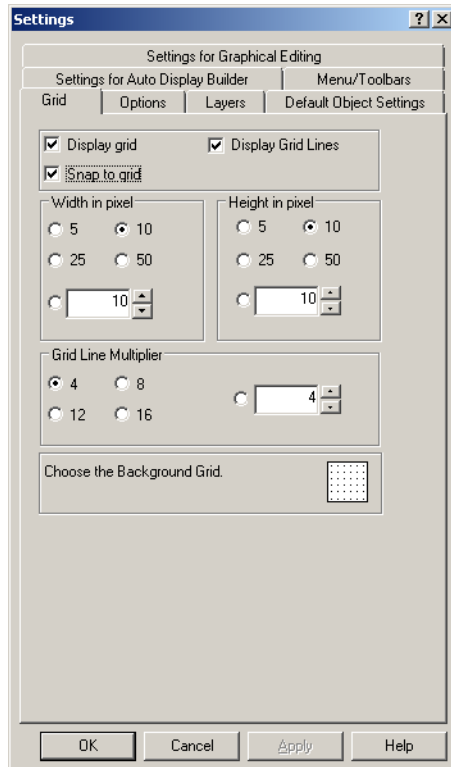
Deleting Library Objects

- Select a library object in the library details area which is to delete.
- Select  from the library structure toolbar or select **Delete** from the context menu.
- Confirm the security inquiry with **OK**.
The selected library object gets deleted.

2.1.6 Defining Basic Settings of the Graphics Editor

The goal of this task is to define various basic settings options for the **Graphics Editor**.

- Open an existing worldmap or create a worldmap.
- Select the **Tools** menu item in the menu bar.
- Click **Settings ...**
The window with the tabs for the various setting options opens.



[rgimm003-140711-01.tif, 1, en_US]

Figure 2-3 Settings Window

- Click the **Grid** tab.
- Select the desired options for the background grid of the worldmap.
- Click the **Options** tab.
- Select the desired options for various program settings.
- Click the **Layers** tab.
- Select the desired options for the visibility of the layers of the worldmap or to rename layers. Select the desired behavior for the layer decluttering of the worldmap.
- Click the **Default Object Settings** tab.
- Select the desired program settings.
- Click the **Settings for Graphical Editing** tab.
- Select the desired options for graphical and technological editing settings.
- Click the **Settings for Auto Display Builder** tab.
- Select the desired settings for substation automatic display building.
- Click the **Menu/Toolbars** tab.
- Select the desired options for customizing menu and toolbars.

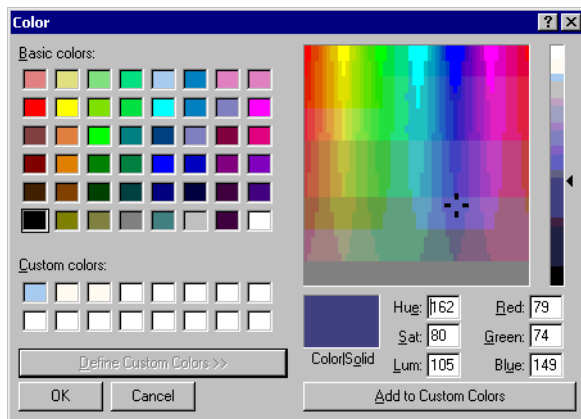
Alternative Operation

- Open the window using the entries **Grid...** or **Layers...** in the **View** menu.

2.1.7 Creating Custom Colors

The goal of this task is to create custom colors using the **Color Palette**.

- Click  in the **Color Palette**.
The **Color** dialog opens.
- Open the color matrix with the **Define Custom Colors >>** button.



[immg020-091208-01.tif, 1, en_US]

Figure 2-4 Color Palette

A color is defined by **RGB (Red Green Blue)** values, which have a range from 0 (dark) to 255 (full intensity). For all worldmaps, a set of predefined colors are available. Precisely define the desired color. The currently selected color is displayed in the preview window **Color/Solid**.

- Click **Add to Custom Colors** to include the color in the user-defined palette.
The new color is entered in the next free box.
User-defined colors can also be replaced with others.
- Click the user-defined custom color before you define and add a new color.
- Click **OK** to confirm.
The new user-defined colors are saved and are available in all worldmaps of the current project.

2.1.8 Customizing the Standard Toolbar

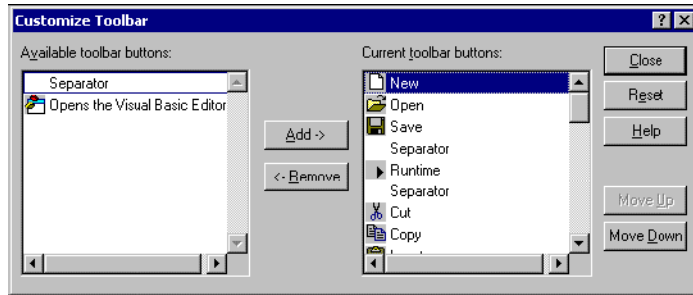
The goal of this task is to change the standard toolbar.

To enable configuration of the standard toolbar:

- Open **Tools > Settings...** in the menu bar.
- Select the **Menu/Toolbars** tab.
- Enable the check boxes of the **Standard Toolbar** area (**Configurable**, **Drag and Drop** with active **ALT** key).

Changing the Standard Toolbar with the Customize Toolbar Dialog

- Keep **ALT** key pressed and double-click the standard toolbar.
The **Customize Toolbar** window opens.
- Add and remove buttons and change the order of the buttons on the toolbar.
- Click **Reset** to restore the original toolbar.



[immg001-301208-01.tif, 1, en_US]

Figure 2-5 Customize Toolbar Dialog

Changing the Standard Toolbar with Drag and Drop

- Hold **ALT** key pressed and drag an icon to the desired position with the mouse.
- Release the mouse button outside the toolbar to remove a button from the toolbar.

Restoring the Standard Toolbar

- Click **View > Toolbars...** in the menu bar.
- Click **Redo** in the **Toolbars** window.



NOTE

- If you have inadvertently removed buttons, simply restore the **Standard** toolbar with the previous operation.
- The standard toolbar is set to configurable for these actions.

2.1.9 Configuring the Control Selection of the Object Palette

The **Controls** tab in the object palette of the **Graphics Editor** contains a selection of controls. These controls are inserted directly into a worldmap.

The selection is changeable as required. Individual controls can be removed from the selection. However, the list can also be supplemented by any ActiveX controls that are registered in the operating system.



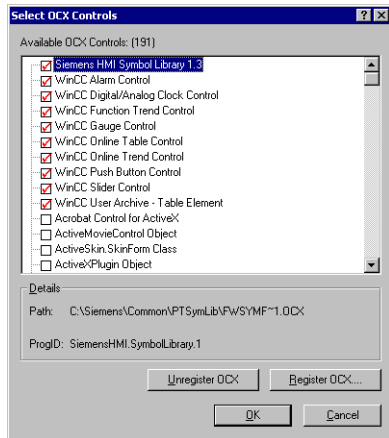
NOTE

The use of ActiveX controls from third-party providers can lead to errors as well as drops in performance or system blocks. The user of the software is responsible for problems caused by the deployment of external ActiveX controls. It is therefore recommended to perform a precise check before deployment.

The goal of this task is to change the available OCX controls in the object palette.

- Right-click the **Controls** tab dialog within the Object Palette.
- In the context menu, select the entry **Add/Remove**.

The **Select OCX Controls** dialog opens.



[immrg120-150109-01.tif, 1, en_US]

Figure 2-6 Select OCX Controls Dialog

The **Select OCX Controls** dialog shows in the area, **Available OCXs (number)** all ActiveX controls that are registered in the operating system. The exact number is the title of the area after the registration is read.

Adding ActiveX Controls to the Object Palette

- Click in the area **Available OCXs** on the square box in front of the name of the desired control.
A red check mark indicates that the control is available in the object palette as soon as you confirm the change with **OK**. You then find the added control in the **Controls** tab of the object palette. The path and the program ID of the selected ActiveX controls is displayed in the area **Details**.

Removing ActiveX Controls from the Object Palette

- In the area **Available OCXs**, click a square box in front of the name of the desired control marked with a red check mark.
The red check mark disappears. The corresponding control is no longer available in the **Controls** tab of the object palette as soon as you confirm the change with **OK**.

Changing the Registration of a Control

Use the buttons **Unregister OCX** or **Register OCX...** to change the registration of a control. For additional information, refer to Changing the Registration of a Control.

2.1.10 Changing the Registration of a Control

ActiveX controls must be configured in the operating system, to be available for applications. Normally the registration is a component of the installation procedure. For troubleshooting, or for upgrading of individual controls a registration or unregistration can be done manually.

In the Graphics Editor the registration is performed in the following dialogs:

- Select OCX Controls** dialog
Use the button **Unregister OCX** and **Register OCX ...**
- Insert a Control** dialog
Use the buttons **Register...** and **Unregister**.

Independent from the different names of the buttons, the procedure for the change of the registration is the same in both dialogs.



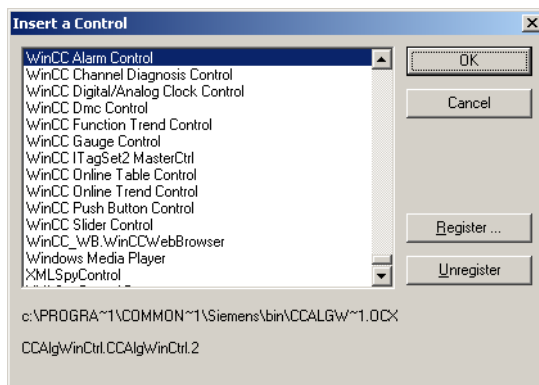
NOTE

Use these buttons to change the settings of the operating system. Changes are executed immediately. Before the unregistration of an ActiveX control, check in which path the related **OCX** file is saved on the hard disk.

The goal of this task is to change the registration of an ActiveX control. The workflow is described here using the **Insert a Control** dialog.

Deregister ActiveX Control

- Insert a control as a smart object **Control** from the Object Palette.
The **Insert a Control** dialog opens.
- Select the desired control in the selection list.



[immg135-140409-01.tif, 1, en_US]

Figure 2-7 Insert a Control Dialog

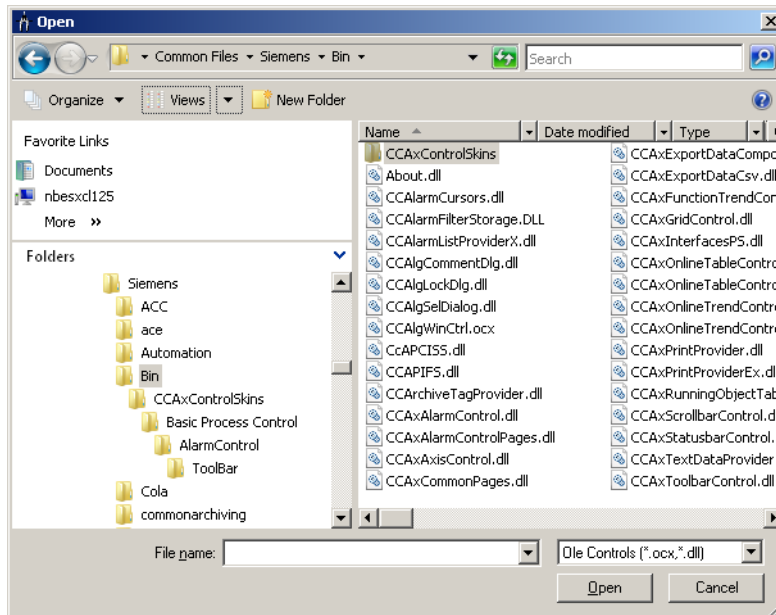
- Check the details under the selection list (path and name).
- Click **Unregister**.
The selected control is de-registered and deleted from the selection list. The related OCX file is maintained in the specified path on the hard disk.

Register ActiveX Control

- Insert a control as a smart object **Control** from the Object Palette.
The **Insert a Control** dialog opens.

- Click **Register...**.

Use the **Open** dialog to open a browser window for the selection of the desired OCX file.



[immrg010-140711-01.tif, 1, en_US]

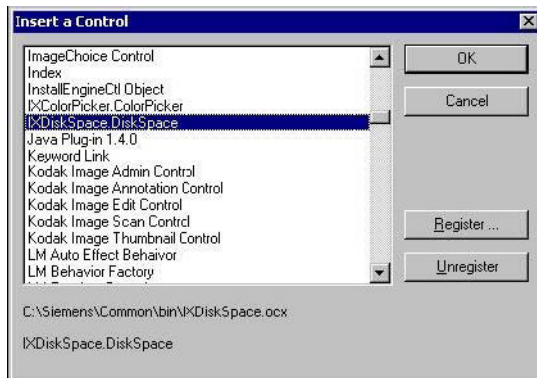
Figure 2-8 Open Dialog

- Select the desired **OCX** file.

During the installation, WinCC controls are copied into the folder ...\\Siemens\\WinCC\\BIN. Spectrum Power 5 controls are copied into the folder ...\\Program Files\\Common Files\\Siemens\\BIN.

- Click **OPEN**.

The selected control is registered in the operating system and displayed in the selection list of the **Insert a Control** dialog.



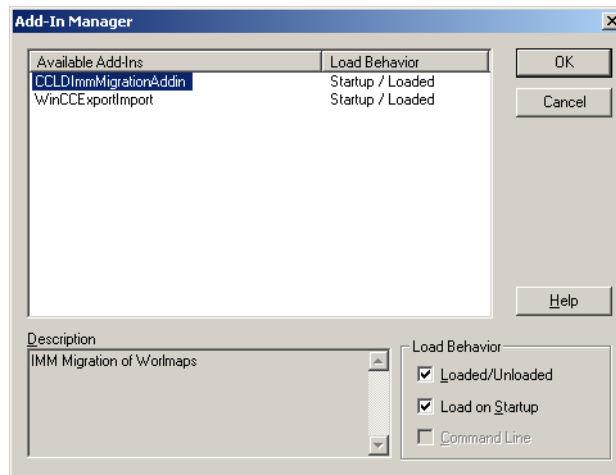
[immrg011-031208-01.tif, 1, en_US]

Figure 2-9 Insert a Control Dialog

2.1.11 Managing Graphics Editor Add-Ins

The goal of this task is to manage available add-ins for the **Graphics Editor**.

- Select **Tools > Add-In Manager** from the **Graphics Editor** toolbar. The **Add-In Manager** window opens.



[immrg001-030809-01.tif, 1, en_US]

Figure 2-10 Add-In Manager Window

- Select the **Add-In** you want to change.
- Select the desired add-in behavior.
- Click **OK**.

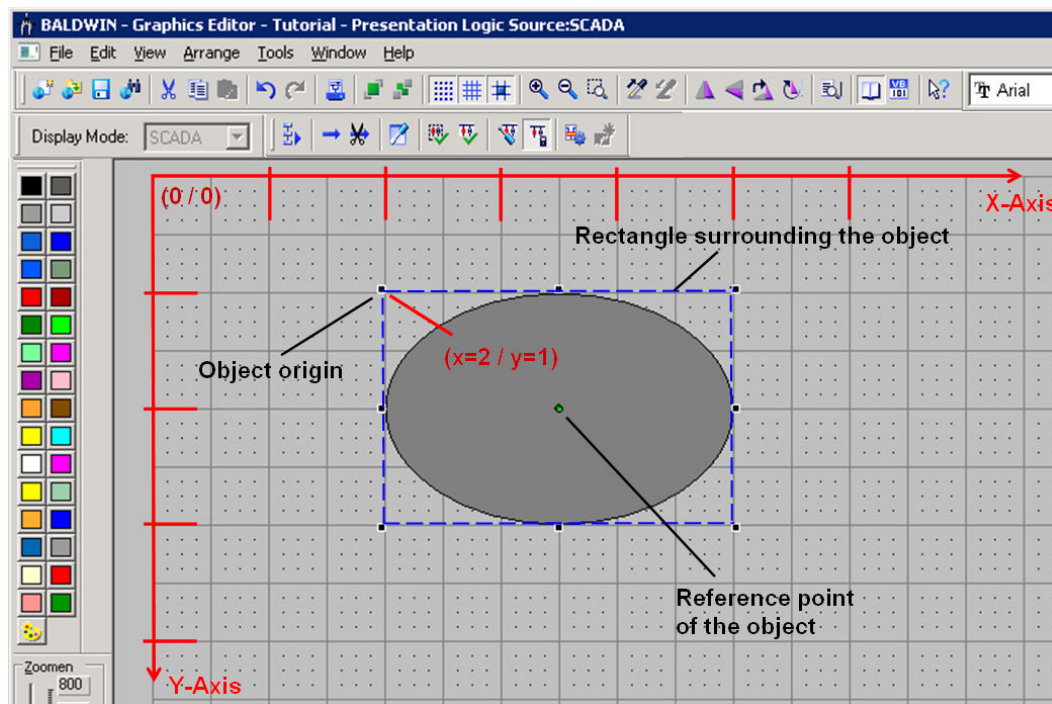
2.2 User Interface Description

2.2.1 Coordinate System of a Worldmap

In the **Graphics Editor**, the basis for setting positions and specifying sizes is a 2-dimensional coordinate system. The 2 axes of the coordinate system, x-axis and y-axis, are perpendicular to one another and intersect at the coordinate origin.

The coordinate origin lies with the coordinates ($X = 0$ and $Y = 0$) in the top left corner of the desktop. Starting from the coordinate origin, the horizontal x-axis runs in positive direction towards the right-hand border of the desktop. The vertical y-axis points in a positive direction towards the lower border of the desktop. The X and Y values of the current position of the mouse pointer are displayed in the status bar of the **Graphics Editor**. The values for X and Y rise when the mouse pointer is moved from the top left to the bottom right on the desktop. The coordinates are shown in pixels.

The coordinates determine the position and size of objects within a worldmap. For example, the attributes **Position X** and **Position Y** sets the position of the object origin. The object origin therefore has the coordinates $X = \text{Position X}$ or $Y = \text{Position Y}$. The values of these attributes describe the distance of the object origin from the coordinate axes.



[imgge007-150313-01.tif, 1, en_US]

Figure 2-11 Coordinate System of a Worldmap

Definition of Origin

The origin is defined as the point of an area or an object that is used as the root for entering position and size. For the creation of worldmaps in the **Graphics Editor**, the following root points are significant:

- **Coordinate Origin**
The coordinate origin is defined by the coordinates: $X = 0$ and $Y = 0$.
- **Worldmap Origin**
The worldmap origin is the top left corner of a worldmap.
Coordinates: $X = \text{Worldmap origin X}$ and $Y = \text{Worldmap origin Y}$.
The Worldmap origin is defined in the **Picture Setting** dialog.

- **Object Origin**
The object origin is the top left corner of the rectangle surrounding the object.
Coordinates: X = Position X and Y = Position Y.

2.2.2 Coordinate System of an Object

The validity of the normal coordinate system is suspended for displays within an object.
The following special forms are used for the internal coordinate system of an object:

- 2D coordinate system of an object.
- 3D coordinate system of an object.

2D Coordinate System of an Object

The 2-dimensional coordinate system of an object differs from that of a worldmap in the direction of the y-axis:

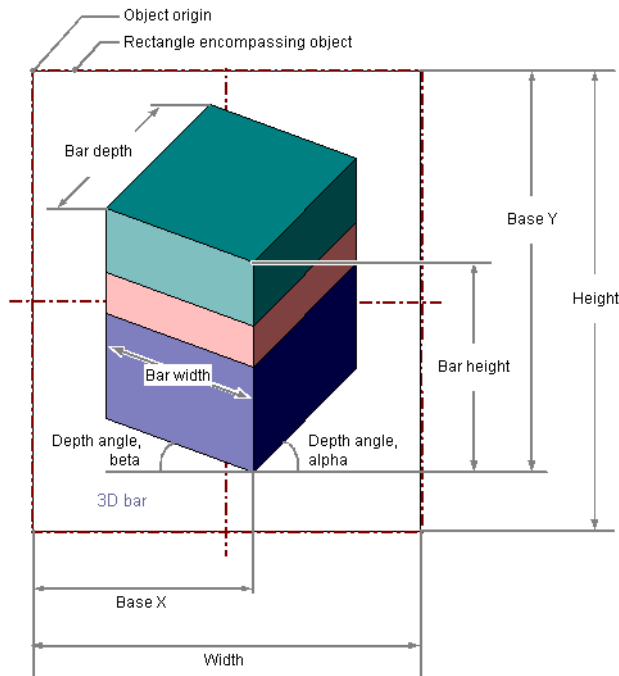
- The positive direction of the y-axis points upward.
- The positive direction of the x-axis points to the right.

3D Coordinate System of an Object

For display in the object **3D bar**, a 3-dimensional coordinate system is used. In addition to the horizontal x-axis and the vertical y-axis, this 3-dimensional coordinate system has a z-axis that points into the display level.

The coordinate system for the spatial representation of the 3D bar is customized with the following attributes:

- **Display axis**
Determines the coordinate axis, where the value scale displayed by the 3D bar runs.
- **Bar direction**
Determines the direction for the **Display axis**, to which the positive axis section points to.
- **Alpha**
Determines the angle for the z-axis pointing into the display level.
- **Beta**
Determines the angle for the x-axis pointing into the display level.

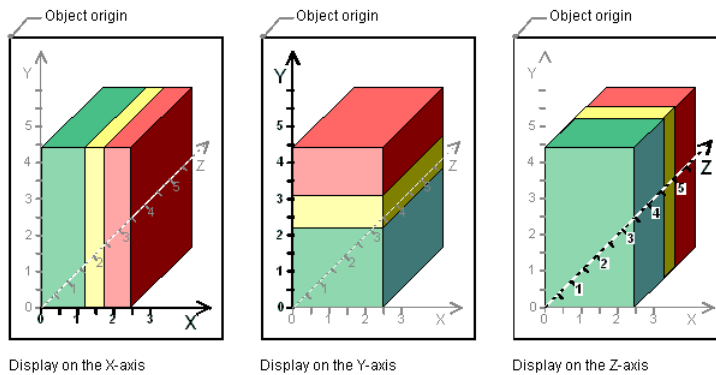


[immrg019-031208-01.tif, 1, en_US]

Figure 2-12 Coordinate system of an Object

Display axis of the 3D bar

Example: "Bar Direction" = Positive, "Alpha" = 45 Degrees, "Beta" = 0 Degrees

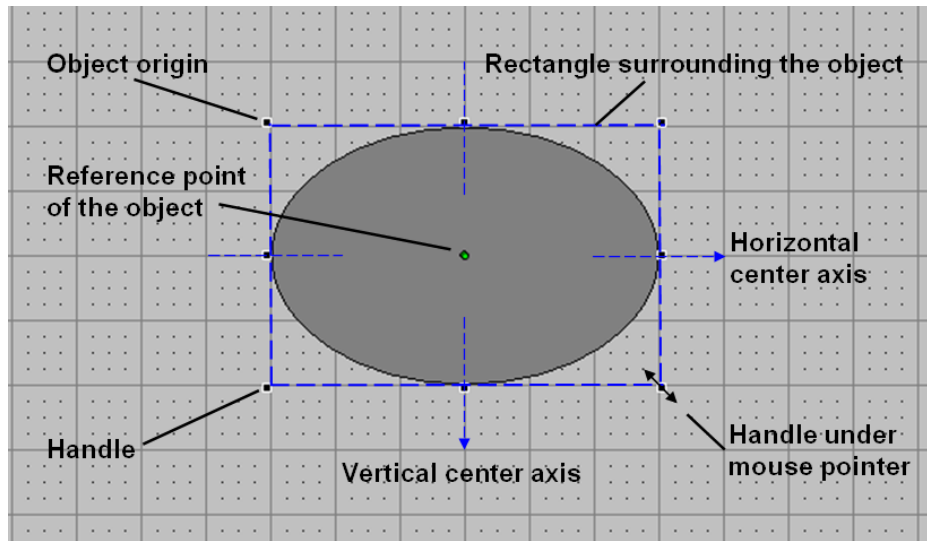


[immrg037-080210-01.tif, 1, en_US]

Figure 2-13 Presentation of 3 Adjustable Display Axes

2.2.3 Rectangle Surrounding the Object

The rectangle surrounding an object is defined as a rectangular frame, which lies on the outer limit of an object. The rectangle surrounding the object is not visible in the **Graphics Editor**. On selection of an object, handles appear. These handles serve as grab points for changing the object size with the mouse. The handles mark the center axes and corner points of the rectangle surrounding the object.



[immrg019-150313-01.tif, 1, en_US]

Figure 2-14 Rectangle Surrounding the Object

The special significance of the rectangle surrounding the object for determining the position is illustrated with an ellipse-shaped object as an example: The position of an object is defined with the position of its origin relative to the coordinate system. The object origin has the coordinates (**Position X** or **Position Y**) and is defined as the upper left corner of the rectangle surrounding the object. The position of the mentioned ellipse-shaped object is set clearly.

In addition, the handles serve as grab points for free rotation with the mouse if the **Free Rotation** mode is enabled. The rotation handles mark the corner points of the rectangle surrounding the object. The rotation of the object happens based on the reference point of the object.

The Handles of the Rectangle Surrounding the Object

The handles of the rectangle surrounding the object appear as soon as an object is selected. By dragging with the mouse, the size of an object can be modified by shifting a handle to a new position. If the mouse pointer is positioned on a handle, it turns into a double arrow.

The alignment of the double arrow indicates the directions in which the handle is moved:

- **Horizontal double arrow**
The width of the object is changed using the handles on the horizontal center axis.
- **Vertical double arrow**
The handles on the vertical center axis are used to change the height of the object.
- **Diagonal double arrow**
The handles on the corners of the object are used to change the height and width of the object.



NOTE

The size of circular objects is changed completely and they have no handles for the center axes.

The Rotation Handles of the Rectangle Surrounding the Object

If the **Free Rotation** mode is enabled, the rotation handles of the rectangle surrounding the object appear as soon as an object is selected. Rotation handles are only showing for the corner points. By dragging with the mouse, the angle of an object is modified by shifting a handle to a new position. If the mouse pointer is positioned on a handle, it turns into a ring-shaped arrow.



NOTE

Resizing of objects is not possible when **Free rotation** mode is enabled.

2.2.4 Reference Point of an Object

The reference point of an object is in the center of the rectangle surrounding the object by default. The reference point itself is not visible in the **Graphics Editor**. On selection of an object, however, the reference point appears. The reference point serves as an anchor point if the **Snap to Grid** mode is enabled. It defines the rotation point, if the **Free Rotation** mode is enabled.

The reference point is shifted to any position within the rectangle surrounding the object by using **<CTRL><SHIFT>**, **left click** on the desired position with the mouse.

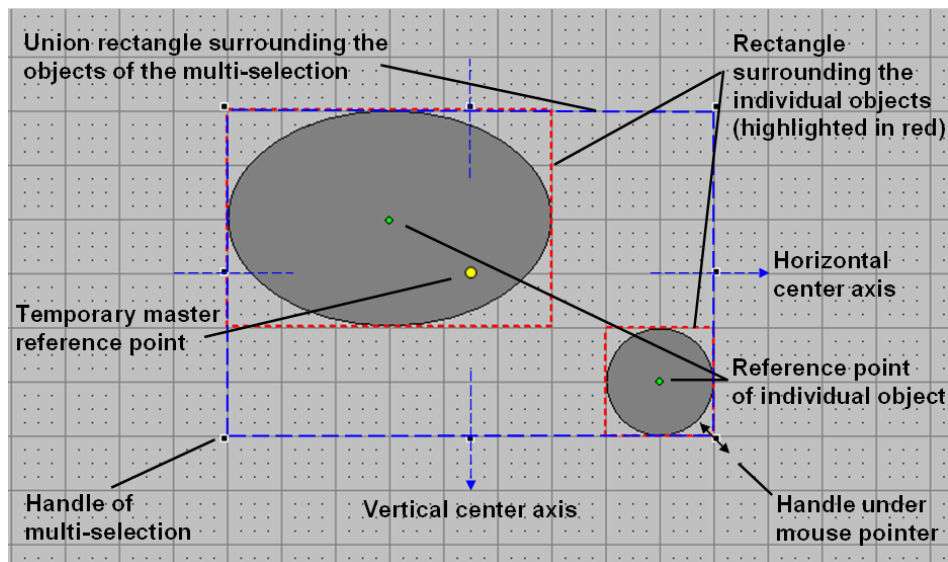
The reference point is also used within alignment functions of the **Graphics Editor**.

2.2.5 Master Reference Point of Objects

The master reference point is in the center of the union rectangle surrounding the objects of a multiple selection by default. It is shown as soon as a multiple selection of objects has been done. If the **Snap to Grid** mode is enabled, the master reference point serves as an anchor point for the multiple selection. If the **Free Rotation** mode is enabled, it defines the rotation point.

The location of the temporary master reference point is shifted to any position within the union rectangle surrounding the objects by using **<CTRL><SHIFT>** **left click** on the desired position with the mouse.

Operations like rotation, mirroring, and alignment use this master reference point.

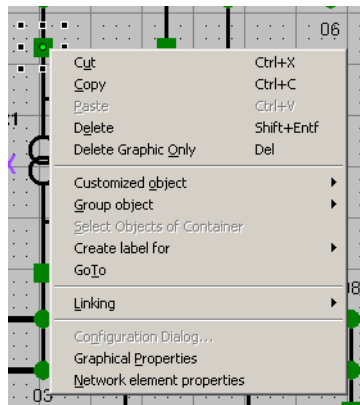


[image008-150313-01.tif, 1, en_US]

Figure 2-15 Union Rectangle Surrounding the Objects of a Multi-Selection

2.2.6 The Context Menu of a Selected Object

The context menu of a selected object gives access to operations regarding this object.



[rgimm004-140711-01.tif, 1, en_US]

Figure 2-16 Context Menu

Cut

Cuts selected graphical objects together with linked IMM domain data and copies them onto the clipboard.

Copy

Copies selected graphical objects together with linked IMM domain data onto the clipboard.

Paste

Pastes copied or cut objects under the current mouse position. Linked IMM domain data is also included in the operation.

Delete

Deletes graphical objects together with linked IMM domain data.

Delete Graphic Only

Deletes only the graphical objects and keeps linked IMM domain data as they are.

Customized Object

For creating, deleting or editing a customized object.

For additional information regarding customized objects, refer to Working with Group or Customized Objects.

Group Object

For grouping objects or ungrouping a group.

For additional information regarding group objects, refer to Working with Group or Customized Objects.

Select Objects of Container

If the selected object is linked to an IMM domain data container, all graphical objects linked to IMM domain data instances within the selected IMM domain data container get selected.

For additional information, refer to Technological Editing.

Create Label for

If the selected object is linked to IMM domain data, a text object is added containing the IMM domain data name.

For additional information, refer to Technological Editing.

GoTo

If the selected object is linked to IMM domain data and is represented also in an additional worldmap, the additional worldmap can be opened.

Linking

Linking Tag Connections and Texts.

For additional information regarding linking objects, refer to Working with Objects.

Configuration Dialog

Some native WinCC objects provide a configuration dialog for specific object settings. Opens the specific configuration dialogs.

Graphical Properties

Opens the **Graphical Properties** window for the selected objects.

Network Element Properties

Opens the **IMM Single Instance Editor** window for the selected object. Is done if the graphical object is linked to IMM domain data.

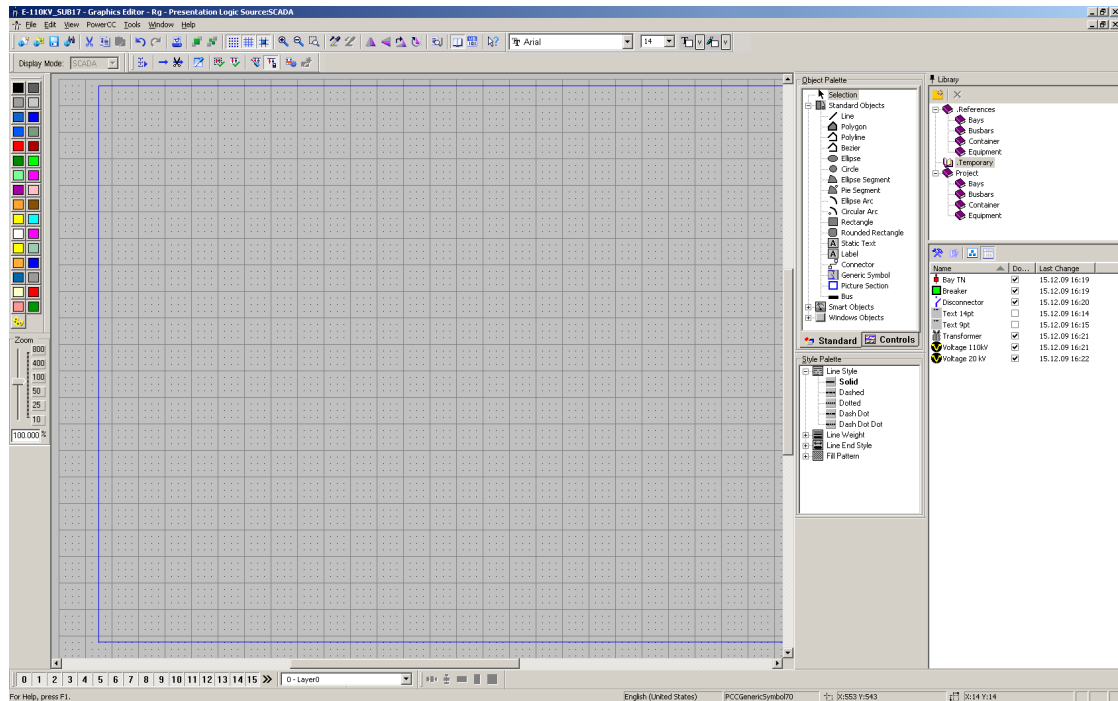
2.2.7 Start Screen of the Graphics Editor

The **Graphics Editor** provides objects and tools for creating worldmaps and allows the user to access all graphical and technological editing functions.

Layout of the Graphics Editor Screen

The **Graphics Editor** is designed according to the windows standard. The **Graphics Editor** has a desktop, toolbar, menu bar, status bar, and various palettes. When **Graphics Editor** is opened, the screen with the default settings appears. As standard, no worldmap is opened automatically.

You can move the palettes and bars as well as the library on the screen and arrange them as you like.



[imgge009-150313-01.tif, 1, en_US]

Figure 2-17 Graphics Editor Framework

The default setting of the **Graphics Editor** is such that all toolbars and palettes including the library are shown, except for the **Tags** palette. To enlarge the desktop, hide toolbars that you do not require.

The Toolbars

- **Objects**
For inserting standard objects, smart objects, Windows objects, and controls.
- **Styles**
For changing the line style and thickness, and fill patterns.
- **Standard**
For creating and saving documents, printing etc.
- **Alignment**
For aligning multiple objects.
- **Colors**
For fast changes to the color of an object.
- **Font**
For changing the font and font size, font color, and border color.
- **Zoom**
For stepless zooming in and out.
- **Status**
For displaying the language setting and coordinates
- **Layers**
For showing and hiding individual layers.
- **Dynamic Wizard**
For making objects dynamic with pre-prepared C Actions.
The **Dynamic Wizard** is not used for Spectrum Power 5 and therefore disabled.

- **Tags**
For fast assignment of tags to native WinCC objects that is made dynamic.
- **TechEd Operations**
For technological editing, opening the single instance editor, invoking presentation logic repair function, searching the worldmap for graphical objects not linked to IMM domain data, and automatic display building of substation displays.
- **Display Mode**
For selecting the data source for presentation logic configuration.



NOTE

The toolbars and palettes are shown or hidden using **View >Toolbars....** Toolbars are movable to any position within the screen.

The Library

The default position of the library is on the right-hand side of the **Graphics Editor**.

The library consists of:

- A title bar with a pin button
- A toolbar which gives access to structure the library with folders and subdirectories
- The tree view of the library structure
- The toolbar of the details area which gives access to library object details
- A details area, which shows additional information for library objects

2.2.8 Standard Toolbar

The default setting is such that the standard toolbar is located beneath the menu bar of the **Graphics Editor**. The buttons of the standard toolbar include both common Windows commands and special buttons of the **Graphics Editor**. For example, save, copy, and open the navigation window.

The Buttons of the Standard Toolbar

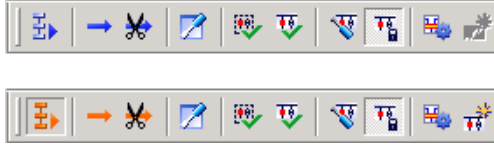
Icon	Function	Key Combination
	Creates and opens a Spectrum Power worldmap.	–
	Opens the Pictures and Sections of Pictures dialog. For example, to open an existing Spectrum Power worldmap.	–
	Saves the active Spectrum Power worldmap. This function is only available if a change to the worldmap has been done.	<CTRL + S>
	Opens the Navigation window. This function is available if a worldmap is opened.	–
	Cuts selected objects together with linked IMM domain data and copies it onto the clipboard. This function is only available if an object is selected.	<CTRL + X>
	Copies selected objects together with linked IMM domain data onto the clipboard. This function is only available if an object is selected.	<CTRL + C>
	Pastes the content of the clipboard into the active Spectrum Power worldmap. Linked IMM domain data is included in the operation. This function is not available if the clipboard is empty.	<CTRL + V>

Icon	Function	Key Combination
	Undoes the last actions (maximum of 30). This function is only available if an action is performed beforehand and not saved. Technological editing function clears the undo buffer.	<CTRL + Z>
	Redoes the last undone action. This function is only available if an action is undone beforehand.	<CTRL + Y>
	Prints the content of the active Spectrum Power worldmap. This function is only available if a picture or worldmap is opened.	<CTRL + P>
	Places the selected object within the layer in the foreground. Objects in the foreground cover objects lying behind them.	–
	Places the selected object within the layer in the background. Objects in the background are covered by objects in the foreground.	–
	Hides and shows the grid in the active window.	–
	Hides and shows grid lines in the active window.	–
	Switches the function Snap to Grid on and off.	–
	Increases the zoom factor in individual steps. The Spectrum Power worldmap is enlarged. This function is only active until the largest zoom factor (800%) is reached.	–
	Reduces the zoom factor in individual steps The Spectrum Power worldmap is displayed as smaller. This function is only active until the smallest zoom factor for the worldmap is reached.	–
	Zooms in on a selected Spectrum Power worldmap area. This area is displayed in the size of the active window.	–
	Copies line and color properties of an object to transfer them to another object.	–
	Assigns properties of an object to another object. This function is only active if properties are copied.	–
	Mirrors a selected object on its vertical center axis defined by its reference point. If multiple objects are selected, the center axis is defined by the master reference point.	–
	Mirrors a selected object on its horizontal center axis defined by its reference point. If multiple objects are selected, the center axis is defined by the master reference point.	–
	Rotates a selected object by 90 degree clockwise around its reference point. If multiple objects are selected the rotation point is defined by the master reference point.	–
	Switches the function Free Rotation mode on and off.	–
	Opens the Graphical Properties window.	–
	Opens the Library window. For example, finished system parts are stored in the library of the Graphics Editor to insert in another worldmap later.	–
	Opens the Visual Basic Editor .	–
	Activates the direct help.	<SHIFT+F1>

2.2.9 Graphical Editing Toolbar

The default setting is such that the graphical and technological editing toolbar is located beneath the standard toolbar of the **Graphics Editor**. The buttons of the **Graphical Editing** toolbar include all necessary tools for graphical and technological editing functions.

Graphical and Technological Editing functions are used within Spectrum Power worldmaps.



[image010-150313-01.tif, 1, en_US]

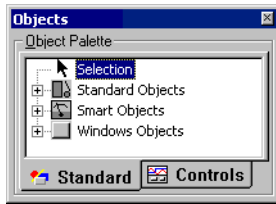
Figure 2-18 Dependent Graphical Editing Toolbar

The Buttons of the Graphical Editing Toolbar

Icon	Function	Key Combination
	Toggles between the modes Technological Editing Off (blue) and Technological Editing On (orange).	–
	Connects the selected objects graphically (blue) or graphically and technologically (orange).	–
	Disconnects the selected object graphically (blue) or graphically and technologically (orange).	–
	Opens the IMM Single Instance Editor for modifying the IMM domain data of the selected instance.	–
	Invokes the presentation logics repair function.	–
	Activates highlighting of selected graphic objects which do not represent any domain object. Highlighting stays active until the button is unchecked again.	–
	Activates highlighting of graphic objects within the complete worldmap which do not represent any domain object. Highlighting stays active until the button is unchecked again.	–
	Display locked objects	–
	Creates the domain data topology for the selected objects.	–
	Creates a substation display based on existing domain data and domain data topology into active worldmap. Some basic automatic layouting is performed.	–

2.2.10 Object Palette

The Object Palette contains different types of objects that are frequently required in worldmaps.



[immg023-031208-01.tif, 1, en_US]

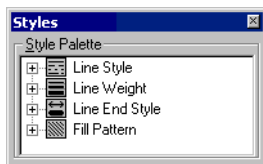
Figure 2-19 Object Palette

The following object groups are available:

- **Standard objects**
For example, line, polygon, ellipse, circle, rectangle, and static text.
- **Smart objects**
For example, application window, picture window, OLE object, I/O field, bars, and status display.
- **Windows objects**
For example, button, check box, option group, and slider object.
- **Controls**
The **Controls** tab contains as standard a selection of the most important ActiveX controls provided by WinCC as well as specific Spectrum Power 5 controls.

2.2.11 Style Palette

The **Style Palette** allows fast changes to line style, line weight, line end style, and fill patterns.



[immg024-031208-01.tif, 1, en_US]

Figure 2-20 Style Palette

The following style groups are available:

- **Line Style**
Contains different line display options such as dashed, dotted etc.
- **Line Weight**
Sets the thickness of the line. The line weight is specified in pixels.
- **Line End Styles**
Enables display of the ends of lines. For example, as arrows or rounded.
- **Fill Pattern**
Provides the possibility to fill closed objects. For example, with a transparent or checkered background.

Displaying the Current Settings

If an object is selected, the style properties of this object are shown in a bold font within the style palette.

2.2.12 Alignment Palette

The functions of the alignment palette are used to process multiple objects together. These functions are called from the **Arrange > Align** menu. The alignment palette contains the following functions for processing the objects in a multiple selection.

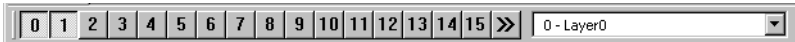


[immrg302-140711-01.tif, 1, en_US]

Buttons	Functions
	Aligning: The selected objects are aligned horizontally or vertically. The alignment axis is defined by the master reference point of the selection. The reference point of the individual objects are aligned in relation to the master reference point.
	Adapting width or height: The selected objects are adapted to each other in width or height. The upper top object of the selection is used as reference.
	Adapting width and height: The selected objects are adapted to each other in height and width. The upper top object of the selection is used as reference.

2.2.13 Layer Palette

To simplify the processing of individual objects in complex worldmaps, the **Graphics Editor** allows you to work with layers. For example, the contents of a worldmap is distributed across up to 32 layers. These layers can be shown or hidden individually. In the default setting, all layers are visible and the active layer is layer 0.



[immrg002-240409-01.tif, 1, en_US]

Figure 2-21 Layer Palette

For example: Layers 0 and 1 are displayed, all other layers are invisible. The active layer is layer 0.

Making Layers Visible or Invisible

Pressed buttons indicate the visible layers. Click a button to make the layer visible or invisible.

Active Layer

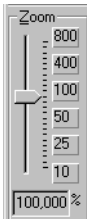
The active layer is set in the list box of the layer palette.

Renaming Layers

The **Settings** menu is used to rename layers. Click on the **Layers** tabbed card.

2.2.14 Zoom Palette

Alongside the buttons for zooming in the standard toolbar, the **Graphics Editor** provides a separate zoom bar. It enables convenient zooming in a worldmap.



[immrg028-031208-01.tif, 1, en_US]

Figure 2-22 Zoom Bar

On the left side of the zoom bar is a slider for finely graded setting of the zoom factor. To the right are buttons with predefined zoom factors. The currently set zoom factor is displayed below the slider in %. Zooming out is restricted to the minimum zoom level during that the whole worldmap fits in the viewport of the Graphics Editor.



NOTE

The minimum and maximum zoom factors are defined individually for all layers. For example, set the detail displays on a separate layer and show it only within a certain zoom factor. To move the visible area in the case of a high zoom factor, use the scroll bars in accordance with the Windows standard.

The **Graphics Editor** provides additional operation options with the scroll wheel:

- **Click** the scroll wheel.
The mouse pointer changes and you can move the visible worldmap area simply by moving the mouse.

2.2.15 Color Palette

Depending on the selected object, the color palette enables fast changes to the line or fill color. It provides 32 standard colors and the bottom button provides the option of selecting a base color or a user-defined color.



[immrg029-140711-01.tif, 1, en_US]

Figure 2-23 Color Palette

A mouse click on a color of the palette has the following effect, for example:

- For closed objects the fill color is changed. For example, rectangle.
- For open objects the line color is changed. For example, polyline.
- For text objects, the background color is changed.

If no object is selected, the selected color is assigned to the background of the active worldmap.

The dialog **Colors** opens with the button . It provides 48 basic colors as well as the possibility to create user-defined custom colors.

2.2.16 Font Palette

The most important text characteristics are changed using the font palette. The font palette allows changes to the font, font size, font color, and line color.



[immrg030-140711-01.tif, 1, en_US]

Figure 2-24 Font Palette

Other text characteristics such as text in italics or bold print are set in the **Graphical Properties** window.



NOTE

The default settings of text characteristics are changed for any text object. Change the default settings using the graphical properties.

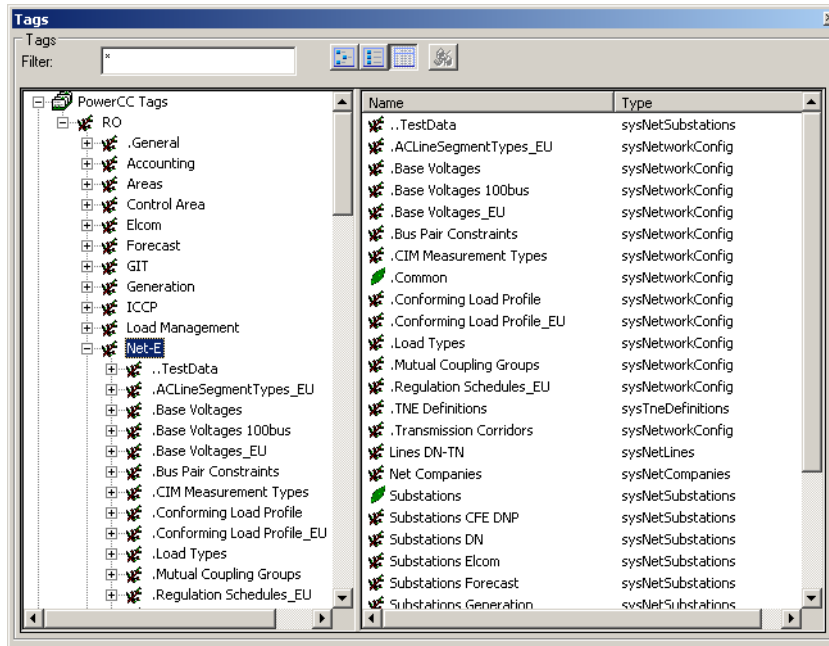
You can also change the default setting using the font palette:

- Select a text object in the object palette.
- Change the properties in the text palette.
- Insert the text object.

The changed text characteristics apply to all other objects of the same type.

2.2.17 Tag Palette

The tag palette enables fast linking of WinCC process tags to objects that are not Spectrum Power 5 specific. The default setting is that the tag palette is hidden.



[immrg023-140711-01.tif, 1, en_US]

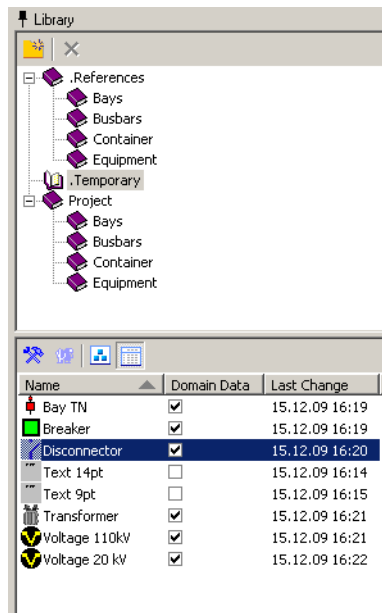
Figure 2-25 Tag Palette

The tag palette has the same layout as an explorer window. The tag palette contains a list of all the tags available in the project as well as a list of the internal tags. Filters are set, the view is changed, and the connections are updated. With the mouse button pressed, the tags in the right-hand window are dragged to attributes of an object in the **Graphical Properties** window.

2.2.18 Library

The library of the **Graphics Editor** is versatile tool for saving and managing graphic objects that are used for creating worldmaps. The library is a container for graphic objects or groups of graphic objects which are reused multiple times within one project or are shared between projects. Objects within the library are created by the user, and imported from an external source or supplied with the **Graphics Editor**. The objects are sorted by subjects within folders and subdirectories. User-defined objects are stored here as copies and made available for multiple use. The names and icons of user-defined objects inserted into the library are freely definable.

The library provides a base set of predefined graphic objects (network components) like transformer, circuit-breaker, etc. You can add objects to this library and the library also allows you to access these graphic objects. You can drag and drop graphic object from the library window onto the drawing section of the **Graphics Editor**. A graphic object in the library also contain IMM domain data associated with.

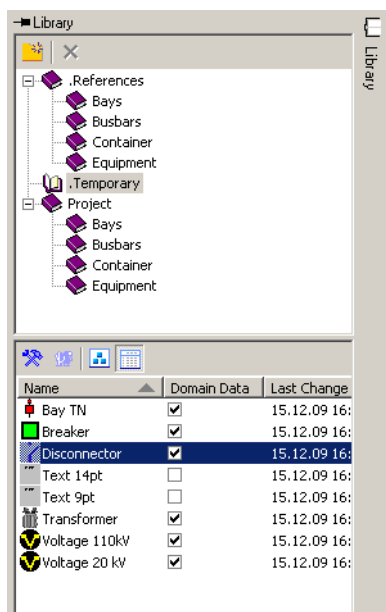


[immrg024-080210-01.tif, 1, en_US]

Figure 2-26 Layout of the Library

The pin button is used to change the appearance of the library.

 	Pin activated or deactivated	Activate: The library window is always shown. Deactivate: The library window is minimized to a library sidebar as soon as the mouse pointer leaves the library. The library window is shown again as soon as the mouse pointer hovers the library sidebar.
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[immg025-080210-01.tif, 1, en_US]

Figure 2-27 Library with Deactivated Pin and Library Sidebar

2.2.19 Library Toolbars

There are 2 library toolbars available. You can make the following settings with the tools in the toolbar of the library structure:

- Folders for sorting library objects by subject are created and deleted in the library structure area.
- Deleting of Library objects.



[immg115-140409-01.tif, 1, en_US]

	New subdirectory	Creates a subdirectory in the selected folder. Folders are renamed or deleted from their context menu.
	Deleting	The Delete command either removes a selected library object from the current library folder or deletes a selected library folder and all its library objects.

You can make the following settings with the tools in the toolbar of the library details:

- The library is extended with user-defined objects.
- The display of the library objects are configurable.



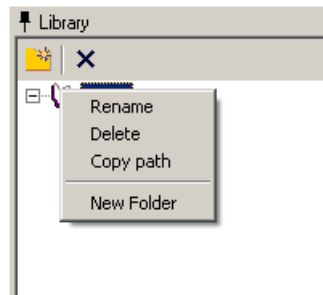
[immg119-140409-01.tif, 1, en_US]

	Add ActiveX Object	The command is used to add an ActiveX object to the selected library folder. The Create new template window opens to specify further settings.
	Add OLE Element	The command is used to add an OLE element to the selected library folder. The Create new template window opens to specify further settings.
	Giant Icons	The Giant Icons command shows the library objects in the current folder as large symbols. No detailed information is shown except for the name of the object and the associated icon for the library objects.

	List	The List command shows the library objects in the current folder as a list with small symbols. Besides the name of the object, the file size and the date of the last modification are displayed as detailed information. For example, to show whether it is only a graphical symbol or whether any domain data is linked with it.
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2.2.20 Library Context Menus

There are 2 library context menus available.

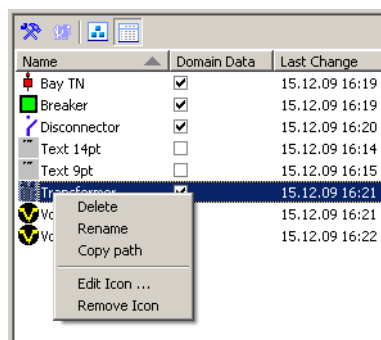


[immg124-140409-01.tif, 1, en_US]

Figure 2-28 Context Menu Within the Library Structure

The operations within the library structure are as follows:

Rename	To rename the selected folder
Delete	To delete the selected folder and all associated library objects
Copy Path	To copy the complete selected folder path within the library structure to the clipboard
New Folder	To create a subdirectory in the selected folder



[immg026-080210-01.tif, 1, en_US]

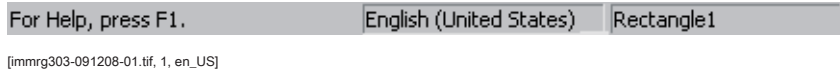
Figure 2-29 Context Menu Within the Library Details

The operations within the library details are as follows:

Delete	To delete the selected library object
Rename	To rename the selected library object
Copy Path	To copy the complete selected library object path within the library structure to the clipboard
Edit Icon	To browse for an existing icon to be displayed for this library object
Remove Icon	To remove the existing displayed icon for this library object

2.2.21 Status Bar

The status bar is laid out in accordance with the Windows standard and it contains useful information.



[immrg303-091208-01.tif, 1, en_US]

- Pressing the **<F1>** key shows tips related to the currently active window.
- The language currently selected in the **Graphics Editor** is displayed.
- The name of the selected object is displayed.
If multiple objects are selected, **Multiple Selection** is displayed.

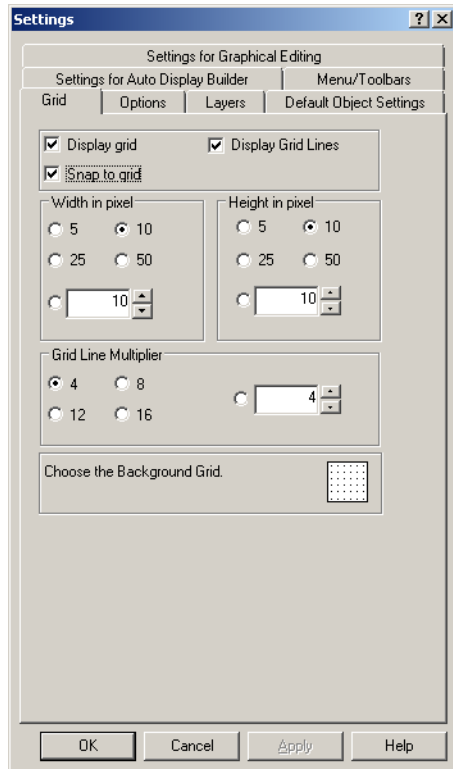


[immrg304-091208-01.tif, 1, en_US]

- Coordinates of the selected object
For example:
X = 0 and Y = 0 means that the object origin of the selected object is located at the top left in the worldmap.
The values are updated automatically during the movement of an object, thus making exact placement easier.
- Size of the selected object
For example:
X = 70 and Y = 40 means that the selected object is 70 pixels wide and 40 pixels high. The values are updated automatically during the size change of an object, thus making exact scaling easier.
- Keyboard settings
 - UF
<CAPS LOCK> is pressed.
The text is entered in capital letters.
 - NUM
<NUM> is pressed.
The numerical keypad is activated.
 - RF
<SCROLL LOCK > is pressed.

2.2.22 Grid Settings

To enable exact work, make a background grid or grid lines appear. If necessary, enable the option **Snap to Grid**. Then, the reference points of all objects that are created or moved are automatically aligned to the grid. Showing the grid and aligning to the grid is set using the standard toolbar. The grid width is set using the basic settings of the **Graphics Editor**. The set grid properties are saved with the worldmap. When a new worldmap is created, the settings of the program are used once again. Showing the grid lines is set using the standard toolbar. The grid line multiplier is set using the basic settings of the **Graphics Editor**.



[rgimm040-140711-01.tif, 1, en_US]

Figure 2-30 Setting Grid Options

Snap to Grid

If the check box is enabled, objects are aligned to the grid points when they are created, inserted, and moved. Objects that are already existent are only aligned to the grid when they are moved. The reference point of the objects is base for snap to grid.

Display Grid

When the check box is enabled, the grid points are shown.

Display Grid Lines

When the check box is enabled, the grid lines are shown.

Width in Pixels

Horizontal spacing of the grid points. Distance between 5 and 100 pixels.

Height in Pixels

Vertical spacing of the grid points. Distance between 5 and 100 pixels.

Grid Lines Multiplier

Spacing of the grid lines in grid points. Distance between 1 and 100 grid points.

Alternative Procedure

All grid and grid line properties are changed in the **Graphical Properties** window of the active worldmap. The color of the grid and grid lines are changed there.

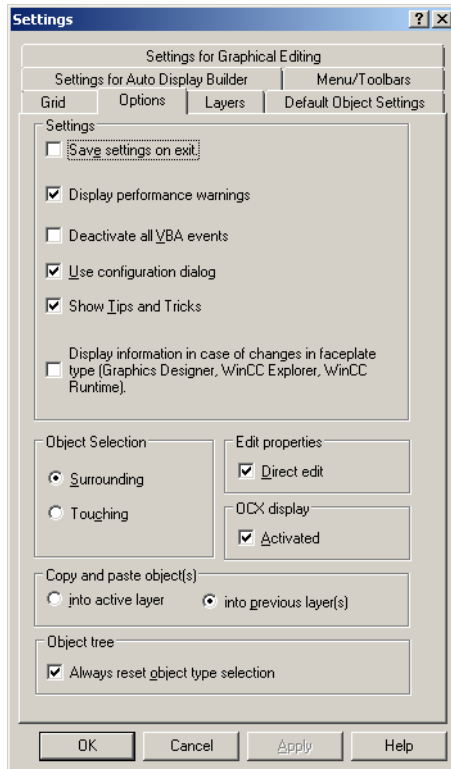


NOTE

The minimum grid spacing that is displayed on the screen is 5 pixels.

2.2.23 Options Settings

Different program settings are changed and saved.



[rgimm004-140711-02.tif, 1, en_US]

Figure 2-31 Setting Options

Save Settings on Exit

Enable the **Save Configuration on Exit** check box to save the settings configuration when you exit from the **Graphics Editor**.

Display Performance Warnings

Cyclic triggers can lead to a high system load. If the check box is enabled, a warning is issued in the event of a system overload as soon as the worldmap is saved. This warning contains the name of the object and the attribute that are made dynamic and has led to the overload. If the check box is disabled, you do not receive a warning, that is, also no indication of the cause of the overload. The default setting is that the check box is enabled.

Deactivate all VBA Events

If the check box is enabled, the configured VBA events to monitor the VBA actions is not triggered. The default setting is that the check box is disabled.

Use Configuration Dialog

If the check box is activated, the **Configuration dialog** window opens automatically if the inserted object has a configuration dialog. Objects with configuration dialog:

- **Smart objects:** Control, I/O Field, Bars, Graphic Object, Status Display, Text List.
- **Windows objects:** Button, Slider Object.

If the check box is not enabled, the objects are inserted with the default settings. Independent of whether the check box is enabled, the configuration dialog is opened at any time using the context menu entry **Configuration Dialog....**

Display Tips and Tricks

If the check box is enabled, every time the **Graphics Editor** is started a tip on creating or optimizing picture or worldmap is displayed. The default setting is that the check box is disabled; no tips and tricks are displayed when the program is started. Not used with Spectrum Power 5 and IMM.

Display information in case of changes in faceplate type

If the check box is enabled, the **Faceplate instances with type change** dialog is shown with all faceplate instances that are to be checked. Not used with Spectrum Power 5 and IMM.

Object Selection

An object is selected by clicking or by drawing a selection frame. This selection frame is also referred to as a lasso.

Specify whether the lasso must fully enclose the objects or whether contact with the lasso is sufficient to select an object. With the option **Surrounding**, the complete object must lie in the selection frame (lasso). With the option **Touching**, it is sufficient that an object touches the selection frame (lasso). The default setting is that the **Surrounding** option is selected.

Edit Properties

Attributes of objects are changed using the Graphical Properties. If a value is entered directly in the **Graphical Properties** window or whether a dialog is opened is specified using the **Direct Edit**. If the check box is enabled, the value of an attribute is changed by double-clicking the attribute in the **Graphical Properties** window. If the check box is not enabled, double-click opens a window for value input. The default setting is that the check box is enabled; digits and text are entered directly in the **Graphical Properties** window.

OCX Display

If the check box is enabled, the representation of ActiveX controls is activated in the **Graphics Editor**. If displaying ActiveX controls is not required, disable the check box. Setting the OCX representation only affects when worldmaps are being created, not in runtime. The default setting is that the representation of ActiveX controls is enabled.

Copying and Pasting Objects

If the option **Into active layer** is selected, copied objects are pasted into the active layer, independent of the layer from which they are copied. If the option **Into previous layers** is selected, copied objects are inserted into the layer from which they were copied. If multiple objects are selected and copied from different layers, each individual object retains the layer to which it is routed.

Always Reset Object Type Selection

If the check box is enabled, the mouse pointer turns back into the selection mouse pointer after an object is inserted. However, if you want to insert multiple objects of the same type, disable the check box. Then, the active object type remains active after an object is inserted, and the same object is inserted multiple times. You then do not have to reselect the object type for each object to be inserted. The default setting is that the check box is enabled.

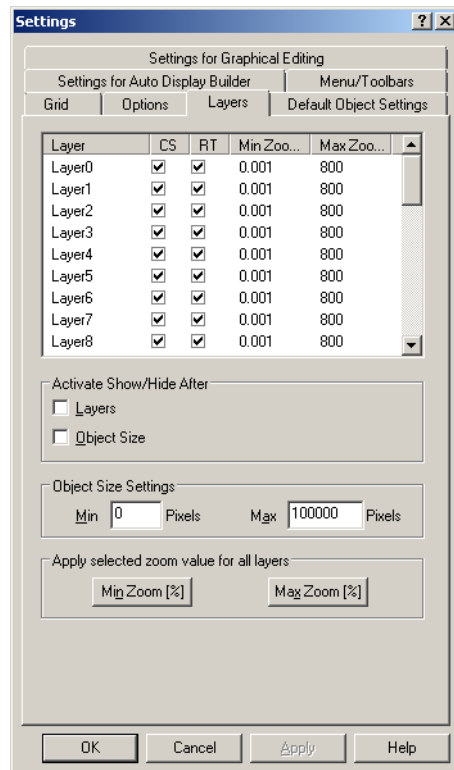
2.2.24 Layers Settings

Make the individual layers of a worldmap visible or invisible. This shows or hides the objects of the various layers. The settings are made independently of one another for Configuration System (CS) and Runtime (RT).

When a worldmap is opened, all layers in the **Graphics Editor** are always visible. The settings for which layers were invisible for the last processing are not saved when you exit from the **Graphics Editor**. The independent visibility of layers in CS and RT has advantages: You can store more information in the worldmap that is visible in runtime.

Whether layers and objects are shown or invisible are made dependent on the current zoom factor. For example, it is possible to show details as of a certain zoom factor and not before. Each layer is assigned its own minimum and maximum zoom. It is also possible to show only objects whose display size lies within a defined range.

Click a layer to rename it.



[rgimm005-140711-01.tif, 1, en_US]

Figure 2-32 Settings - Layers

Layers Window

Make the individual layers of a worldmap visible or invisible. When the check box is enabled, the corresponding layer is visible. The active layer always remains visible in the **Graphics Editor**.

Setting Zoom Ranges for Individual Layers

Click the digits in the columns **Min Zoom** or **Max Zoom** to set the zoom range in which the contents of a layer are displayed. The zoom range is specified in %. Minimum value is 0.001%, maximum value is 800%. Activate the **Activate - Layers** check box.

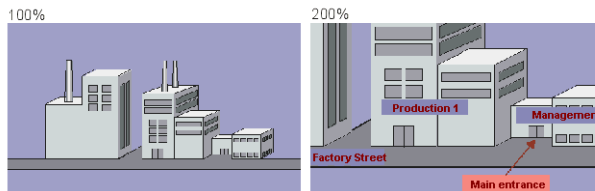


NOTE

If you have set the zoom ranges for the layers and want to zoom into the worldmap when processing, disable the **Activate - Layers** check box. The set zoom ranges for the layers are disabled, but are retained, and all the details of the worldmap are viewed.

Example

The coarse display lies on Layer 8, the detail display of an area lies on Layer 10. Layer 8 is displayed in the zoom range between 80% and 800%. Layer 10 is displayed in the zoom range between 100% and 800%. With this setting, the details of Layer 10 are only shown when the user zooms in to the worldmap by over 100%.



[immrg014-091208-01.tif, 1, en_US]

Setting the Size Range for the Display of Objects

If objects are only displayed in the range between a minimum and maximum display size, define the size range. The display size is independent of the object size: An object is shown within the zoom factors that show the object in the set size. Enable the **Object Size** check box and enter the minimum and maximum size in pixels. Confirm this input with **OK**, only objects in this display size are displayed.



NOTE

If you have set the display size and want to zoom into the worldmap when processing, disable the **Activate - Object Size** check box. The object size settings are disabled, but are retained, and all the details of the worldmap are viewed.

Adopting Selected Zoom Value for All Layers

If you want to change the zoom range for all layers, set it in the upper window for a layer and click **MinZoom** or **MaxZoom** button. The layer from which the zoom ranges are adopted must be selected. Enable the **Activate - Layers** check box to receive a display of the settings.

2.2.25 Default Object Settings

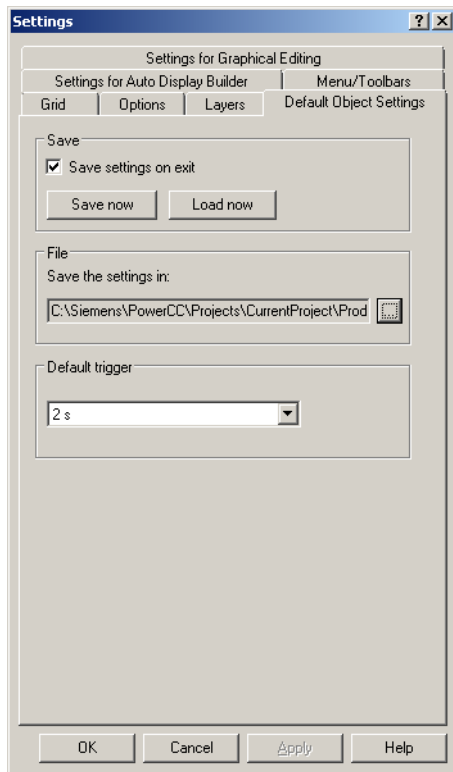
In the **Graphics Editor**, the various object types have default properties. If an object from the object palette is inserted in a worldmap, the object adapts to these default settings.

Working with Default Objects

The default setting for the individual object types are adapted as required. Ideally, the default settings are adapted before creating a worldmap. Hence, the extent of later adaptations is limited. For example, frequently used objects.

All the information on default settings for object types is contained in a file of the format PDD. As standard, the **Default.pdd** file is in the graphic folder **GraCS** of each project. Use the **Default Objects Settings** tab to select the following options for default settings of object types:

- The default settings are saved immediately or also automatically on exiting from the **Graphics Editor**.
- The default settings of another project are loaded into the current project.
- The file name and the directory path for storing the PDD file is changed.



[rgimm006-140711-01.tif, 1, en_US]

Figure 2-33 Settings - Default Object Settings

Saving Default Settings of Objects

- In the field **File**, click .
The **Default Objects** dialog opens.
- Select the PDD file to save the changes and confirm your selection with **Open**.
By default, the PDD file of a project is stored in the graphic folder **GraCS**.
- In the **Save** field, click **Save Now**.
The changed default settings are saved in the selected PDD file.
- Enable the **Save Configuration on Exit** check box to save the default object configuration when you exit from the **Graphics Editor**.

Loading Default Settings of Objects

- In the field **File**, click .
The **Default Objects** dialog opens.
- Select the PDD file, to load, and confirm your selection with **Open**.
By default, the PDD file of a project is stored in the graphic folder **GraCS**.
- Click **Load Now**.
The default objects configuration is imported from the selected PDD file into the current project.

- Now save the loaded default settings in the current project directory: In the field **File**, click .
The **Default Objects** dialog opens.
- Select the PDD file of the current project and confirm the selection with **Open**.
- Click **Save Now** in the **Save** field.
The imported default settings of the objects are saved in the PDD file of the current project.



NOTE

The default settings for objects do not apply across all projects. The changed default settings only apply to the worldmaps of the currently opened WinCC project. The basic settings of the **Graphics Editor** are not changed here.

Changing the Default Trigger

The set default trigger specifies the update cycle in runtime for native WinCC objects. Each attribute of a native WinCC object that is made dynamic is assigned an individual update cycle. The value specified for the default trigger is the default update cycle for all native WinCC objects. For example, to have many objects updated in a cycle of 2 sec, set the default trigger to 2 s. You are then no longer required to change the trigger setting for tag connection of the individual attributes.

Selecting the Default Trigger

- Open the list box and select a value **Picture Cycle**.
A cyclic trigger is used as the trigger. The cycle time is defined by the object property **Update Cycle** of the **Picture** object. This cycle provides the option of defining the cycles of all the actions, tag connections, and dynamic dialogs used in a worldmap centrally.
- **Window Cycle**
A cyclic trigger is used as the trigger. The cycle time is defined by the object property **Update Cycle** of the **Picture Window** object. This value applies to all actions, tag links, and dynamic dialogs, which are configured with the trigger type **Window Cycle**.
- **Upon Change**
This trigger updates the object every time a change is made. If external tags, there is a brief delay; internal tags lead to an immediate update.
- **Time Values (250 ms to 1 h)**
This trigger updates the object in the set cycle.
- **User cycle (1 to 5)**
This trigger updates the object in the defined cycle. Not used within Spectrum Power and IMM.



NOTE

The shorter the set update cycles, the higher the system load. To prevent an overload, the update does not take place more often than necessary.

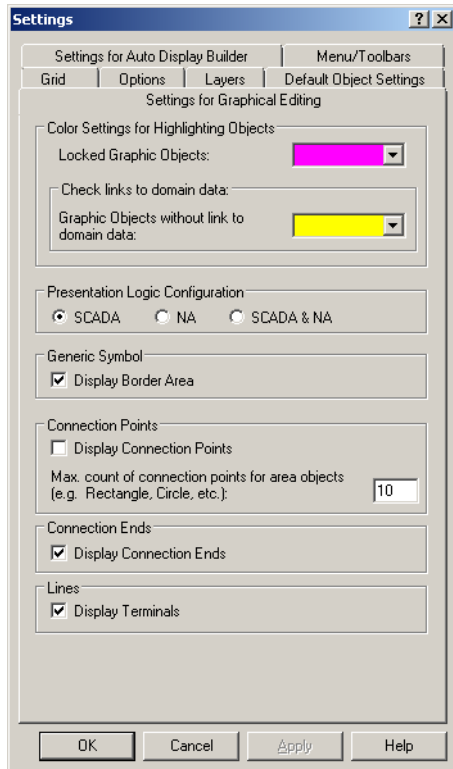


NOTE

The default trigger is used only for native WinCC functionality. Updating of Spectrum Power 5 functionality is spontaneous.

2.2.26 Graphical Editing Settings

Different settings for graphical and technological editing are changed and saved. The settings are made only for the Configuration System (CS).



[rgimm007-140711-01.tif, 1, en_US]

Figure 2-34 Settings - Graphical Editing

Locked Graphic Objects Highlighting

The background color of the locked graphical objects is selected from the opened color palette. Locked graphics highlighting is only used within the interlock-based job model.

Not linked Graphics Highlighting

The background color of objects that are not linked to IMM domain data is selected from the opened color palette. Not linked objects are detected during validation request for the selected worldmap area or the complete worldmap.

Presentation Logic Configuration

The possible data source for presentation logic configuration is selected.

Show border Area for Generic Symbols

If the check box is enabled, the border area for generic symbols are permanently shown in the worldmaps. Alternatively, only the main symbol is permanently shown. Nevertheless, the zones are always shown if the generic symbol is selected.

Show Connection Points

If the check box is enabled, the connection points for the objects are shown permanently in the worldmap. Alternatively, the connection points are only shown if the object gets connected. According to the definitions for a BusbarSection, the graphical object bus always shows a yellow frame if the check box is enabled.

Maximum Count of Connection Points

The maximum number of connection points are set here. The connection point is a connectable region within a graphical object. Only at these locations a connector connects.

Show Connection Ends

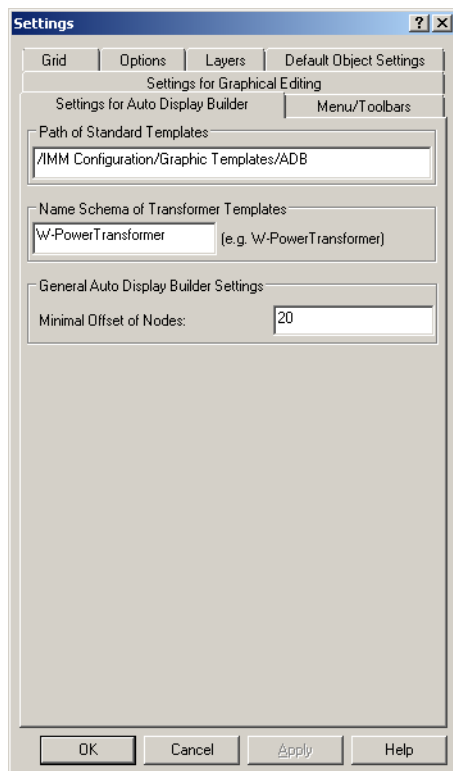
If the check box is enabled, connection ends for the connector objects are shown if they are not yet connected. Alternatively, no special connection ends for free connections are shown.

Show Terminals for Lines

If the check box is enabled, the terminal number (first and second terminal) is shown for lines linked to an ACLineSegment in IMM domain data. Alternatively, the terminal number is not shown.

2.2.27 Automatic Display Building Settings

Different settings for automatic display building are changed and saved.



[immrg033-140711-01.tif, 1, en_US]

Figure 2-35 Settings – Automatic Display Building

Path to the Default Symbols

The default symbols for Automatic Display Building are defined and stored within a default folder of the library. The configured folder in the above figure corresponds to the library folder with name **ADB** and is located in the root of the library hierarchy.

Name Schema of Transformer Templates

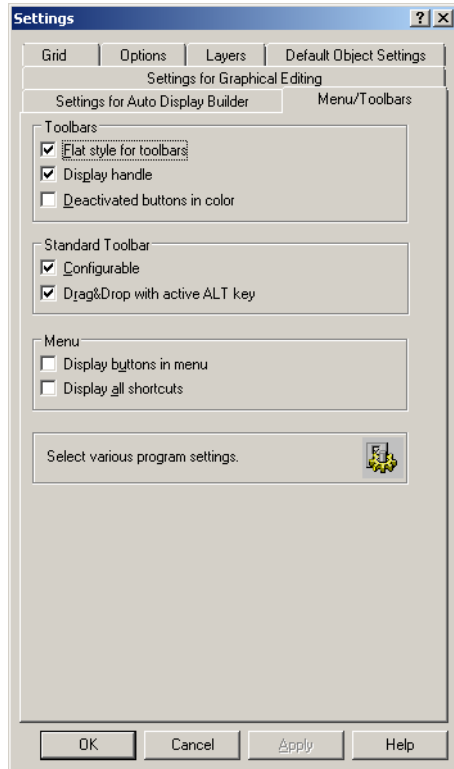
It defines the name schema for the complex default symbols of Power Transformers. In this release, the Auto Display Builder can only handle complex templates for Transformer Windings underneath Power Transformer.

General Auto Display Builder Settings

It defines the minimum offset of 2 nodes within a substation display and therefore influences the layout of the substation.

2.2.28 Menu and Toolbar Settings

The view of the menu and toolbars of the **Graphics Editor** are changed. Buttons are either shown as 3 dimensional or flat. Keyboard shortcuts can be displayed in the menus. Whereas the menus always comprise all entries, configure the standard toolbar as required. The settings that are made here are program settings that are independent of individual worldmaps and are retained when you exit from the program.



[rgimm008-140711-01.tif, 1, en_US]

Figure 2-36 Settings - Menu and Toolbars

Flat Style for Toolbars

If the check box is enabled, all buttons are shown as flat. Alternatively, they are shown as 3 dimensional.

Display Move Handles

If the check box is activated, handles are displayed beside or above toolbars. These handles are used to grip the toolbar, to move on the screen. When the check box is disabled, the handles are not shown.

Deactivated Buttons in Color

If the check box is enabled, both active and inactive buttons are displayed in color. Alternatively, only the active buttons are displayed in color.

Standard Toolbar - Configurable

If the check box is enabled, the standard toolbar is configurable, and the **Drag and Drop with active ALT key** check box is enabled. Alternatively, the toolbar is not configurable, but it remains movable. For additional information on configuration dialog for the standard toolbar, refer to Changing the Standard Toolbar.

Standard Toolbar - Drag and Drop with Active ALT Key

If the check box is enabled, the icons are shiftable within the standard toolbar. Hold the **ALT** key and drag the icon to the desired position.

Display Buttons in the Menu

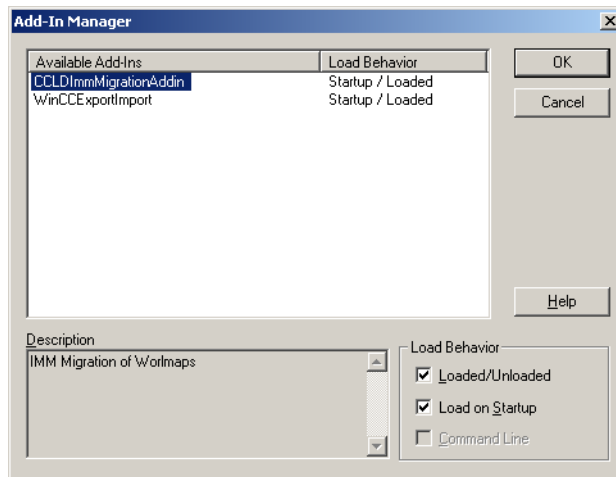
If the check box is enabled, the icons of the standard toolbar are shown to the left of the menu items. Alternatively, only the text appears in the menus.

Displaying All Shortcuts

If the check box is enabled, all the possibilities for input using the keyboard are displayed to the right of the menu items. Alternatively, only the most important keyboard shortcuts appear in the menus. For example, copying an object in the clipboard is made with 2 different keyboard shortcuts: In accordance with the Windows standard, by pressing **<CTRL + C>** or by pressing **CTRL** key and **0** on the numerical keypad.

2.2.29 Add-In Manager

The **Add-In Manager** registers an add-in, load, or unload it, and set its load behavior. Add-Ins are customized tools that adds capabilities to the **Visual Basic** development environment of the **Graphics Editor**. If you close only the visible portions of an add-in, its forms disappear from the screen, but the add-in is still present in memory. The add-in object itself stays always resident in memory until the add-in is disconnected through the **Add-In Manager** dialog.



[immrg001-030809-01.tif, 1, en_US]

Figure 2-37 Add-In Manager Window

Add-Ins List

The available add-ins are listed together with their configured load behavior.

Description

Displays a description of what the add-in does.

Load Behavior

Loaded or or Unloaded loads or unloads the selected add-in.

Load on Startup loads the selected add-in on startup of the **Graphics Editor**.

Command Line loads the selected add-in when the development environment is started from the command prompt or from a script.

3 Working with Worldmaps

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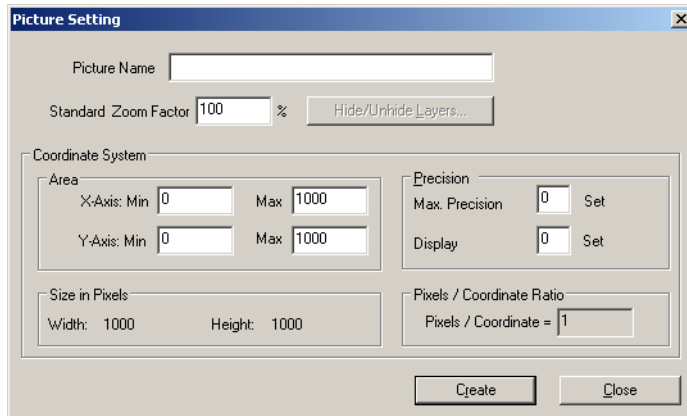
3.1 Operational Steps

3.1.1 Creating a Worldmap

The goal of this task is to create a worldmap.

- Select **New Network Graph** in the **File** menu.
- Alternatively, click  within the toolbar.

The **Picture Setting** window opens.



[immrg187-140711-01.tif, 1, en_US]

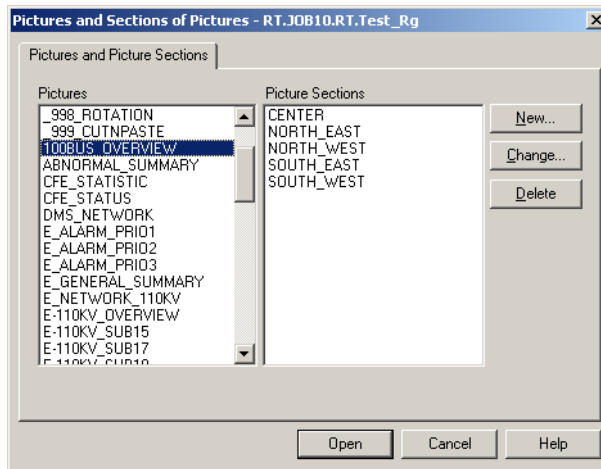
Figure 3-1 Picture Setting Window

- Define a new worldmap name and the standard zoom factor for that worldmap.
The worldmap name is definite within the system.
- Select the dimensions of the coordinate system for that worldmap.
The **Max. Precision** value defines the logarithm of the scaling factor. The **Pixels/Coordinate** ratio based on the value is displayed in the window.
The **Display** value defines the number of decimal places for the coordinates shown on the user interface. The higher the scaling factor, the more decimal places need to be selected. This configuration is helpful for geo-schematic displays.
- Click **Create**.
The new worldmap opens on the drawing section of the **Graphics Editor**. The empty worldmap with the specified standard zoom factor and **Pixels/Coordinate** ratio is displayed. The worldmap contains a picture section object spanning over the complete worldmap as default.

Alternative Operation

- Select **Manage Network Graphs** in the **File** menu.

- Alternatively, click  within the toolbar.
The **Pictures and Sections of Pictures** window opens.



[immrg189-150409-01.tif, 1, en_US]

Figure 3-2 Pictures and Settings of Pictures Window

- Click **New...** to open the **Picture Setting** window.
- Select the desired settings and click **Create**.
The new worldmap is shown in the **Pictures** section of the window.
- Click **Open** to open the new worldmap on the drawing section of the **Graphics Editor**.

3.1.2 Saving a Worldmap

The goal of this task is to save worldmaps that are currently opened in the **Graphics Editor**.

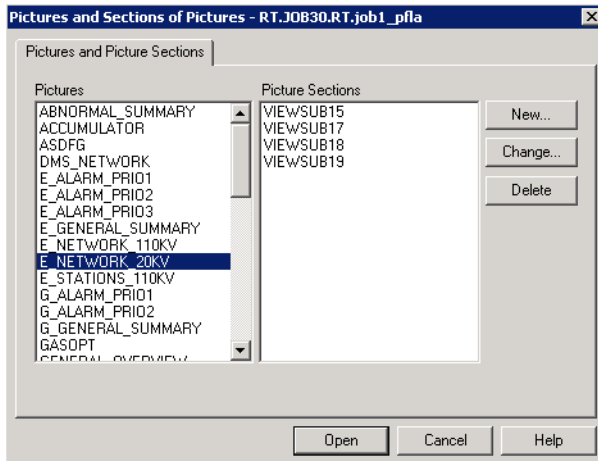
- Save the currently active worldmap using the button  or with the menu item **File > Save**.
- Alternatively, use the keyboard shortcut **<CTRL+ S>** key.
- Select **Save As...** in the **File** menu to save the active worldmap under a new worldmap name.
A new worldmap is created based on the currently active one.
- Select **Save All** in the **File** menu to save all the open worldmaps.

3.1.3 Opening an Existing Worldmap or Worldmap View

The goal of this task is to open an existing worldmap or worldmap view in the **Graphics Editor**.

Worldmap

- Select **Open Worldmap** from the **Graphics Editor** toolbar.
The **Pictures and Section of Pictures** window opens.



[immrg008-111208-01.tif, 1, en_US]

Figure 3-3 Worldmap

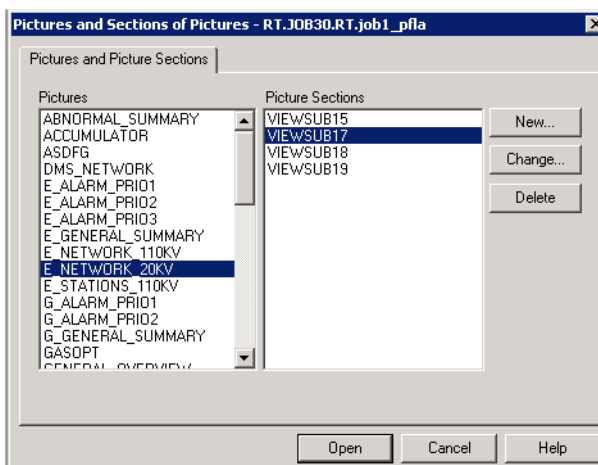
- Select the desired worldmap within the **Pictures** area and click **Open**.
The selected worldmap opens on the drawing section of the **Graphics Editor**. The worldmap with the specified zoom factor is displayed.

Alternative Operation

- Double-click the desired worldmap within the **Pictures** area of the **Pictures and Sections of Pictures** dialog to open the worldmap.
- Alternatively, use the keyboard to jump within the **Pictures** area to the desired worldmap, character by character.

Worldmap View

- Select **Open Worldmap** from the **Graphics Editor** toolbar.
The **Pictures and Sections of Pictures** window opens.
- Select the desired worldmap to see the defined views within the **Picture Sections** area.



[immrg009-111208-01.tif, 1, en_US]

Figure 3-4 Worldmap View

- Select the desired view of the selected worldmap within the **Picture Sections** area and click **Open**.
The selected worldmap view opens on the drawing section of the **Graphics Editor**. It displays the selected worldmap view completely. Hence, the view-specific coordinates and zoom factor are considered.



NOTE

By nature, the displayed content of the worldmap view is according to the presentation settings of the **Graphics Editor** and differs from the presentation at runtime.

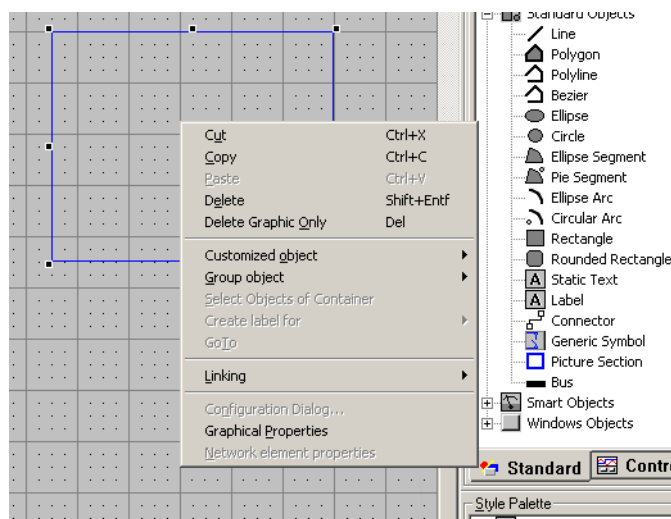
Alternative Operation

- Double-click the desired worldmap view within the **Picture Sections** area of the **Pictures and Sections of Pictures** dialog to open the worldmap view of the selected worldmap.
- Alternatively, use the keyboard to jump within the **Picture Sections** area to the desired worldmap view, character by character.

3.1.4 Creating a Worldmap View

The **Graphics Editor** provides the creation of worldmap views which allows to open a worldmap using predefined areas. The goal of this task is to create a worldmap view.

- Open a worldmap.
- Select the **Picture Section** object from the **Object Palette** of the **Graphics Editor** and paste it onto the drawing section.
The graphical representation of the **Picture Section** object is a rectangle.
- Specify an area for the worldmap view by modifying the size of this rectangle.
- Right-click the created **Picture Section** object.



[immg038-140711-01.tif, 1, en_US]

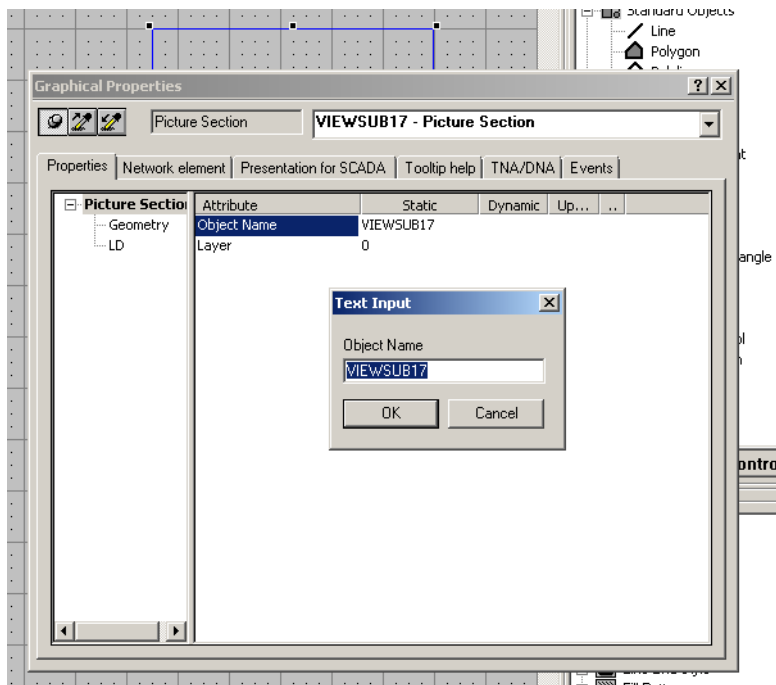
Figure 3-5 Context Menu of the Picture Section Object

The object context menu opens.

- Click the menu item **Graphical Properties** on the context menu.
The **Graphical Properties** window of the **Picture Section** object opens.
- Double-click the **Object Name** attribute in the **Graphical Properties** window.
The **Text Input** box opens.

- Enter the desired view name and click **OK**.

The name of the worldmap view is displayed in the **Graphical Properties** window.



[immrg039-140711-01.tif, 1, en_US]

Figure 3-6 Graphical Properties Window

- Save the worldmap.

The corresponding worldmap is opened using the predefined view within the **Picture Sections** area of the **Pictures and Sections of Pictures** dialog. In this case, the specified area of the worldmap is displayed on the drawing section of the **Graphics Editor**.

- Alternatively, double-click the Picture Section object to open the Graphical Properties window.



NOTE

During creation of a new worldmap, one view is automatically created covering the entire worldmap. This view receives the worldmap name as view name per default.

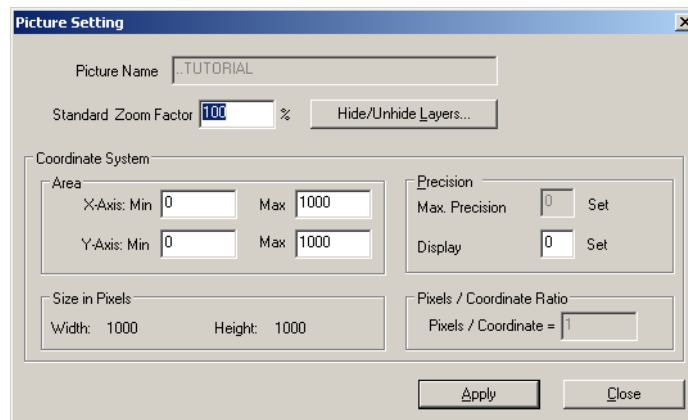
3.1.5 Resizing an Existing Worldmap

The goal of this task is to resize an existing worldmap

- Select **Open Worldmap** from the **Graphics Editor** toolbar.
The **Pictures and Sections of Pictures** window opens.
- Select the corresponding worldmap.

- Click **Change....**

The **Picture Settings** window of the selected worldmap opens.



[immg193-140711-01.tif, 1, en_US]

Figure 3-7 Picture Settings Window

- Modify the minimal or maximal values of the x-axis and y-axis.
- Click **Apply**.
The worldmap size is changed according to the specified coordinate values.



NOTE

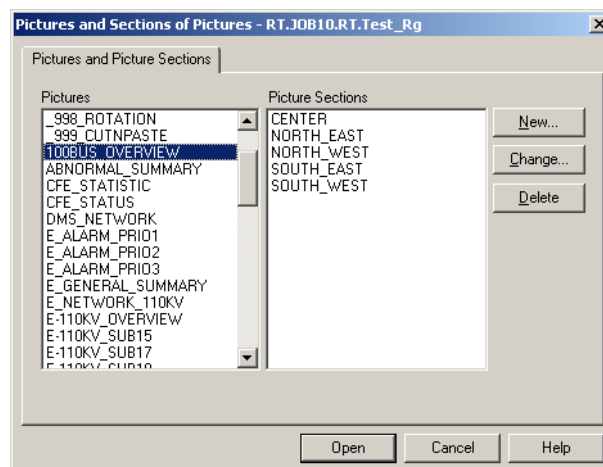
Resizing the worldmap to a smaller area is only possible, if the current graphical objects can still be displayed, thus no information gets lost unexpectedly,
The scaling factor and therefore the **Pixel/Coordinate** ratio of the worldmap cannot be changed.

3.1.6 Deleting a Worldmap

The goal of this task is to delete a worldmap in **Graphics Editor**.

- Close the worldmap that you want to delete.
- Select **Open Worldmap** from the **Graphics Editor** toolbar.

The **Pictures and Sections of Pictures** window opens.



[immg194-150409-01.tif, 1, en_US]

Figure 3-8 Pictures and Sections of Pictures Window

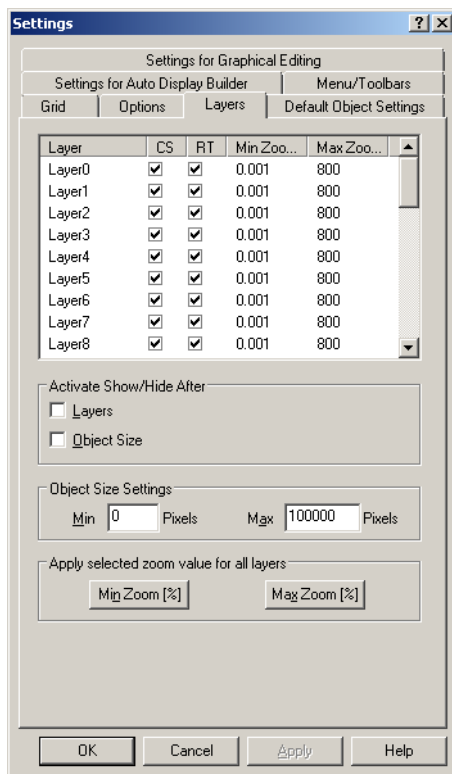
- Select the worldmap to be deleted and click **Delete**.

- Confirm the security inquiry with **OK**.
The worldmap gets deleted.

3.1.7 Modifying the Layers of a Worldmap

The goal of this task is to define or modify various settings for the layers of a worldmap.

- Open an existing worldmap.
- Select the **Tools** menu item in the menu bar.
- Click **Settings...**
The window with the tabs for various setting options opens.
- Click the **Layers** tab to select the desired options for the visibility, decluttering, and naming of the layers of the worldmap. For additional information, refer to [2.2.24 Layers Settings](#)



[rgimm005-140711-02.tif, 1, en_US]

Figure 3-9 Settings - Layers

- Click **OK** or **Apply** to confirm the settings for the worldmap.

Alternative Operation

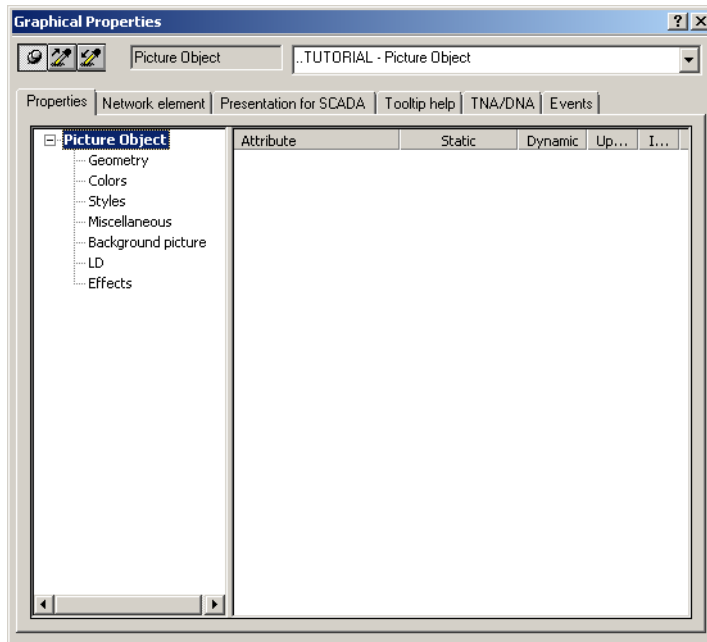
- Use the **Hide/Unhide Layers...** button in the **Picture Settings** dialog to select the desired options for the visibility, decluttering and naming of the layers.

3.1.8 Modifying Important Properties of a Worldmap

The goal of this task is to modify important properties of a worldmap.

- Open an existing worldmap.
- Open the **Graphical Properties** window by clicking  in the toolbar.

As an alternative, the **Graphical Properties** window is opened using the context menu or by selecting **Graphical Properties** in the **View** menu.



[immg042-140711-01.tif, 1, en_US]

Figure 3-10 Graphical Properties – Picture Object

- Check whether the properties of the worldmap are displayed:
The entry **Picture Object** must be displayed in the object type field at the top of the window.
The worldmap name is shown in the list box at the right.
- Click **Properties** or **Event** tab to change the attributes.



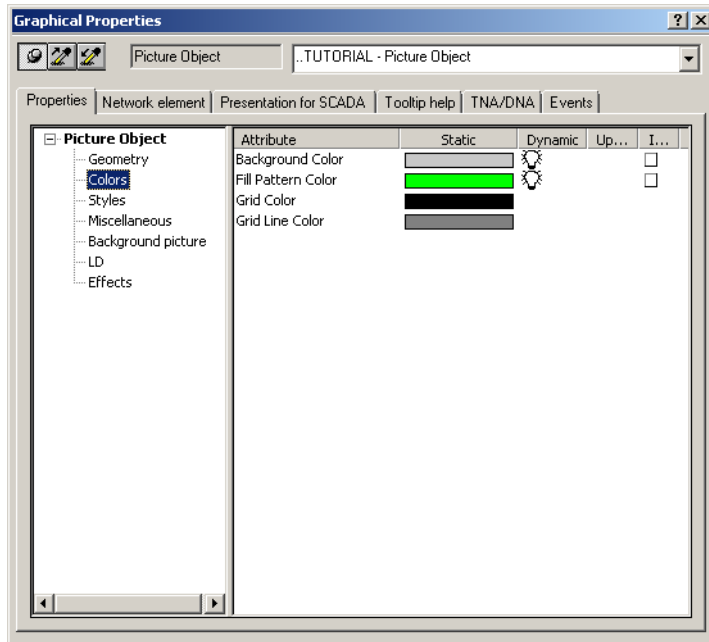
NOTE

If the properties of a worldmap are changed, make certain that no objects are selected.

The most important properties to be considered are:

Worldmap Colors

- Select the **Colors** property group within the **Properties** tab to change the attributes. For example, **Background Color**.

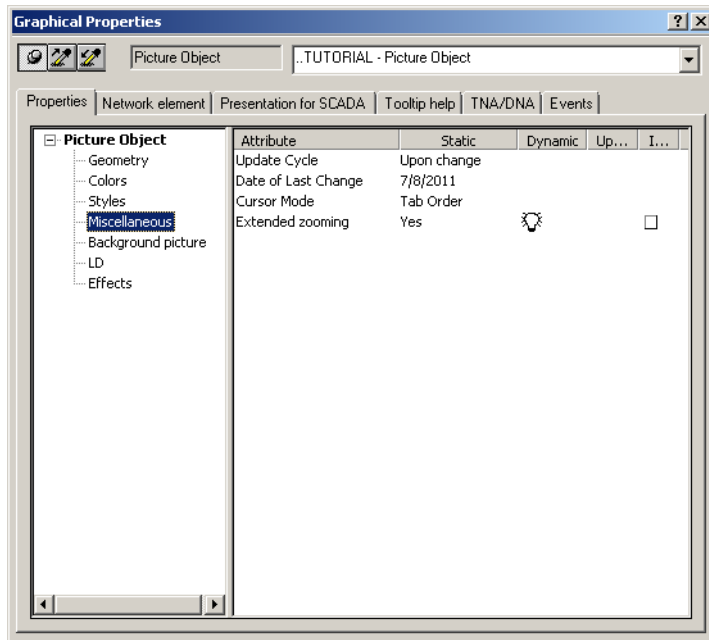


[immrg043-140711-01.tif, 1, en_US]

Figure 3-11 Graphical Properties – Picture Object, Colors Property Group

Worldmap Zooming with Mouse Wheel

- Select the **Miscellaneous** property group within the **Properties** tab to change the attributes.



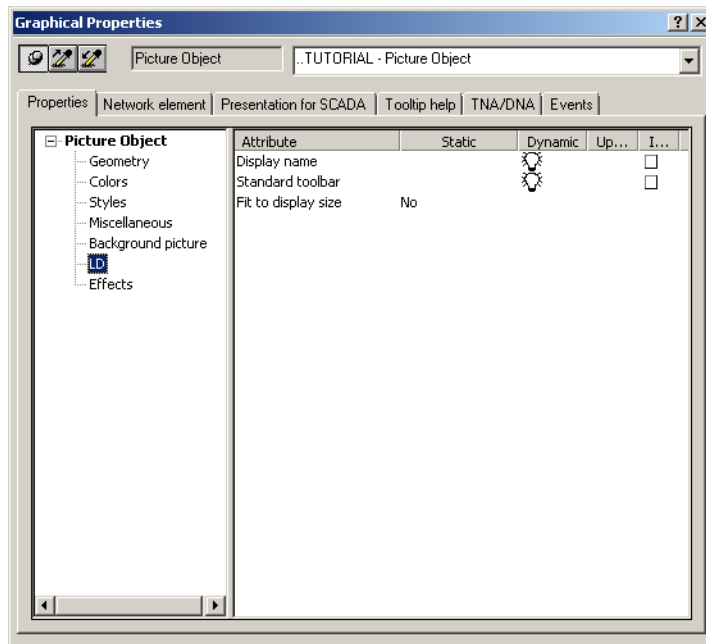
[immrg044-140711-01.tif, 1, en_US]

Figure 3-12 Graphical Properties – Picture Object, Miscellaneous Property Group

If **Extended Zooming** is enabled, worldmap zooming at runtime is performed using the mouse wheel.

Worldmap Name and Automatic Screen Resolution Adaptation

- Select the **LD** property group within the **Properties** tab to change the attributes. For example, **Fit to display size**.



[rgimm010-140711-01.tif, 1, en_US]

Figure 3-13 Graphical Properties – Picture Object, LD Property Group

If a name is specified for the **Display name** attribute, this name is used in the title bar.



NOTE

Within Spectrum Power 5, worldmap views are considered during display callup at runtime. That means that changes within the **LD** property group of a worldmap are not considered at runtime. You need to make the required changes for the worldmap view.

3.1.9 Assigning Display Access Rights to a Worldmap

Worldmaps are configured to take care of available **Area of Responsibilities (AoR)**. During runtime, if you do not have the required AoR, the worldmap including all worldmap views cannot be opened.

The goal of this task is to assign an AoR to a worldmap.

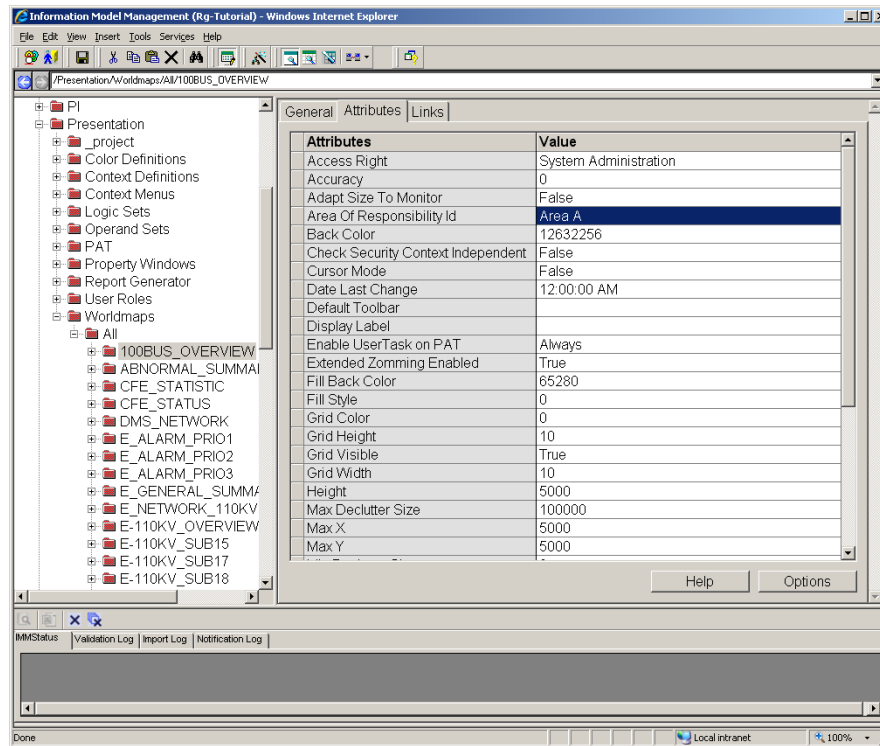


NOTE

To assign an AoR to a worldmap, the **IMM Single Instance Editor** is used. For additional information on the **Single Editor**, refer to Reference Guide Information Model Management (IMM).

- Select the desired worldmap within IMM.

Worldmaps are stored under the path **/Presentation/Worldmaps/All/**



[immrg006-070809-02.tif, 1, en_US]

Figure 3-14 IMM Single Instance Editor

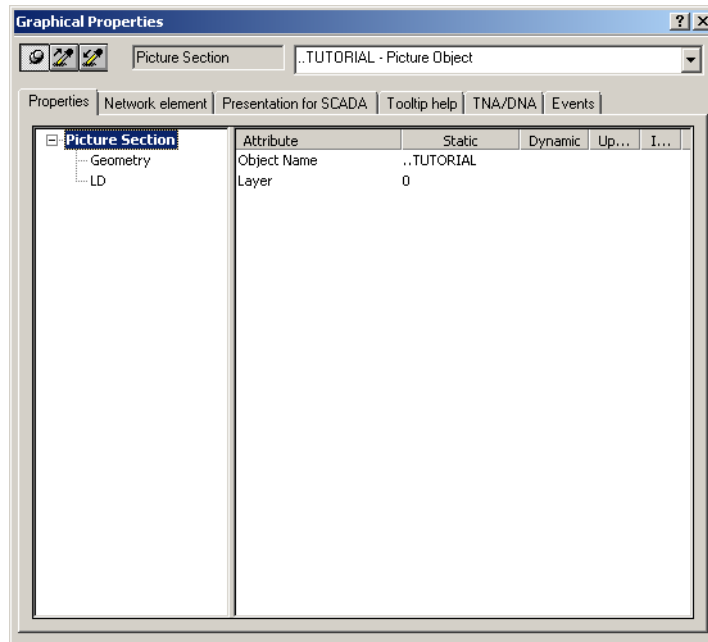
- Modify the attribute **Area Of Responsibility Id** within the **Attributes** tab of the **Single Line Editor** as required.

3.1.10 Modifying Important Properties of a Worldmap View

The goal of this task is to modify important properties of a worldmap view.

- Open an existing worldmap.
- Open the **Graphical Properties** window by clicking  in the toolbar.

As an alternative, the **Graphical Properties** window is opened using the context menu or by selecting **Graphical Properties** in the **View** menu.



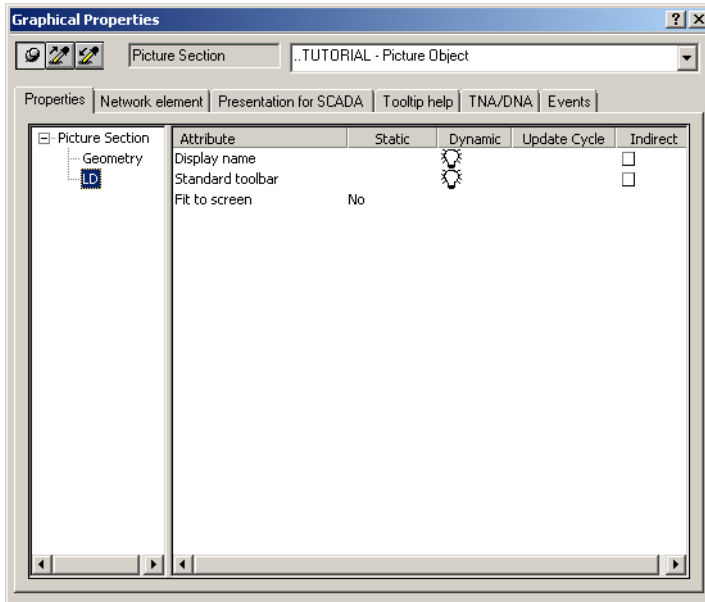
[sc_rgimm011_07052014, 1, en_US]

Figure 3-15 Graphical Properties - Picture Section

- Check whether the properties of the worldmap view are displayed:
The entry **Picture Section** is displayed in the object type field at the top of the window.
The worldmap view name is shown in the list box at the right.
- Click **Properties** or **Event** tab to change the attributes.
The most important properties to be considered are:

Worldmap View Name, Standard Toolbar, and Automatic Screen Resolution Adaptation

- Select the **LD** property group within the **Properties** tab to change the attributes.



[immrg047-140711-01.tif, 1, en_US]

Figure 3-16 Graphical Properties – Picture Section, LD Property Group

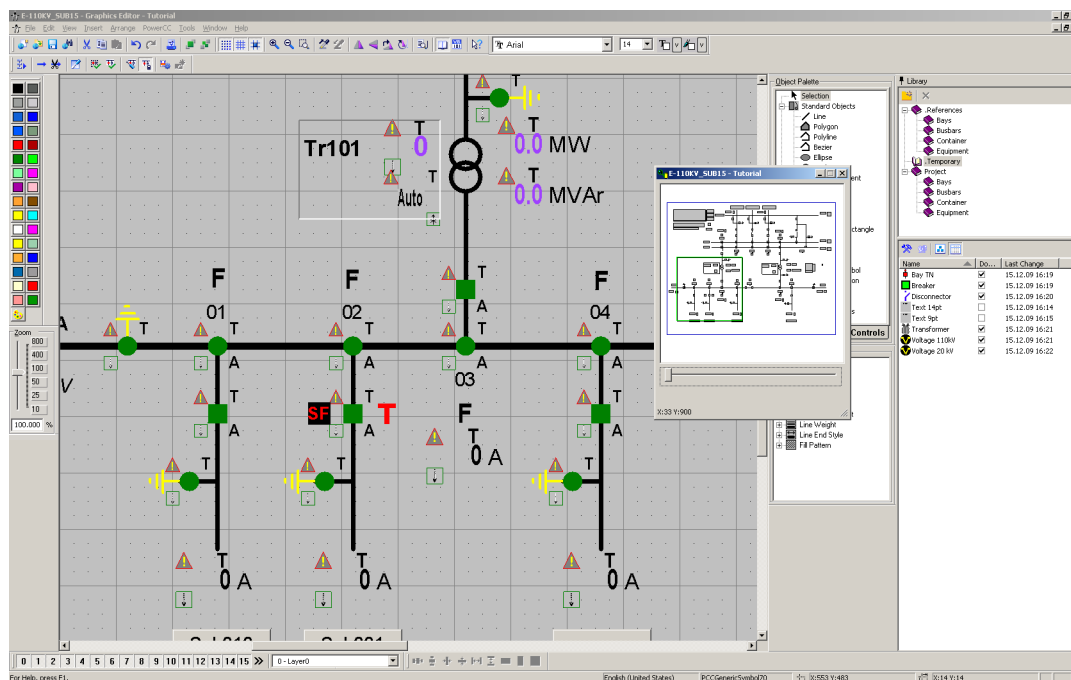
If a name is specified for the **Display name** attribute, this name is used in the title bar at runtime. The **Standard toolbar** entry specifies the toolbar for that view at runtime.

If the **Fit to display size** property is enabled, the call up of the worldmap view at runtime opens in full size, independent of the screen resolution specified for the particular monitor screen.

3.1.11 Navigating through a Worldmap with the Navigation Window

The goal of this task is to display a specific segment of a worldmap using the **Navigation** window.

Click  the toolbar button to open the **Navigation** window.



[image001-150313-01.tif, 1, en_US]

Figure 3-17 Navigation of a Worldmap

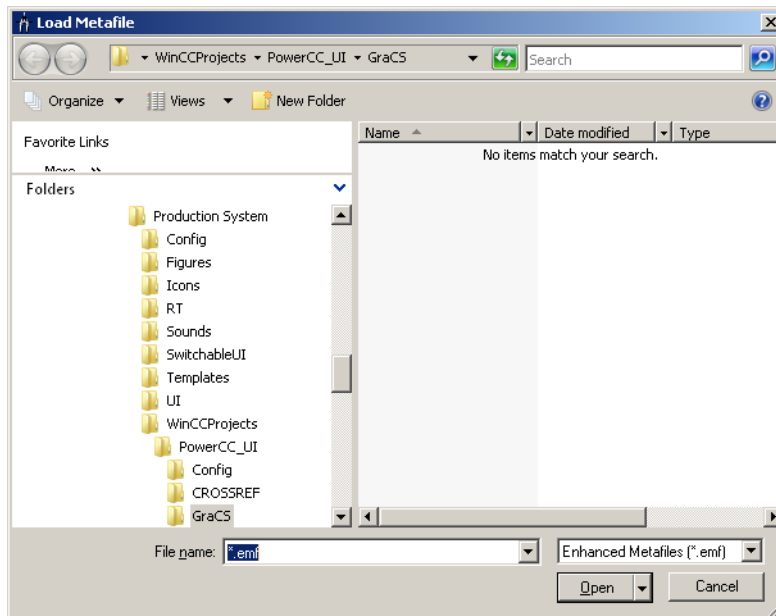
- The navigator window opens. The green rectangle represents the displayed segment in the associated main window.
- To scroll to a different part of the single-line diagram, move the green rectangle in the navigator window by using drag and drop.
- To enlarge or reduce the size of the rectangle, press and hold the **CTRL** key and use the scroll wheel of the mouse.
- To zoom the display of the navigator window, use the slider.

3.1.12 Importing Graphics

The goal of this task is to import graphics in **Graphic Editor**.

Enhanced Meta Files (EMF), Windows Meta Files (WMF), and Native WinCC Graphic Files (PDL)

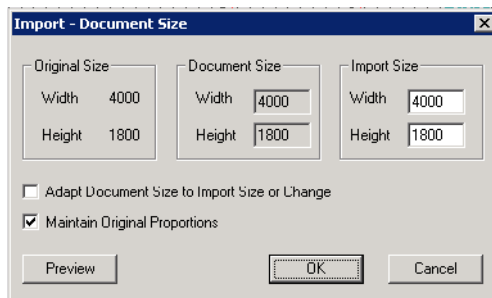
- Either activate an unused layer or make certain that there are no objects in the upper left of the active layer. This is covered by the imported graphic later. You can also create a worldmap to import and edit graphics.
- Open the **File** menu and select the **Import...** entry.
A new native WinCC display opens.
The **Load Metafile** window opens.



[immrg098-140711-01.tif, 1, en_US]

Figure 3-18 Load Metafile Dialog

- Select the graphics file type – EMF, WMF, or PDL.
- In the file list, double-click the graphic to be imported.
The **Import - Document size** window opens.



[immrg099-090109-01.tif, 1, en_US]

Figure 3-19 Import - Document Size

- Select the desired import size.
- Confirm the import of the graphic by clicking **OK**.
The graphic is inserted in the native WinCC display.
- Copy the imported graphic into the desired worldmap.
If necessary, select all the imported graphics vectors and group them.

**NOTE**

Referenced graphic files must be copied to the **GraCS** graphic folder for the current WinCC project to be available on a destination computer.

Spectrum Power 5 Worldmap Graphic (XDF) or (CIM-RDF)

IMM provides data exchange of all engineering data, including graphical data in eXchange Data Format (XDF) and Common Information Model – Resource Description Format (CIM-RDF). IMM supports complete, partial, and incremental data exchange.

For additional information on graphical data exchange and import in CIM-RDF or XDF format, refer to Reference Guide Information Model Management (IMM).

**NOTE**

The preferred way of importing worldmaps within the Spectrum Power IMM is using XDF and CIM-RDF formats.

3.1.13 Exporting Display Data

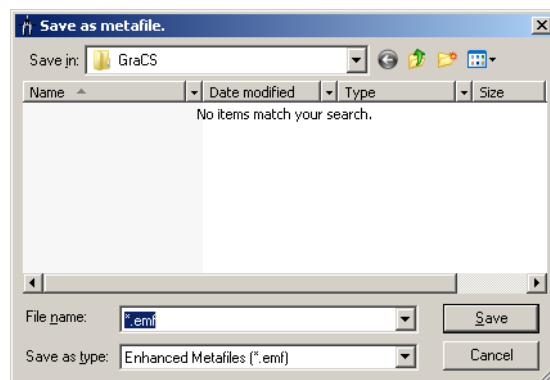
The goal of this task is to export graphics.

Enhanced Meta Files (EMF) and Native WinCC Graphic Files (PDL)

- Select the object to be exported. If you do not select an object, the entire worldmap is exported.

Open the **File > Export...** menu.

The **Save as metafile** window opens.



[immrg018-140711-02.tif, 1, en_US]

Figure 3-20 Export Dialog

- Define the target directory, the file name, and file format.
- Click **Save** to export the selected objects or the complete worldmap into the corresponding file.

Spectrum Power 5 Worldmap Graphic (XDF) or (CIM-RDF)

IMM provides data exchange of all engineering data, including graphical data in eXchange Data Format (XDF) and Common Information Model – Resource Description Format (CIM-RDF). IMM supports complete, partial, and incremental data exchange.

For additional information on graphical data exchange and export in CIM-RDF or XDF format, refer to Reference Guide Information Model Management (IMM).

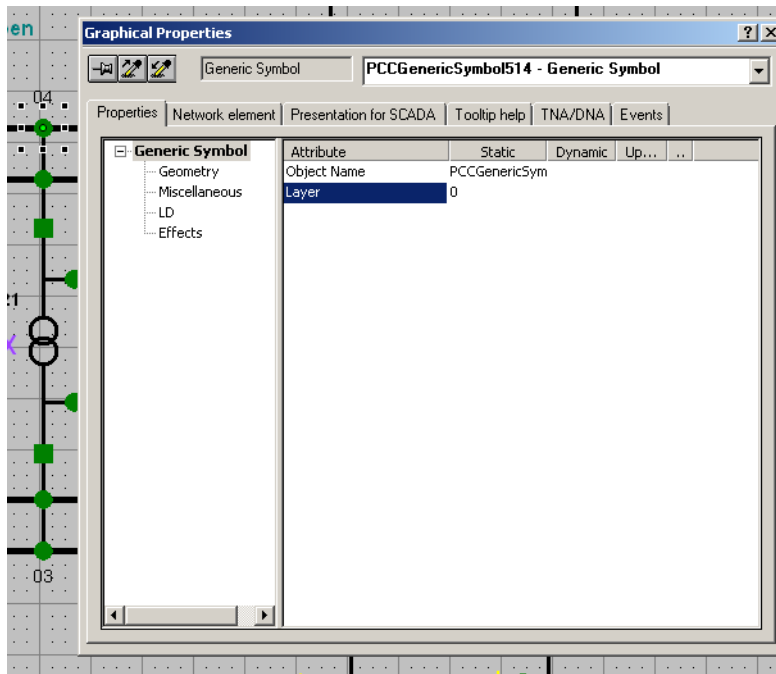
**NOTE**

The preferred way of exporting worldmaps within the Spectrum Power IMM is using XDF and CIM-RDF formats.

3.1.14 Assigning a Graphical Object to a Layer of the Worldmap

The goal of this task is to assign a graphical object to a certain layer in **Graphics Editor**.

- Select an object and open the **Graphical Properties** window.



[immrg049-140711-01.tif, 1, en_US]

Figure 3-21 Graphical Properties Window

- Select the object type within the **Properties** tab.
- Double-click the layer attribute and enter the number of the desired layer.

3.1.15 Transferring Objects to Another Worldmap

The goal of this task is to transfer graphic objects from one worldmap to another worldmap.

- Press **<CTRL+X>** to copy the selected object to the clipboard.

The cut object remains visible in the worldmap.

Alternatively, click  in the standard toolbar or the entry **Cut** in the context menu of the selected object, or in the **Edit** menu of the **Graphics Editor**.

- Open the worldmap into which the object in the clipboard is pasted.
- Press **<CTRL+V>**.

The cut object is deleted within the original worldmap.

The current content of the clipboard is inserted into the active worldmap under the mouse position and selected.

Alternatively, click  in the standard toolbar or use the entry **Paste** in the context menu or in the **Edit** menu.

Alternative Operation

- Select the object to be transferred and drag it with the left mouse button depressed to another worldmap.



NOTE

The described workflow is valid for non technological objects (graphical objects without linked to domain data). In principle, that is also true for technological objects (graphical objects with linked domain data). Hence, additional steps are needed to be performed. For additional information on transferring technological graphical objects, refer to [1.2.4 Technological Editing](#).

3.1.16 Copying Objects to Another Worldmap

The goal of this task is to copy graphic objects into another worldmap.

- Press **<CTRL+ C>** to copy the selected object and to transfer it to the clipboard. Alternatively click  in the standard toolbar use the entry **Copy** in the context menu of the selected object or in the **Edit** menu of the **Graphics Editor**.
 - Open the worldmap into which the object in the clipboard should be pasted.
 - Press **<CTRL+V>**.
The current content of the clipboard is inserted into the active worldmap under the mouse position and selected.
Alternatively, click  in the standard toolbar or use the entry **Paste** in the context menu or in the **Edit** menu.
 - Repeat this step to insert further copies of the contents.
-



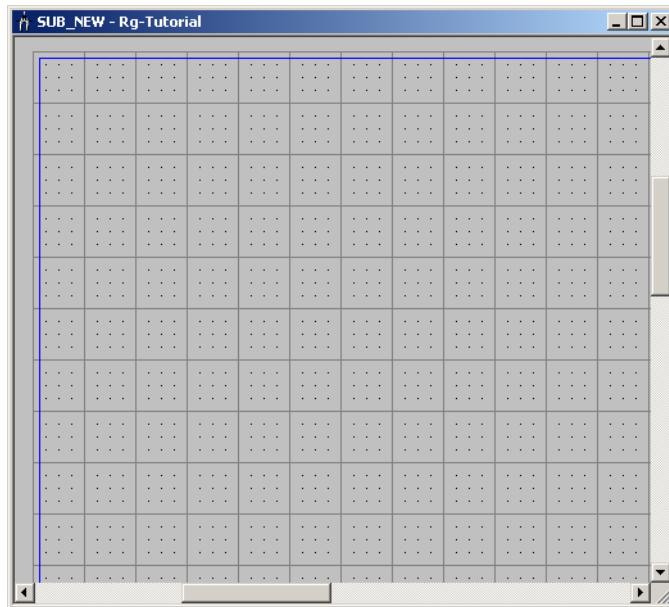
NOTE

The described workflow is valid for non technological objects (graphical objects without linked to domain data). In principle, that is also true for technological objects (graphical objects with linked domain data). Hence, additional steps are needed to be performed. For additional information on copying technological graphical objects, refer to [1.2.4 Technological Editing](#).

3.2 User Interface Description

3.2.1 Worldmaps

In the **Graphics Editor**, a worldmap is a file in the form of a sheet of drawing paper. The size of the drawing sheet can be adjusted. A drawing sheet has 32 layers which is used to improve the organization of the drawing. The complete process is distributed over several individual worldmaps, which are linked together. Retrievals of other applications and files are included in a worldmap. The more complex the configured process is, the more detailed the planning.



[immrg001-070809-01.tif, 1, en_US]

Figure 3-22 Drawing Worldmaps

Project Planning Notes

- Plan the structure of the network diagrams.
How many worldmaps are required in which hierarchy? For example, substations are shown in individual worldmaps, which are then merged into an overview worldmap. Overview worldmaps even uses a condensed presentation of what is shown in individual worldmaps.
- Plan the navigation within a worldmap and between the individual worldmaps.
Operating sequence, buttons, and keyboard shortcuts are uniform in all worldmaps.
- Create project specific simple library elements with and without domain data and default settings. For example, Circuit Breaker, Ground Disconnecter ...
- Create project specific complex library elements based on the simple ones. For example, Transformator, Bay, Substation ...
- Use these library elements as template for each individual worldmap.
- Use the program functions.
Worldmaps are efficiently created by using library templates, copying, and transferring properties and by working with layers and multiple worldmaps.
- Each worldmap is displayed during the design phase to avoid changes in the worldmap layout.
The worldmap is of the same size as displayed on the target computer at runtime.
- Use only fonts that are available on the target computer to avoid changes in the worldmap layout.



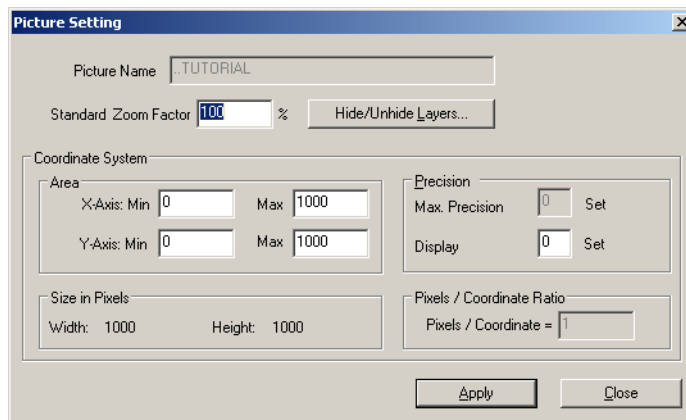
NOTE

The total number of graphical objects within a worldmap does influence the display call up and zooming/panning performance. This is valid for both, in the design time with the **Graphics Editor** as well as within the Spectrum Power Control Center runtime environment.

It is recommended not to exceed the total number of 25.000 graphical objects per worldmap. Split the worldmap in case of reaching this amount into several worldmaps to stay within the specified performance requirements of Spectrum Power.

3.2.2 Worldmap Dimension Settings

The worldmap dimensions and the worldmap scaling factor are defined in the **Picture Settings** window.



[immg193-140711-02.tif, 1, en_US]

Figure 3-23 Picture Settings Window

Picture Name

The name of the worldmap is used as an identifier and is definite within the system. The specified worldmap name is seen in the **Pictures and Sections of Pictures** window and within the IMM tree under the path / **Presentation/Worldmaps/All/**.

Standard Zoom Factor

Defines the default zoom factor when the worldmap is opened within the **Graphics Editor**. The zoom factor is specified in %. Minimum value is 0.001%, maximum value is 800%.

Hide/Unhide Layers

Opens the **Layers** window to define the visibility, decluttering and naming of the layers for the worldmap. Refer to Layer Decluttering Settings.

Coordinate System - Area

Defines the range and the zero point of the coordinate system of the worldmap. Minimum value has to be between -1.999.000 and +1.000, maximum value has to be between 0 and 2.000.000 for each axis.

Coordinate System - Precision

Max. Precision defines the scaling factor of the coordinate system of the worldmap. The factor has to be from 0 to 3 and specifies the logarithm pixel/coordinate ratio.

Display defines the number of decimal places for coordinates shown on the user interface. The factor has to be from 0 to 8.

Coordinate System – Size in Pixels

Shows the worldmap size in pixels based on the entries made for the coordinate system **Area** and **Precision**.

Coordinate System – Pixel/Coordinate Ratio

Shows the worldmap pixel/coordinate ratio based on the entry made for the **Max. Precision**.

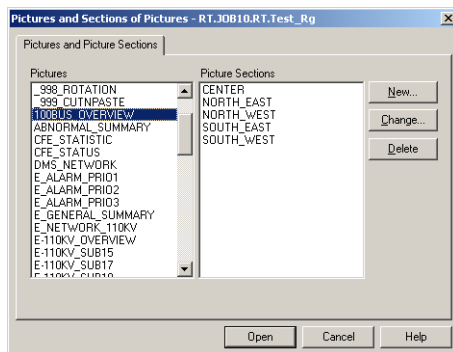


NOTE

The scaling factor of the coordinate system (**Max. Precision**) cannot be changed once it is defined for the worldmap.

3.2.3 Managing a Worldmap and Worldmap Views

Worldmaps and worldmap views are managed with the **Pictures and Sections of Pictures** window.



[immg003-070409-01.tif, 1, en_US]

Figure 3-24 Pictures and Sections of Pictures Window

Pictures Area

The existing worldmaps are shown. A worldmap name is selectable.

Picture Sections Area

Existing worldmap views of the selected worldmap in the pictures area are shown in the pictures area. A worldmap view is selectable. Every worldmap contains at least one worldmap view that is automatically created when worldmap is created.

Open a Worldmap or Worldmap View

The button **Open** allows you to open the selected worldmap or worldmap view in the **Graphics Editor**.

New Worldmap

The button **New...** allows you to create a worldmap. The **Picture Setting** window opens to specify the worldmap name and dimensions.

Change Worldmap

The button **Change...** allows you to modify the standard zoom factor and the axis of the coordinate system of the selected worldmap. It also allows you to change the number of decimal places for coordinates shown on the user interface. The **Picture Setting** window opens to specify the new settings.

Delete Worldmap or Worldmap Views

The button **Delete...** allows you to delete a selected worldmap or worldmap view. Deleting a worldmap deletes all graphical representations within the worldmap. Linked domain data is not affected by this operation.



NOTE

Deleting a worldmap view within the **Pictures and Sections of Pictures** window is not recommended, as you do not have a graphical presentation of that worldmap view.

Open the worldmap and delete the corresponding **Picture Section** object instead.



NOTE

Creating a worldmap view is done within the opened worldmap by inserting a **Picture Section** object. The name of the worldmap view is defined in the **Graphic Properties** window of the selected Picture Section object.

After specifying the name, it is shown in the pictures sections area of the **Pictures and Sections of Pictures** window.

3.2.4 Worldmap Manipulations

Open and Save

Worldmaps are opened and saved in **Graphics Editor** in the usual Windows manner.

Resize

Worldmap is resized according to the requirements within the **Picture Setting** window.

Copy

To transfer the settings that you have made to another worldmap, save the old worldmap under a different name. Alternatively, you can change the contents of the new worldmap.

Rename

A worldmap is renamed within the IMM environment only.

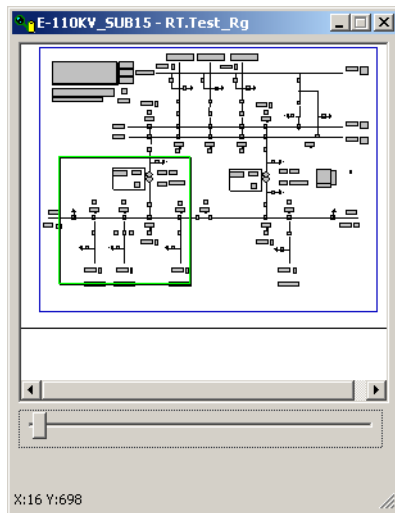
For additional information, refer to Reference Guide Information Model Management (IMM).

Delete

Deleting a worldmap is done using the **Pictures and Sections of Pictures** window.

3.2.5 Navigation Window

The **Navigation** window allows the user to conveniently maneuver quickly in large worldmap diagrams. You can simultaneously view a subset of the worldmap along with the entire worldmap. The **Navigation** window shows how the subset of information in the worldmap window located in a confined area and identified by a green border in the **Navigation** window. Whatever is included in the worldmap window is shown in the green box of the **Navigation** window.



[immrg202-150409-01.tif, 1, en_US]

Figure 3-25 Navigation Window

Navigation Techniques

After the **Navigation** window is opened, standard MS Windows functionality is supported and the user manipulates both views as follows:

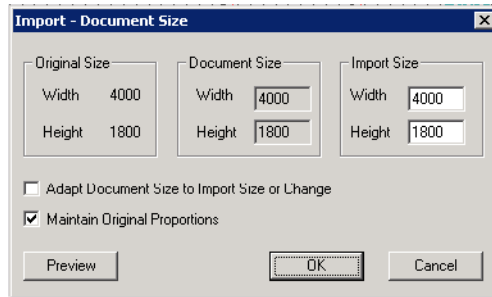
- Both windows are resized as desired.
- Whatever is included in the worldmap window is automatically replicated within the area (green box) in the navigation window.
- When the scroll bars are used in the worldmap window to move vertically or horizontally, the area (green box) in the navigation window mimics the movements.
- When zooming is used in the worldmap window, the area (green box) in the navigation window automatically adjusts to replicate the exact information subset.
- When the area (green box) is moved vertically or horizontally, the worldmap window automatically displays the same information subset.
- The zoom slider at the bottom of the navigation window allows the user to magnify or reduce the worldmap image in the navigation window.

3.2.6 Graphics Import

Graphics, which were created with other programs, are inserted in **Graphics Editor** as graphic objects, OLE objects, or editable graphics. An editable graphic must be a vector graphic saved in either the EMF or WMF format. When these graphics are imported, they are broken down into individual vectors, which are then interpreted by the **Graphics Editor**. For example, individual polygons. Within **Graphics Editor**, graphics can also be created, exported from or imported into a worldmap. However, in this case, dynamic settings, and multiple object-specific properties are lost as they are not supported by the graphic formats.

File formats, like GIF, JPEG, etc. must be saved as EMF or WMF file formats to import them onto the drawing section of the **Graphics Editor**.

After specifying the graphical file to be loaded, the **Import – Document Size** window opens to define the size within the worldmap.



[immrg099-090109-01.tif, 1, en_US]

Figure 3-26 Import - Document Size

Original, Document, and Import Size

All sizes are indicated in pixels for zoom level 100%:

- The **Original Size** is the initial size of the graphic within the selected file.
- The **Document Size** is the size of the worldmap the graphic imported.
- The **Import Size** is the size of the graphic after importing into the native WinCC display.

Adapt Document Size

Adapt document size to import size. If the check box is activated, the document size is changeable. The fields in the document size area become active and the values of the document size are adapted to the values of the import size.

Maintain Original Proportions

If the check box is activated, it is certain, that the graphic keeps its original proportions, even if the import size is changed. If you want to allow distortion of the imported graphic, remove the tick from the check box.

Preview

The selected graphic is inserted into a native WinCC display for a preview. If you then change the import size, the preview is not updated automatically. In this case, click **Preview** again.



NOTE

When using EMF file, which are not included with WinCC, it is possible that resources are lost in runtime. If resources are lost, it causes the system to behave in an undefined manner. The user assumes full responsibility for the use of such data. Before such use, it is recommended to check if the system is running securely.



NOTE

Referenced graphic files must be copied to the **GraCS** graphic folder for the current WinCC project to be available on a destination computer.



NOTE

Imported graphics are placed ungrouped at the top left on the native WinCC display. Copy the imported graphic into the desired worldmap. If you need an imported graphic frequently, save it in the library.

IMM provides data exchange of all engineering data, including graphical data in eXchange Data Format (XDF) and Common Information Model – Resource Description Format (CIM-RDF). IMM supports complete, partial, and incremental data exchange.

For additional information on graphical data exchange and import in CIM-RDF or XDF format, refer to Reference Guide Information Model Management (IMM).



NOTE

The preferred way of importing worldmaps within the Spectrum Power IMM is using XDF and CIM-RDF formats.

3.2.7 Export Display Data

Graphics are exported from **Graphics Editor** in the Enhanced Metafile (EMF) file format. However, in this case, dynamic settings and multiple object-specific properties are lost. They are not supported by the graphic format.

It is also possible to export graphics in the native WinCC PDL format. However, it is only possible to export an entire worldmap in the PDL format and not individual objects.



NOTE

Check all exported worldmaps, which you export for further processing in other programs, for both completeness and interpretation errors.

Special Features

- Different programs do not use the same graphic filters. It is possible that other programs are not able to open a graphic that was exported from **Graphics Editor**. Furthermore, fonts and object information are sometimes incorrectly interpreted by these programs.
If the display must appear precisely as it is in **Graphics Editor**, but your program displays it incorrectly, solve this problem by making a screenshot.
- When exporting graphics created with the **Graphics Editor**, all object properties dynamics of the individual objects are lost.
- Misinterpretations can occur within **Graphics Editor**. For example, while an exported polygon is still a polygon after it has been imported, the functionality of a connector is lost when it is exported. After it is imported, it is shown as a polyline. This misinterpretation is caused by the fact that the EMF format does not save all the functionality of **Graphics Editor**.

Spectrum Power Worldmap Graphic (XDF) or (CIM-RDF)

IMM provides data exchange of all engineering data, including graphical data in eXchange Data Format (XDF) and Common Information Model – Resource Description Format (CIM-RDF). IMM supports complete, partial, and incremental data exchange.

For additional information on graphical data exchange and export in CIM-RDF or XDF format, refer to Reference Guide Information Model Management (IMM).



NOTE

The preferred way of exporting worldmaps within the Spectrum Power IMM is using XDF and CIM-RDF formats.

3.2.8 Worldmap Properties

In **Graphics Editor**, a worldmap is treated as an object and is called a **Picture Object**. It has properties and is dynamized. The properties and dynamics of the worldmap can be adjusted at any time. If no object has been selected in the worldmap, the **Graphical Properties** window displays the properties of the **Picture Object** that equals to the worldmap.

Display access rights by assigning an Area of Responsibility (AoR) to a worldmap ensures that only runtime users having the right permissions can open a worldmap and view it at runtime. The AoR is assigned to the

worldmap within the **IMM Single Instance Editor**. For additional information, refer to Reference Guide Information Model Management (IMM).

3.2.9 Working with Layers

In the **Graphics Editor**, a worldmap consists of 32 layers in which the graphical objects are inserted. The position of an object in a worldmap is set when it is assigned to a layer. Objects of the layer 0 are located at the screen background, while objects of the layer 32 are located in the foreground.

Objects are always added on the active layer, and quickly moveable to another layer. An objects assignment to a layer is changed using the layer attribute in the **Graphical Properties** window.

In addition, it is also possible to change the positioning of the objects with respect to each other within a layer. 2 functions are available for this purpose in the **Arrange > Within the Layer** menu. When creating a worldmap, the objects of a level are by default arranged in the order in which they were configured. The object inserted first lies at the back of the level, each additional object is inserted single position toward the front.

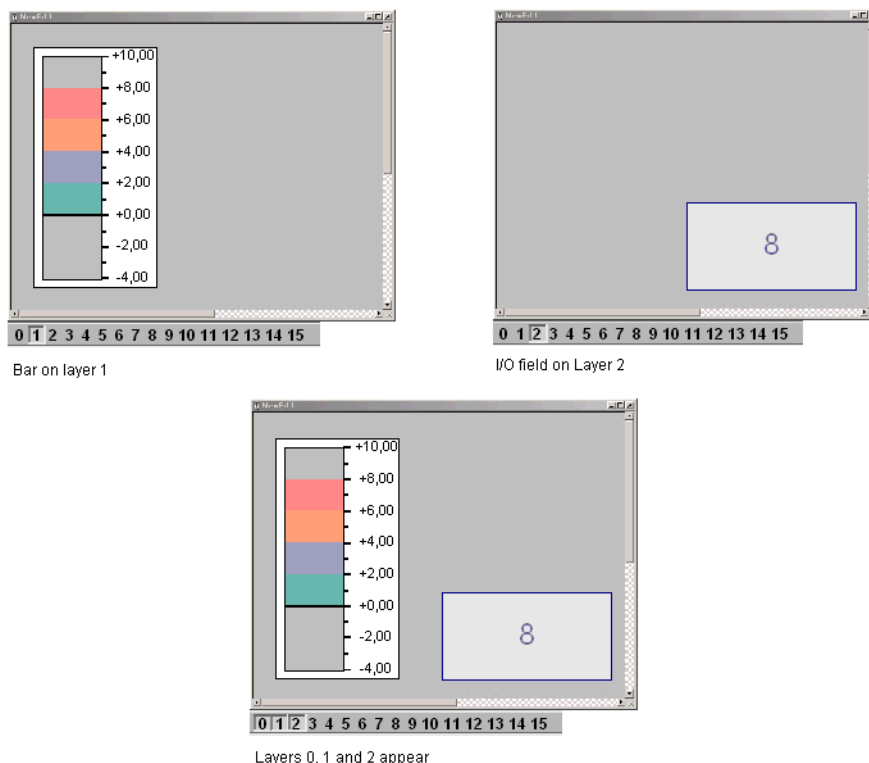
Layer Techniques - Principles

When a worldmap is opened, all 32 layers of the worldmap are displayed by default. The layer palette is used to hide all but the active layer. In this manner, it is possible to concentrate on editing the objects of the active layer.

Layers are useful when preparing worldmaps that contain many different types of objects. For example, you could place all the bar objects on layer 1 and all the I/O Field objects on layer 2.

If you later decide to change the font color of all the I/O fields, display the layer 2 only. Then select all the objects on this layer. The time-consuming selection of individual I/O fields scattered throughout the worldmap is thus unnecessary.

Working with layers

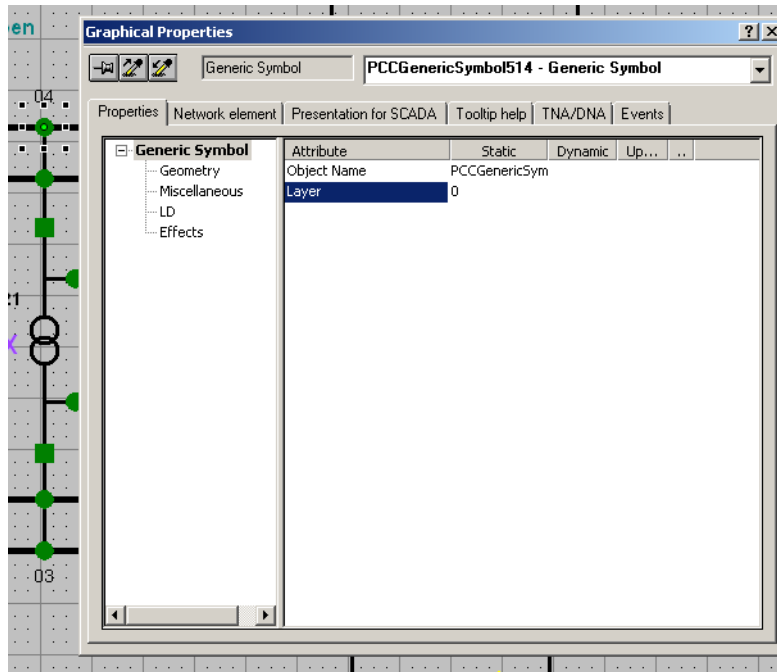


[immrg022-091208-01.tif, 1, en_US]

Figure 3-27 Working with Layers

Assign a Layer

The **Graphical Properties** window is used to assign the layer in which an object is placed.



[immrg050-140711-02.tif, 1, en_US]

Figure 3-28 Graphical Properties Window

Multiple Selection

Using multiple selection you can move several objects from different layers to a single common layer. For example, there are objects on layers 2, 5, and 7, which are moved to Layer 9.

Hide all the layers except layers 2, 5, and 7, select the objects and then assign Layer 9 to the object type **Multiple Selection** in the **Graphical Properties** window. All the objects are moved to Layer 9.

Layer Palette

With the aid of the layer palette, it is possible to display only those layers that are used for editing a worldmap. You can hide up to 31 layers. The active layer remains visible.

3.2.10 Multiple Worldmaps

3.2.10.1 Transfer Worldmap Properties to Another Worldmap

Except for the display of the grid and the grid width, the properties of a worldmap are transferred to other worldmaps using the **Pipette**.

3.2.10.2 Transfer Objects to Another Worldmap

With **Cut** and **Paste**, you can cut-out a selected object and paste it from the operating systems clipboard. Pasting from the clipboard, it can be copied into any worldmap. The cut object remains visible in the source worldmap as long as it is pasted onto the target worldmap. The mouse pointer position determines the paste location.

Cut and paste using **Drag** and **Drop** from one worldmap to another is supported too.

Multiple Selection

It is possible to cut-out several selected objects and paste them into another worldmap.

3.2.10.3 Copy Objects to Another Worldmap

With **Copy** and **Paste**, a selected object is copied to the clipboard and pasted from there into any worldmap. The advantage of copying to the clipboard is that the object can be inserted several times, and into different worldmaps. Drag and drop from one worldmap to another is supported too. The mouse pointer position determines the paste location.

Copy and paste using **Drag** and **Drop** for graphical objects linked to IMM domain data from one worldmap to another is supported too.

Multiple Selection

It is possible to simultaneously copy several selected objects and paste them into another worldmap.

4 Working with Objects

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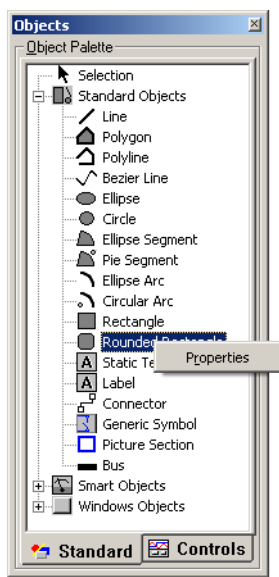
4.1 Operational Steps

4.1.1 Changing the Default Settings of Object Types

The goal of this task is to change default settings of object types according to specific needs.

The section shows the general procedure to view the default setting for the object types. Changing the default properties is based on changing the individual attributes of an object.

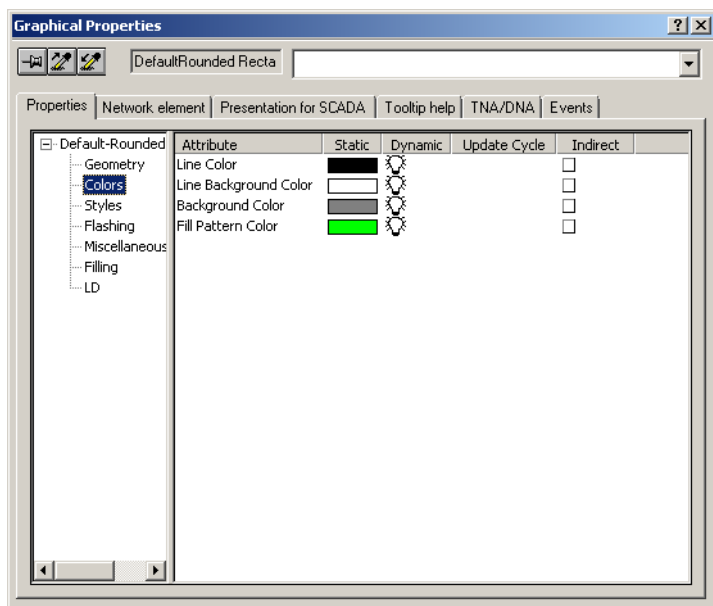
- Right-click in the object palette on the object type whose default setting you want to change.



[immge001-260510-01.tif, 1, en_US]

Figure 4-1 Object Palette

- Select the entry **Properties** in the context menu.
The window **Graphical Properties** opens.
The name of the default object type is displayed in the toolbar of this window.



[immrg051-190711-01.tif, 1, en_US]

Figure 4-2 Graphical Properties Window

- Select the property group in the **Properties** tab that contains the attribute you want to change to a new default value.
- Double-click the desired attribute in the **Attribute** column.
A dialog to enter the new value opens. You can enter the new value directly or the value immediately toggles to the new value.
- Alternatively, double-click on the value in the **Static** column to enter the new value directly or the value immediately toggles to the new value.
- After confirmation or immediately in case of toggle cells, the **Static** column shows the new default value.



NOTE

The default setting for the object types applies across the project. The changed default settings only apply to the images of the currently opened WinCC project. The basic settings of the **Graphics Editor** are not changed.

4.1.2 Basic Static Operations

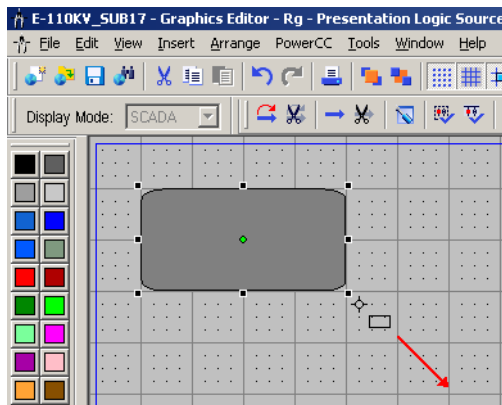
4.1.2.1 Inserting an Object from the Object Palette

The goal of this task is to insert an object from the object palette into a worldmap.

As an example, the standard object, rounded rectangle is used here to show the general procedure on how to insert an object in a worldmap. Some object types require additional steps. You can learn more about these additional steps in the detailed description of the individual objects.

- Click the **Rounded Rectangle** standard object in the **Object Palette**.
- Place the mouse pointer at the position in the worldmap where you want to insert the rounded rectangle. The mouse pointer changes into a crosshair with an object symbol attached.
- Drag the rounded rectangle to the desired size.

As soon as you release the mouse, the **Rounded Rectangle** is added.



[jimmrg051-190711-02.tif, 1, en_US]

Figure 4-3 Inserting and Sizing an Object

When inserting an object, the object name is automatically assigned with a continuous number followed by the name of the used object type. You can change the object name using the **Object Name** attribute in the **Properties** tab of the **Graphical Properties** window.

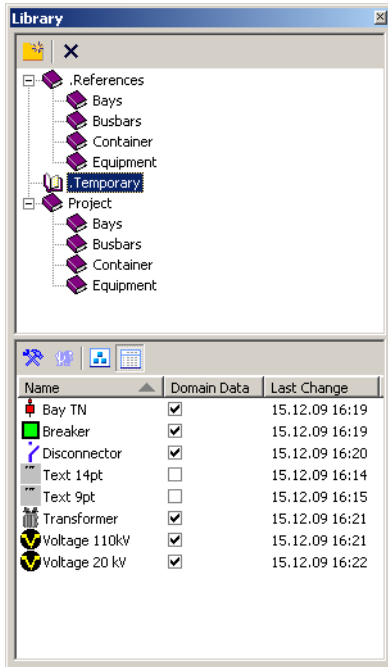
Alternative Operation

- Double-click the required object in the object palette. The new object is inserted close to the left upper corner of the viewport (the current visible area of the worldmap). The properties of the new object correspond to the default settings for the selected object type.

4.1.2.2 Inserting an Object from the Library

The goal of this task is to import library objects into a worldmap.

- In the default toolbar of the **Graphics Editor** click the icon  if it is not yet selected. The **Library** window opens.



[rgimm019-180210-01.tif, 1, en_US]

Figure 4-4 Layout of the Library

- Select a subdirectory of the library in the library structure area. The library objects in this subdirectory are shown in the library objects details area.
- Select a library object and drag it with the mouse to the desired position in the worldmap. When you release the mouse button, the graphical insertion procedure is finished.
- Later additional configuration might be required for that new object especially if linked domain data is included.

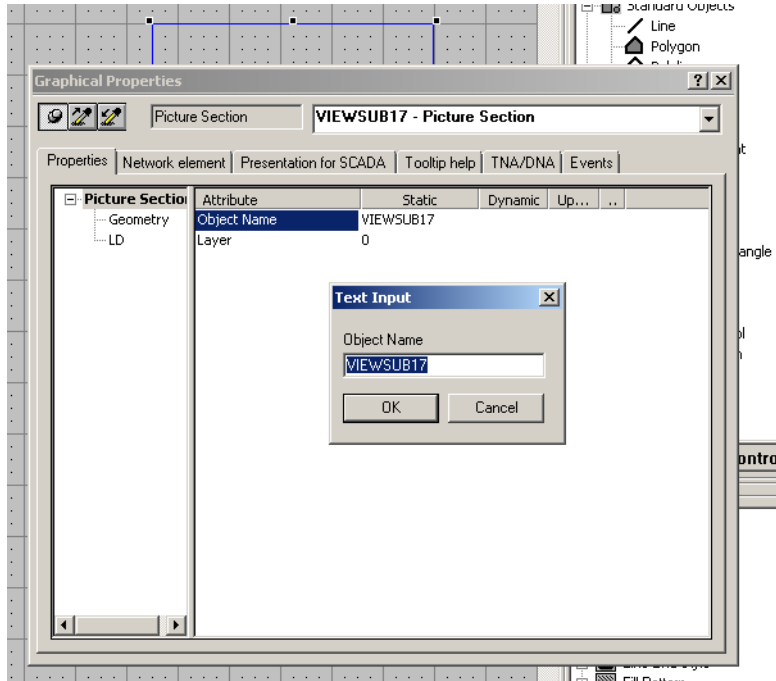
Basically the same mechanisms as inserting from the **Object Palette** are provided.

For additional information regarding linked domain data and technological editing, refer to chapter Technological Editing.

4.1.2.3 Renaming Objects

The goal of this task is to rename an existing object in a worldmap. The name of the object is used, for example, in configuring C actions to call the object.

- Select an object and open the **Graphical Properties** window.
- Select the object type within the **Properties** tab.



[immg016-190711-01.tif, 1, en_US]

Figure 4-5 Graphical Properties Window

- Double-click the object name attribute.
The **Text Input** dialog opens.
- Enter the new name.
- Click **OK** to confirm your entries.



NOTE

Special characters in object names are refused. The name has to comply the rules of naming IMM instances. For additional information, refer to Reference Guide Information Model Management (IMM).

4.1.2.4 Selecting an Object

The goal of this task is to select an existing object in a worldmap to change the properties of an object.

- Position the mouse pointer on the required object.
- Click the object.
The handles as well as the reference point of the object are displayed.

Alternative Operation

- Open the **Graphical Properties** window.
- On the toolbar of the window, open the list box to see all objects within the worldmap.
- Select the desired object within the list box.
The worldmap is moved to show the desired object in the center of the viewport.
The handles as well as the reference point of the object are displayed.

4.1.2.5 Selecting Multiple Objects

The goal of this task is to change properties of several objects in a worldmap at once. Therefore all the objects to be changed must be selected.

Multiple Selection Using < SHIFT >

- Hold down < **SHIFT** > key.
- With the mouse, click the objects you want, one after the other.
The handles of the rectangle surrounding the multiple objects appear. Within a multiple selection, the rectangle surrounding the objects are displayed with a red dashed line together with the individual reference points. A master reference point is shown in the center of the union rectangle surrounding the objects. The mouse pointer changes to a crosshair with arrow points when it is positioned within the union rectangle surrounding the objects.



NOTE

If an object was selected unintentionally, it can be removed from the multiple selection by again clicking the object while holding down the < **SHIFT** > key.

Alternative Operation

The multiple selection can also be made by dragging with the mouse. The procedure depends on the setting for the type of object selection:

- **Surrounding object selection**
All objects inside the displayed frame are selected.
- **Touching object selection**
All objects touched by the frame are selected, in addition to those inside of it.

For additional information on defining the type of object selection, refer to [2.2.23 Options Settings](#).

The procedure described here is for the surrounding object selection type.

- Position the mouse pointer on the worldmap close to one of the required objects.
- Drag a frame around the desired objects while holding down the mouse button.
- As soon as you release the mouse button, the rectangle surrounding the objects is displayed with a red dashed line together with the individual reference points. A master reference point is shown in the center of the union rectangle surrounding the objects.



NOTE

You can combine both alternatives for selecting multiple objects in any order.



NOTE

<**CTRL-a**> or **Edit > Select All** in the menu bar selects all objects within the worldmap in the current activated layers. Even not visible objects within the current viewport are selected.

4.1.2.6 Positioning Objects

The goal of this task is to reposition an existing object or a multi selection of objects already placed within a worldmap.

- Position the mouse pointer over the selected object or an object within a multi selection.
The mouse pointer changes to a crosshair with arrowheads.
- Click the object and hold the left mouse button pressed.
The mouse pointer changes into an arrow with an outline icon (small rectangle with broken line).

- Move the mouse pointer.
The objects are displayed and show a possible new position for the objects. The objects itself initially keep its original position.
- Release the mouse button.
The objects are moved to the position previously displayed. The mouse pointer again changes into a crosshair with arrow ends.

Alternative Operation

- Move the selected objects using the arrow keys for mouse pointer control.
- Alternatively, change the value for the geometric **Position X** and **Position Y** attributes in the **Graphical Properties** window.



NOTE

In the settings of the **Graphics Editor**, if the **Snap to grid** function is activated, the object can only be positioned with the mouse according to the settings for the grid. In that case the reference point for a single object or the master reference point of multi selected objects are moved according to the grid points.

Respecting Snap to Grid

If the snap to grid function is activated make sure that the reference point or the master reference point in case of a multi selection is correctly set on a grid point.

- Select a single object or select multiple objects.
- Move the reference point with **<CTRL+SHIFT+LButtonDown>** to the desired position within the rectangle surrounding the objects.
- Reposition the single object or the multi selection of objects.



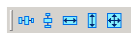
NOTE

Shifting the reference point of a single object makes the new reference point permanent.
Shifting the master reference point of a multi selection is temporary as long as the multi selection exists.

4.1.2.7 Aligning Multiple Objects

The goal of this task is to arrange existing objects already placed within a worldmap.

- Select at least 2 objects.
- In the **Alignment Palette**, click the required icon or on the **Arrange / Align** menu select the required entry for the aligning operation.



[rgimmge3-240611-01.tif, 1, en_US]

Figure 4-6 Alignment Palette

The alignment or the size of the selected objects changes accordingly. The master reference point of the multi selection is used for the operations.

4.1.2.8 Resizing (Scaling) an Object

The goal of this task is to resize an existing object or a multi selection of objects already placed within a worldmap.

- Position the mouse pointer over a handle of the selected object or a handle of the rectangle surrounding the objects of a multi selection. The mouse pointer changes to double arrow. The alignment of the double arrow indicates the directions in which you can move the handle:
 - **Horizontal double arrow**
The handles on the horizontal center axis can be used to change the width of the object.
 - **Vertical double arrow**
The handles on the vertical center axis can be used to change the height of the object.
 - **Diagonal double arrow**
The handles on the corners of the object can be used to change the height and width of the object.
- Use the mouse to drag the handle to the position you want.



NOTE

In the settings of the **Graphics Editor**, if the **Snap to Grid** function is activated, the object can only be scaled with the mouse according to the settings for the grid. In that case the reference point for a single object or the master reference point of multi selected objects are moved according to the grid points.

Alternative Operation

- Change the values for the geometric **Width** and **Height** attributes in the **Graphical Properties** window.

Respecting Snap to Grid

If the snap to grid function is activated make sure that the reference point or the master reference point in case of a multi selection is correctly set on a grid point.

- Select a single object or select multiple objects.
- Move the reference point with **<CTRL+SHIFT+LButtonDown>** to the desired position within the rectangle surrounding the objects.
- Resize the single object or the multi selection of objects.



NOTE

Shifting the reference point of a single object makes the new reference point permanent.
Shifting the master reference point of a multi selection is temporary as long as the multi selection exists.

4.1.2.9 Mirroring the Objects

The goal of this task is to mirror existing objects already placed within a worldmap on its horizontal or vertical center axis.

- In the **Standard toolbar** click the icon .
Selected object gets mirrored on its horizontal center axis defined by the reference point.
- In the **Standard toolbar** click the icon .
Selected object gets mirrored on its vertical center axis defined by the reference point.

Alternative Operation

- Use the entries **Horizontal** and **Vertical** on the **Arrange / Mirror** menu.



NOTE

It is possible to mirror several selected objects simultaneously. The mirroring occurs on the horizontal or vertical middle axis defined by the master reference point.

Respecting Snap to Grid

If the snap to grid function is activated make sure that the reference point or the master reference point in case of a multi selection is correctly set on a grid point.

- Select a single object or select multiple objects.
- Move the reference point with **<CTRL+SHIFT+LButtonDown>** to the desired position within the rectangle surrounding the objects.
- Mirror the single object or the multi selection of objects.



NOTE

Shifting the reference point of a single object makes the new reference point permanent.
Shifting the master reference point of a multi selection is temporary as long as the multi selection exists.

4.1.2.10 Rotating the Objects Clockwise

The goal of this task is to rotate existing objects already placed within a worldmap around their reference point.

- In the **Standard toolbar** click the icon .

Selected object gets rotated clockwise around its reference point. It rotates 90 degrees.

Alternative Operation

- Use the entry **Rotate** in the **Arrange** menu.



NOTE

- It is possible to rotate several selected objects simultaneously. The objects are rotated around their master reference point.
- Make sure that the reference point or the master reference point are correctly set.
- Some objects cannot be rotated by nature.

4.1.2.11 Rotating the Objects in Free Rotation Mode

The goal of this task is to rotate existing objects already placed within a worldmap freely around their reference point.

- In the **Standard Toolbar** click the icon .
- The free rotation mode gets enabled.
The rotation handles of the rectangle surrounding the object appear.
- Position the mouse pointer over a rotation handle of the selected object.
The mouse pointer changes into a ring shaped arrow.
- Use the mouse pointer to drag the handle to the position you want.
The object is rotated around its reference point.



NOTE

- It is possible to rotate several selected objects simultaneously. The objects are end rotated around their master reference point.
- Make sure that the reference point or the master reference point are correctly set.
- Some objects cannot be rotated by nature.

4.1.2.12 Deleting Objects

The goal of this task is to delete existing objects including linked domain data already placed within a worldmap.

- Press **<SHIFT + DEL>** to remove the selected object including linked domain data from the worldmap.



NOTE

If graphical objects are linked to domain data, the domain data in IMM gets also deleted. To prevent deleting linked domain data use **Delete Graphic Only** on the context menu of the object. Additional steps may be needed to be performed. Refer to Technological Editing.

Alternative Operation

- Use the entry **Delete** on the context menu of the object or on the **Edit** menu.



NOTE

It is possible to delete several selected objects simultaneously.

4.1.2.13 Deleting Objects Only Graphically

The goal of this task is to delete existing objects already placed within a worldmap.

- Press **** to remove the selected object from the worldmap.



NOTE

If graphical objects are linked to domain data, the domain data in IMM is not deleted and stays as they are. Refer to Technological Editing.

Alternative Operation

- Use the entry **Delete Only Graphic** on the context menu of the object or on the **Edit** menu.



NOTE

It is possible to delete several selected objects simultaneously.

4.1.2.14 Cutting and Pasting Objects

The goal of this task is to cut and transfer existing objects together with possible linked domain data already placed within a worldmap and paste it elsewhere.

- Use the keyboard shortcut **<CTRL + X>** to cut the selected object from the worldmap and copy it to the clipboard.
The cut object remains visible in the worldmap.
- Use **<CTRL + V>** to paste the current contents of the clipboard in the active worldmap under the mouse position.
The former cut object is deleted within the original position and inserted at the paste position.

Alternative Operation

- Click  in the standard toolbar or use the entry **Cut** in the context menu or in the **Edit** menu.
- Click  in the standard toolbar or use the entry **Paste** on the context menu or on the **Edit** menu.



NOTE

It is possible to cut several selected objects simultaneously.



NOTE

If graphical objects are linked to domain data, the domain data is cut and pasted.
For additional information on the steps to be performed, refer to Technological Editing.

4.1.2.15 Copying Objects

The goal of this task is to copy selected object together with possible linked domain data into a worldmap.

- Use **<CTRL + C>** to copy the selected object from the worldmap to the clipboard.
- Use **<CTRL + V>** to copy the current contents of the clipboard in the active worldmap under the mouse position.

When inserting a copied object, an object name is automatically assigned. The name of the object is a continuous number followed by the name of the used object type. You can change the object name using the object name attribute.

- Repeat this step to insert further copies of the contents.

Alternative Operation

- Click  in the standard toolbar or use the entry **Copy** on the context menu or on the **Edit** menu.
- Click  in the standard toolbar or use the entry **Paste** on the context menu or on the **Edit** menu.



NOTE

It is possible to copy several selected objects simultaneously.



NOTE

If graphical objects are linked to domain data, the domain data gets also copied.
For additional information on the steps to be performed, refer to Technological Editing.

4.1.2.16 Changing the Position of an Object Within a Layer

The goal of this task is to bring an object within a worldmap in front or back of other overlapping objects.

Use one of the following shortcuts to change the position of the selected object:

- **<CTRL> and <+>**
Bring to front
The selected object is positioned in front of all the other objects contained in the worldmap layer.
- **<CTRL> and <->**
Send to back
The selected object is positioned behind all the other objects contained in the worldmap layer.

Alternative Operation

- Select the required entry on the **Arrange > Within the Layer** menu to change the position of the selected object.

- In the standard toolbar, click , to position the selected object in the front.
- In the standard toolbar, click , to position the selected object in the back.



NOTE

It is possible to change the position of several selected objects simultaneously. This modification always applies to the reference object relative to the position of the unselected objects. An individual position behind the reference object is assigned to each selected object.



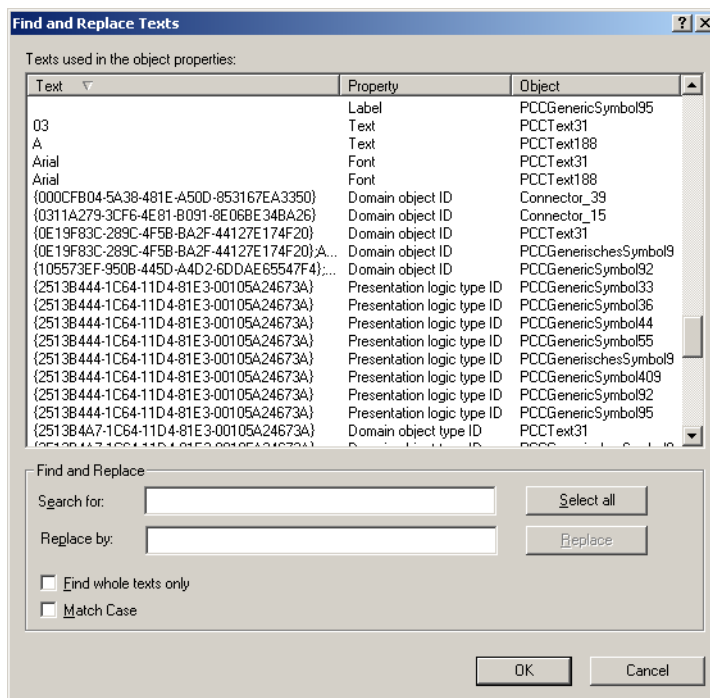
NOTE

Multiple objects of the **Control** type generally are in the foreground.

4.1.2.17 Changing the Text Contents of Objects

The goal of this task is to rename configured text contents of attributes for objects in a worldmap.

- Select all the objects whose text contents you want to change.
If you do not select an object, all the objects in the active worldmap are evaluated.
- Select **Linking >Texts...** in the **Edit** menu or from the context menu of the selection or of the active worldmap.
The **Find and Replace Texts** window opens.



[immrg181-140409-01.tif, 1, en_US]

Figure 4-7 Find and Replace Texts

- Click the column name according to texts, properties, or objects in the three-columned upper area of the dialog to sort the list.
- In the **Text** column, select the text contents to be changed.
- Click the text once to activate the edit function.
- Enter the new text contents and confirm by pressing **<ENTER>**.

You can select certain text contents using the **Find** function in the lower area of the dialog:

- Click **Select All** if you want to change the text contents displayed.
- Enter the text or parts of text you want to search for in the search field.
You can also activate the options **Find whole texts only** and **Match case**. All text contents which do not meet the search criteria are hidden.

You can replace the selected text contents using the **Replace** function in the lower area of the dialog:

- Enter the new text in the field and click **Replace**.



NOTE

The **Replace** function replaces only the part of the text contents displayed in the **Find** field.



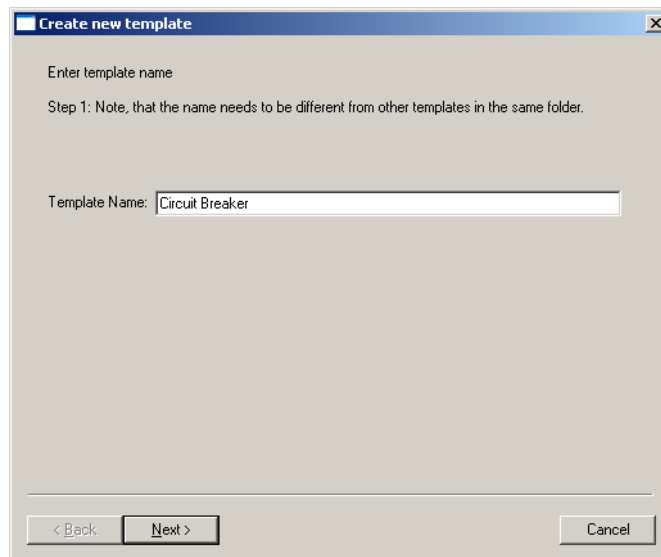
NOTE

Grouped objects can only be linked if there are no grouped objects in the group itself.

4.1.2.18 Storing Objects Into the Library as Templates

The goal of this task is to import objects from a worldmap into the library of the **Graphics Editor**. The library objects can be used afterwards as templates for the worldmap editing.

- In the default toolbar of the **Graphics Editor** click  if it is not yet selected. The **Library** window opens.
- Select or create a subdirectory of the library in the library structure area. The library objects in this subdirectory are shown in the library objects details area.
- Select the graphical object that you want to add to the selected folder in the active worldmap.
- Hold down the mouse button and drag the selected object into the library details area. When you release the mouse button, the **Create new template** window to define the name of the template is opened.



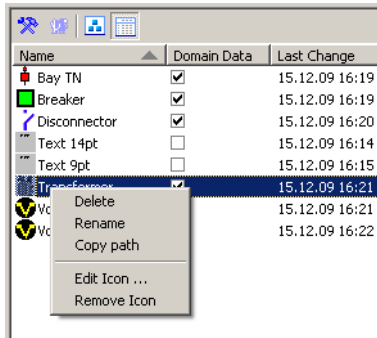
[immg128-140409-01.tif, 1, en_US]

Figure 4-8 Create New Template - Name of Template

- Define an unambiguous name for the new template within the library subdirectory and click **Next >**.
- Later additional configuration is required for that new template if also linked domain data is included. If so, the second step of the **Create new template** window for further configuration issues opens automatically.

For additional information regarding technological editing, refer to Technological Editing.

- The library object gets created.
- If you want to assign a user-defined icon for the new library object, select **Edit Icon...** in the context menu of the new library object.



[immrg018-110210-02.tif, 1, en_US]

Figure 4-9 Library Element - Context Menu

The **OPEN** menu to browse for a new icon appears.

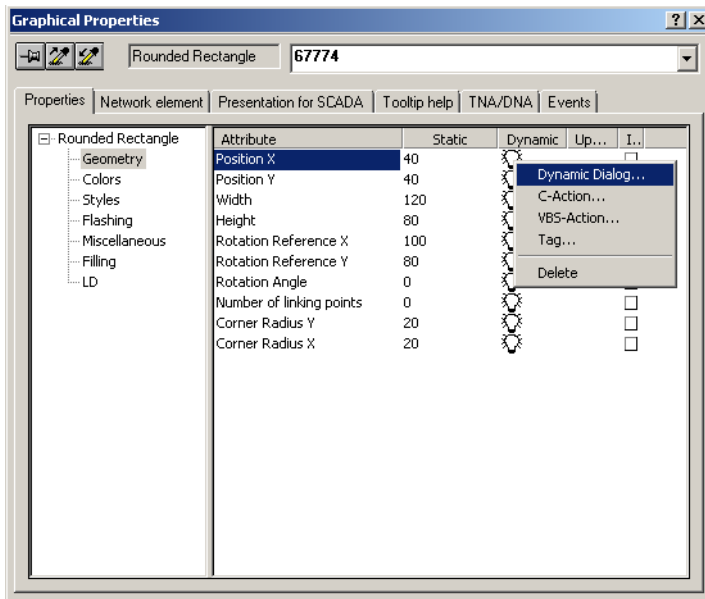
- Browse for the icon, select the desired icon and confirm with **Open**.

4.1.3 Basic Dynamic Operations

4.1.3.1 Making Attributes Dynamic

The goal of this task is to make attributes for objects in a worldmap dynamic.

- Open the **Graphical Properties** window and select the **Properties** tab.
- Select the property group containing the attribute to be made dynamic.
- Right-click the dynamic icon of the required attribute in the **Dynamic** column.



[rgimm021-190711-01.tif, 1, en_US]

Figure 4-10 Graphical Properties – Dynamic Attributes

- Select the required dynamics type on the context menu.
The associated dialog opens.
There is a brief description of the dynamics types in this chapter.



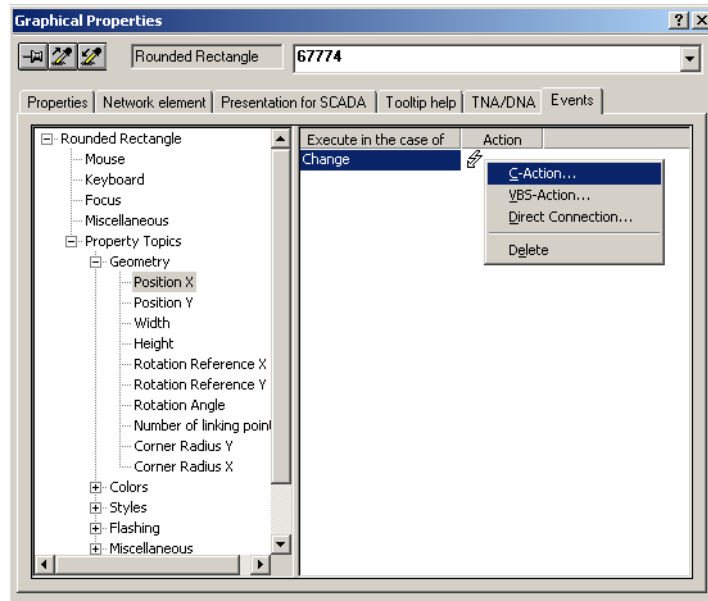
NOTE

This configuration is native WinCC functionality and does not have that importance within the Spectrum Power 5 environment.

4.1.3.2 Configuring Events

The goal of this task is to define events for objects in a worldmap to trigger VBS actions or C actions.

- Open the **Graphical Properties** window and select the **Events** tab.
- Select the event group containing the event to be configured.
- Right-click the action icon of the required event in the **Action** column.



[rgimm022-190711-01.tif, 1, en_US]

Figure 4-11 Graphical Properties – Dynamic Actions

- Select the required dynamics type on the context menu.
The associated dialog opens.
There is a brief description of the action types in this chapter.



NOTE

This configuration is native WinCC functionality and does not have that importance within the Spectrum Power 5 environment.

4.1.3.3 Configuring a Dynamic Dialog

The goal of this task is to define events for objects in a worldmap to trigger VBS actions or C actions.

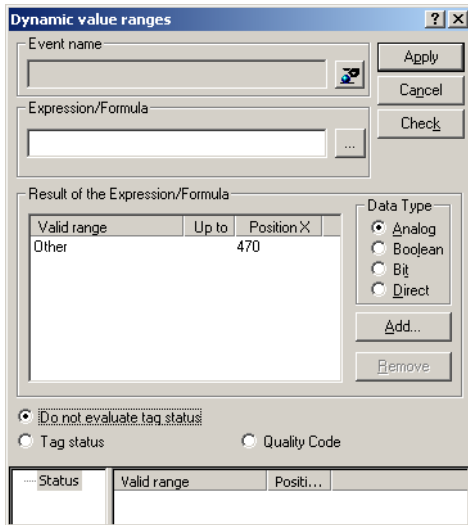
A dynamic dialog is used to achieve high runtime performance when making attributes dynamic. A C action is automatically generated from the dynamic dialog. This can however be extended later. The advantage of higher runtime performance is then lost.



NOTE

This configuration is native WinCC functionality and does not have that importance within the Spectrum Power 5 environment.

- Open the **Graphical Properties** window and select the **Properties** tab.
 - Select the property group containing the attribute to be made dynamic.
 - Right-click the dynamic icon of the required attribute in the **Dynamic** column.
 - Select the **Dynamic Dialog...** entry in the context menu.
- The **Dynamic value ranges** window opens.



[immrg182-140409-01.tif, 1, en_US]

Figure 4-12 Dynamic Value Ranges

- Configure the dynamics as required.
- Click  to open the **Change trigger** dialog.
- Click the button next to Expression/Formula to create the formula from the configured tags, global script functions, and operators.
- Click **Check** to check the formula for errors.
- Select the data type for the evaluation of the event.
- Activate the check box **Tag status** to view the current values of the tags in the formula.
- Click **Apply** to store your configuration.

4.1.3.4 Configuring a C Action

The goal of this task is to define a C action for objects in a worldmap.

A C action is used to make attributes and events dynamic. The powerful script language ANSI-C opens almost endless options for dynamics. However, the runtime performance achieved is lower than with other dynamic types. Therefore, before using a C action check whether the required dynamics can be implemented with another dynamics type.



NOTE

This configuration is native WinCC functionality and does not have that importance within the Spectrum Power 5 environment.



NOTE

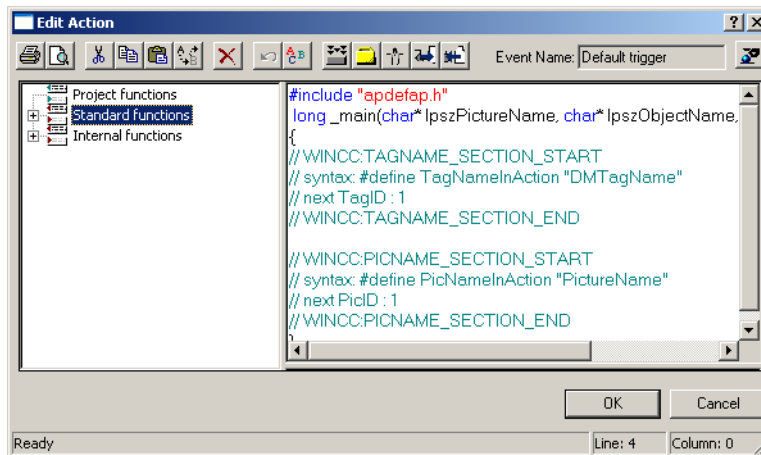
The configuration of C Actions is not supported for the Web Navigator. It is recommended to use VBScript on Web Navigator instead.

Attribute

- Open the **Graphical Properties** window and select the **Properties** tab.
- Select the property group containing the attribute to be made dynamic.
- Right-click the dynamic icon of the required attribute in the **Dynamic** column.
- Select the **C Action...** entry from the context menu.
The **Edit Action** window opens.

Event

- Open the **Graphical Properties** window and select the **Events** tab.
- Select the event group containing the event to be configured.
- Right-click the action icon of the required event in the **Action** column.
- Select the **C Action...** entry from the context menu.
The **Edit Action** window opens.



[immg183-140409-01.tif, 1, en_US]

Figure 4-13 Edit Action

- Configure the dynamics as required.
- Click  to open the **Change trigger** dialog.
- Edit the program code of the C action in the editor window.
- Double-click the required function to configure the contained parameters with the **Set Parameters** dialog.
- Click **OK** to insert the function with the new values at the position of the mouse pointer in the editor window to confirm the new parameters.

4.1.3.5 Configuring a VBS Action

The goal of this task is to define a VBS action for objects in a worldmap.

A VBS action is used to make attributes and events dynamic. The script language of a VBS action is Visual Basic.



NOTE

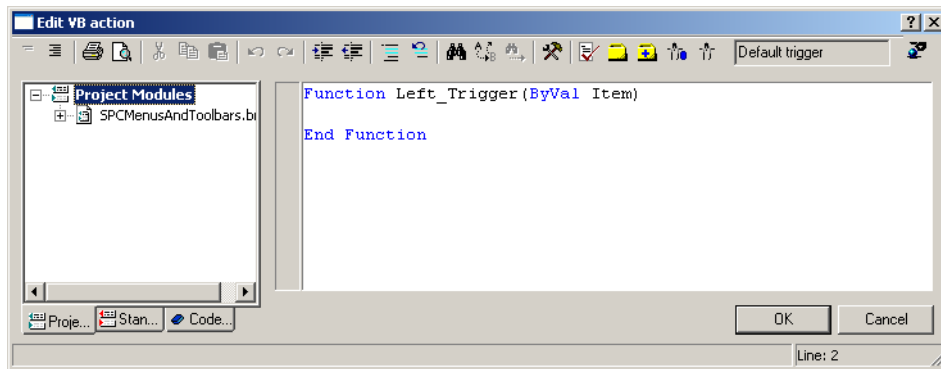
This configuration is native WinCC functionality and does not have that importance within the Spectrum Power 5 environment.

Attribute

- Open the **Graphical Properties** window and select the **Properties** tab.
- Select the property group containing the attribute to be made dynamic.
- Right-click the dynamic icon of the required attribute in the **Dynamic** column.
- Select the **VBS Action...** entry from the context menu.
The **Edit VB action** window opens.

Event

- Open the **Graphical Properties** window and select the **Events** tab.
- Select the event group containing the event to be configured.
- Right-click the action icon of the required event in the **Action** column.
- Select the **VBS Action...** entry from the context menu.
The **Edit VB Action** window opens.



[immg217-150409-01.tif, 1, en_US]

Figure 4-14 Edit VB Action

- Configure the dynamics as required.
- Click  to open the **Change trigger** dialog.
- Edit the program code of the VBS action in the editor window.
- You can use the already available modules to create the VBS actions.
- Click **OK** to insert the function with the new values at the position of the mouse pointer in the editor window to confirm the new parameters.

4.1.3.6 Configuring a Tag Connection

The goal of this task is to define a Tag Connection for objects in a worldmap.

Attributes can be made dynamic using a tag connection. Connecting an attribute to a process tag for example enables the change of the measured value to be displayed graphically.



NOTE

This configuration is native WinCC functionality and can only be used in a limited way within the Spectrum Power 5 environment.

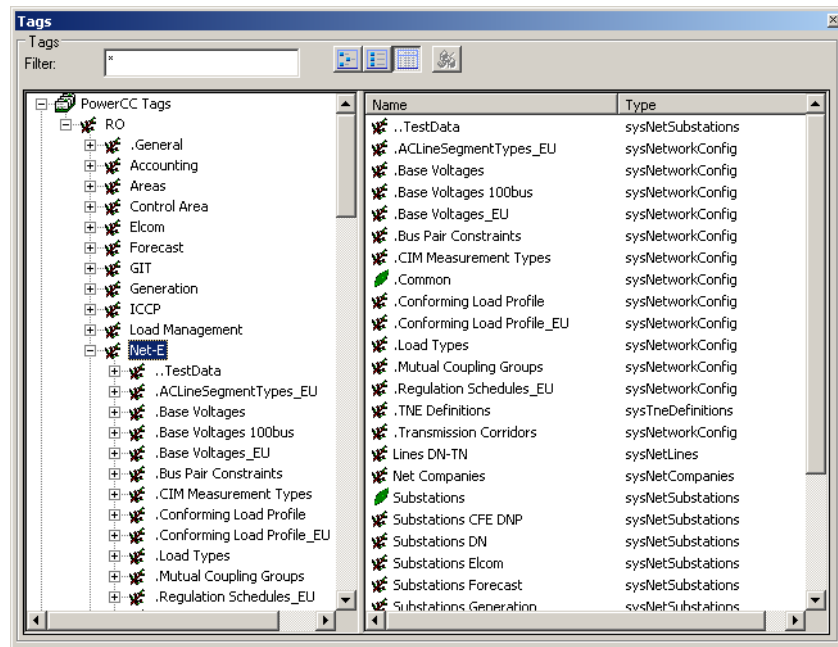
WinCC and PowerCC Tags can only be read during Runtime. Writing data or setting values is not possible within the Spectrum Power 5 environment.

Tags can be assigned to certain objects (for example, bar chart, slider) that are not Spectrum Power 5 specific to use values as dynamic data source. For this purpose set the flag **WinCC tag enabled** to true for the

respective measurement in the **Information Model Management** to determine that this measurement can be used as tag.

- Open the **Graphical Properties** window and select the **Properties** tab.
- Select the property group containing the attribute to be made dynamic.
- Right-click the dynamic icon of the required attribute in the **Dynamic** column.
- Select the **Tag...** entry from the context menu.

The **Tags** window opens.



[immg184-190711-01.tif, 1, en_US]

Figure 4-15 Tags Window

- Configure the dynamics as required.
- Click the + or - icons to open or close directories and subdirectories.
The tags available for the selected entry are shown in the tag display.
 - Click the respective column heading to sort the tags.
- Select the required filter. An asterisk means that no filter is used or no filter has been defined.
- Open the desired tag group in the left-hand section of the window.
- Select the desired tag from the right-hand section of the window.
- Confirm your selection with **OK**.

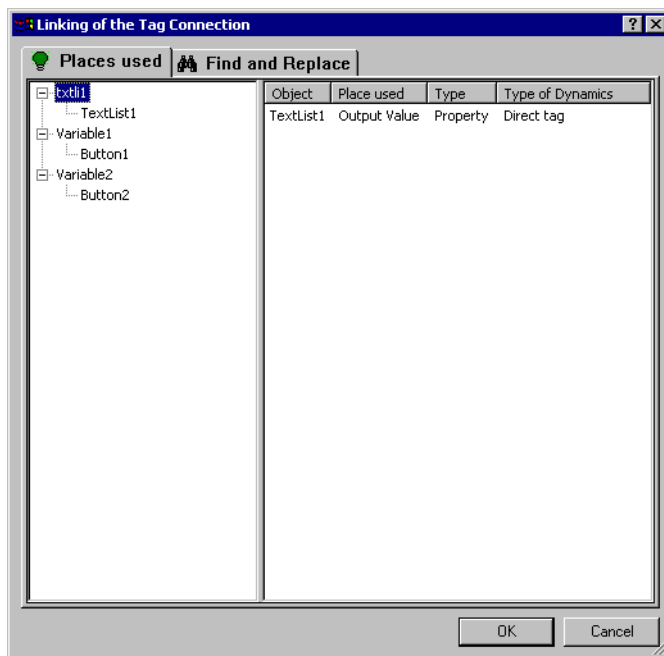
The dialog **Tags** is closed. The selected tag is assigned to the selected object and you can continue with the configuration.

4.1.3.7 Linking the Tag Connection of Objects

The goal of this task is to reconfigure the tag connection for all selected objects in a worldmap.

- Select all the objects whose tag connection contents you want to link.
- Select **Linking >Tag Connection** from the objects context menu of the selection or on the **Edit** menu of the **Graphics Editor**.

The **Linking of the Tag Connection** window opens.

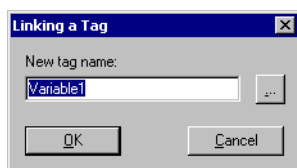


[immrg023-101208-01.tif, 1, en_US]

Figure 4-16 Linking of the Tag Connection

Tag Selection

- Select a tag or object. The detail view shows the current tag connections.
- Right-click the required tag connection to call the linking command.
The **Linking a Tag** dialog opens.

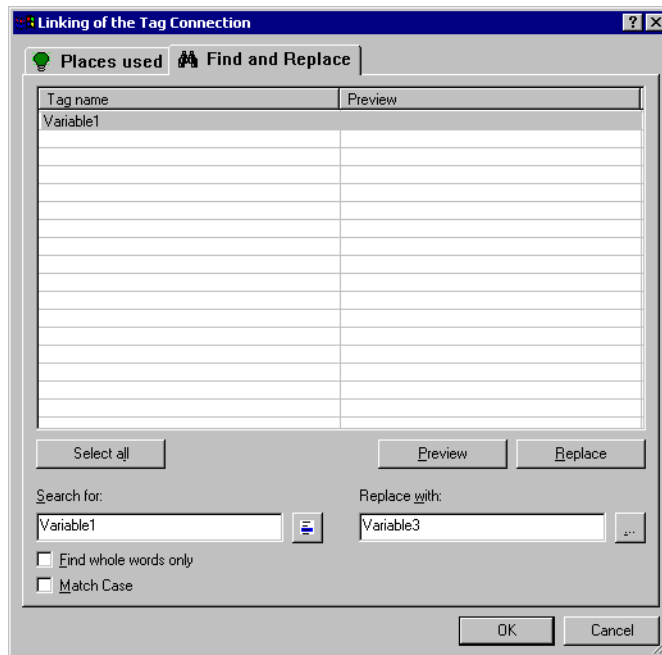


[immrg019-161208-01.tif, 1, en_US]

Figure 4-17 Linking a Tag

- Enter the name of the new tag or click the associated button to select a tag.

Linking Tags by Searching and Replacing



[immrg020-161208-01.tif, 1, en_US]

Figure 4-18 Linking of the Tag Connection - Find and Replace Tab

- Click **Select All** if you want to link the texts displayed.
- Select a tag name and click , if you only wish to change the connections to this tag. All remaining tag names are hidden.
- Enter the tag names or parts of names you want to search for in the search field. You can also activate the options **Find whole words only** and **Match case**. All tags which do not meet the search criteria are hidden.

In the right area of the **Find and Replace** tab, you can enter the name of the new tag with which the selected tag names should be replaced.

Alternatively you can click the associated button to select the required tag.

To replace a tag:

- Click **Preview** to view and check the planned replacement.
- Click **Replace** to link the selected tag connections as in the preview.



NOTE

The **Replace** function replaces only the part of the tag names contents displayed in the **Find** text box of the area.



NOTE

Grouped objects can only be linked if there are no grouped objects in the group itself.

4.1.3.8 Configuring a Direct Connection

The goal of this task is to define a direct connection for objects in a worldmap.

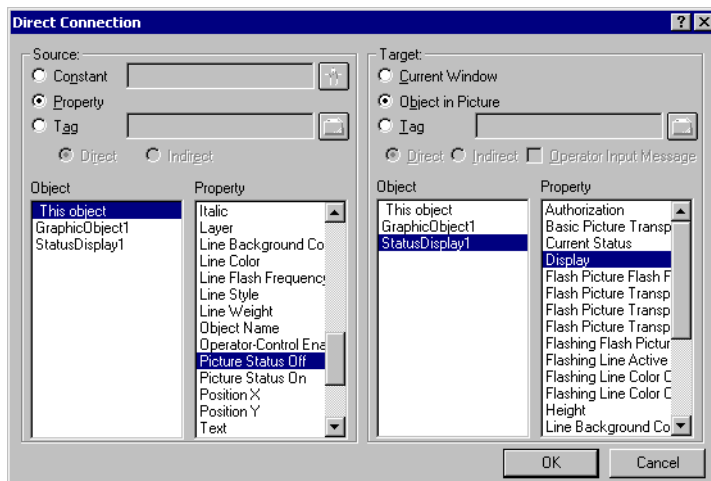
Events can be made dynamic using a direct connection. A direct connection offers the quickest dynamics in the worldmap and achieves the highest runtime performance. However the direct connection can only be used within a worldmap and only single connections can be created.



NOTE

This configuration is native WinCC functionality and does not have that importance within the Spectrum Power 5 environment. It can only be used in a limited way. WinCC and PowerCC Tags can only be read during Runtime. Writing data or setting values is not possible within the Spectrum Power 5 environment.

- Open the **Graphical Properties** window and select the **Events** tab.
 - Select the event group containing the event to be configured.
 - Right-click the action icon of the required event in the **Action** column.
 - Select the **Direct Connection...** entry from the context menu.
- The **Direct Connection** window opens.



[immrg024-101208-01.tif, 1, en_US]

Figure 4-19 Direct Connection

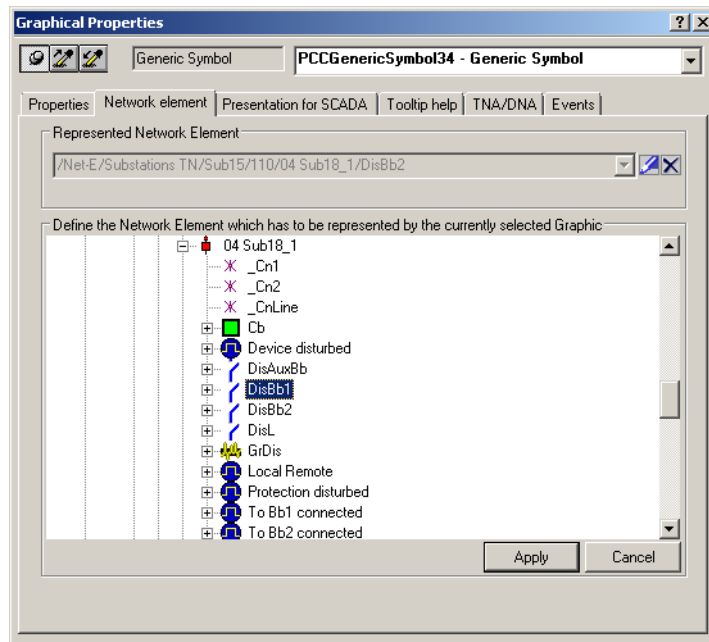
- Configure the dynamics as required.
 - Select a constant, object property or a tag as the data source.
 - Select **Current** window, **Object in Picture** or **Tag** as the target, which takes the value of the data source.
- Confirm your selection with **OK**.

4.1.3.9 Linking to Spectrum Power Domain Data

The goal of this task is to link graphic objects in a worldmap to existing Spectrum Power **Domain Data**. The connection of Spectrum Power objects is combined within the **Graphical Properties** window. Specific properties (linked domain object, representation, etc.) of an object can be defined within associated tabs of the **Graphical Properties** window.

- Double-click an object to display the **Graphical Properties** window.
- Alternatively, right click an object and select **Graphical Properties** to display the **Graphical Properties** window.

- Select the **Network Element** tab. The network domain object tree is displayed.



[immg023-190711-01.tif, 1, en_US]

Figure 4-20 Graphical Properties Window - Network Element

- Select the desired domain data object in the tree that shall be linked and click **Apply**.
- Click the **Presentation** tab to choose the desired presentation logic type for runtime and click **Apply** to confirm.
- Click the **TNA / DNA** tab to choose Spectrum Power 5 application data sources if SCADA is not used as data source at runtime.
- Click the **Tooltip Help** tab to configure the tooltip that is to be presented at runtime if the mouse hovers over the object.



NOTE

Not every object within the object palette can be linked to Spectrum Power domain data.



NOTE

Every change in the mentioned tabs forces the **Graphics Editor** to save the entire worldmap. This is needed to guarantee the integrity of graphical and domain data. The undo and redo buffer of the **Graphics Editor** is automatically reset in this case.

For additional information on Spectrum Power objects and technological editing, refer to Technological Editing.

4.1.4 Creating Dynamic Spectrum Power Linkages

4.1.4.1 Overview

Dynamic Spectrum Power linkages are defined by dragging and dropping standard graphic objects (like, Generic Symbol, Bus, etc.) into the drawing pane of Graphics Editor, and linking these objects to existing domain data objects and, finally, if applicable, defining the presentation logic and further configurations.



NOTE

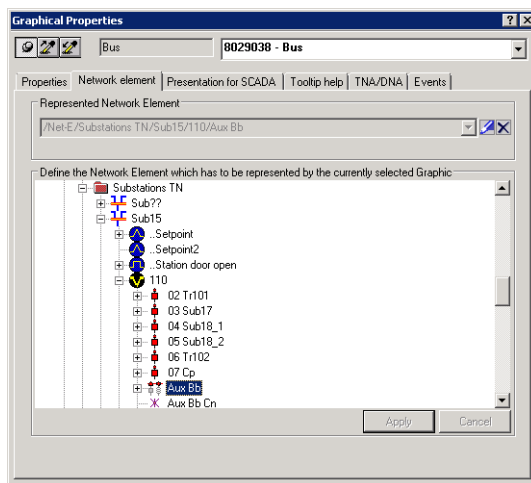
As long as there are no library templates available, you have to perform those steps to create the base for meaningful templates that can be reused afterwards for further worldmap editing. Templates provide you much more comfort and usability during technological editing while defining new or mapping graphic and domain data instances.

For additional information on Spectrum Power objects and technological editing, refer to Technological Editing.

4.1.4.2 Example: Busbar Linkage Creation

The goal of this task is to represent a busbar section already existing in the Spectrum Power **Domain Data** using the standard object **Bus**.

- Drag a bus object from the object palette into the worldmap.
- Draw the bus and modify certain properties (for example, Line Style and Line Weight) as desired.
- Double-click the bus representation to display the **Graphical Properties** window.
- Alternatively, right click on the object and select **Graphical Properties** to display the **Graphical Properties** window.
- Go to the **Network Element** tab, expand the tree view, and select the desired busbar section within the domain data hierarchy.



[rgimme1-050711-01.tif, 1, en_US]

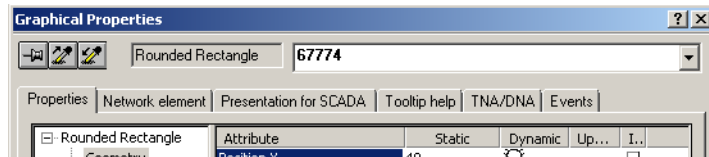
Figure 4-21 Network Element - Bus

- Click **Apply** to establish the link between the graphical representation and the domain data object. The **Represented Network Element** field shows the path of the linked domain data instance.
- Set other properties in the **Tooltip help** tab if required.

4.1.5 Properties of an Object

4.1.5.1 Working with the Graphical Properties Window

The goal of this task is managing the various options of the **Graphical Properties** window. The options are provided within the toolbar of the **Graphical Properties** Window.



[immrg024-190711-02.tif, 1, en_US]

Figure 4-22 Elements of the Toolbar in the Graphical Properties Window

Activating or Deactivating the Pin

- On the toolbar of the **Graphical Properties**, click . The button is activated and displays . The **Graphical Properties** window is pinned on top so that it is always in the foreground in the **Graphics Editor**. It remains visible even if another object on the worldmap is selected.
- On the toolbar of the **Graphical Properties**, click . The button is deactivated and displays . The **Graphical Properties** window is closed as soon as the graphical object on the worldmap is deselected.

Transferring Object Properties

- Select the object whose graphical properties you want to copy.
- In the standard toolbar, click . The graphical properties of the selected object are copied.
- Select the object to which you want to assign the copied properties.
- In the standard toolbar, click . The selected object is displayed with the new graphical properties, but without changing its geometric attributes. With different object types, only those attributes which both object types possess are changed.



NOTE

Except for the user object, you can copy the properties of the **Properties** tab of any object using the eyedropper. You can also transfer the copied properties to any object such as to a multiple selection, a group, or a user object.

These buttons are in the **Standard Toolbar** and in the **Graphical Properties** toolbar. Alternatively, these functions can be selected using the **Edit > Properties** menu.



NOTE

It is possible to transfer the value of a selected attribute to other attributes using the eyedroppers in the **Graphical Properties** window as well.

Calling Direct Help for the Graphical Properties Window

- In the title bar of the **Graphical Properties** window click .
- Click the window element for which you want to call the direct help. When available, a brief description of the element is shown.

4.1.5.2 Opening the Graphical Properties Window

The goal of this task is to demonstrate the various ways of opening the **Graphical Properties** window. There are 4 ways to open the **Graphical Properties** window:

- In the **Standard Toolbar**, click the icon .
- Double-click the selected object.
- Choose the **Graphical Properties** option from the context menu of a selected object.
- Select **Graphical Properties** on the **View** menu of the **Graphics Editor**.



NOTE

If you want to change several objects, you can pin the display of the **Graphical Properties** window.

4.1.5.3 Changing an Attribute

The goal of this task is to change an object or a multiple selection of objects property by assigning a new value to accompanying attribute.

- Open the **Graphical Properties** window.
- On the **Properties** tab, select the required property group.
- Double-click the desired attribute.

A dialog to input or select the new value opens or the value is automatically changed by double-clicking.

Attribute Change Using the Value Input Dialog

- Enter a new value in the **Value Input** dialog.
 - Click **OK** to confirm.
- Alternatively, you can directly enter the new value by double-clicking the **Static** column.

Attribute Change Using the Text Input Dialog

- Click the **Text** box to enter the new text in the **Text Input** dialog.
 - Click **OK** to confirm.
- Alternatively, you can directly enter the new value by double-clicking the **Static** column.

Attribute Change Using a Switchover

- If there are 2 values available for an attribute, for example the choice between **Yes** and **No**, each double-click causes a switchover between these 2 values.
- Alternatively you can cause the switchover by double-clicking the **Static** column.

Attribute Change Using a Selection Dialog in the Form of Buttons

- Click the button to set the new value.
- Alternatively you can select the new value in the list box field after double-clicking the **Static** column.

Attribute Change Using a Selection Dialog for Selecting a Color

- Select one of the 16 standard colors in the selection dialog or open the palette with the customized colors.
 - Define additional colors in this palette.
- Alternatively you can select the color selection by double-clicking the **Static** column.

Attribute Change Using a Selection Dialog for Selecting a Picture

- Select a picture from the picture selection or open a selection window using the **Find** button.
 - Click **OK** to confirm.
- Alternatively, you can directly enter the name of the new picture by double-clicking the **Static** column.

4.2 User Interface Description

4.2.1 Introduction

Objects of the Object Palette

Objects in the **Graphics Editor** are predefined graphic elements that enable efficient creation of worldmaps. All objects can be easily inserted into a worldmap from the object palette.

Table 4-1 The **Standard** tab of the object palette provides the objects in the following object groups:

Standard Objects	Smart Objects	Windows Objects
Line	Application Window	Button
Polygon	Picture Window	Check Box
Polyline	Control	Option Group
Bezier Line	OLE Element	Round Button
Ellipse	I/O Field	Slider Object
Circle	Bar	
Ellipse Segment	Graphic Object	
Pie Segment	Status Display	
Ellipse Arc	Text List	
Circular Arc	3D Bar	
Rectangle	Group Display	
Rounded Rectangle	Status Display (Extended)	
Static Text	Analog Display (Extended)	
Label		
Connector		
Picture Section		
Generic Symbol		
Bus		



NOTE

Line presentation in the **Graphics Editor** when you zoom more than 100%:

If you zoom more than 100%, all instances of a dotted line with line thickness 1 are shown as a straight line.

This behavior is dictated by the graphics system of Microsoft. It does not happen if the line thickness is more than 1 or zoom is less than 100%.

In the **Graphics Editor**, the various object types have default properties. When they are inserted the objects import these default properties (except for individual geometric properties). After insertion the properties of an object can be modified. In the same way, the default settings for the object types can be modified as required.

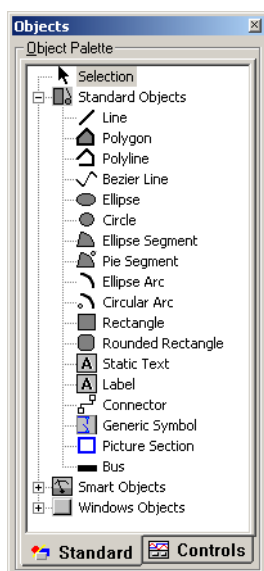
Combined Objects

The objects of the object palette can be combined by creating a group or a user object from a multiple selection. In addition, a multiple selection or a combined object can be added to the library. It can be made available as a library object for reuse in other worldmaps or projects.

4.2.2 Standard Objects

The standard objects include geometric shapes and static text. The geometric shapes draw the individual elements of the worldmaps.

For example, static text can be used for labels.



[image033-310510-01.tif, 1, en_US]

Figure 4-23 Standard Objects

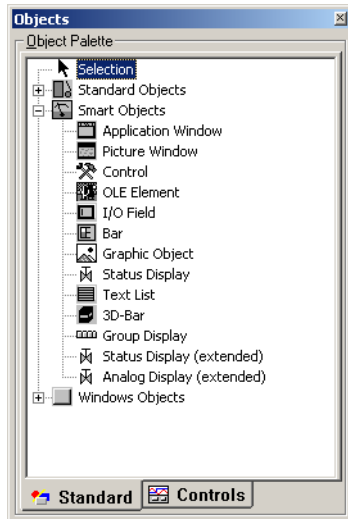
Overview

Icon	Object	Description
	Line	The line is an open object. The length and angle of a line are determined by the height and width of the rectangle around the object. The line ends, for example, can be shown as arrows or points.
	Polygon	The polygon is a closed object that can be filled with a color or pattern. A polygon can have any number of corners. The corners are numbered in their sequence of creation and can be modified individually or deleted.
	Polyline	The polyline is an open object. Even if the start and finish point have the same coordinates, the area cannot be filled. A polyline can have any number of corners. The corners are numbered in their sequence of creation and can be modified individually or deleted. The line ends of a polyline can for example be shown as arrows or points.
	Bezier line	The bezier line is an open object. Even if the start and finish point have the same coordinates, the area can not be filled. A bezier line can have any number of corners. The corners are numbered in their sequence of creation and can be modified individually or deleted. The layout between the corner points can be changed. The line ends of a bezier line can for example be shown as arrows or points.
	Ellipse	The ellipse is a closed object that can be filled with a color or pattern. The height and width of an ellipse can be modified as desired to allow it to be aligned horizontally or vertically.

Icon	Object	Description
	Circle	A circle is a closed object that can be filled with a color or pattern. A circle can be resized.
	Ellipse Segment	The ellipse segment is a closed object that can be filled with a color or pattern. The height and width of an ellipse segment can be modified as desired to allow it to be aligned horizontally or vertically.
	Pie Segment	The pie segment is a closed object that can be filled with a color or pattern. A pie segment can be resized.
	Ellipse Arc	The ellipse arc is an open object. The height and width of an ellipse arc can be modified as desired to allow it to be aligned horizontally or vertically.
	Circular Arc	The circular arc is an open object. A circular arc can be resized.
	Rectangle	The rectangle is a closed object that can be filled with a color or pattern. The height and width of a rectangle can be modified as desired to allow it to be aligned horizontally or vertically.
	Rounded Rectangle	The rounded rectangle is a closed object that can be filled with a color or pattern. The height and width of a rounded rectangle can be modified as desired to allow it to be aligned horizontally or vertically. The corners of a rounded rectangle can be rounded as much as desired.
	Static Text	The field for static text is a closed object that can be filled with a color or pattern. The static text is entered into a field of any desired size. One-line or multiline text can be input for the designed language.
	Label	The label is a field for text and a closed object that can be filled with a color and pattern. The text is automatically derived from the Spectrum Power domain object name in IMM that is linked to.
	Connector	The connector is a linear object whose ends can be to the connection points of other objects. The number and arrangement of the connection points depend on the specific object type. If connected objects are moved, the connectors length and direction are automatically adapted and the connection is retained.
	Generic Symbol	The generic symbol is a special object representing graphics of any Spectrum Power domain object.
	Picture Section	A picture section defines a view within a worldmap. Specific properties as well as the worldmap toolbar can be configured.
	Bus	The bus is a special object representing the electrical busbars. Connectors can connect to the bus at any position.

4.2.3 Smart Objects

Smart objects offer the option of building complex system worldmaps. The smart objects include items such as various windows, fields, and bars, which offer various dynamic options.



[immrg223-150409-01.tif, 1, en_US]

Figure 4-24 Smart Objects

Overview

Icon	Object	Description
	Application Window	The application window is an object that can be provided from applications of the global script and the logging system. In runtime, these applications open an application window, transfer information and enable operation. The size and properties which an application window accepts in runtime are defined in the Graphics Editor .
	Picture Window	The picture window offers the option of displaying other pictures that were created with the Graphics Editor in the current picture. For example, with dynamics the contents of a picture window can be continuously updated. The size and properties that a picture window has in runtime are defined in the Graphics Editor .
	Control	The object control provides the option to integrate system process control and monitoring elements into a worldmap. Controls are prefinished objects such as for example alarm windows, measuring windows, selection dialogs, or buttons. ActiveX controls, WinCC controls, and controls from other manufacturers are available if they are registered in the operating system. These are changed as required and dynamically integrated into the process. The size and properties which a control accepts in runtime are defined in the Graphics Editor .
	OLE Element	The OLE element enables files created in other programs to be inserted into a worldmap. Therefore all OLE element registered in the operating system can be integrated. The size and properties which an OLE element accepts in runtime are defined in the Graphics Editor . No changes can be made to OLE elements in runtime.
	I/O Field	The I/O field can be defined as a text box, an output field, or a combined input/output field. The following data formats are available: Binary, decimal, string, and hexadecimal. Limiting values such as Hidden Input or Accept on complete input can also be specified. The size and properties which an I/O field accepts in runtime are defined in the Graphics Editor .

Icon	Object	Description
	Bar	The bar offers the option of displaying values graphically. The values can also be displayed in a combined view as graphics with freely definable number scale. The size and properties which a bar accepts in runtime are defined in the Graphics Editor .
	Graphic Object	The graphic object enables graphics created in other programs to be inserted into a worldmap. Graphics and pictures of the following formats can be inserted: EMF, WMF, DIB, and BMP. The size and properties which a graphic object accepts in runtime are defined in the Graphics Editor .
	Status Display	The status display offers the option to display almost any number of different states of an object. The states are implemented using tags whose value corresponds to the respective state. The states are displayed via the assigned pictures. The size and properties which a status display accepts in runtime are defined in the Graphics Editor .
	Text List	The test list offers the option of assigning specific values to a text. The text list can be defined as an input list, an output list or as a combined input/output list. The following data formats are available: Decimal, Binary, and Bit.
	3D Bar	The 3D-bar graph enables values to be displayed graphically three-dimensionally. The size and properties that a 3D bar assumes in runtime, are defined in the Graphics Editor . The 3D bar is only available if the basic process control optional package was installed with WinCC.
	Group Display	The group display enables the current states of certain message types to be displayed by hierarchy. 4 buttons allow messages to be represented and operated. Using a group display, it is possible for example to implement a quick change to represent an error source. The size and properties that a group display assumes in runtime, are defined in the Graphics Editor . The group display is only available if the basic process control optional package was installed with WinCC.
	Status Display (extended)	Extended Status Display
	Analog Display (extended)	Extended Analog Display derived from status display.

4.2.4 Windows Objects

The Windows objects are elements familiar from Windows applications:

A round button can also be designed. The objects can be modified and made dynamic in many ways. They enable you to operate process events and to control processes.



[immrg226-150409-01.tif, 1, en_US]

Figure 4-25 Windows Objects

Overview

Icon	Object	Description
	Button	The button enables process operation. It is used for example to acknowledge messages. A button can have one of the 2 status: • On • Off A button is integrated into the process by providing the corresponding attributes with dynamics.
	Check Box	The check box enables the operator to select several options and to check the corresponding check boxes. It is also possible to check boxes by default so that the operator only changes the predefined value as required. A check box is integrated into the process by providing the corresponding attributes with dynamics.
	Option Box	The option box enables the operator to select one of the presented options and to activate the corresponding options field. It is also possible to activate one of the options fields by default so that the operator only changes the predefined value as required. An option box is integrated into the process by providing the corresponding attributes with dynamics.
	Round Button	The round button enables process operation. For example, it is used to acknowledge messages or to navigate in runtime. A round button can have one of 3 status: • On • Off • Deactivated It is also possible to define whether a round button is displayed as pressed and whether it latches down when pressed. A round button is integrated into the process by making the corresponding attributes dynamic
	Slider	The slider is a slide controller and can be used to control processes. For example, it can be used for infinite variable changes to a value. A slider is integrated into the process by providing the corresponding attributes with dynamics.

4.2.5 Default Setting of Object Types

The various object types have default properties. If an object from the **Object Palette** is inserted in a worldmap, the object adapts to these default settings. The default setting for the individual object types can be adapted as required. Ideally, the default settings are adapted before you begin creating a worldmap. In this way, the extent of later adaptations, for example for frequently used objects, can be limited.

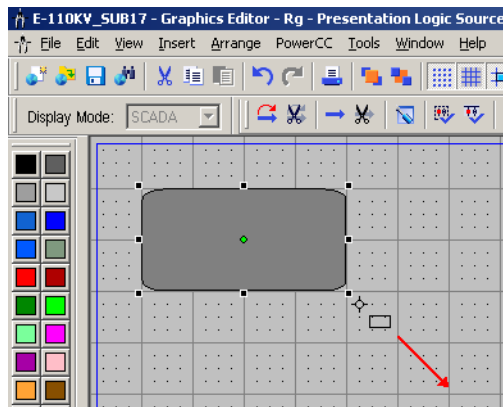
For additional information on backing up default settings, refer to Graphics Editor Basics.

4.2.6 Basic Static Operations

4.2.6.1 Insert Object from the Object Palette

To create a worldmap, the required objects must be inserted from the object palette into a worldmap. These objects are then dynamically linked suitably to a process to use the worldmap to control and monitor processes. The objects, which form a process and a subprocess in a worldmap are inserted from the **Graphics Editor Object Palette**.

When they are inserted, the objects import their default properties (except for individual geometric properties). After insertion the properties of an object can be modified as required.



[immrg008-190711-01.tif, 1, en_US]

Figure 4-26 Inserting and Sizing an Object

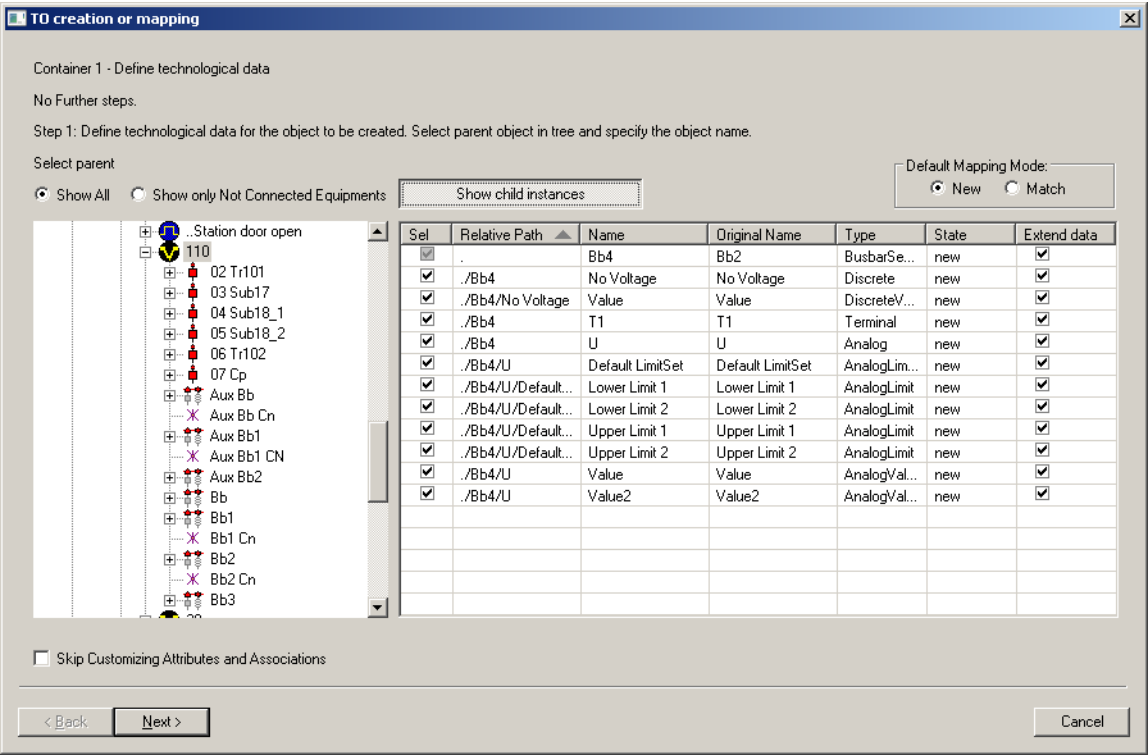
4.2.6.2 Insert Object from the Library

Library objects can be added to a worldmap by dragging them with the mouse.

When a library object is added to a worldmap it uses the properties stored in the template. After insertion the properties of an object can be modified as required.

Basically the same mechanisms as inserting from the **Object Palette** apply.

Dragging library objects with linked domain data opens the **TO creation or mapping** window automatically for further configuration issues.



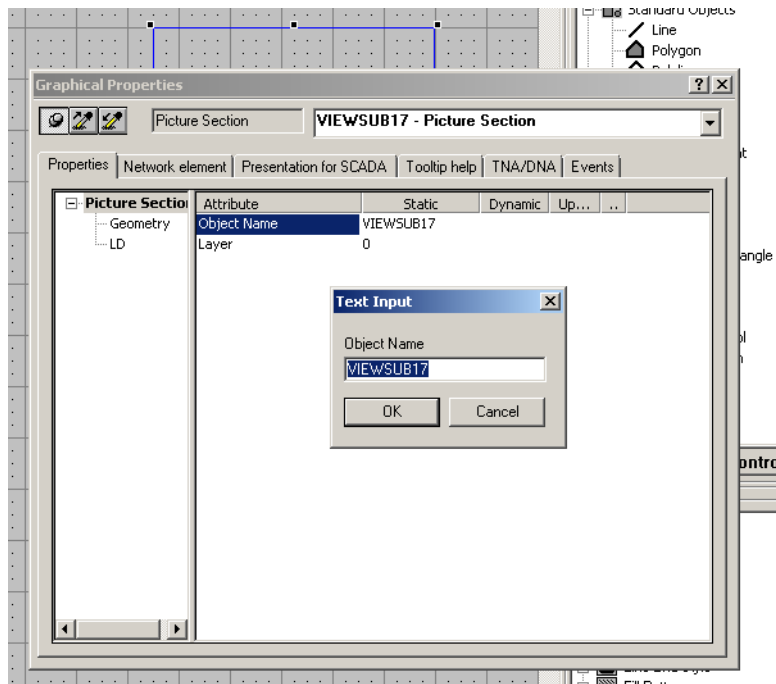
[rgimm001-190711-01.tif, 1, en_US]
Figure 4-27 TO Creation or Mapping Window



NOTE
For additional information regarding Spectrum Power objects and technological editing, refer to Technological Editing.

4.2.6.3 Rename Objects

The **Object Name** attribute specifies the name of an object in the worldmap. The object name is unique within a worldmap. The name of the object is used for example in configuring C actions to call the object. When an object is inserted, the object name assigned as standard is a sequential number followed by the name of the used object type. This name can be changed using the **Object Name** attribute within the **Graphical Properties** window.



[rgimm002-190711-01.tif, 1, en_US]

Figure 4-28 Graphical Properties Window



NOTE

Avoid special characters in object names as these object names are used in scripts. The name has to comply the rules of naming IMM instances. For additional information, refer to Reference Guide Information Model Management (IMM).

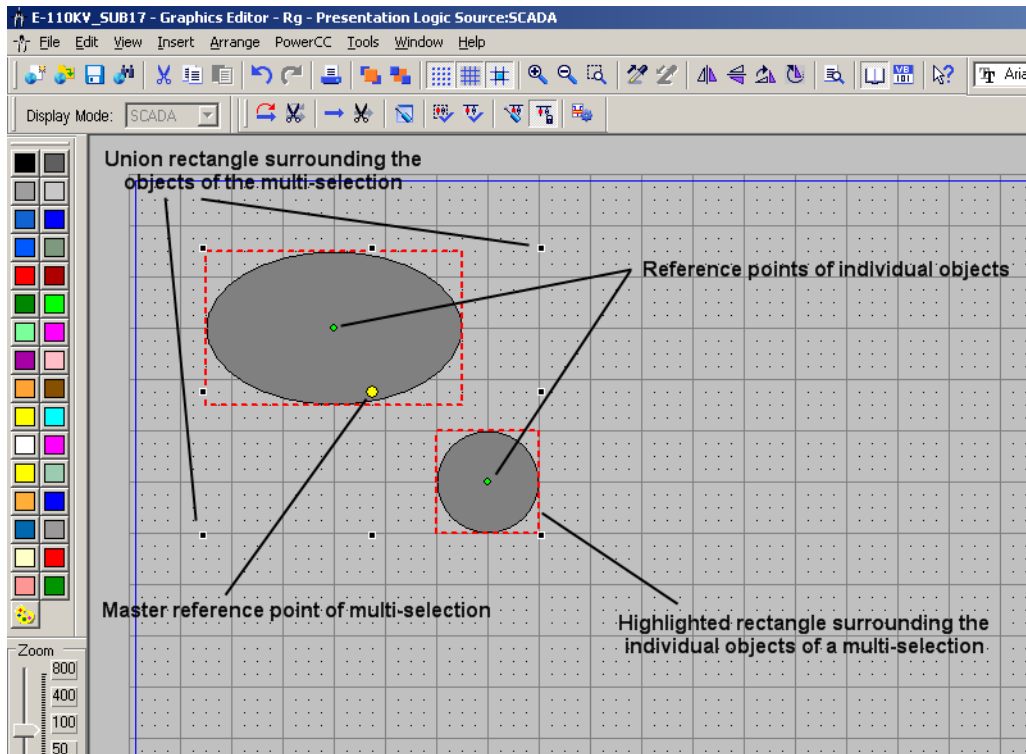
4.2.6.4 Multiple Selection of Objects

During a multiple selection, the graphical properties attributes held by at least one of the selected objects are displayed in the **Graphical Properties** window. The value of an attribute is only displayed, however, if it is the same for all selected objects.

Along with selection frames and the master reference point, a multiple selection has 2 characteristics which play an important role. For example, for the common alignment of the selected objects. These characteristics are however not displayed visibly in the **Graphics Editor**.

Selection Frame of a Multiple Selection

The selection frame encloses all objects in a multiple selection, comparable to the rectangle surrounding individual objects. The selection frame is known as the union rectangle surrounding the objects. The rectangle surrounding the individual objects within the selection are displayed with a red dashed line.



[rgimm003-190711-01.tif, 1, en_US]

Figure 4-29 Selection Frame of a Multiple Selection

The position and size of the union selection frame are dependent on the position of the selected objects. The midpoint of the selection frame corresponds to the common midpoint of the selected objects. Within the midpoint the master reference point is shown. The limit line of the selection frame touches the outer edges of the objects which have the greatest distance from the common midpoint.

The union selection frame is not visible. After finishing the multiple selection, the handles of the union selection frame, the master reference point and the rectangles surrounding the individual objects will be shown.

The master reference point can be interactively shifted within the union selection frame to optimally support snap to grid related functions.

Reference Object of a Multiple Selection

During a multiple selection, one of the selected objects is set as a reference object. For example, if the same width function is selected in the alignment palette, then all selected objects are set to the same width as the reference object.

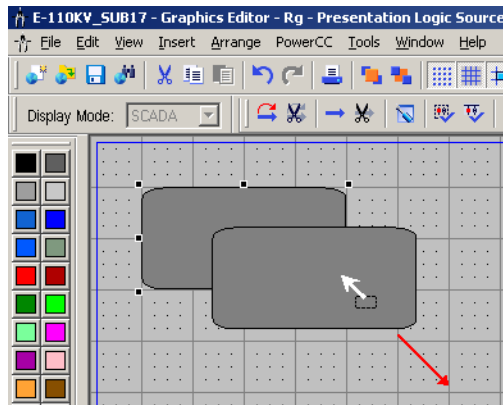
With a multiple selection, the upper top object is used as that reference.

4.2.6.5 Objects Position

The position of an object is defined by the coordinates of the object origin. The left upper corner of the rectangle surrounding the object is called the object origin.

Objects can be positioned by:

- Dragging the object to a new position.
- Moving the selected object with the aid of the arrow keys for mouse pointer control.
- Changing the **Position X** and **Position Y** attributes.



[rgimm004-190711-01.tif, 1, en_US]

Figure 4-30 Objects Position

The old position of the object is shown together with the dragged object as long as the mouse is not released.

Multiple Selection

It is possible to move several selected objects simultaneously.

4.2.6.6 Multiple Objects Alignment

Using the functions of the alignment palette, the objects of a multiple selection can be arranged together in the worldmap. For additional information on these functions, refer to [2.2.12 Alignment Palette](#).

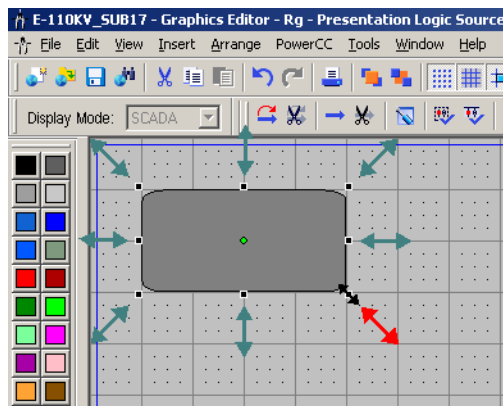


[rgimmge4-240611-01.tif, 1, en_US]

Figure 4-31 Alignment Palette

4.2.6.7 Object Scale

Objects within a worldmap can be scaled. The geometry of the rectangle surrounding the object defines the size. The rectangle surrounding the object is symbolized when selecting an object by showing the selection markers.



[rgimm005-190711-01.tif, 1, en_US]

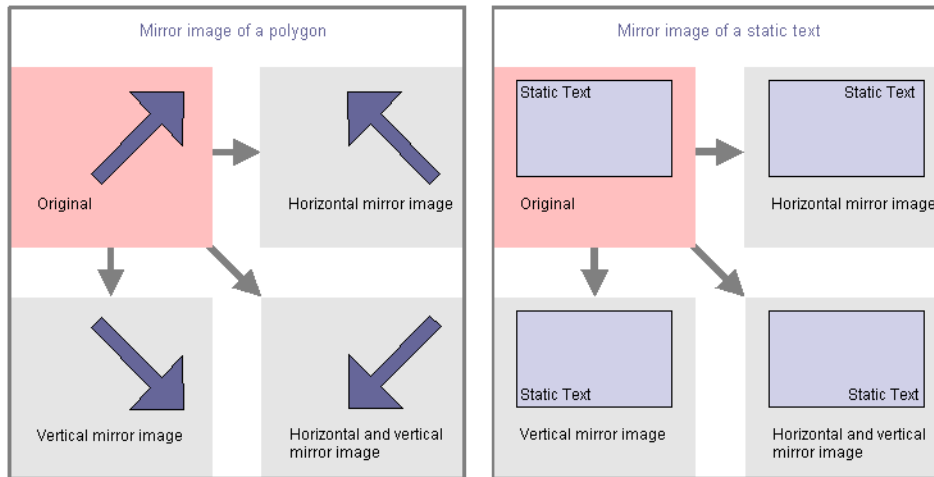
Figure 4-32 Objects Scale

Multiple Selection

The objects must be grouped to scale several selected objects simultaneously using the mouse.

4.2.6.8 Mirror Objects

Objects can be mirrored on their horizontal or vertical axis defined by the reference point. The mirroring of an object can cause its geometric attributes to change. Mirroring can also change the alignment of elements contained in the object such as text or check boxes.



[immrg015-101208-01.tif, 1, en_US]

Figure 4-33 Mirror Objects

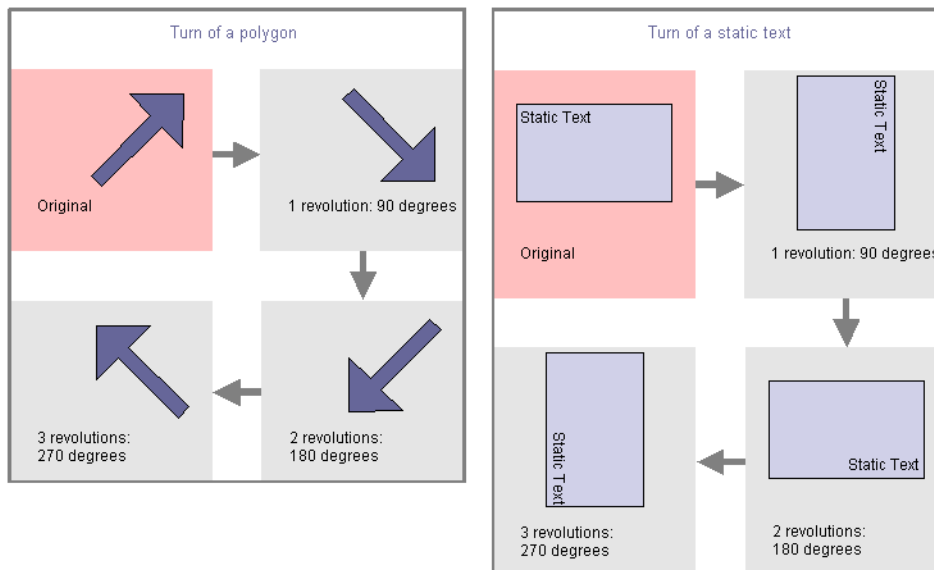
Multiple Selection

It is possible to mirror several selected objects simultaneously. The mirroring occurs on the horizontal or vertical axis defined by the master reference point.

4.2.6.9 Rotate Objects Clockwise

Objects can be rotated around their reference point.

The rotation is clockwise and in steps of 90 degrees. The rotation of an object can cause its geometric attributes to change. Rotation can also change the alignment of elements in an object such as a text.



[immrg016-101208-01.tif, 1, en_US]

Figure 4-34 Objects Rotation Clockwise

Multiple Selection

It is possible to rotate several selected objects simultaneously. The objects are each rotated around their master point.

4.2.6.10 Free Rotation of Objects

Objects can be rotated freely around their reference point if the free rotation mode is enabled in the **Standard Toolbar** of the **Graphics Editor**.

The rotation is performed by means of the rotation handles and dragging with the mouse. The rotation of an object can cause its geometric attributes to change. Rotation can also change the alignment of elements in an object and as a text.

Multiple Selection

It is possible to rotate several selected objects simultaneously. The objects are each rotated around their master reference point.

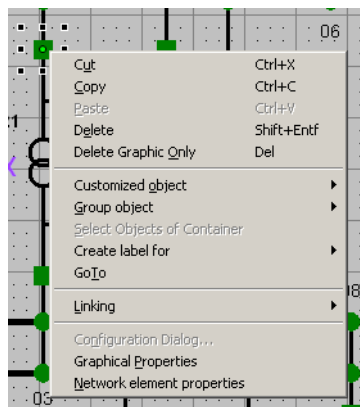
4.2.6.11 Delete Objects

The **Delete** command removes a selected object including linked domain data from a worldmap.



NOTE

To prevent deleting linked domain data use **Delete Graphic Only** on the context menu of the object. For additional information on steps to be performed, refer to Technological Editing.



[rgimm006-190711-01.tif, 1, en_US]

Figure 4-35 Context Menu – Delete Graphic Only

Multiple Selection

It is possible to delete several selected objects simultaneously.

4.2.6.12 Cut and Paste Objects

The **Cut** command copies a selected object including linked domain data into the clipboard of the operating system. The object itself remains visible in the worldmap.

The **Paste** command inserts the current contents of the clipboard into the active worldmap under the mouse position. The former cut object is removed within the original position and inserted at the paste position.



NOTE

If graphical objects are linked to domain data, the domain data gets also cut and pasted. For additional information on steps to be performed, refer to Technological Editing.

Multiple Selection

It is possible to cut and paste several selected objects simultaneously.

4.2.6.13 Copy Objects

The **Copy** command copies a selected object together with possible linked domain data into the clipboard of the operating system. The advantage of copying to the clipboard is that object can be inserted several times, and into different worldmaps.

The **Paste** command inserts the current contents of the clipboard into the active worldmap as often as required.



NOTE

If graphical objects are linked to domain data, the domain data gets also copied.
For additional information on steps to be performed, refer to Technological Editing.

Multiple Selection

It is possible to copy several selected objects simultaneously.

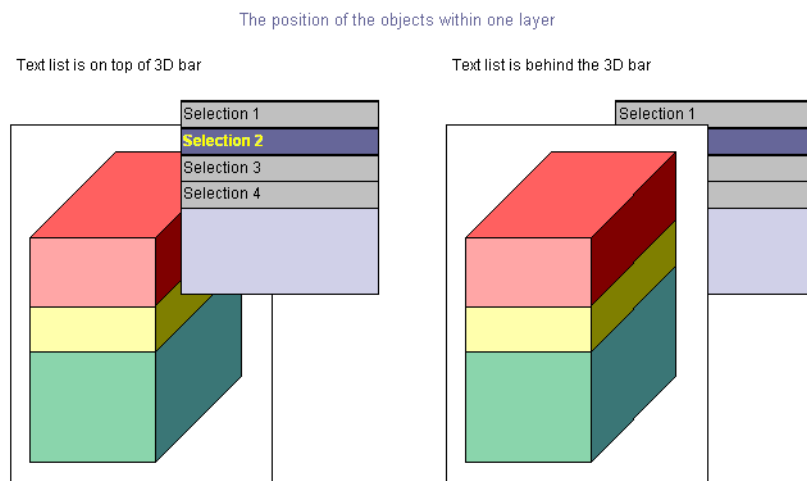


NOTE

When inserting a copied object, an object name is automatically assigned. The name is a sequential number followed by the name of the used object type.

4.2.6.14 Layer Position of an Object

By default, the objects are inserted in a layer in the level predefined for the respective object type. The position of the objects to one another can also be changed within a layer. This is important for example, if several objects overlap over each other.



[immg017-101208-01.tif, 1, en_US]

Figure 4-36 Layer Position of an Object

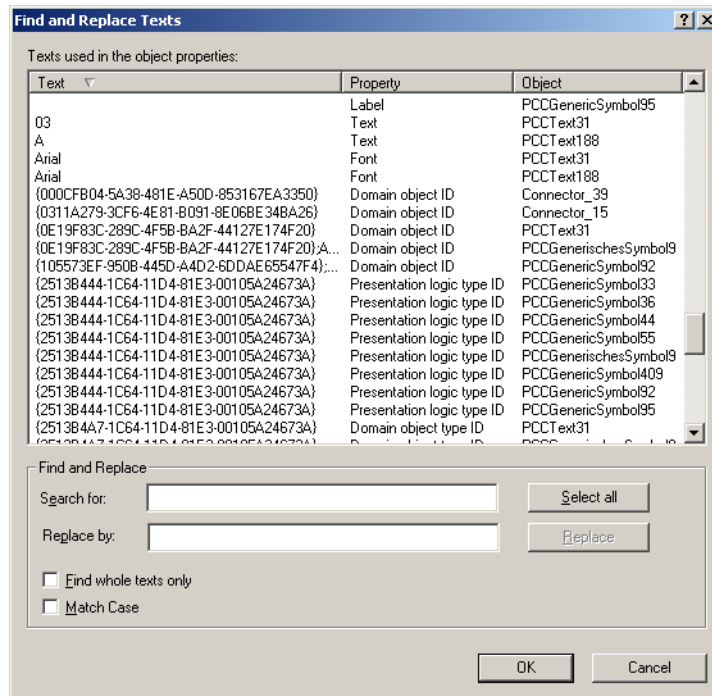
When creating a worldmap the objects of a layer are by default arranged in the order in which they were configured. The object inserted first lies at the back of the layer, each additional object is inserted single position toward the front.

Multiple Selection

It is possible to change the position of multiple selected objects simultaneously.

4.2.6.15 Text Contents of Objects

The **Linking > Texts...** command is used to reconfigure the text contents of attributes for all selected objects. All configured texts are listed in a dialog and can be edited directly or using the **Find** and **Replace** functions.



[immrg234-150409-01.tif, 1, en_US]

Figure 4-37 Find and Replace Texts

Sort Text Contents and Directly Edit Text Contents

The three-columned upper area of the dialog lists the texts configured in the objects selection and the associated attributes and objects. To sort the list, click the column name according to texts, properties, or objects. Selected texts in the **Text** column can be directly edited after clicking it.

Find

You can select certain text contents using the **Find** function in the lower area of the dialog.

- **Select All** allows you to change the text contents displayed.
- Enter the text or parts of text you want to search for in the search field.
You can also activate the options **Find whole texts only** and **Match case**. All text contents which do not meet the search criteria are hidden.

Replace

You can replace the selected text contents using the **Replace** function in the lower area of the dialog.

- Enter the new text in the field and click **Replace**.



NOTE

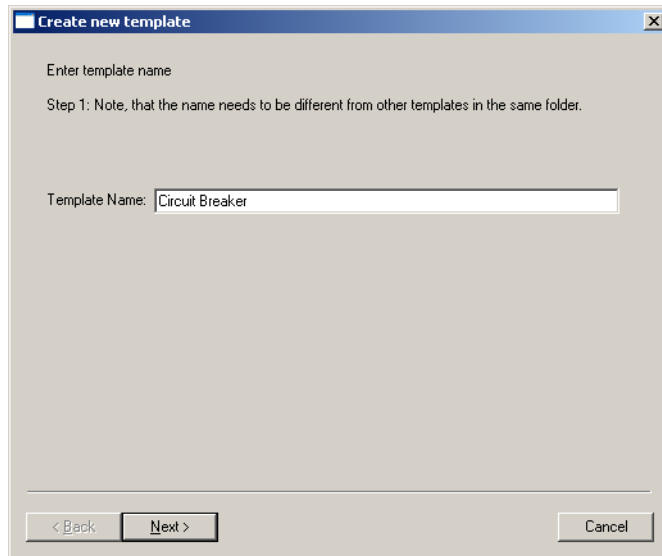
The **Replace** function replaces only the part of the text contents displayed in the **Find** field.

4.2.6.16 Store Objects into the Library as Templates

The objects in a worldmap can be added to the library by dragging them with the mouse. The library objects are sorted by subjects by creating folders and subdirectories. A new name can be assigned to the imported library object to simplify for later use.

The Library can be extended with user-defined objects to make them available for other projects. These objects may or may not be linked with domain data.

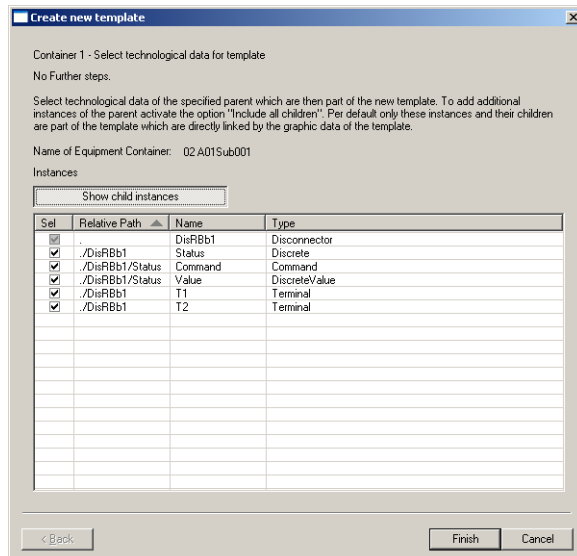
Dragging objects into the library opens the **Create new template** window to define the name of the template.



[immrg128-140409-01.tif, 1, en_US]

Figure 4-38 Create New Template - Name of Template

Dragging objects with linked domain data into the library opens as second step of the **Create new template** window for further configuration issues.



[rgimm007-230210-01.tif, 1, en_US]

Figure 4-39 Create New Template - Linked Domain Data



NOTE

If graphical objects are linked to domain data, the domain data gets also inserted into the library.
For additional information on steps to be performed, refer to Technological Editing.

4.2.7 Basic Dynamic Operations

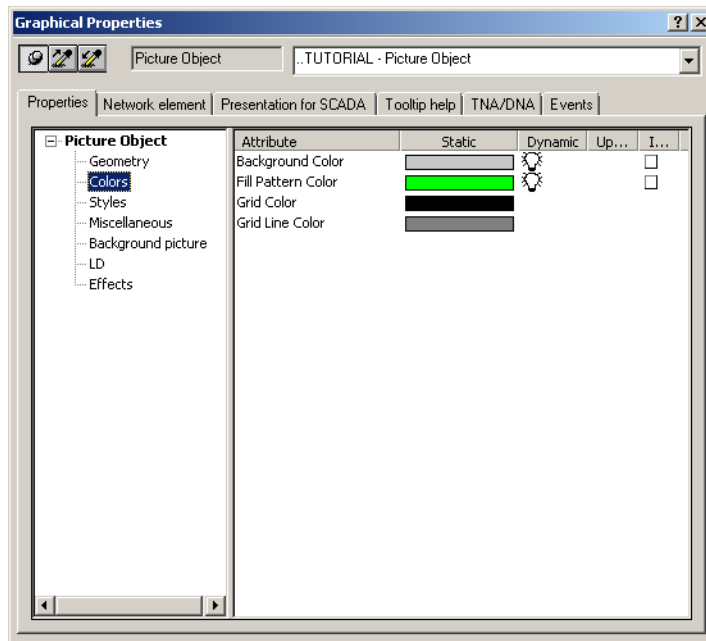
4.2.7.1 Graphical Properties Window

To create worldmaps it is important to adapt the static values of attributes of a graphics object to for example define the shape, appearance, position, or operability of an object.

The **Graphical Properties** window allows however worldmaps to be made dynamic. By linking the attributes with dynamic dialogues, C actions or tags, the properties of an object can be dynamically adapted to the requirements of the processes to be displayed. More important is to link the dynamics within the Spectrum Power 5 environment by configuring the tabs **Network Element**, **Presentation**, **TNA / DNA**, and **Tooltip Help**.

The **Graphical Properties** window represents the central dialog for changing the object properties.

The properties of an object are described by many attributes. An object property can be changed by assigning a new value to its attributes. Depending on the attribute type, the change of an value is made using different procedures and dialogs.



[rgimmge7-190711-01.tif, 1, en_US]

Figure 4-40 Graphical Properties Window

The **Graphical Properties** window can be pinned on top of the **Graphics Editor**. The position and size of the window can be changed as desired. The Graphical Properties window is subdivided into **Elements** toolbar and the various tabs. You can find more about these elements in their detailed description.



NOTE

For additional information regarding Spectrum Power objects and technological editing, refer to Technological Editing.

4.2.7.2 Toolbar in the Graphical Properties Window

You can make the following settings with the toolbar in the **Graphical Properties** window:

- The object properties can be pinned on top in the **Graphics Editor**.
- **Graphical Properties** can be transferred.
- You can select individual objects of the active worldmap.
For example, the toolbar can have the following appearance for a rectangle with the object name Rectangle 1.



[immrg026-101208-01.tif, 1, en_US]

Figure 4-41 Toolbar in the Graphical Properties Window

Table 4-2 Elements of the Toolbar in the Graphical Properties Window

Icon	Name	Function
	Pin activated/deactivated	Deactivate: The Graphical Properties window is closed as soon as the mouse is clicked outside the windows. Activated: The Graphical Properties window stays in the foreground at all times. Different objects can be edited without having to open the window again.
	Eyedropper Copy properties	The properties of a selected object are copied.
	Eyedropper Assign properties	The previously copied properties are transferred to the selected object without changing its geometric attributes. For different object types, only those attributes which both object types have are transferred.
	Object type	Displays the object type of the selected object. With several objects, here displays whether it is a Multiple selection, a Group or a User object.
	Object selection	Displays the name and the type of the selected object. A different object can be selected in the list box field. The list box field shows the names and types of all objects in the active worldmap.



NOTE

It is also possible to transfer the value of a selected attribute to other attributes using the eyedroppers in the **Graphical Properties** window.

Font Styles of the Object Selection

Different font styles particularly highlight dynamic objects and the type of dynamics in the object selection.

The following font styles are used:

- **Normal**
No dynamics were configured for the object. The object is not the target of a configured direct connection.
- **Bold**
At least, one dynamics was configured for the object. However, the object is not the target of a configured direct connection.
- **Italics**
No dynamics were configured for the object. The object is the target of at least one configured direct connection.
- **Bold and italic**
At least, one dynamics was configured for the object. The object is the target of at least one configured direct connection.

4.2.7.3 Properties Tab in the Graphical Properties Window

In the **Graphical Properties** window, the **Properties** tab represents the central dialog to change static values of a selected object.

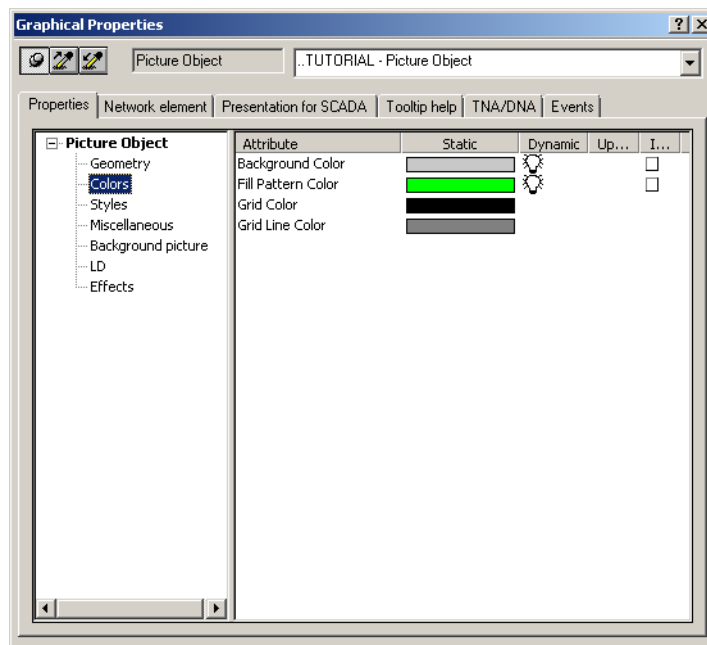
The **Properties** tab contains all the attributes of a selected object or a multiple selection of objects. The properties of an object are described by many attributes. An object property can be changed by assigning a new value to its attributes.

The attributes are divided into property groups, such as geometry or color. The type and number of the available property groups and attributes are dependent on the type of the selected objects.

For example, the **Font** property group is only shown for the object types which can be displayed with text. With a multiple selection, it is possible that individual property groups for one specific object type are summarized to a shared property group.

All attributes of the summarized property groups available in this group are contained under the **Customized label**. An alternative to changing the attributes in the **Graphical Properties** window: you can adjust the objects with the mouse and keyboard or by using the toolbar and palettes. However, in this way only certain object properties, such as the basic geometric size, color and line style, can be changed.

The **Properties** tab can for example have the following appearance.



[rgimm012-190711-01.tif, 1, en_US]

Figure 4-42 Properties Tab

The width of both section of the windows and the columns in the attribute display can be changed by moving the vertical separation lines.

Properties Folder

The left area shows the selected object with its property groups in the form of a directory structure. The selected object is displayed as a directory. With a multiple selection or a user object, this folder only receives the shared property groups of the contained individual objects. With a selected group the shared property groups are listed first and then, as a subdirectory, the individual objects with their property groups.

It is not possible to change the entries in the property folder. Directories and subdirectories can be opened or closed by clicking the + or - icons. The attributes available for the selected entry are shown in the attribute display.

Attribute Display

The right-hand area contains all the attributes that are available for the entry. The attribute display is divided into columns in which the static values of the selected object are displayed. The displayed values can be changed by double-clicking or opening the context menu in the respective column.

Table 4-3 Columns of the Attribute Display

Column	Function	Description
Attribute	Attribute name	All attributes from the selected property group that are available for the selected object are shown. The name of the attribute cannot be changed. The static value of the attribute can be changed by double-clicking the name of the attribute.
Static	Static value of the attribute	The current value of the attribute is shown for the selected object. The value is displayed as a number, text, or graphic display, depending on the type of attribute. The static value of the attribute can be changed by double-clicking the attribute name.
Dynamic	Dynamization of attribute	The type of dynamics is displayed and marked by one of the following icons: • White bulb = no dynamics • Green bulb = dynamics with a tag • Red lightning = dynamics via a dynamic dialog • Green lightning with attached abbreviation VB = dynamics with a VBS action • Green lightning with attached abbreviation C = dynamics with a C action • Yellow lightning with attached abbreviation C = dynamics with C action that has not yet been translated The dynamics of the attribute can be changed. Right-clicking opens a context menu on which the required dynamics can be set.
Update Cycle	Update cycle of the attribute	The setting for the update cycle is displayed if dynamics of the attribute are set. The update cycle of the attribute can be changed by double-clicking on the value.

Column	Function	Description
Indirect	Indirect attribute addressing	An attribute can be directly or indirectly made dynamic: • Direct: The attribute is made dynamic directly with the tag content. • Indirect: The attribute is linked to a tag of the type string, which references another tag. The attribute is made dynamic with the contents of the referenced tag. Indirect addressing of the attribute can be activated by double-clicking the check box if the attribute is made dynamic with a tag.

Dynamic Attributes

By linking the attributes with dynamic dialogues, VBS actions, C actions, or tags, the properties of an object can be dynamically adapted to the requirements of the processes to be displayed.



NOTE

Within the Spectrum Power 5 environment, dynamics are configured by means of linked domain data. Linked domain data are visible within additional tabs of the **Graphical Properties** window and not displayed by the preceding described dynamics icons.

There are attributes, which cannot be made dynamic because a change in runtime has no effects. They have no dynamics icon.

Font Styles to Display Dynamics and Events

Dynamics and events are particularly highlighted with the aid of different font styles.

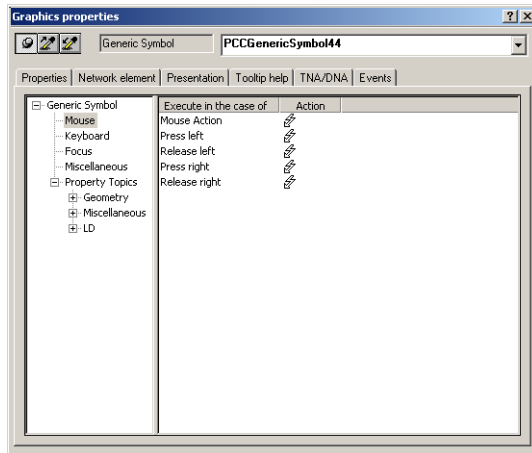
The following font styles are used:

- **Bold**
As soon as a dynamics or event has been configured for an attribute of the selected object, this attribute is displayed in bold in the attribute screen. The associated property group in the property folder and the selected object in object selection of the toolbar are also displayed in bold.
- **Italics**
If a direct connection was configured for an event, the target of the direct connection is displayed in italics in the attribute screen. The target object is also displayed in italics in the object selection of the toolbar.
- **Bold and italic**
If an object is the target of a direct connection as well as dynamic itself, the attribute in the attribute screen and the object in the object selection are displayed in bold and italics.

4.2.7.4 Events Tab in the Graphical Properties Window

In the **Graphical Properties** window the **Event** tab represents the central dialog for configuring events.

- The **Event** tab in the **Graphical Properties** window, for example, can have the following appearance for a generic symbol with the object name PCCGenericSymbol44.



[rgimm013-230210-01.tif, 1, en_US]

Figure 4-43 Events Tab

The width of the both section of the windows and the columns in the event display can be changed by moving the vertical separation lines.

Event Folder

The left area shows the selected object in the form of a directory structure. This folder shows entries such as for example Mouse, Keyboard, Focus, and Miscellaneous with which events for the operation of the entire object can be configured.

The **Property Topics** displays all property groups of the selected object. An event can also be configured for each individual attribute of a property group. If there are several individual objects in the selected object, the structure is repeated for each individual object. It is not possible to change the entries in the property folder. Directories and subdirectories can be opened or closed by clicking the + or - icons. The configurable events for the selected entry are shown in the event display.

Event Display

The right-hand area contains all the events that are available for the entry. The event display is divided into 2 columns in which the events and the linked actions are displayed. The displayed actions can be changed by double-clicking or opening the context menu in the **Action** column.

Table 4-4 Columns of the Event Display

Column	Function	Description
Execute in the case of	Event type	All events are displayed which are available for the selected object and assigned to the entry selected in the event directory. The name of the event cannot be changed. The configured action can be changed by double-clicking the name of the attribute.

Column	Function	Description
Action	Action selection	The action, which is run when the event occurs, is displayed. The selected action is marked by one of the following symbols • White lightning = no action for the event • Blue lightning = There is an action for this event via a direct connection • Green lightning with attached abbreviation VB = There is a VBS action for this event • Green lightning with attached abbreviation C = There is a C action for this event • Yellow lightning with attached abbreviation C = There is a not yet translated C action for this event The configured action can be changed by double-clicking the column Execute with or by retrieving the context menu in the column Action.

Triggering Events

Event	Run At	Description
Mouse	Mouse Action	Is triggered if the mouse pointer is on the object when the mouse button is pressed and released.
Mouse	Press left/right	Is triggered when pressing the mouse button directly on the object.
Mouse	Release left/right	Is triggered when releasing the mouse button for the object the mouse pointer was over when the mouse button was pressed.
Keyboard	Press	Is triggered when pressing a key on the keyboard. The keys < F10 > and < ALT+PRINT > must not be used for process operation.
Keyboard	Release	Is triggered when releasing a key on the keyboard. The keys < F10 > and < ALT+PRINT > must not be used for process operation.
Focus	Focus change	Occurs during focus reception, caused by C action, VBS action, or selection of an object using (TAB sequence) or mouse click
Miscellaneous	Object change	Occurs if at least 1 object attribute changes.
Object attributes	Change	Most object attributes have the Change event. It is used to be able to react specifically to the change of a certain attribute. The event occurs if the attribute value changes. The action linked to this event is logged on individually. With closing the worldmap, all actions logged on are logged off individually. This leads to an increased system load. To keep the system load low, this event type be used sparingly and only there where it is necessary to react to the change. This is the case for example when inputting a value in the I/O field.

Events Configuration

By linking the events with direct connections, VBS actions or C actions, certain actions can be triggered in runtime by the operation of an object.



NOTE

This configuration is native WinCC functionality and does not have that importance within the Spectrum Power 5 environment.

There are events, which cannot be made dynamic because a change in runtime has no effects. They have no action icon.

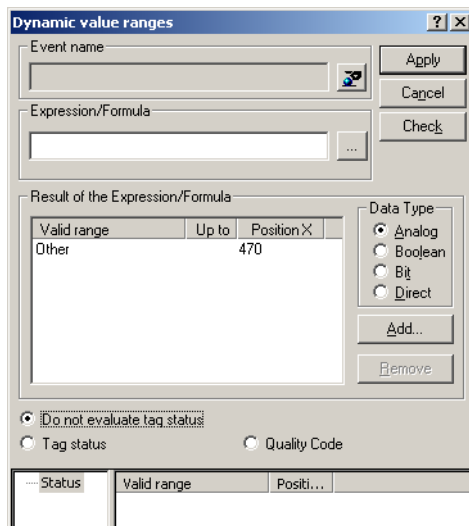
4.2.7.5 Dynamic Dialog

A dynamic dialog is used to achieve high runtime performance when making attributes dynamic. A C action is automatically generated from the dynamic dialog. It can be extended later. The advantage of higher runtime performance is then lost. A dynamic dialog is created with the **Dynamic Value Ranges** dialog.



NOTE

This configuration is native WinCC functionality and does not have that importance within the Spectrum Power 5 environment



[immrg235-150409-01.tif, 1, en_US]

Figure 4-44 Dynamic Value Ranges Dialog

Event Name

The **Event Name** field displays the name of the selected trigger event which defines the update cycle for the value to be calculated. The trigger event is configured using the **Trigger** dialog. There is a choice of the events **Tag**, **Standard cycle**, **Picture cycle**, and **Window cycle**, which can be configured and renamed as required.

The button  opens the **Change Trigger** dialog.

Expression / Formula

Specifies the formula with which the new value of the attribute is calculated in runtime.

- The button on the right allows to create the formula from the configured tags, global script functions, and operators.
- The button **Check** checks the formula for errors.

Result of the Expression / Formula

This field displays the calculated value or range of value. The evaluation of the event depends on the set data type.

Data Type

Selects the data type for the evaluation of the event. The options **Analog**, **Boolean**, **Bit**, and **Direct** are available.

Evaluate Status of Tags

Activate this check box to view the current values of the tags in the formula.

4.2.7.6 C Action

A C action is used to make attributes and events dynamic. The powerful script language ANSI-C opens almost endless options for dynamics. However, the runtime performance achieved is lower than with other dynamic types. Therefore, before using a C action check whether the required dynamics can be implemented with another dynamics type. A C action is created with the **Edit Action**.



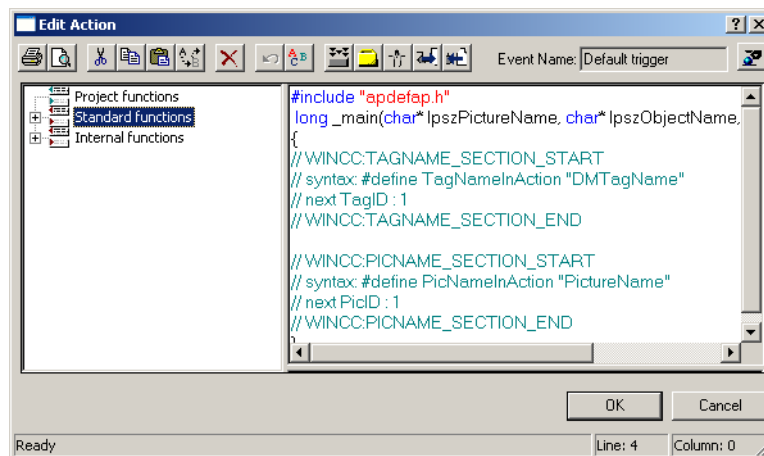
NOTE

This configuration is native WinCC functionality and does not have that importance within the Spectrum Power 5 environment.



NOTE

The configuration of C Actions is not supported for the Web Navigator. It is recommended to use VBScript on Web Navigator instead.



[immrg236-150409-01.tif, 1, en_US]

Figure 4-45 Edit Action Dialog

Table 4-5 Edit Action Dialog Toolbar

Icon	Name	Description
	Create Action	Checks the program codes of the C action for errors
	Tag Selection	Selects tags, which are evaluated in the C action
	Selection of Picture	Selects the picture of the PDL format, which are evaluated in the C action
	Import an Action	Imports an existing C action. C actions are saved in the ACT file format.
	Export an Action	Exporting the new or modified C action. C actions are saved in the file format ACT

Event Name

The **Event Name** field displays the name of the selected trigger event, which defines the update cycle for the value to be calculated.

The trigger event is configured using the **Trigger** dialog. There is a choice of the events **Tag**, **Standard cycle**, **Picture cycle**, and **Window cycle**, which can be configured and renamed as required.

The  button opens the **Change trigger** dialog.

Editor Window

The editor window shows and allows to edit the program code of the C action. Functions, tags, and pictures can be inserted and modified at the respective current mouse pointer position.

Function Selection

The function selection shows the **Global Script** functions available in the project in the form of a directory tree. You can use these functions to create the C actions.

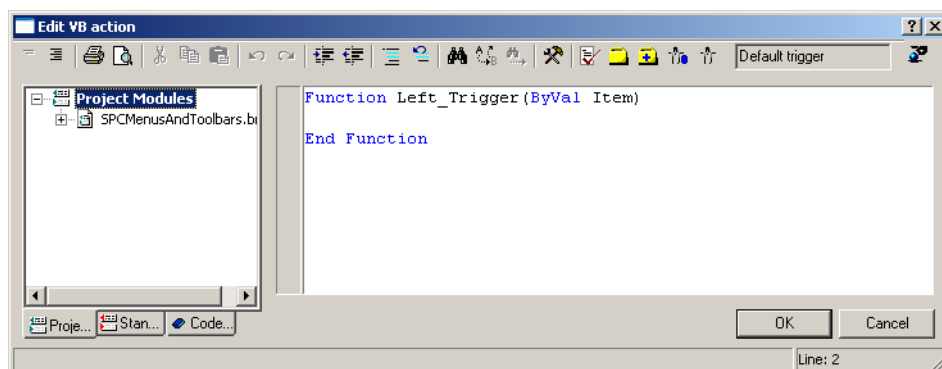
4.2.7.7 VBS Action

A VBS action is used to make attributes and events dynamic. The script language of a VBS action is Visual Basic. A VBS action is created with the **Edit VBS Action** dialog.



NOTE

This configuration is native WinCC functionality and does not have that importance within the Spectrum Power 5 environment.



[immrg237-150409-01.tif, 1, en_US]

Figure 4-46 Edit VBS Action

Toolbar

The toolbar of the **Edit VBS Action** dialog also contains buttons in addition to the familiar icons. The composition of the icons on the toolbar varies according to the types of attribute or event for which a VBS action was configured.

Table 4-6 Toolbar of Edit VBS Action

Icon	Name	Description
	Single view	Displays the currently edited module
	Full view	Displays all modules
	Comment	Changes marked program lines into commentary lines
	Uncomment	Deletes the commentary marking of a program line

Icon	Name	Description
	Setting Bookmarks	Marks a program line for further editing
	Delete Bookmark	Deletes the marking of a program line
	Next Bookmark	Jumps from the current mouse pointer position to the next bookmark
	Previous bookmark	Jumps from the current mouse pointer position to the previous bookmark
	Syntax Check	Checks the program codes of the VBS action for errors
	Tag Selection	Selects tags, which should be evaluated in the VBS action
	Tag dialog with extended return parameter	Selects tags, which should be evaluated in the VBS action The selection is made via a dialog with extended return parameter
	Selection of Picture	Selects the picture of the PDL format, which should be evaluated in the C action

Event Name

The **Event Name** field displays the name of the selected trigger event which defines the update cycle for the value to be calculated.

The trigger event is configured using the **Trigger** dialog. There is a choice of the events **Tag**, **Standard cycle**, **Picture cycle** and **Window cycle**, which can be configured and renamed as required.

- The  button opens the **Change trigger** dialog.

Editor Window

The editor window shows and allows to edit the program code of the VBS action. Visual Basic modules, tags, and pictures can be inserted and modified at the respective current mouse pointer position.

Module Selection

The module selection displays Visual Basic modules already available in the form of a directory tree. There are 3 tabs in which Project modules, Standard modules, and Code templates are made available. You can use these modules to create the VBS actions.

4.2.7.8 WinCC Tag Connection

Attributes can be made dynamic using a tag connection. Connecting an attribute to a process tag for example enables the change of the measured value to be displayed graphically.

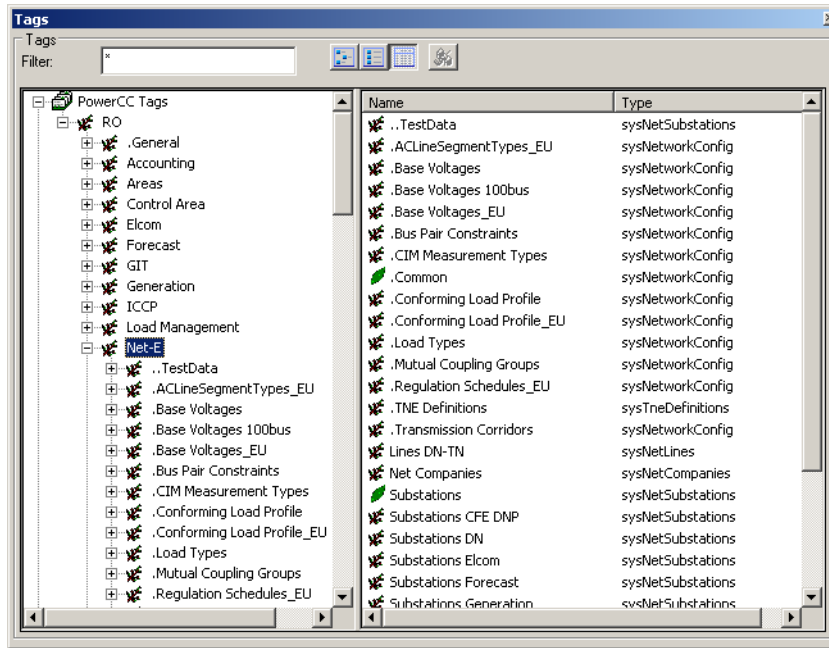


NOTE

This configuration is native WinCC functionality and can only be used in a limited way within the Spectrum Power 5 environment.

WinCC and PowerCC Tags can only be read during Runtime. Writing data or setting values is not possible within the Spectrum Power 5 environment.

Tags can be assigned to certain objects (for example, bar chart, slider) that are not Spectrum Power 5 specific to use values as dynamic data source. For this purpose set the flag **WinCC tag enabled** to true for the respective measurement in the **Information Model Management** to determine that this measurement be used as tag. All tags available in the project can be selected in the **Tags** dialog.



[immg238-190711-01.tif, 1, en_US]

Figure 4-47 WinCC Tag Connection

Filter

If there are many configured tags, the search procedure can take some time. For instance, for 15,000 tags the search procedure takes about a minute.

By specifying a filter, you can narrow the scope of the search and reduce the duration of the search procedure. If you configure tags with the aid of a tag prefix, you can achieve a significant increase in efficiency.

Tag Groups

The selected tag group complete with its subdirectories is displayed in the left-hand area in the form of a directory tree.

The tags available for the selected entry are shown in the tag display.

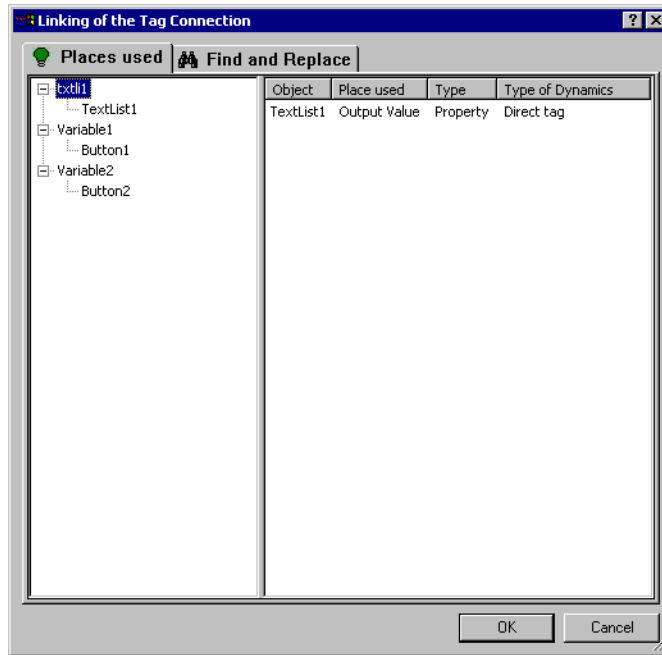
Tag Display

The right-hand area contains all the tags that are available for the entry.

The tag display is divided into 4 columns showing the tag name, type and parameter, and the point in time at which the tag was last changed. Click the respective column heading to sort the tags.

4.2.7.9 Link the Tag Connection of Objects

The Linking command is used to reconfigure the tag connection for all selected objects. The tag connections configured for the selection are listed in a dialog and can be linked directly or with the aid of the **Find and Replace** functions.



[immrg023-101208-01.tif, 1, en_US]

Figure 4-48 Linking of the Tag Connection

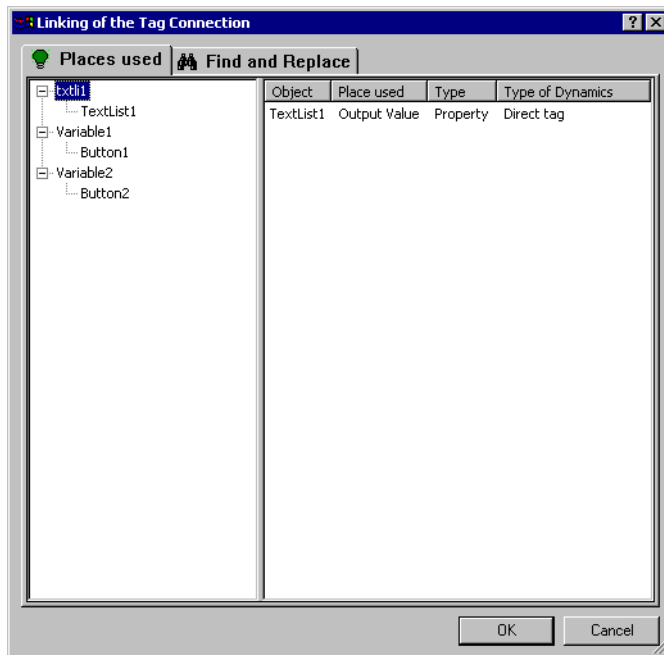
Tag Selection

The left area of the **Places Used** tab shows a list of the tags configured in the object selection. All objects are assigned to these tags in the form of a directory tree, which contains the dynamics of the respective tag.

Details

The right area of the **Places Used** tab shows the current tag connections for the selected tag or object. You can link these tag connections. A right-click on the required tag connection opens the **Linking** dialog.

Linking Tags by Searching and Replacing



[immrg023-101208-01.tif, 1, en_US]

Figure 4-49 Linking of the Tag Connection

Find

The left area of the Find and Replace tab shows the names of all tags configured in the object selection. There are 3 search options available to select certain tag names:

- Select **All** if you want to link the texts displayed.
- Click the  button, if you only wish to change the connections to the selected tag. All remaining tag names are hidden.
- Enter the tag names or parts of names you want to search for in the search field. You can also activate the options **Find whole words only** and **Match case**. All tags which so not meet the search criteria are hidden.

Replace

In the right area of the **Find and Replace** tab, you can enter the name of the new tag with which the selected tag names should be replaced.

Alternatively you can click the associated button to select the required tag.

- The **Preview** button provides viewing and checking the planned replacement.
- The **Replace** button links the selected tag connections as shown in the preview.



NOTE

The **Replace** function replaces only the part of the tag names contents displayed in the **Find** text box of the area.



NOTE

Grouped objects can only be linked if there are no grouped objects in the group itself.

4.2.7.10 Direct Connection

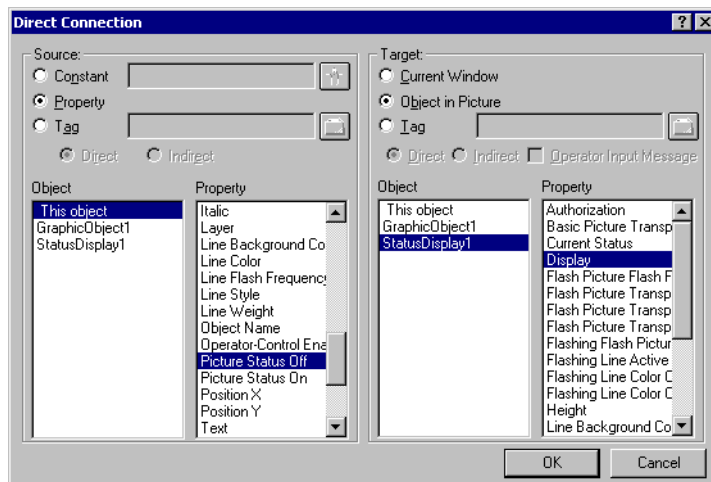
Events can be made dynamic using a direct connection. A direct connection offers the quickest dynamics in the worldmap and achieves the highest runtime performance. However the direct connection can only be used within a worldmap and only single connection can be created. A direct connection is created with the **Direct Connection** dialog.



NOTE

This configuration is native WinCC functionality and does not have that importance within the Spectrum Power 5 environment. It can only be used in a limited way.

WinCC and PowerCC Tags can only be read during Runtime. Writing data or setting values is not possible within the Spectrum Power 5 environment.



[immrg024-101208-01.tif, 1, en_US]

Figure 4-50 Direct Connection

Source

You can select a constant, object property or a tag as the data source.

Constant

Click the associated button or select a picture of the PDL format, which provides the constant.

Object Property

Select the object and an attribute of the object whose value is used as the data source.

Tag

Click the associated button and select the required tag. Define whether the update of the tags is done permanently (directly) or only when called (indirectly).

Target

Select **Current** window, **Object in Picture** or **Tag** as the target, which takes the value of the data source.

Current Window

Select this option if you want to assign the value of the data source to the active worldmap.

Object in Picture

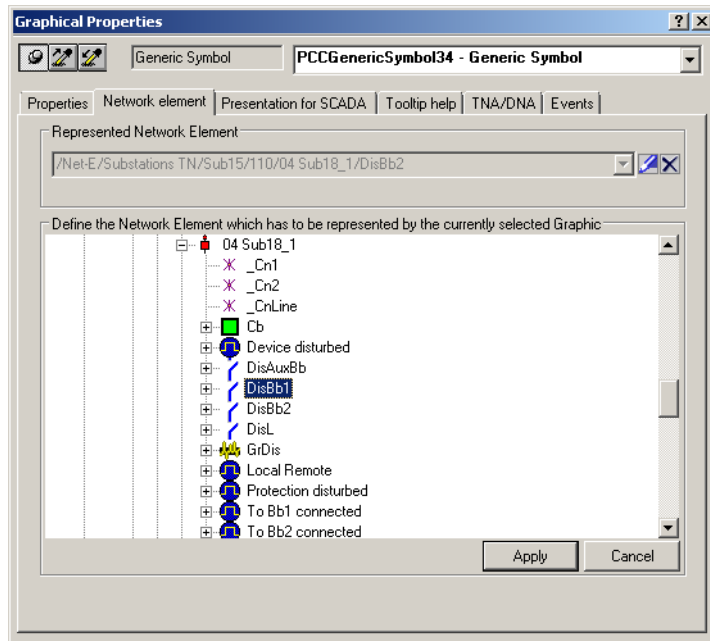
Select the object and attribute of this object to which you want to assign the value of the data source.

Tag

Click the associated button and select the required tag, which takes the value of the data source. Define whether the update of the tags is done permanently (directly) or only when called indirectly.
If necessary, you can activate the output of an operator message.

4.2.7.11 Network Element Tab in the Graphical Properties Window

In this tab the domain object can be selected, to which the graphic objects are assigned. Select the desired domain object by clicking the respective item in the tree.

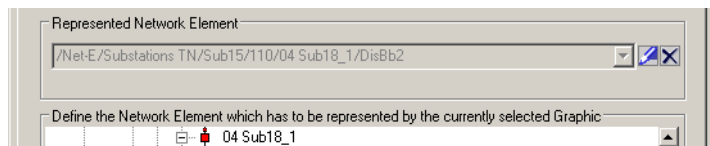


[rgimm008-190711-01.tif, 1, en_US]

Figure 4-51 Graphical Properties Window - Network Element

Represented Network Element field shows the path of the network element selected or, for existing links, the path of the domain data object the graphic object is linked to.

Clicking on  enables the field so that user can modify the technological address alternatively.  undoes user entry and deletes the link.

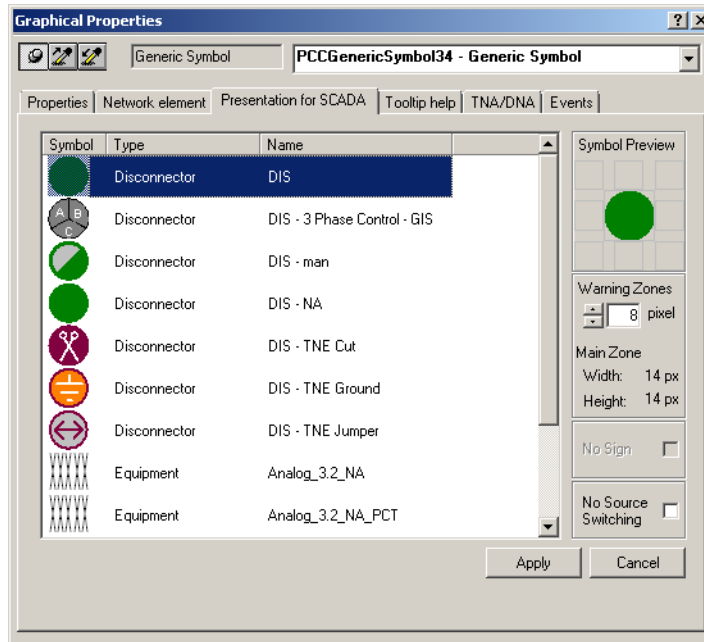


[rgimmge5-190711-01.tif, 1, en_US]

Figure 4-52 Edit Technological Address

4.2.7.12 Presentation Tab in the Graphical Properties Window

In the **Presentation** tab, an object is associated with a presentation logic when applicable. A presentation logic basically describes the decision table and determination of figure and color (including blinking rate). EMF files represent the figures as well as the color.



[rgimm009-190711-01.tif, 1, en_US]

Figure 4-53 Graphical Properties Window - Presentation

In the **Presentation** tab, presentation types for SCADA and NA values are defined. At first, select the value type (SCADA or NA) within the Display Mode toolbar. Then click the desired item in the list of presentation types available for this object.

The option **No Source Switching** prohibits value source switching at runtime for that particular value. The preview of the selected presentation logic is shown on the top right side of the presentation types list.



NOTE

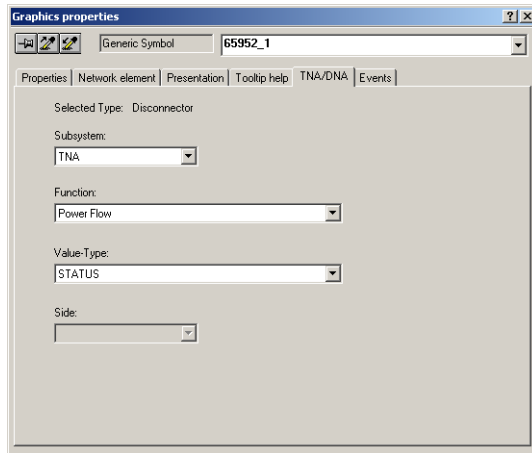
For additional information regarding Spectrum Power presentation logics, refer to Reference Guide Information Model Management (IMM).

The warning zone dimension of the graphical object can be specified in the **Warning Zone** area.

The **No sign** check box is used for analog value representation.

4.2.7.13 TNA/DNA Tab in the Graphical Properties Window

In this tab the application that is used as data source for the object can be selected if SCADA is not used as data source. Select the application, function, etc. from the respective lists.

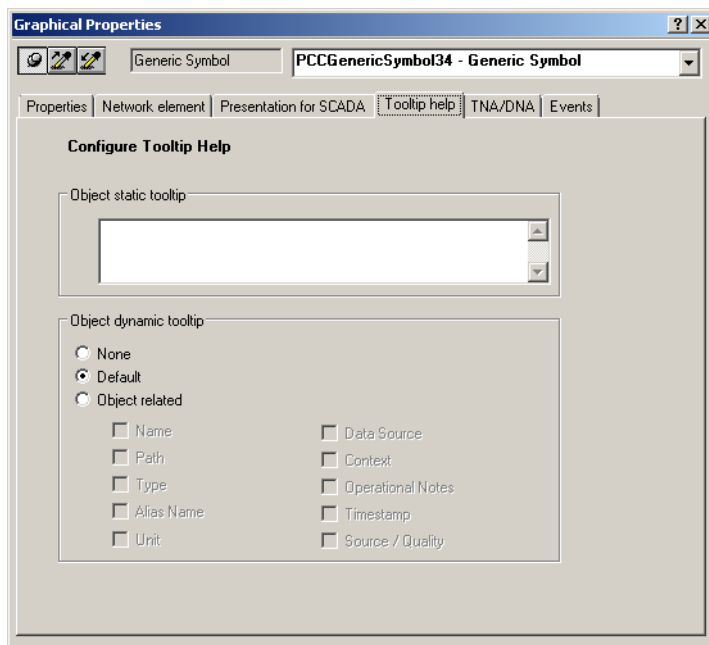


[rgimm010-230210-01.tif, 1, en_US]

Figure 4-54 Spectrum Power 5 Graphical Properties Window - TNA / DNA

4.2.7.14 Tooltip Help Tab in the Graphical Properties Window

In this tab the tooltip help presented at runtime for a certain object can be selected. Select the options you want to have during runtime.



[imgmge034-190711-01.tif, 1, en_US]

Figure 4-55 Spectrum Power 5 Graphical Properties Window - Tooltip Help

4.2.8 Property Groups and Attributes

4.2.8.1 Introduction

On the **Properties** tab, of the **Graphical Properties** window all property groups are displayed which are available for the selected object. If one of these properties is selected, the associated attributes for the selected object are listed in the attribute display. The properties of an object can be changed by assigning a new value to the associated attributes.

4.2.8.2 Geometry Property Group

The **Geometry Property** group contains attributes, which you can use to change the geometric properties of objects. The **Geometry Property** group is available for all objects. However, the available attributes are dependent of the object type.

Table 4-7 Attributes in the Geometry Property Group

Attribute	Function
Current value X	Horizontal position of the currently selected corner point, relative to the image origin. (Unit: Pixels)
Current value Y	Vertical position of the currently selected corner point, relative to the image origin. Unit: Pixels)
Alpha	Depth angle
Start angle	Start angle for the display of segment or arc objects. (Unit: Degree)
Number of corners	Number of corner points
Number of boxes	Number of available list boxes
Alignment	Direction of movement of the slider control (Value: horizontal or vertical)
Bar Width	Width of the 3D bar (Unit: Pixels)
Bar Height	Height of the 3D bar (Unit: Pixels)
Bar direction	Direction of the positive bar axis (top, bottom, left, right) bar
Bar depth	Depth of the 3D bar (Unit: Pixels)
Base X	Horizontal distance from the right bar edge to the left object field edge
Base Y	Vertical distance from the lower bar edge to the upper object field edge
Beta	Beta for the 3D effect (Unit: Degree)
Width	Distance between the left and right borderline of an object (Unit: Pixels)
Button 1-8 width	Individual width for each of the 8 buttons (1-8) (Unit: Pixels)
Display axis	Coordinate axis on which the measured values are displayed as 3D bars
Corner radius X	Radius of the corner rounding (Unit: Percentage)
Corner radius Y	Radius of the corner rounding (Unit: Percentage)
End angle	End angle for the display of segment or arc objects (Unit: Degree)
Window width	Distance between the left and right borderline (Unit: Pixels)

Attribute	Function
Window height	Distance between the top and bottom border line (Unit: Pixels)
Same size	Display type of the 8 buttons
Height	Distance between the top and bottom border line of an object (Unit: Pixels)
Index	Number of the currently selected corner points
Position X	Horizontal distance of the object origin from the image origin (Unit: Pixels)
Position Y	Vertical distance of the object origin from the image origin (Unit: Pixels)
Radius	Radius of a circular object (Unit: Pixels)
Radius X	Horizontal radius of an elliptical object (Unit: Pixels)
Radius Y	Vertical radius of an elliptical object Unit: Pixels)
Rotation reference X	Horizontal coordinates of the reference point about which an object rotates in runtime (Unit: Percentage)
Rotation reference Y	Vertical coordinates of the reference point about which an object rotates in runtime (Unit: Percentage)
Rotation angle	Rotation angle, by which the position of an object deviates from its configured initial position following rotation around the reference point (Unit: Degree)
Angle settings	Type of depth display for the 3D effect
Zoom Factor	Initial zoom factor when calling up a worldmap view at runtime
Number of linking points	Number of connection points for attaching a connector to that object.

4.2.8.3 Colors Property Group

The **Colors Property** group contains attributes which you can use to change the color settings of objects. The **Colors Property** group is available for all objects except **Generic Symbol**, **Picture Section**, **Application** windows, **Control**, OLE element, and **Picture** windows. However, the available attributes are dependent of the object type.

Table 4-8 Attributes in the **Color Property Group**

Attribute	Function
3D Border Color	Color of the left and upper part of the 3D border
3D Shadow Color	Color of the right and lower part of the 3D border
List Background Color	Background color of the non-selected entries in the selection list
List Font Color	Font color of the non-selected entries in the selection list
Bar Color	Color of the bar to display the current value
Bar Background Color	Color of the bar background
High Limit Color	Color of the upper and right arrow key

Attribute	Function
Low Limit Color	Color of the lower and left arrow key
Fill pattern Color	Color of the fill pattern
Background Color	Fill color of the object
Button Color	Color of the regulator
Line Color	Color of the line
Line Background Color	Background color of the line
High Limit Background Color	Color of the upper and right slider surface
Border Color	Color of the border
Border Background Color	Background color of the border
Font color	Color of the text in an object
Selection Font Color	Color of the font in the selected entry of the selection list
Selection Background Color	Background color in the selected entry of the selection list
Scale Color	Color to display the scale and label
Trend Color	Color of the trend display which displays the trend of the measured value in the form of a small arrow at the lower end of the bar
Dividing Line Color	Color of the dividing lines in the selection list
Dividing Line Background Color	Background color of the dividing lines in the selection list
Low Limit Background Color	Color of the lower and left slider surface
Grid Color of Quality Code: Bad	Grid color in case of values with bad quality
Grid Color of Quality Code: Uncertain	Grid color in case of values with uncertain quality
Background flashing color (OK, Simulation)	Background flashing color
Font flashing color (OK, Simulation)	Font flashing color
Background Color (OK)	Fill color of the object
Font Color (OK)	Font color of the object
Background Color (Simulation)	Fill color of the object
Font Color (Simulation)	Font color of the object

4.2.8.4 Styles Property Group

The **Styles Property** group contains attributes which you can use to change the style for the appearance of objects. The **Styles Property** group is available for all objects except **Generic symbol**, **Picture Section**, **Application** windows, **Control**, OLE element and **Picture** windows. However, the available attributes are dependent of the object type.

Table 4-9 Attributes in the Styles Property Group

Attribute	Function
3D Border Weight	Line weight of the 3D border (Unit: Pixels)
Bar Pattern	Pattern to display the bar surface, for example solid or broken
Border weight	Width of the border (Unit: Pixels)
Latch Down	Saves the last selected switch positioned Pressed or Not pressed)
Box Alignment	Arrangement of the buttons relative to the describing text
Fill pattern	Pattern for the display of background areas, for example solid or hatched
Pressed	Start position of the switch
Background	Display of the background
Light Effect	Display of the sides of a 3D bar with different brightness
Line style	Display type of a line, for example dotted or dashed. It also defines line end and corner shapes.
Line End Style	Display type for the end points of a line.
Line Weight	Thickness of a line (Units: Pixel)
Border Style	Display type of a border line, for example dotted or broken
Border Width	Thickness of a border (Unit: Pixel)
Dividing Line style	Display type of the dividing lines of a selection list, for example dotted or broken
Dividing Line weight	Thickness of the separation lines in a selection list (Unit: Pixels)
Windows Style	Adaptation of the object style to the default settings on the operating system
Draw border inside	Draw border inside the object
Bevel	Defines the bevel effect (off, up, or down)
Display as drop-down list box	Display object as drop-down list box

4.2.8.5 Font Property Group

The **Font Property** group contains attributes which you can use to change the appearance of text in objects. The **Font Property** group is only available for the following objects. Static text, label, I/O field, bar, text list, group display, analog display (extended), button, check box, and radio box. However, the available attributes are dependent of the object type.

Table 4-10 Attributes in the Font Property Group

Attribute	Function
Bold	Text is displayed in boldface

Attribute	Function
Index	Numbering of the individual label fields Check Box, Radio Box Italic Text is displayed in italics
Italic	Text is displayed in italics
Text orientation	Horizontal or vertical arrangement of the text in the object
Text	The object text
Underline	Text is displayed underlined
X alignment	Horizontal alignment of the text in the object
Y alignment	Vertical alignment of the text in the object
Font	Font of the text
Font size	Font size of the text in points
Format	Format of the text

4.2.8.6 Flashing Property Group

The **Flashing Property** group contains attributes with which you can change the flashing of object elements. The settings of the **Flashing Property** group can only be seen in runtime. The **Flashing Property** group is available for all objects except **Generic Symbol**, **Picture Section**, **Application** windows, **Picture** windows, **Control**, OLE element, and 3D bar. However, the available attributes are dependent of the object type.

Table 4-11 Attributes of the Flashing Property Group

Attribute	Function
Flashing Background Active	Activates flashing of the background
Flashing Line Active	Activates flashing of the line
Flashing Border Active	Activates flashing of the border
Flashing Text Active	Activates flashing of the font
Flash Frequency	Frequency for the flashing of the group display
Background Flash Frequency	Frequency for the flashing of the background
Line Flash Frequency	Frequency for the flashing of the power frequency for the flashing of the border
Border Flash Frequency	Frequency for the flashing of the border
Text Flash Frequency	Frequency for the flashing of the text
Flashing Background Color Off	Color for the flash state Off
Flashing Background Color On	Color for the flash state On
Flashing Line Color Off	Color for the flash state Off
Flashing Line Color On	Color for the flash state On
Flashing Border Color Off	Color for the flash state Off
Flashing Border Color On	Color for the flash state On
Flashing Text Color Off	Color for the flash state Off
Flashing Text Color On	Color for the flash state On
Flashing	

4.2.8.7 Miscellaneous Property Group

The **Miscellaneous Property** group contains attributes which you can use to change the miscellaneous settings of objects. The **Miscellaneous Property** group is available for all objects except Picture Section. However, the available attributes are dependent of the object type.

Overview

Table 4-12 Attributes in the Miscellaneous Property Group

Attribute	Function
Update Cycle	Type and frequency of the update
Display	Activates the display of an object
Display options	Specifies if a button is assigned to a graphic text, or both
Operator-Control Enable	Release to operate an object
Operator input message	Defines whether a message is output following operation
Operator Activities Report	Defines whether the activities of an operation should be logged
Operation Steps	Defines how many steps the slider is moved on the slider surface when the mouse is clicked
User Value 1-4/ Status Value 1-4	Any 32-Bit value which for example can be used for a script evaluation
Authorization	User-specific authorization to operate an object
Picture Name	Defines the name of the picture to be shown
Picture Status Off	Defines the picture displayed when a button is not pressed
Picture Status On	Defines the picture displayed when a button is pressed
Bit Pattern Group display	System-internal Output Value
Cursor Control	Defines whether the Alpha mouse pointer automatically jumps to the next field of the TAB order following input in a field
Immediate Input	Defines whether when jumping to the object it should change directly to input mode
Extended Operation	Defines whether the slider jumps to the minimum or maximum value when the mouse is clicked the slider surface
Change Color	Defines whether the color changes segment by segment or for the entire bar when the limiting value is reached
Adapt Size to monitor	Permits adaptation of the window size to the monitor size
Sizeable	Permits the operator to change the window size
Hotkeys	Defines a key or keyboard shortcut as an alternative operation to a mouse click
Hysteresis	Permits display with hysteresis
Hysteresis Range	Defines the hysteresis in percentage of the display area

Attribute	Function
Maximum value	Absolute value to display the largest value
Can Be Maximized	Permits the enlargement of a window to the maximum screen size
Minimum value	Absolute value to display the smallest value
Average Value	Permits an average value over the last 15 values
Zero Point Value	Absolute value to display the zero point
Process driver connection	Default values for the display in runtime if the associated tag is not linked or updated
Acknowledgment Mask	During display of the group display object, the events Tolerance Low, Tolerance High, and Operator Request are automatically identified as being acknowledged.
Border	Display of a window with or without a border
Adapt Border	Dynamic adaptation of the field border to the text size
Group Relevant	Defines the consideration to form a group display
Group Value	Collect value of the current state of the active message classes
Can Be Closed	Permits a window to be closed
Server name	Display under which name an embedded object is registered in the operating system
Server Prefix	Name of the server from which the picture to be shown is called
Configured Languages	Displays the languages configured for the selected object
Text Reference	Defines whether the language-dependent assignment texts are managed in the text library or directly in the object
Title	Activates the title bar of a window
Tool Tip Text	Describing text which appears in runtime as soon as the mouse pointer is positioned on the object
Trend	Permits the display of the trend of a measured value
Heading	Headline displayed in the title bar of the picture window
Tag Prefix	Name extension prefixed on all tag in the picture
Movable	Permits a window to be moved
Foreground	Defines whether a window is pinned on top
Menu / Toolbar Configuration	Configuration of the Menu/Toolbar in the picture window
Use group value for display	If used, you can assign pictures for the individual alarm statuses
Outgoing state visible	Defines whether an outgoing state is visible or not
Tag name	Is used for graphic object wizard
Tag	Is used for graphic object wizard
Tag type	Is used for graphic object wizard
Value	Linking with any Analog/Text tag
Value as Text	Defines whether a text tag is linked instead of a formatted analog value

Attribute	Function
Simulation	Defines linking with any tag which is used for simulation
Simulation Bit	Shows the bit position of the linked simulation tags.

4.2.8.8 Filling Property Group

The **Filling Property** group contains attributes, which you can control the dynamic filling of objects. The **Filling Property** group is available for objects. Polygon, ellipse, circle, ellipse segment, pie segment, rectangle, Rounded rectangle, static text, label, graphic object, button, check box, option group, round button, and slider. However, the available attributes are dependent of the object type.

Table 4-13 Attributes of Filling Property Group

Attribute	Function
Dynamic filling	Permits dynamic filling of objects with closed border line
Fill Level	Height of the object filling with closed border line as a percentage of the object height

4.2.8.9 Picture Property Group

The **Picture Property** group contains attributes with which you can change the properties of the picture to be displayed. The **Picture Property** group is only available for the object **Graphic Object**. However, the available attributes are dependent of the object type.

Table 4-14 Attributes of Picture Property Group

Attribute
Picture
Picture Transparent Color
Picture Transparent Color On

4.2.8.10 Pictures Property Group

The **Pictures Property** group contains attributes with which you can change the properties of the pictures to be displayed for different states. The **Pictures Property** group is only available for the object round button. However, the available attributes are dependent of the object type.

Table 4-15 Attributes of Pictures Property Group

Attribute	Function
Picture Off Transparent Color	For the Off state, defines which color of the picture is set to Transparent
Picture Off Transparent Color On	Permits the Transparent Color function for the Off state
Picture Deact. Transparent color	For the Deactivated state, defines which color of the picture is set to Transparent
Picture Deact. Transparent Color On	Permits the Transparent Color function for the Deactivated state
Picture On Transparent Color	For the On state, defines which color of the picture is set to Transparent
Picture On Transparent Color On	Permits the Transparent Color function for the On state
Picture Status Off	Picture displayed for the Off state
Picture Status Deactivated	Picture displayed for the Deactivated state
Picture Status On	Picture displayed for the On state

4.2.8.11 Axis Property Group

The **Axis Property** group contains attributes with which you can change the properties of the bar. The **Axis Property** group is only available for the object bar.

Table 4-16 Attributes of Axis Property Group

Attribute	Function
Axis section	Distance between the main tick marks (value difference between 2 adjacent axis labels)
Alignment	Orientation of the scale relative to the bar (top/bottom or left/right)
Bar Scaling	Type of scale divisions (for example linear or logarithmic)
Label Each	Each Number of labeled main tick marks (each one, every second one, every third one ...)
Exponent display	Permits number representation with exponents
Large Tick Marks	Style for the display of the main tick marks of a scale (bold or normal)
Length of Large Tick Marks	Length of the main tick marks of a scale. The length of the trend tick marks is 50% of the value set here (Unit: Pixels)
Digits to the Right of the Decimal Point	Number of digits to the right of the decimal point to display numerical values on the scale
Zero Point	Position of the zero point relative to the distance of the scale end values (Unit: Percentage)
Only Large Tick Marks	Defines whether only the main tick marks or only the trend tick marks are displayed on the scale.
Scale	Permits the display of a scale to display the values
Scale Marks	Number of bar segments subdivided by the main tick marks of the scale
Digits to the Left of the Decimal Point	Number digits to the left of the decimal point to display numerical values on the scale

4.2.8.12 State Property Group

The **State Property** group contains attributes with which you can change the properties of the status display. The **State Property** group is available for the objects Status display, Status display (extended) and Analog display (extended). However, the available attributes are dependent of the object type.

Table 4-17 Attributes of Status Property Group

Attribute	Function
Current status	Status set to display a basic picture or a flash picture
Flash Picture	Picture to be displayed as a flash picture
Flash Picture Transparent Color	Defines which color of the flash picture is set to transparent
Flash Picture Transparent Color On	Permits the Transparent Color function for the flash picture
Flashing Flash Picture Active	Activates flashing of the flash picture
Flash picture flash frequency	Frequency for the flashing of the flash picture

Attribute	Function
Basic Picture	Picture to be displayed as basic picture
Basic Picture Transparent Color	Defines which color of the basic picture is set to Transparent
Basic Picture Transparent Color On	Permits the Transparent Color function for the basic picture
Index	Shows the index of currently selected state
Highest Index	Shows the highest index of all configurable alarm and status combinations
Bit Selection 0-3	Indicates the status tag defined
Status Word Bit 0-3	Indicates the bit position of the selected tag for bit 0 of the status value
Priority Bit 16-31	Indicates the priority of the bit for the alarm evaluation

4.2.8.13 Limits Property Group

The **Limits Property** group contains attributes with which you can set limiting values and the monitoring them. The **Limits Property** group is only available for the following objects. I/O field, bar, and 3D bar. However, the available attributes are dependent of the object type.

Table 4-18 Attributes of Limits Property Group

Attribute	Function
Type (AH, WH, TH, RH4, RH5, AL, WL, TL, RL4, RL5)	Type of display for the specified limiting value (percent or absolute)
Bar color (0 - 10)	Color the 3D bar takes when the specified limit is reached
Bar color (AH, WH, TH, RH4, RH5, AL, WL, TL, RL4, RL5)	Color the bar takes when the specified limit is reached
Limit (0 - 10)	Value of the specified limit
Limit Marker	Permits the display of the limiting values on the scale
High limit value	Upper limit to display a value in the I/O field
Limit (AH, WH, TH, RH4, RH5)	Value of the specified upper limit
Low limit value	Lower limit to display a value in the I/O field
Limit (AL, WL, TL, RL4, RL5)	Value of the specified lower limit
Monitoring (0 - 10)	Activates monitoring of the specified limit
Monitoring (AH, WH, TH, RH4, RH5, AL, WL, TL, RL4, RL5)	Activates monitoring of the specified limit

4.2.8.14 Output / Input Property Group

The **Output / Input Property** group contains attributes with which you can set the properties of input and output values. The **Output / Input Property** group is only available for the following objects. I/O field, text list, check box, and radio box. However, the available attributes are dependent of the object type.

Table 4-19 Attributes of Output / Input Property Group

Attribute	Function
Number of visible Lines	Defines how many lines the selection list should Text list contain
Output format	Format indication for the display of the output value

Attribute	Function
Output value	Default values for the display in runtime if the associated tag is not linked or updated
Bit Number	Number of the relevant bit in the output value
Data Format	Data type of the field content (binary, decimal, string, hexadecimal)
Field type	Defines the function type of the field
List type	Data type of the field content (binary, decimal, bit)
Clear on Invalid Input	Defines whether the field content is automatically deleted, if input is incorrect
Clear on New Input	Defines whether the existing field contents are deleted when the I/O Field text box is activated
Selected boxes	Informs the system about which fields are active
Selected Box	Informs the system about which field is activated
Apply on Exit	Defines whether the specified data are applied on exiting the field, I/O Field and Text List
Apply on Full	Defines whether the field is automatically exited and the data applied simultaneously as soon as the predefined number of characters have been entered
Hidden Input	Defines whether the input is visible immediately or replaced by *
Assignments	Assignment between the output value and the output value to be actually output

4.2.8.15 Message Types Property Group

The **Message Types Property** group contains attributes with which you can set the display of messages depending on message type and message class. The **Message Types Property** group is only available for the objects Group display and Analog display (extended). However, the available attributes are dependent of the object type.

Table 4-20 Attributes of Message Types Property Group

Attribute	Function
Display Text	Label of buttons depending on the displayed message type
Went Out Unacknowledged – Background Flashing	Permits flashing background when a message goes out unacknowledged
Went Out Unacknowledged – Background Color Off	Background color for the flash state Off
Went Out Unacknowledged – Background Color On	Background color for the flash state On
Went Out Unacknowledged – Text Flashing	Permits flashing font when a message goes out unacknowledged
Went Out Unacknowledged – Text Color Off	Color of the text for the flash state Off
Went Out Unacknowledged – Text Color On	Color of the text for the flash state On

Attribute	Function
Came in – Background flashing	Permits flashing background when a message comes in
Came in – Background Color Off	Background color for the flash state Off
Came in – Background Color On	Background color for the flash state On
Came in – Text flashing	Permits flashing text when a message comes in
Came in – Text Color Off	Color of the text for the flash state Off
Came in – Text Color On	Color of the text for the flash state On
Came in acknowledged – Background flashing	Permits flashing background when a message comes in acknowledged
Came in acknowledged – Background Color Off	Background color for the flash state Off
Came in acknowledged – Background Color On	Background color for the flash state On
Came in acknowledged – Text flashing	Permits flashing text when a message comes in acknowledged
Came in acknowledged – Text Color Off	Color of the text for the flash state Off
Came in acknowledged – Text Color On	Color of the text for the flash state On
Message Type	Defines the message type (for example AL, AH, WL, etc.) for which the attributes of the Message Types property group are configured

4.2.8.16 Lock Property Group

The **Lock Property** group contains attributes with which you can change the properties to display locked measurement points. The **Lock Property** group is only available for the object Group display.

Table 4-21 Attributes of Lock Property Group

Attribute	Function
Lock Display	Permits viewing a locked measurement point
Lock Display Text	Label of the buttons for a locked measurement point
Lock Background Color	Background color of the buttons for a locked measurement point
Lock Text Color	Font color of the buttons for a locked measurement point

4.2.8.17 Connected Objects Property Group

The **Connected Objects Property** group contains attributes with which you can change the connection of 2 objects using the connector object. The **Connected Objects Property** group is only available for the object connector.

Table 4-22 Attributes of Connected Objects Property Group

Attribute	Function
Object name of the source object	Defines the object which is connected to the beginning of the connector
Connection point index of the source object	Defines for the beginning at which connection point of the source object the connection is established
Object name of the target object	Defines the object which is connected to the end of the connector

Attribute	Function
Connection point index of the target object	Defines for the end at which connection point of the target object the connection is established
Change Orientation	Turns the definition from the Start and End of the connector with the current assignments around.
Connection Type	Defines whether an automatic or simple direct connection should be created between the objects.
Automatic Positioning	

4.2.8.18 Assignment Property Group

The **Assignment Property** group contains attributes you use to assign message types to buttons of the group display object. The display of message types is configured in the Message Types property group. The **Assignment Property** group is only available for the Group display object.

Table 4-23 Attributes of Assignment Property Group

Attribute	Function
Message Types for Button 1	Specifies one or several message events for the first button of the group display object.
Message Types for Button 2	Specifies one or several message events for the second button of the group display object.
Message Types for Button 3	Specifies one or several message events for the third button of the group display object.
Message Types for Button 4	Specifies one or several message events for the fourth button of the group display object.
Message Types for Button 5	Specifies one or several message events for the fifth button of the group display object.
Message Types for Button 6	Specifies one or several message events for the sixth button of the group display object.
Message Types for Button 7	Specifies one or several message events for the seventh button of the group display object.
Message Types for Button 8	Specifies one or several message events for the eighth button of the group display object.
Domain object ID	GUID of the referenced Domain Object Instance
Domain object type ID	GUID of the referenced Domain Object Type
Presentation Logic ID	GUID of the referenced Presentation Logic Instance
Presentation Logic type ID	GUID of the referenced Presentation Logic Type
Data source	Specifies the data source (SCADA or NA)
Use standard data source	Yes: use object specific source No: use window specific source
Data source ID	Information about NA data source containing eventtype, subsystem, function number, side and original event type
Tooltip type	Specifies the tooltip type
Label	Specifies the label text

4.2.8.19 LD Property Group

The **LD Property** group contains attributes you use to assign the linking and connection to the Spectrum Power 5 domain. In addition, special Spectrum Power 5 functionality is also included here. Typically the properties of the **LD Property** group is filled automatically within the editing workflows by other means. The **LD Property** group is available for all Windows objects, and Graphic object, Status display and Status display (extended). However, the available attributes are dependent of the object type.

Table 4-24 Attributes of LD Property Group

Attribute	Function
Domain object ID	GUID of the referenced Domain object Instance
Domain object type ID	GUID of the referenced Domain object Type
Presentation Logic ID	GUID of the referenced presentation logic Instance
Presentation Logic type ID	GUID of the referenced presentation logic type
Data source	Specifies the data source (SCADA or NA)
Use standard data source	Yes: use object specific source No: use window specific source
Data source ID	Information about NA data source containing event-type, subsystem, function number, side and original event type
Tooltip type	Specifies the type of tooltip
Label	Specifies the label text
Zone size	Specifies the zone height for generic symbol
Zone width	Specifies the zone width for generic symbol
Square	Show object always square
Display name	Name of the display to be shown at runtime
Standard toolbar	Defines the toolbar of the object at runtime
Fit to screen	Defines that at display, call up at runtime shall open in full size regarding the monitor used.

4.2.8.20 Zoom Property Group

The **Zoom Property** group contains attributes to describe the object behavior during zoom operations of a worldmap. The behavior is described in the Zoom property group. The **Zoom Property** group is only available for the object Static Text and Label. However, the available attributes are dependent of the object type.

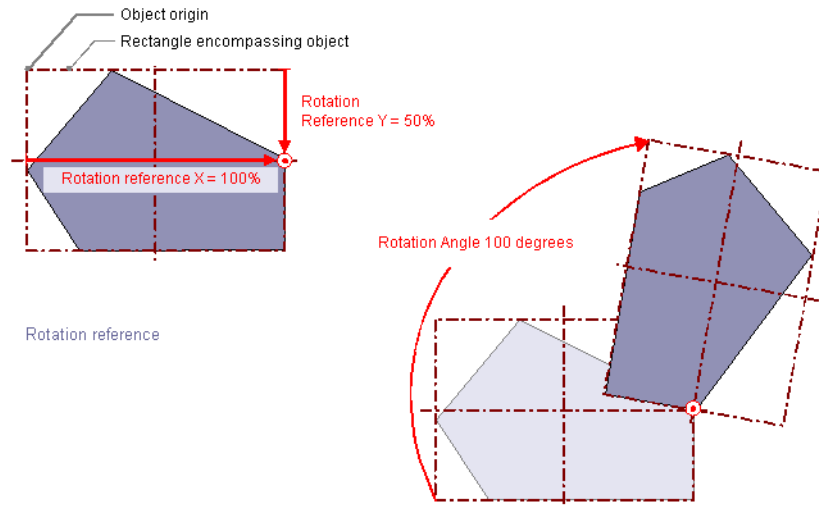
Table 4-25 Attributes of Zoom Property Group

Attribute	Function
Scalable	Defines non-linear zooming behavior or linear zooming behavior for the object
Linear	Defines whether vanishing ranges are to be considered
Lower visibility limit	Defines lowest zoom factor when visible in % of zoom factor
Upper visibility limit	Defines highest zoom factor when visible in % of zoom factor
Anchor point	Defines the Anchor Point for non-linear zooming behavior

4.2.9 Special Runtime Settings

4.2.9.1 Dynamic Rotation of Objects at Runtime

A runtime rotation around the reference point can be configured for objects of the type line, polygon, polyline, or static text. The rotation of an object is only visible in runtime. The coordinates of the reference point and the configured start position are defined by the attributes Rotation reference X, Rotation reference Y, and Rotation angle in the **Geometry property** group.



[immg029-101208-01.tif, 1, en_US]

Figure 4-56 Dynamic Rotation of Objects at Runtime

Rotation Reference X, Rotation Reference Y

The attribute Rotation reference X and Rotation reference Y define the horizontal and vertical distance of the reference point from the object origin. The values are specified in percent. The object width or object height correspond to a value of 100%. The reference point value can be outside the selection rectangle. This means that both negative values and values higher than 100% are possible.

Rotation Angle

The Rotation Angle attribute defines the rotation of an object around the reference point. The Rotation Angle is specified in degrees. The configured start point corresponds to a value of 0°. The position of an object deviates from its configured start position by the value of the Rotation Angle attribute.

4.2.9.2 Display of Windows at Runtime

The Application Window and **Picture** Window objects are displayed in runtime as a window in a worldmap. The properties of this window display can be changed as required with the attributes described here. Some of these attributes are available for other object types too.

Adapt Size

The Adapt Picture attribute specifies whether a picture window adapts to the size of the embedded picture in runtime.

Sizeable

The Sizeable attribute specifies whether the size of a window can be changed in runtime.

Border

The Border attribute specifies whether a window is displayed with a border.

Scroll Bars

The scroll bars attribute specifies whether a picture window is displayed with a scroll bar if necessary. A scroll bar is only visible in runtime.

Can Be Closed

The Can Be Closed attribute specifies whether a window can be closed in runtime.

Title

The Title attribute specifies whether a window is displayed in runtime with or without a title bar.

Heading

The Heading attribute specifies which text appears in the title of a picture window.

Movable

The Movable attribute specifies whether a window can be moved in runtime.

Foreground

The Foreground attribute specifies whether a window lies in the foreground. This setting is independent of the layer to which the window is assigned.

4.2.10 Logging and Operator Authorizations

In WinCC to operate a worldmap you can define user-specific authorization for the contained objects. Each Operation can be logged and sent to the message system. The associated settings can be changed with the attributes described here.

Display

The Display attribute specifies whether an object is displayed in runtime.

Operator-Control Enable

The Operator-Control Enable attribute specifies whether an object can be operated in runtime.

Operator Input Message

The Operator Input Message attribute specifies whether a message is displayed after an operation. If the attribute Operator Input Message has been set to yes, a message is generated upon operation, sent to the message system and archived. Using the message system, a message may be output in a message line, for example.

Operator Actions Report

The Operator Actions Report attribute specifies whether the reason for an operation is to be logged by the operator. The operator actions report is archived in the message system.

Mouse Pointer Control

The **Cursor Control** attribute specifies whether the alpha mouse pointer automatically jumps to the next field in the tab sequence after an entry has been made in a field. This function enables rapid input in different fields without having to use the tab key to jump.

Keyboard Shortcuts

As an alternative to operation with the mouse, a button can also be triggered using the keyboard. To do so, a keyboard shortcut must be set.

4.2.11 Quick Object Configuration

The **Graphics Editor** has configuration dialogs for quick configuration of individual objects. The important features of an object can be set in the configuration dialogs.

If the use of configuration dialogs is activated in the **Options** tab of the **Tools > Settings...** menu, the associated dialog is opened when an object is inserted. The configuration dialog can also be opened with the context menu of the object.

The contents of the configuration dialogs are different for different objects.
Configuration dialogs are available for the following objects:

- I/O field
- Bar
- Graphic object
- Status display
- Text list
- Button
- Slider
- Status display (extended)
- Application window
- Control
- OLE Element

Short Instruction of the Configurable Features

Features	Description	Object Type
Updating	Specifies the frequency with which the display of the displayed value is updated. Enter the desired interval or select an interval from the list box.	I/O Field, Bar, Status Display, Text List, Slider
Alignment	Specifies whether the slider is moved horizontally or vertically.	Slider
Bar direction	Specifies the direction of the coordinate axis to which the maximum value of bar points.	Bar
Operation	An authorization can be assigned for operation of the button. A keyboard shortcut can also be defined, which allows operation with a key or keyboard shortcut. You can find the detailed description in the Button chapter.	Button
Bit Position	A state can also be assigned to a certain bit position on the tag. Click the arrow key to select a bit position.	Status Display
Picture Selection with Preview	The picture selection shows all pictures in the graphics directory GraCS of the current WinCC project. The picture selected in the picture selection is displayed as a preview. Select a picture or click the Browse... button to insert additional pictures from the picture selection. See the Select Pictures section for the detailed description.	Graphic Object, Status Display
Change Picture on mouse click	An event can be assigned for the selected button with this function. If the button is clicked with the mouse, the display of the worldmap specified here is opened. See the Select Pictures section for the detailed description.	Button
Field type	Specify whether the selected object is used for input, output or input and output of values.	I/O Field, Text List

Features	Description	Object Type
Limits	Enter values as Max. value and Min. value to define the 2 ends of the slider. You can also define the number of operation steps for the slider. The number of operation steps specifies the number of increments by which the slider is moved when the mouse is clicked the slider.	Bar, Slider
Font	Change font options for viewing button labels and display values. Click one of the buttons to open the associated dialog.	I/O Field, Text List, Button
Text	Enter a text as a button label, which for example describes its function. Only single text line is available in the configuration dialog. Multiline text can be input by changing the Text attribute in the object properties window.	Button
Tag	The selected object can be set dynamically by embedding a tag. With an I/O field and text list, the value of the tags can be displayed as output or changed by an input depending on the selected field type.	I/O Field, Bar, Status Display, Text List, Slider
Status list	The status list shows the linked pictures and the specified flash frequency for all configured states. States can be inserted and every setting can be changed with a context menu. Pictures of the picture selection can be referenced by dragging them with the mouse to the required position in the state list.	Status Display

5 Technological Editing

5.1	Operational Steps	190
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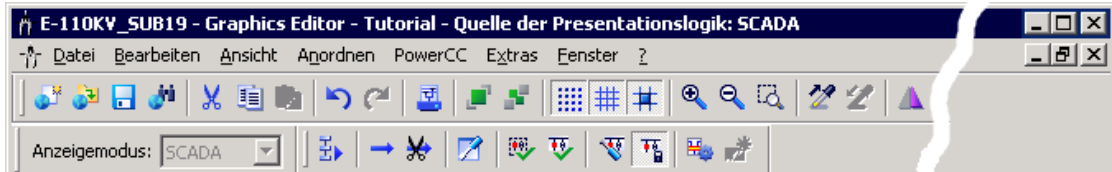
5.1 Operational Steps

5.1.1 Switching between the Graphical Editing Modes

The goal of this task is to switch the graphical editing mode between graphical and technological.

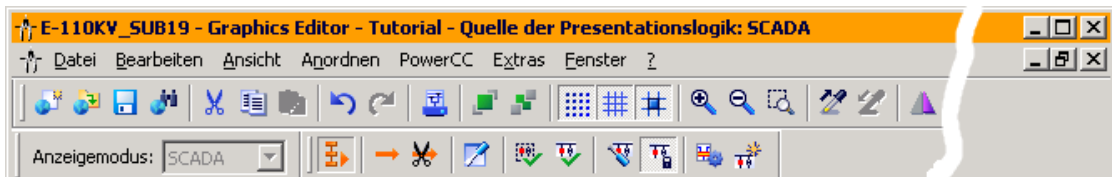
- Open a worldmap.
- Verify the current graphical editing mode.

The titlebar of the Graphics Editor and the Icons in the graphical editing toolbar display the current mode.



[image012-150313-01.tif, 1, en_US]

Figure 5-1 Titlebar and Graphical Editing Toolbar in Graphical Mode



[image013-150313-01.tif, 1, en_US]

Figure 5-2 Titlebar and Graphical Editing Toolbar in Technological Mode

- Use  or  in the graphical editing toolbar to switch to the desired graphical editing mode. The new graphical mode is displayed accordingly.

5.1.2 Creating New Domain Data Using Library Templates or Existing Graphics

The goal of this task is to create graphically new domain data using library templates with linked domain data or already existing graphics within a worldmap with linked domain data.

- Open a worldmap.
- Make sure that the **technological** editing mode is switched on.

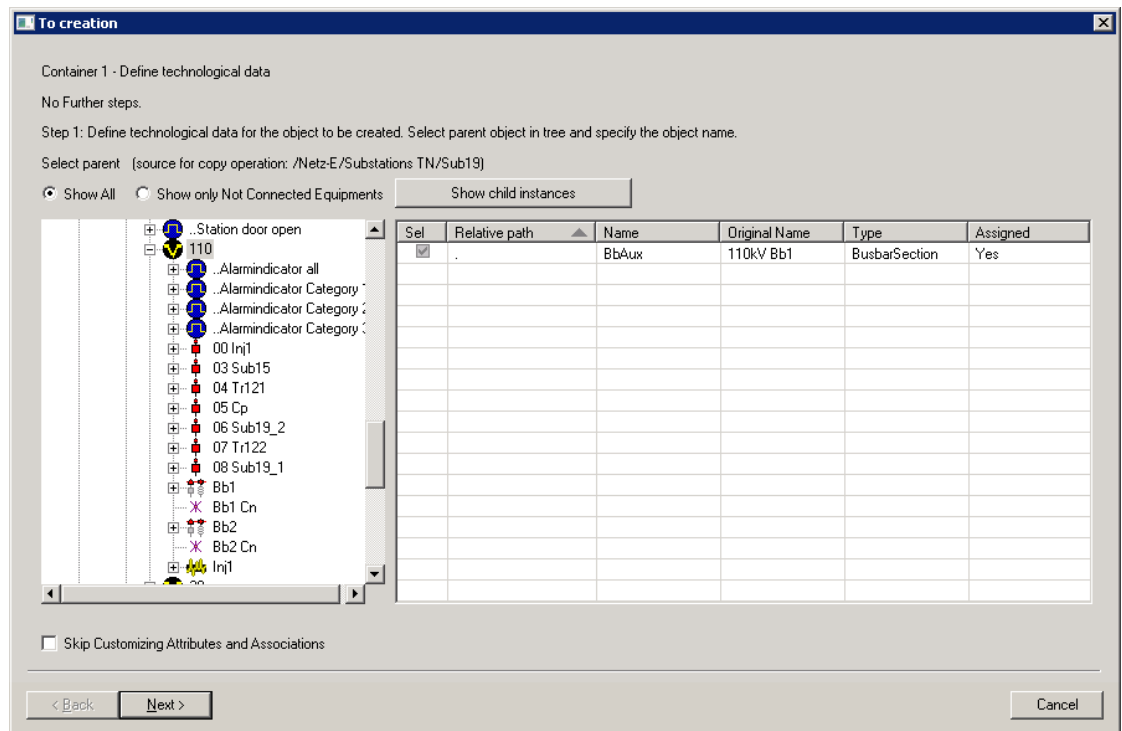
Using Library Templates

- Select a library object with linked domain data from the library and drag and drop it onto the drawing section of the Graphics Editor. The new graphic objects are displayed within the worldmap. The **TO creation** window opens.

Using Existing Graphics with Linked Domain Data

- Select at least 1 object with linked domain data within an opened worldmap.
- Use **<CTRL + C>** or click  in the standard toolbar to copy the selected objects from the worldmap to the clipboard.

- Use **<CTRL + V>** or click  in the standard toolbar to paste the copied content of the clipboard in the active worldmap under the mouse position.
- The copied graphic objects are displayed within the worldmap. The **TO creation** window opens.



[rgimm007-150313-01.tif, 1, en_US]

Figure 5-3 TO Creation Window



NOTE

The number of steps to be performed and the amount of domain data objects included in the domain data list are dependent on the library object used or the copied objects.

- Define the hierarchy where the domain data has to be created by selecting the parent container in the tree.
- Specify a name for the domain data objects in the domain data list within the column **Name**.



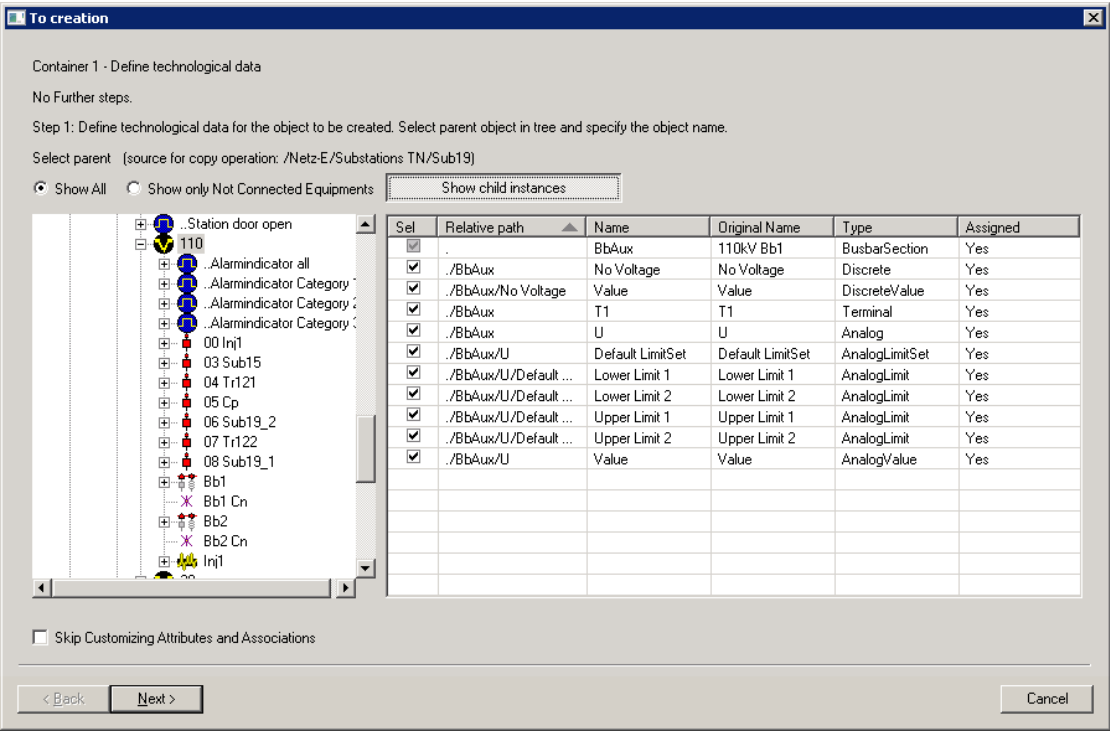
NOTE

In order to create new domain data objects, make sure that **Yes** is displayed in the domain data list within the column **Assigned** for all domain data instances. Otherwise the specified domain data gets not created.

- Click **Next >** if there are additional steps to be performed.
Repeat this step until all steps and definitions are done and the button **Next >** is not longer visible.
- Click **Finish** to create the specified domain data.
The **TO creation** window closes.

Show Child Instances

To get an overview about the included child instances, click **Show child instances**.



[image016-150313-01.tif, 1, en_US]

Figure 5-4 TO Creation Window – Show Child Instances



NOTE

The amount of domain data objects included in the domain data list are dependent on the library object used or the copied objects.

- If desired, specify names for the child domain data objects in the domain data list within the column **Name**.
- If you do not want to create all child domain data objects, uncheck the hook within the column **Sel** for those specific domain data objects.



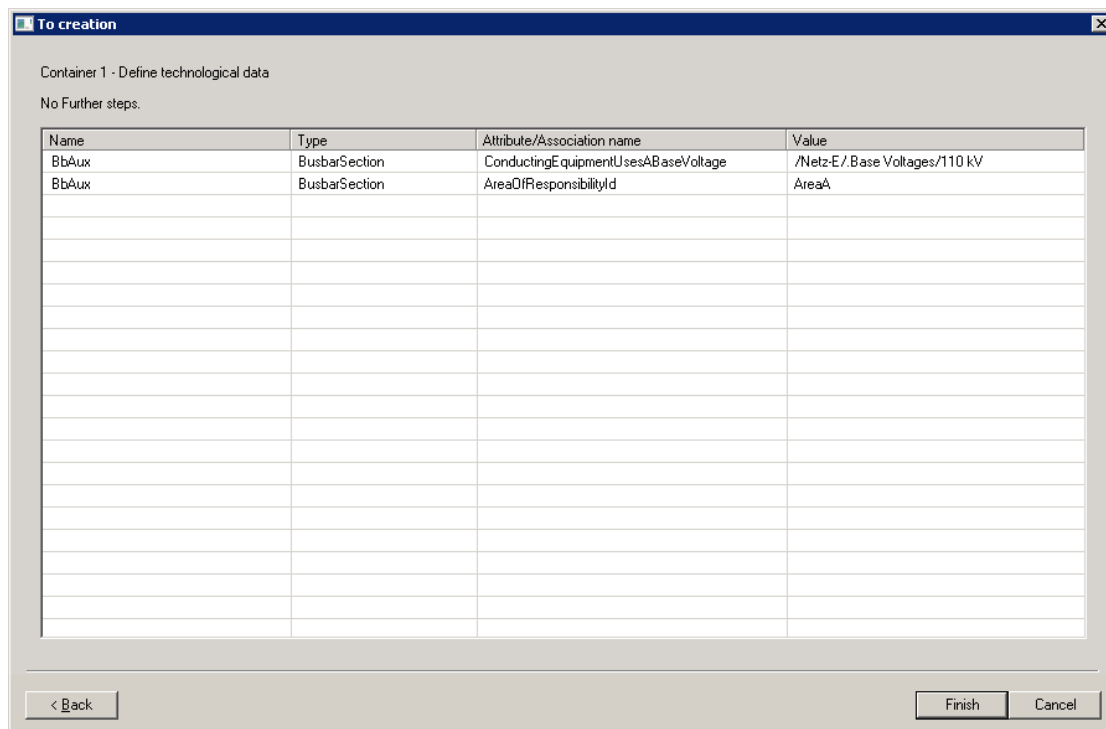
NOTE

The child instance objects within the domain data list may have also additional hierarchies and subhierarchies. If you uncheck a domain data object that includes a subhierarchy, all domain data instances within the subhierarchy are automatically unchecked too.

- Click **Show Child instances** again to hide all child instances within the domain data list.

Customizing Attributes and Associations

You can customize a defined fix set of attributes and associations if **Skip Customizing Attributes and Associations** is not checked within a dedicated step after clicking **Next >**.



[Image017-150313-01.tif, 1, en, US]

Figure 5-5 TO Creation Window – Customize Attributes and Associations

NOTE

The amount of domain data objects and customization possibilities included in the domain data list are dependent on the library object used or the copied objects.

- If desired, specify the values for the domain data objects in the domain data list within the column **Value**.
 - Click **Next >** if there are additional steps to be performed.
 - Click **Finish** to create the specified domain data with the specified values.
- The **TQ creation** window closes.

5.1.3 Mapping New Graphic Objects to Existing Domain Data Using Library Templates or Existing Graphics

The goal of this task is to map new graphic objects to existing domain data using library templates with linked domain data or already existing graphics within a worldmap with linked domain data.

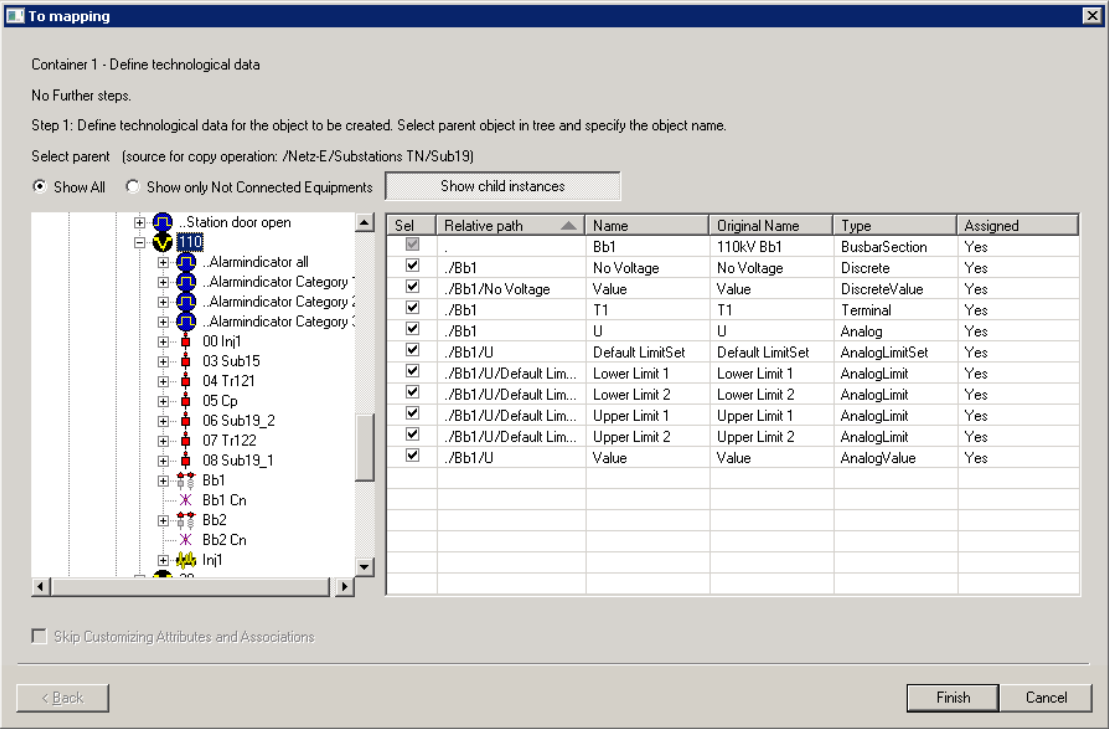
- Open a worldmap.
- Make sure that the **graphical** editing mode is switched on.

Using Library Templates

- Select a library object with linked domain data from the library and drag and drop it onto the drawing section of the Graphics Editor. The new graphic objects are displayed within the worldmap. The **TO mapping** window opens.

Using Existing Graphics with Linked Domain Data

- Select at least 1 object with linked domain data within an opened worldmap.
- Use <CTRL + C> or click  in the standard toolbar to copy the selected objects from the worldmap to the clipboard.
- Use <CTRL + V> or click  in the standard toolbar to copy the current content of the clipboard in the active worldmap under the mouse position.
- The copied graphic objects are displayed within the worldmap.
The TO mapping window opens.



[image018-150313-01.tif, 1, en_US]
Figure 5-6 TO Mapping Window



NOTE

The number of steps to be performed and the amount of domain data objects included in the domain data list are dependent on the library object used or the copied objects.

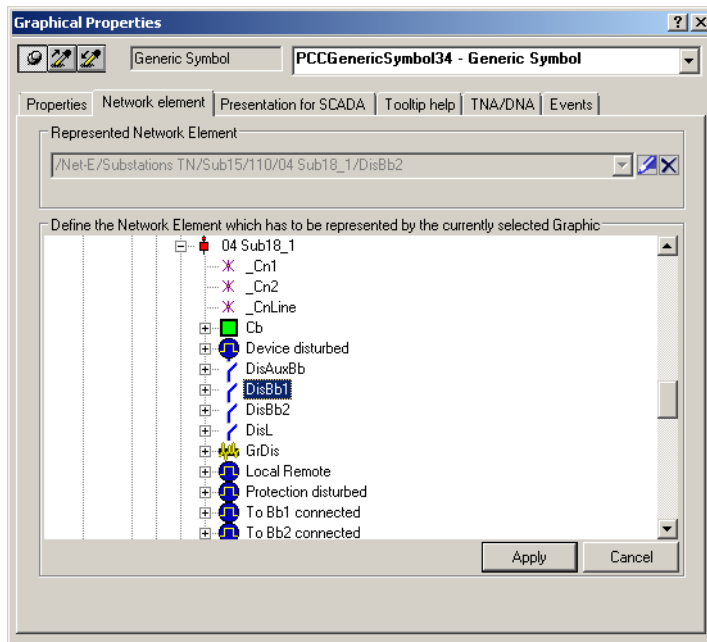
- Select the domain data or parent container in the tree that is to be mapped according the domain data list.
- If desired, specify names for the child domain data objects in the domain data list within the column **Name**.



NOTE

In order to fully map the graphic objects to the existing domain data objects, make sure that **Yes** is displayed in the domain data list within the column **Assigned** for all domain data instances. Otherwise your specified domain data is not fully mapped to already existing domain data within the selected parent container.

- Select the **Network element** tab.



[rgimm56-180711-01.tif, 1, en_US]

Figure 5-8 Graphical Properties – Network Element

- Select the desired domain data instance within the tree.
- Alternately, enter the complete path into the list box after the field has been enabled for input using .
- Click **Apply** to create the link between the selected graphical object and domain data instance.



NOTE

The domain data object types that can be linked depend on the selected graphical object.

- To remove a link to a domain data instance, delete the complete link using  and click **Apply**.
- To assign a new link to a different domain data instance, select the desired domain data instance within the tree and click **Apply**.
The former link gets automatically deleted.



NOTE

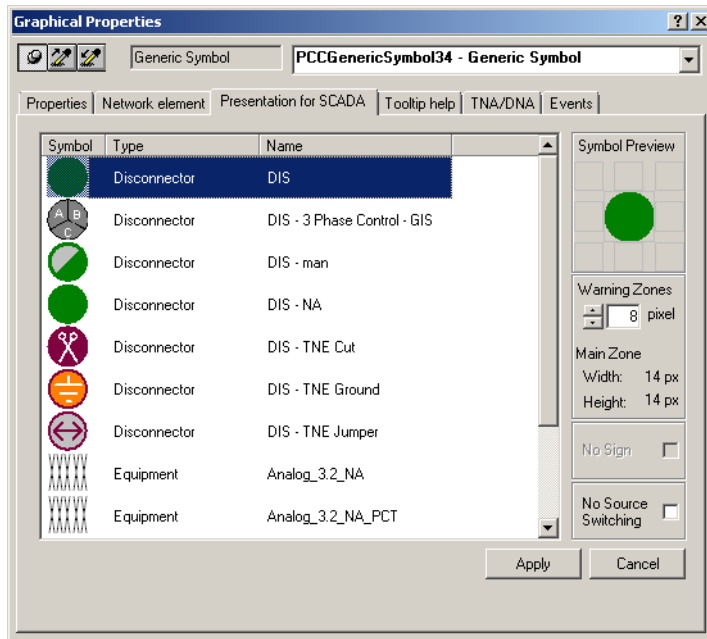
Linking graphical objects to domain data containers adds additional functionality within the graphical and technological editing workflows:

- Select **Select Objects of Container** within the context menu of a selected object linked to a domain data container to select and highlight all graphical objects within the worldmap representing domain data objects within this container.
- Use this graphical object for cut and paste or copy and paste operations to copy all subsequent domain data objects independent on the fact that they are graphically represented.
- Use this graphical object to create templates within the **Library** that contain domain data objects that do not necessarily have a graphical representation.
- Use this object and the **Select Objects of Container** operation for simple validation purposes within the active worldmap.
For example, verify the hierarchy within the domain data with the graphical objects configured.

5.1.5 Selecting the Presentation Logic for Already Linked Graphic Objects

The goal of this task is to select the presentation logic for existing graphic objects within a worldmap that are already linked to existing domain data using the **Graphical Properties** window.

- Open a worldmap.
- Double-click the desired object.
The **Graphical Properties** window opens.
- Select the **Presentation** tab.
In the **Presentation** tab, different presentation types are determined for SCADA and NA values by selecting the appropriate option using the **Display Mode** toolbar.



[rgimm008-180711-01.tif, 1, en_US]

Figure 5-9 Graphical Properties – Presentation

- Select the desired presentation logic from the list of available presentation types depending on the linked domain data object.
- Select the desired options.
- Click **Apply**.



NOTE

The selection of presentation logics is only possible for (GO) **Generic Symbol** objects. A domain data object (TO) needs to be linked already.

- To assign a new presentation logic, select the desired one from the list and click **Apply**.



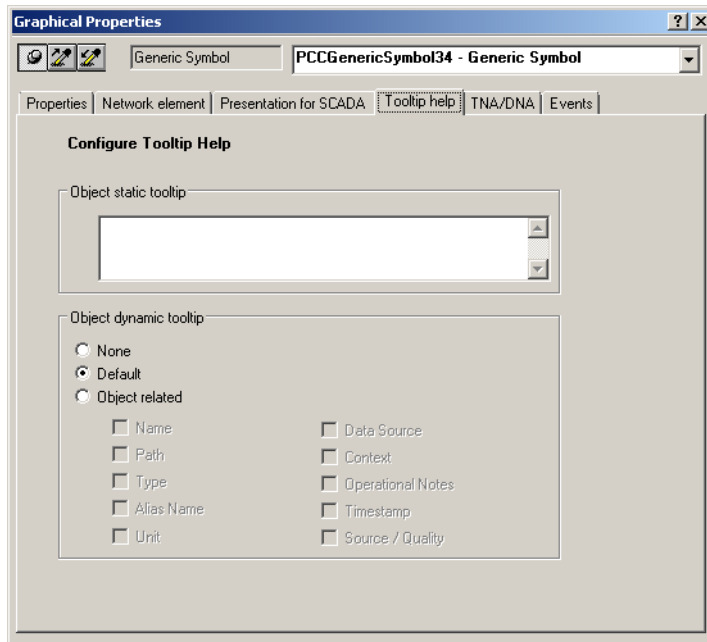
NOTE

A linked (TO) domain data object can have multiple (GO) presentations with different presentation logics within the worldmap or different worldmaps.

5.1.6 Configuring the Tooltip for Already Linked Graphic Objects

The goal of this task is to configure the tooltip for existing graphic objects within a worldmap that are already linked to existing domain data using the **Graphical Properties** window. The tooltip is visible at runtime.

- Open a worldmap.
- Double-click the desired object.
The **Graphical Properties** window opens.
- Select the **Tooltip help** tab.
In the **Tooltip help** tab, different configuration types are supported.



[image003-180711-02.tif, 1, en_US]

Figure 5-10 Graphical Properties – Tooltip Help

- Select the desired configuration for dynamic object related tooltip or edit a static tooltip.



NOTE

Default object tooltips are configured in the IMM in AC Data set.

5.1.7 Adding a Dynamic Text Label

The Graphics Editor provides dynamic text labels within the worldmap. The text is automatically derived from the name of the linked domain data instance within IMM. If the name is changed within the IMM domain data, the displayed text is automatically changed.

The goal of this task is to add a dynamic text label and link it to a domain data object.

- Open a worldmap.
- Select a graphical object that is already linked to domain data.

- Open the context menu of the selected object and select **Add new Label** and the desired source of the name out of the displayed possibilities.
A new graphical object of type **Label** gets created. The text displayed corresponds to the name of the chosen domain data instance.
- Alternately, use the **Label** object of the **Object Palette** and link the graphical object by means of the **Graphical Properties** window using the tab **Network** element.



NOTE

The text displayed is automatically derived from the linked domain data instance name. Renaming of the domain data instance by means of IMM does not immediately change the displayed text within the Graphics Editor.

The functionality assures that the runtime user interface displays always the correct and actual name for the domain data instance after activating the IMM job. The text within the Graphics Editor gets automatically updated after opening the corresponding worldmap for further editing purposes.

5.1.8 Linking Graphic Objects to Domain Data Container

Linking graphical objects to domain data containers (for example: **Substation**, **VoltageLevel**, **Bay** and so on) adds additional functionality within the graphical and technological editing workflows.

The goal of this task is to add a graphic object and link it to a domain data container.

- Open a worldmap.
- Select a graphical object in the standard object palette and insert it in the worldmap.
- Link the graphical object by means of the **Graphical Properties** window using the tab **Network element**.



NOTE

The type of the used graphical object needs to allow linking to a domain data container.

Supported Features

- Select **Select Objects of Container** within the context menu of a selected object linked to a domain data container to select and highlight all graphical objects within the worldmap representing domain data objects within this container.
- Use this graphical object for cut and paste or copy and paste operations within the **technological editing mode** to copy all subsequent domain data objects independent on the fact that they are graphically represented.
- Use this graphical object to create templates within the **Library** that contain domain data objects that do not necessarily have a graphical representation.
- Use this object and the **Select Objects of Container** operation for simple validation purposes within the active worldmap.
For example, verify the hierarchy within the domain data with the graphical objects configured.

5.1.9 Creating and Manipulating the Topology of the Power System Network

The Graphics Editor provides multiple ways to connect or disconnect graphic objects topologically:

- Manually connecting or disconnecting objects using **Connector** objects.
Connector objects are graphical objects that can represent domain data ConnectivityNodes.

- Automatically connecting or disconnecting objects using associated graphical editing toolbar buttons.
Connector objects are automatically created and linked to **ConnectivityNodes**.
- Manually create the topology through **ConnectivityNodes** using the icon  in the graphical editing toolbar.



NOTE

If the domain data topology between the graphical objects does not exist before the operations, a new **ConnectivityNode** is created and the **Terminals** of the domain data represented with the graphical objects are linking to this **ConnectivityNode**.

If the domain data topology between the graphical objects exists already before the operations, the already existing **ConnectivityNode** is used and linked.



NOTE

If a fully connected **Connector** gets disconnected at one end, the topological relationship within the domain data gets also removed.

Non existing Connector Object

The goal of this task is to connect 2 graphic objects topologically that are already linked to domain data representing equipments.

- Open a worldmap.
- Make sure that the **technological** editing mode is switched on.
- Select 2 graphical objects to be topologically connected.

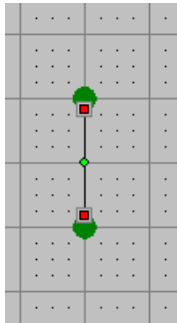
The multiselection is shown.



[immge004-270510-01.tif, 1, en_US]

Figure 5-11 Multiselection of 2 Graphical Objects

- Use the  icon from the Graphical Editing toolbar to create the topological connection for the selected objects.



[image005-270510-01.tif, 1, en_US]

Figure 5-12 Topologically Connected Objects

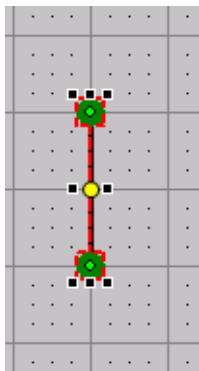
- Alternately, use the **Connector** object of the **Object Palette** and drag each end point of the **Connector** object on top of the linking points of the graphical objects to be connected topologically.
- The connection is shown graphically.
If the connection has been successfully established, the connection points for the **Connector** object are displayed in red, otherwise the connection points are still displayed in green color.
- The **Connector** object is linked to a ConnectivityNode of the domain data.

Existing Connector Object

The goal of this task is to create the topology of graphically connected objects that are already linked to domain data representing equipments.

- Open a worldmap.
- Make sure that the **technological** editing mode is switched on.
- Select two graphical objects that are graphically connected and between which the domain data topology has to be created.

The multiselection is shown.



[image021-150313-01.tif, 1, en_US]

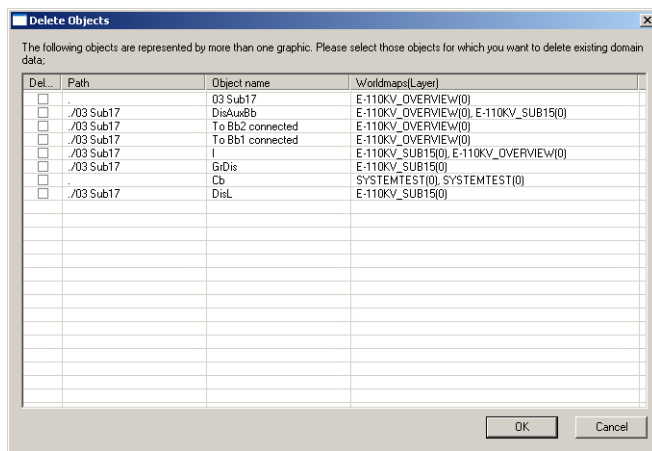
Figure 5-13 Multiselection of two Graphical Connected Objects

- Use the  icon from the Graphical Editing toolbar to create the topological connection for the selected objects.
- The **Connector** object is linked to a ConnectivityNode of the domain data.

5.1.10 Deleting Domain Data Using Deletion of Graphic Objects Linking this Domain Data

The goal of this task is to delete domain data by deleting graphic objects linked with the domain data.

- Open a worldmap.
- Make sure that the **technological** editing mode is switched on.
- Select a graphic object and delete it.
For Example, with **<SHIFT + DEL>**.
- The graphic object including the linked domain data object gets deleted immediately.
- If this linked domain data object is represented in any other worldmap or even in the same but another layer, a dialog is shown to take care of these graphical objects as well.



[immrg010-110210-02.tif, 1, en_US]

Figure 5-14 Delete Objects

- Select those objects for which you want delete the domain data although it is represented at least once more with a graphical object.
- Click **OK** to delete the selected domain data instances.

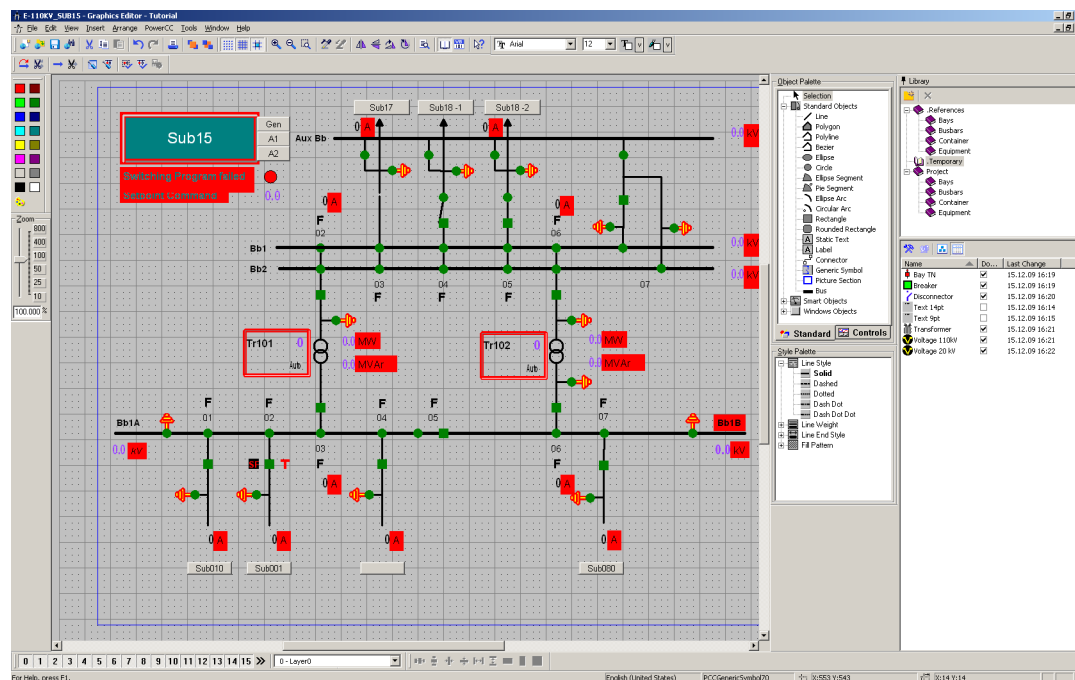
If you allow deleting the domain data object, remove the remaining orphan graphical objects manually within the worldmaps mentioned in this dialog.

5.1.11 Identifying Graphical Objects not linked to Domain Data

The goal of this task is to identify graphical objects that are not linked to domain data objects.

- Open a worldmap.
- Click  within the Graphical Editing toolbar to verify the selected objects.
- Click  within the Graphical Editing toolbar to verify the complete worldmap.

Graphical objects without linked domain data are highlighted according to the settings done in **Settings for Graphical Editing** using the **Settings** window.



[rgimm009-190210-01.tif, 1, en_US]

Figure 5-15 Verify Worldmap - Highlighted Graphical Objects

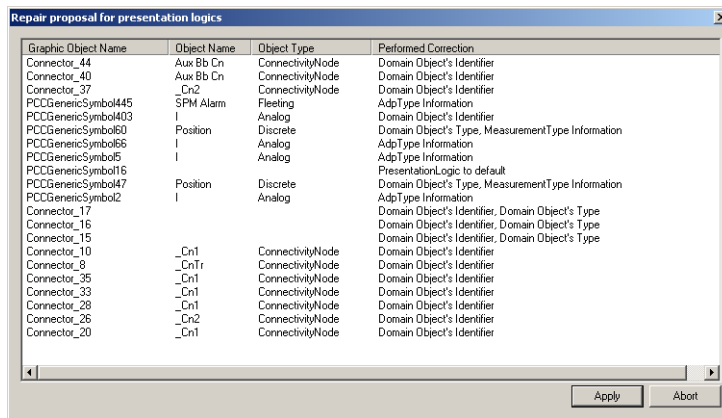
Highlighting is done as long as the Graphical Editing toolbar button is kept active.

5.1.12 Performing a Presentation Logic Correction

The goal of this task is to repair presentation logics assignments or label text assignments within a worldmap. This task is necessary at least after importing worldmaps to create all necessary data for the correct presentation at runtime. Nevertheless, if any presentation problem at runtime happens, perform this task and activate the changes to get the problems resolved.

- Open a worldmap.
- Click  within the Graphical Editing toolbar to initialize the presentation logic correction for the complete worldmap.
- Confirm the query to close the worldmap with **YES**.
If unsaved changes within the worldmap exist, the worldmap is automatically saved before closing.

- The **Repair proposal for presentation logic** window opens.



[immrg012-110210-01.tif, 1, en_US]

Figure 5-16 Repair Proposal for Presentation Logic Window

- Click **Apply** to perform all proposed corrections.
 - Alternately, click **Abort** to refuse all proposed corrections or if nothing is required to repair.
- The **Repair proposal for presentation logic** window closes. The repaired worldmap opens again.

5.1.13 Automatically Creating a New Substation Display Based on Existing Domain Data

The goal of this task is to create a new substation display within a new worldmap based on the existing domain data topology.

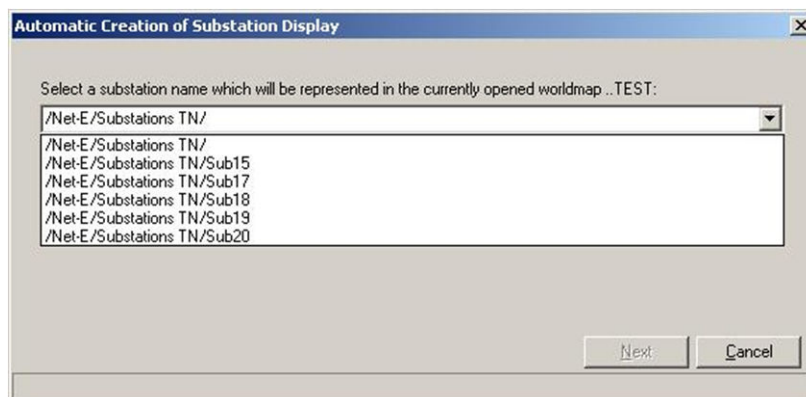


NOTE

Automatic display building produces proper displays only for correct domain data models by nature. In case of inaccurate or non-standard (in terms of CIM) domain data, the automatically created displays will look strange.

- Create a new worldmap and open it.
- Click  within the Graphical Editing toolbar to initialize the automatic display building.

The **Automatic Creation of Substation Display** window opens.



[immrg013-110210-01.tif, 1, en_US]

Figure 5-17 Automatic Creation of Substation Display Window

- Select the desired domain data substation within the list box.

- Click **Next >** to start the display creation.

A user feedback is given within the window because dependent on the amount of network elements within the selected substation, the creation takes some seconds.

If the creation is finished, the substation layout is presented within the worldmap. Otherwise problems are shown within the **Errors** dialog.

5.2 User Interface Description

5.2.1 Introduction

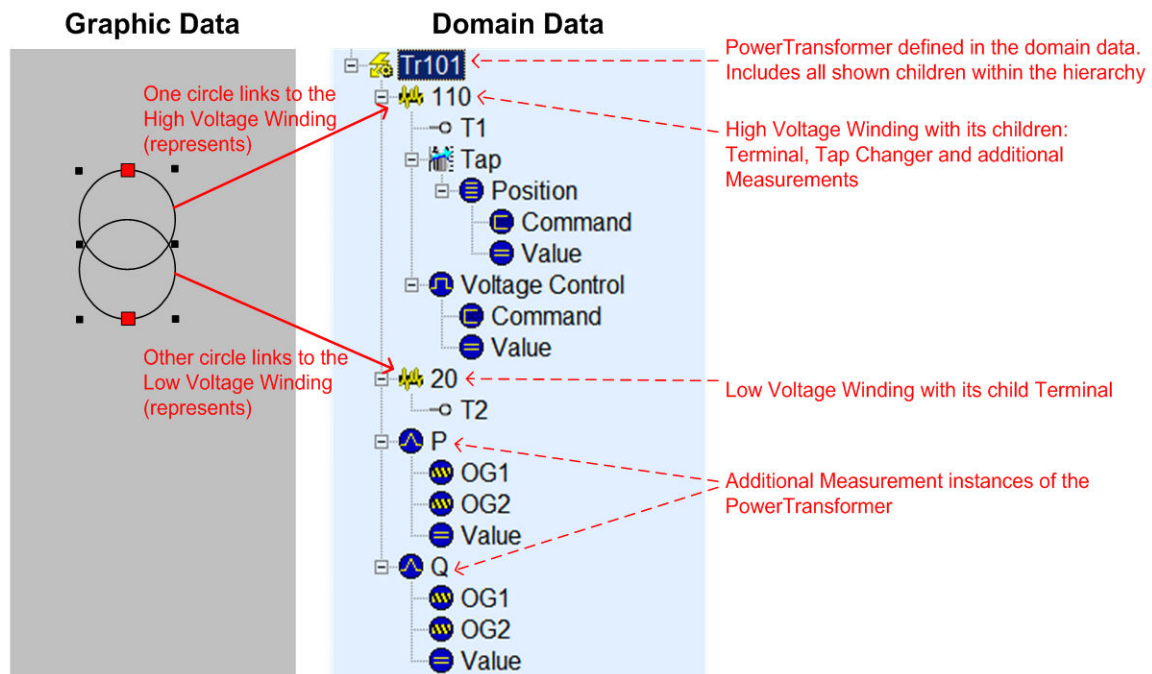
Spectrum Power provides a domain topology model using terminal and connectivity node classes, which describe the technological object or instance data model. This domain topology is the base for all Spectrum Power Control Center applications.

Graphical Editing

There are basically 2 different workflows to describe the domain data and the related domain topology:

- One is editing the domain data directly in the IMM Editors of the IMM User Interface. These editors are used for engineering the available field resources and the topology of the network.
For additional information on IMM, refer to Reference Guide Information Model Management (IMM).
- The other one is using the graphic data which describes the graphic objects (GO) within the worldmaps. This data model is used and created by the Graphics Editor. It contains types for graphical primitives, like rectangles, circles, but also more specific ones designed for Spectrum Power 5.

The idea of graphical and technological editing is to create, update and delete the domain data in a graphical way while constructing the worldmaps. For this purpose the user gets additional functionality within the Graphics Editor. The Graphical Editing creates the link between the instances of the graphic data to instances of the domain data and therefore builds the bridge between both.



[immrg022-090210-01.tif, 1, en_US]

Figure 5-18 Example: Link between Graphical Data (GO) to IMM Domain Data (TO)

Technological Editing is designed for editing an electrical power system network. All graphical **Standard Objects**, except the **Picture Section**, of the **Object Palette** can technologically link to certain objects within the domain data of any CIM subtype of FieldResource or Measurement. Further these domain data objects are called throughout the chapter as technological objects (TO).



NOTE

Creation of topological relationships with the functions for technological editing between the field resources for other domains like gas or water networks is not provided because the Common Information Model (CIM) does not define how to model the equivalent of the topology of power system networks.

With graphical and technological editing, execute the following use cases in the Graphics Editor to manipulate the domain data:

- Create new domain data using drag and drop from the **Object Palette** or from the **Library**.
- Create new domain data using copy and paste within the worldmap.
- Link graphic objects to already existing domain data.
- Create and manipulate the topology of the electrical power system network.
- Delete domain data using deletion of graphic objects linking to this domain data.
- Support creation of library templates that contain domain data and graphic data.



NOTE

To achieve consistent engineering data, after each performed workflow the engineering data has to be persisted. Therefore the Graphics Editor executes a saving operation after a graphical or technological editing workflow. That causes the deletion of the undo stack.

Automatic Display Building

The Automatic Display Building functionality creates for the electrical equipment and topology modeled within existing domain data graphical network diagrams (worldmaps).

The idea of automatic display building is to create graphical representations of the selected domain data hierarchy within worldmaps that can be used within the Spectrum Power 5 runtime system. For this purpose the representation of the electrical equipments has to be similar to manually edited worldmaps. The layout algorithm considers the correct positioning of the graphic objects (GO) within the worldmap according to the domain data topology and hierarchy.

The user gets additional functionality within the Automatic Display Building, as the layout depends on the topology modeled within the existing domain data. Wrong topology models can be detected. Automatically created displays can be modified by means of graphical editing.

Automatic display building produces proper displays only for correct domain data models by nature. In case of inaccurate or non-standard (in terms of CIM) domain data, the automatically created displays will look strange.



NOTE

The existing functionality is the 1st implementation version and serves as a base framework for future enhancements to be planned for next releases of the **Graphics Editor**. It currently supports:

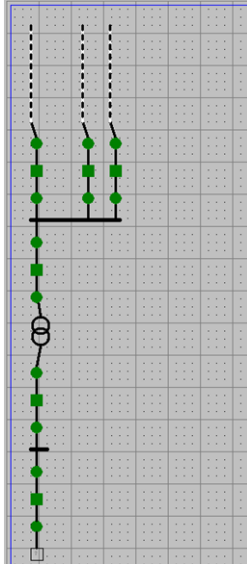
- Initial layout of substation displays in empty worldmaps
- Vertical layout within the substation displays
- Simple constructs for default symbols
- Limited layouting for **BusbarSections** constructs like couplings or auxiliary buses.
- **TransformerWindings** underneath the **PowerTransformer**
- Consideration of the CIM types **ConnectivityNode (CN)** and subtypes of **ConductingEquipment**. Only to-pologically relevant domain data is graphically represented.

The graphical representation for the considered domain data objects are derived from a defined folder within the **Library**. All templates within this folder are used as default symbols (GO) for the specified domain data instance type. These templates are created and maintained by means of graphical editing and library func-

tionality. If there is no template specified for an existing domain data object type, as fallback strategy, the **Generic Symbol** is used.

The current layout algorithm supports layout calculation for a schematic, vertical substation display. The layout calculation respects the user's definition of a minimum distance between 2 graphical objects. The layout starts on the top left side of the worldmap and positions the graphical symbols downwards and rightwards. Within the resulting substation display the GO are position orthogonally to each other.

The layout calculation considers in first place the topology of the involved domain data objects and as second criterion the base voltage (BaseVoltage) of these objects. The layout algorithm positions AC Lines with the highest BaseVoltage at the top of the display. At the bottom the AC Lines, Generating Units and Energy Consumers with the lowest BaseVoltage are positioned. Furthermore the topological dependent constructs within the substation are laid out together.



[immrg002-100210-02.tif, 1, en_US]

Figure 5-19 Example: Simple Substation Layout with 1 PowerTransformer and 2 BaseVoltages

All supported equipments and connectivity nodes are laid out, even those which are not part of the substation topology. All irrelevant or topologically not connected domain data objects are positioned on the left side of the display – one underneath the other.

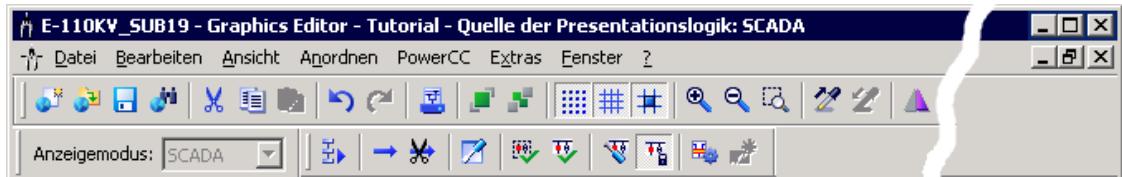
5.2.2 Graphical Editing Modes

The Graphics Editor supports two graphical editing modes in order to support the two different basic editing workflows regarding maintenance of the domain data and related domain data topology:

- **Graphical Editing Mode**
This mode is best when domain data and domain data topology exists already and the graphics need to be mapped. The existing domain data might be imported or copied within the IMM user interface. Basically this is the main use-case when migrating system data from own or foreign systems. The main criterion in this mode is that existing domain data and domain data topology shall not be modified by graphical operations without explicit awareness of the user.
- **Topological Editing Mode**
This mode is best when domain data and domain data topology shall be created or modified while drawing single-line diagrams. Basically this is the main use-case when the graphical interface of the Graphics Editor shall be the primary user interface to maintain data. The main criterion in this mode is that the graphics need to show the detailed domain data topology in order to create or modify the domain data and domain data topology without frequent user dialogs.

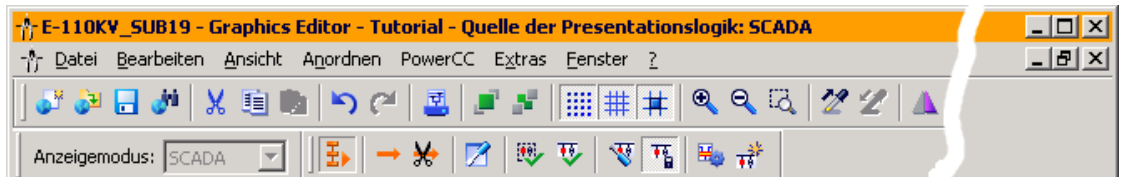
Visibility of the selected Modes

The currently selected mode is clearly visible with the color code of the Graphic Editor Title Bar and the Graphical Editing Toolbar.



[image012-150313-01.tif, 1, en_US]

Figure 5-20 Titlebar and Graphical Editing Toolbar in Graphical Mode



[image013-150313-01.tif, 1, en_US]

Figure 5-21 Titlebar and Graphical Editing Toolbar in Technological Mode

Feature Differences by Mode Selection

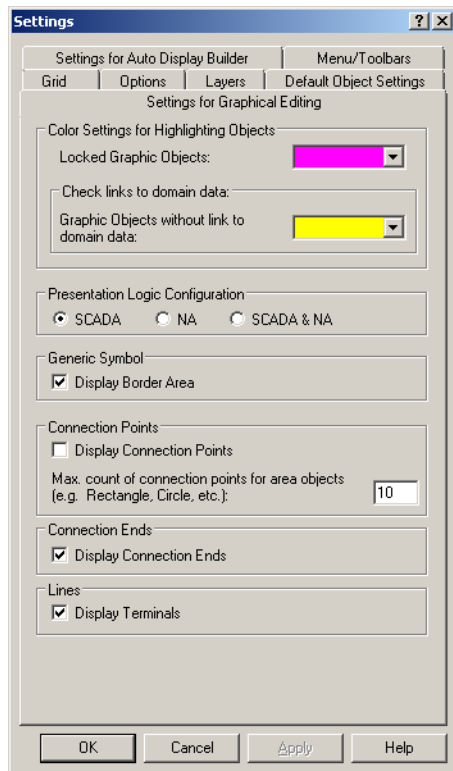
The graphical editing mode selection influences the behavior of connector objects and the behavior of template and copy/cut and paste use-cases.

Feature	Graphical Editing Mode	Technological Editing Mode
Mode Selection Feedback	- Title bar color scheme "Normal" - Icon in graphical toolbar for mode change not pressed - Icons in graphical toolbar for mode change, create connection and delete connection in color scheme "Normal"	- Title bar color scheme "Attention" - Icon in graphical toolbar for mode change pressed - Icons in graphical toolbar for mode change, create connection and delete connection in color scheme "Attention"
Connect Icon clicked in graphical editing toolbar	Creates a graphical connection. A new Connector object gets created between the two selected graphical objects. No Connectivity-Node gets linked.	Creates a topological connection. A new Connector object gets created between the two selected graphical objects. A Connectivity-Node gets linked. The Connectivity Node as well as the domain data topology gets either be created or an existing Connectivity Node describing the domain data topology will be used.
Disconnect Icon clicked in graphical editing toolbar	Disconnects a connection graphically. An existing Connector object gets removed between the two selected graphical objects. If the Connector was linked to a ConnectivityNode the domain data topology gets not touched and remains.	Disconnects a connection graphically and topologically. An existing Connector object gets removed between the two selected graphical objects. If the Connector was linked to a ConnectivityNode it gets removed as well in the domain data (except for a valid Busbar ConnectivityNode).

Feature	Graphical Editing Mode	Technological Editing Mode
Create Domain Data Topology Icon clicked in graphical editing toolbar	Not available (button disabled).	Creates the domain data topology for the selected graphical objects. A ConnectivityNode gets linked to existing connected Connector objects within the multi selection. The Connectivity Node as well as the domain data topology gets either be created or an existing Connectivity Node describing the domain data topology will be used.
Connect objects manually using a Connector	Creates a graphical connection.	Creates a graphical and topological connection.
Disconnect objects manually using a Connector	The Connector is detached from the graphical object. If the Connector is linked to a ConnectivityNode the link and the domain data topology gets not touched and remains.	The Connector is detached from the graphical object. If the Connector is linked to a ConnectivityNode the link gets not touched, the domain data topology gets involved.
Delete a Connector manually	Deletes a Connector graphically. If the Connector was linked to a ConnectivityNode the domain data topology gets not touched and remains.	Choice between a topologically or graphically only deletion of the Connector . Clicking behaves similar than for Graphical Editing Mode . Clicking <SHIFT + DEL> removes the Connector graphically and topologically. If the Connector was linked to a ConnectivityNode it gets removed as well in the domain data (except for a valid Busbar Connectivity-Node).
Connector auto-connect	Creates a graphical connection if a Connector with open ends lies on a connection point after dragging a multi selection.	Creates a graphical and topological connection if a Connector with open ends lies on a connection point after dragging a multi selection.
Template (with domain data) usage	Template data can be matched to existing domain data and domain data topology only. Domain data and domain data topology gets not modified at all.	Template data can be used to create new domain data and domain data topology only. Existing domain data and domain data topology can't be used at all.
Copy & Paste of existing graphics linked to domain data	Copied data can be matched to existing domain data and domain data topology only. Same behavior as Template usage.	Copied data can be used to create new domain data and domain data topology only. Same behavior as Template usage.
Cut & Paste of existing graphics linked to domain data	Just moving of graphical objects. The pasted graphical objects keep their links to the existing domain data.	Pasted data can be used to create new domain data and domain data topology while deleting the old domain data and domain topology.

5.2.3 Settings for Graphical Editing

Different settings for graphical and technological editing are changed and saved via the **Settings** window.



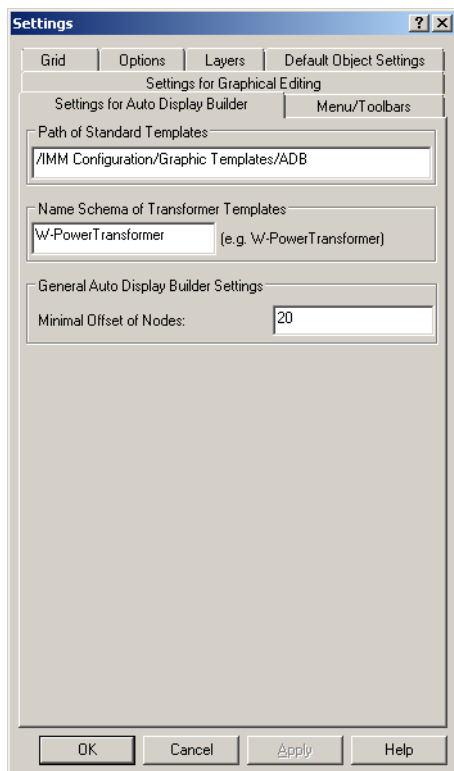
[rgimm013-180711-01.tif, 1, en_US]

Figure 5-22 Settings Window - Graphical Editing

For additional information, refer to Graphics Editor Basics.

5.2.4 Settings for Automatic Display Building

Different settings for automatic display building are changed and saved via the **Settings** window.

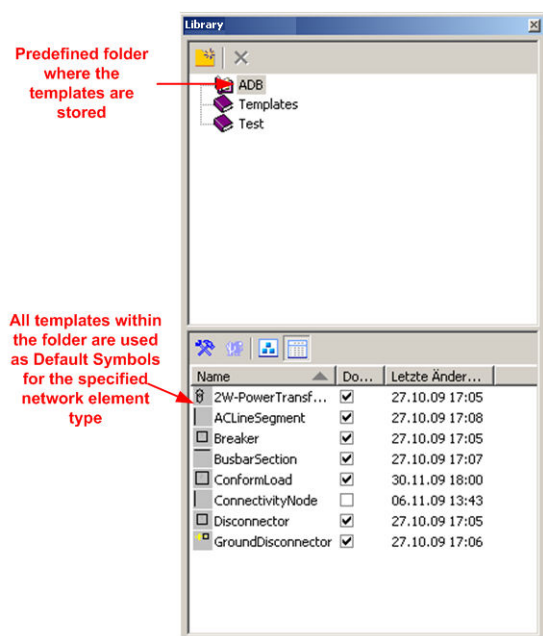


[rgimm014-180711-01.tif, 1, en_US]

Figure 5-23 Settings Window – Automatic Display Builder

Path to Default Symbols

The default symbols for automatic display building are defined and stored within a default folder of the IMM AC dataset. The configured folder in the screenshot above corresponds to the library folder with name **ADB** and is located in the root of the library hierarchy.



[immr006-100210-01.tif, 1, en_US]

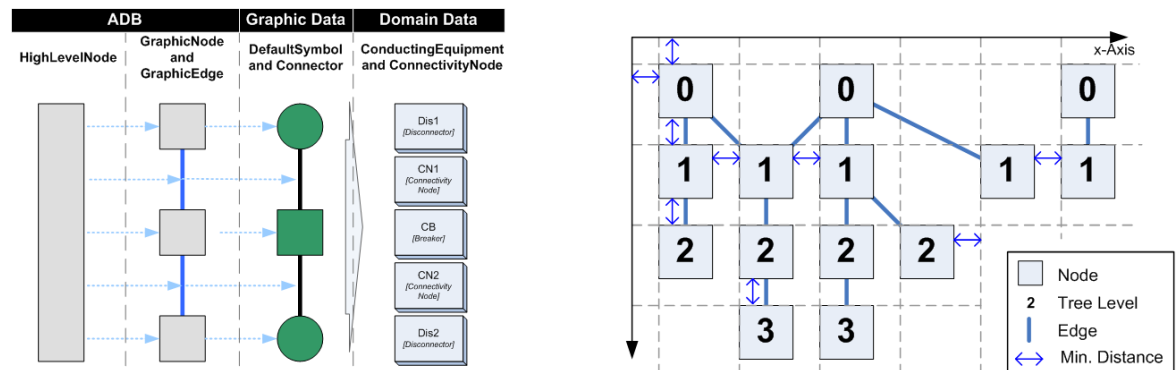
Figure 5-24 Example for Templates used for Automatic Display Building

Name Schema of Transformer Templates

Defines the name schema for the complex default symbols of Power Transformers. In this release, the Auto Display Builder can only handle complex templates for **TransformerWindings** underneath **PowerTransformer**. The graphical representation of PowerTransformers is dependent on the Transformer Winding constellations, for example it has 2 windings or 3 windings. For every constellation a separate template has to be created to be able to differentiate between both types. The common name of the templates has to match with the name given in the **Settings** window. For each winding the number has to be the first character as displayed in the figure above. In this example, a three-winding transformer has to have the name **3W-PowerTransformer**.

Minimal Distance of Nodes

Defines the minimum distance in pixels of 2 graphic nodes within a substation display and therefore influences the layout of the substation.

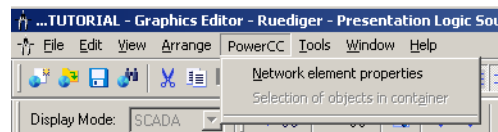


[immrg007-100210-01.tif, 1, en_US]

Figure 5-25 Algorithm of the Layout Positioning and Minimum Offset

5.2.5 Graphical and Technological Editing Functions within the PowerCC Menu

Some additional graphical and technological editing functions are available within the PowerCC menu in the **Graphics Editor** menu bar.



[immrg014-180711-03.tif, 1, en_US]

Figure 5-26 PowerCC Menu

Network Element Properties

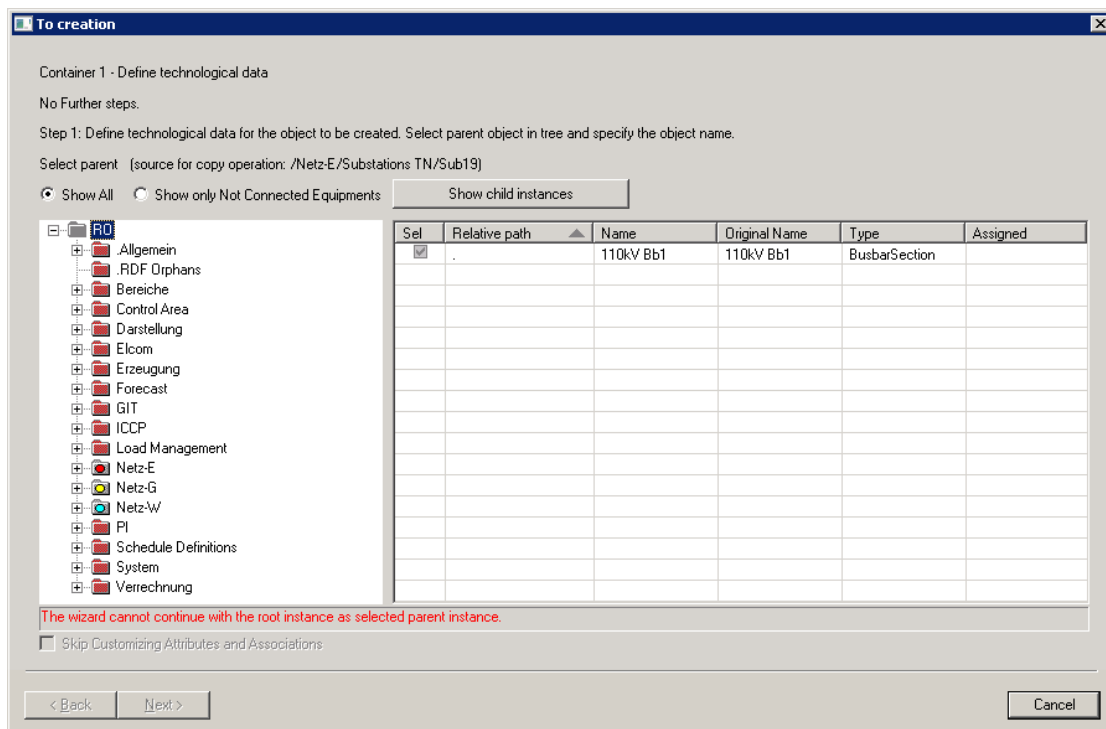
Opens the **IMM Single Instance Editor** for modifying the IMM domain data of the selected graphical object.

Select Objects of Container

If the selected graphical object is linked to an IMM domain data container, all graphical objects linked to IMM domain data instances within the selected IMM domain data container get selected.

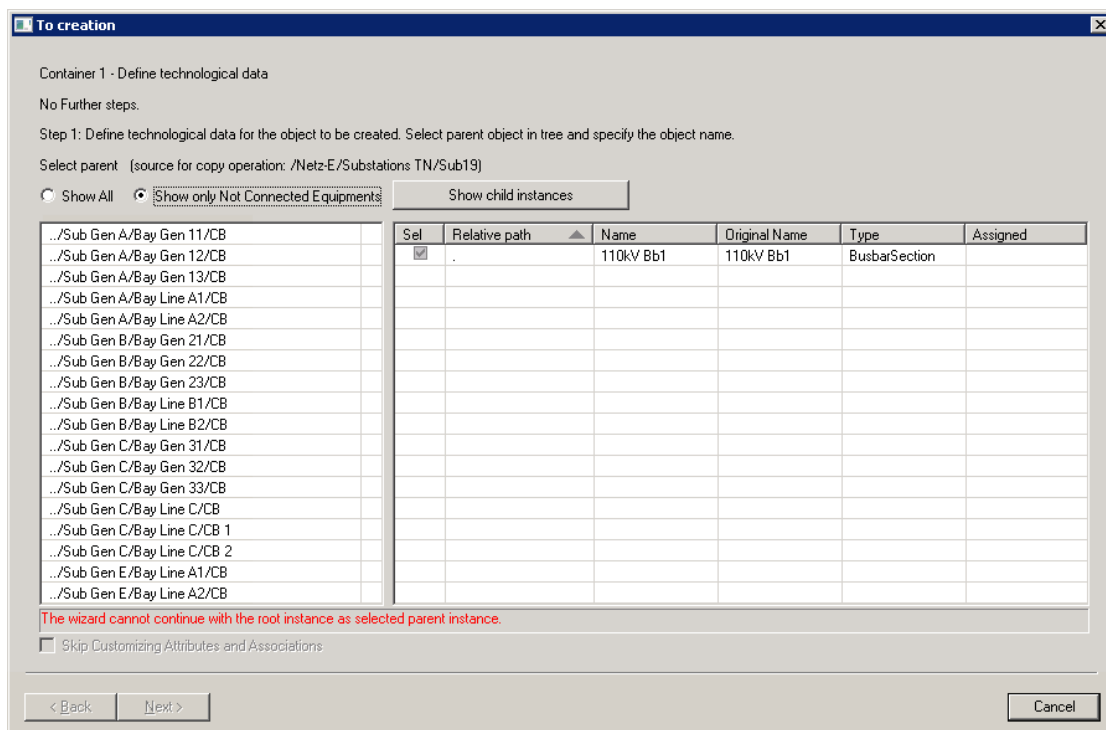
5.2.6 Technological Object Creation Window

The creation of new domain data based on existing graphics or templates linked to domain data is performed using the **TO creation** window.



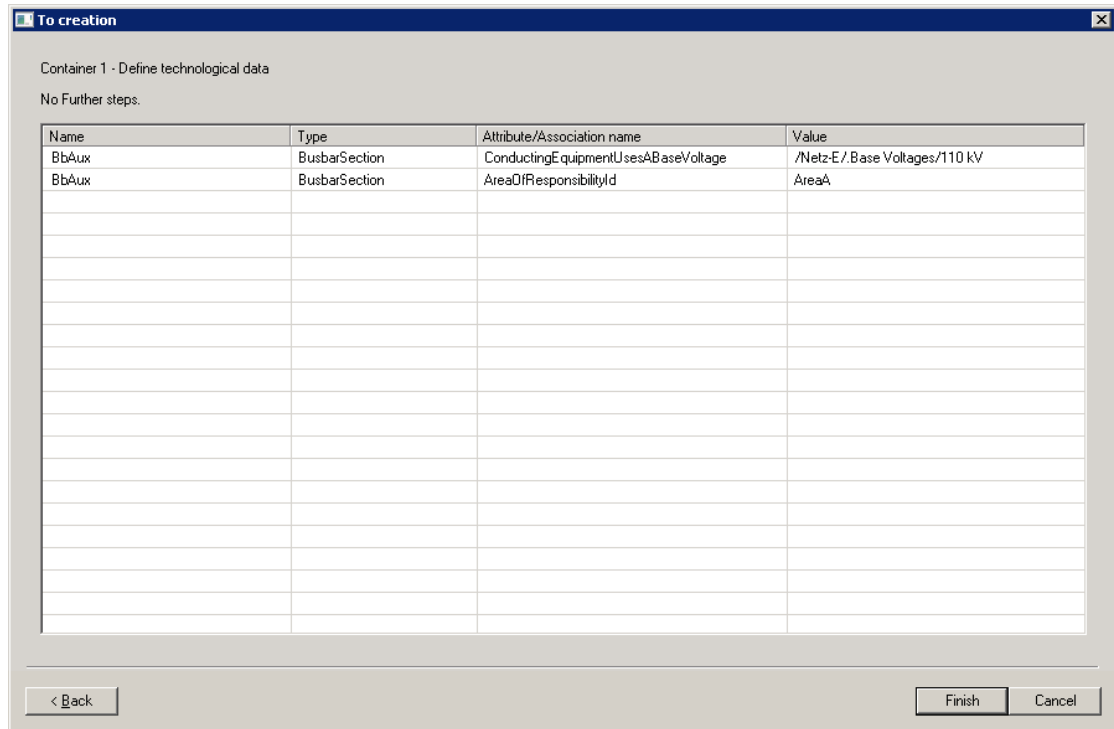
[imgge022-150313-01.tif, 1, en_US]

Figure 5-27 TO Creation Window - Show All



[imgge023-150313-01.tif, 1, en_US]

Figure 5-28 TO Creation Window – Show only Not Connected Equipment



[immge017-150313-01.tif, 1, en_US]

Figure 5-29 TO Creation Window – Show Customize Attributes and Associations

Title Area

In the title area the intention what needs to be done is mentioned. Follow the instructions to finish the current step. If more than one step has to be performed this is also indicated.

Select parent

- **Show All** shows the already existing domain data in the form of a directory structure. Directories and subdirectories can be opened and closed by clicking the + or – icons.

Domain Data List

The left area shows the associated domain data object within the currently performed workflow. It is structured within a tabular form. Click the column header of the desired column to sort the content of the tabular form. The columns are dependent of the current step.

- Column **Sel** indicates the domain data objects that are included within the operation. You can exclude domain data objects by means of the hook. If you deselect domain data objects that include subhierarchical domain data objects, all subhierarchical domain data objects get automatically deselected. You can not deselect the greyed out selection boxes. Those domain data objects are linked with the already created graphical objects.

Deselected objects are no longer considered within the workflow and will not be created.

- Column **Relative Path** indicates the hierarchical position of the domain data object based on the selected parent.
- Column **Name** specifies the domain data names that are to be created. The names are editable.
- Column **Original Name** includes the domain data names that are stored in the library template. Within the copy and paste workflow, the names of the copied domain data are visible.
- Column **Type** identifies the CIM type of the domain data object.

- Column **Assigned** shows what will happen with the domain data instance.
Yes means that all domain data including all child instances contained within the step will be created.
Partial means that not all domain data contained within the step will be created. This happens if at least 1 child instance has been manually deselected.
No means that the domain data has been unchecked in the column **Sel**.
- Column **Attribute/Association name** identifies the attributes or association names that can be customized during the workflow. Depending of the CIM type, the following attributes and associations can be customized for new domain data instances: Area of Responsibility (AoR), Base Voltage and Member of Company.
- Column **Value** allows specifying the value for the displayed attribute and association.

Show child instances

If additional child instances are available the **Show child instances** button displays all associated child instance domain data objects in the Domain Data List according to the currently performed workflow.

Error Area

The error area gives you the reasons if something can not be performed with the actual selection for the current step. As long as errors are displayed, the domain data list column **Assigned** is empty.

Customizing Step

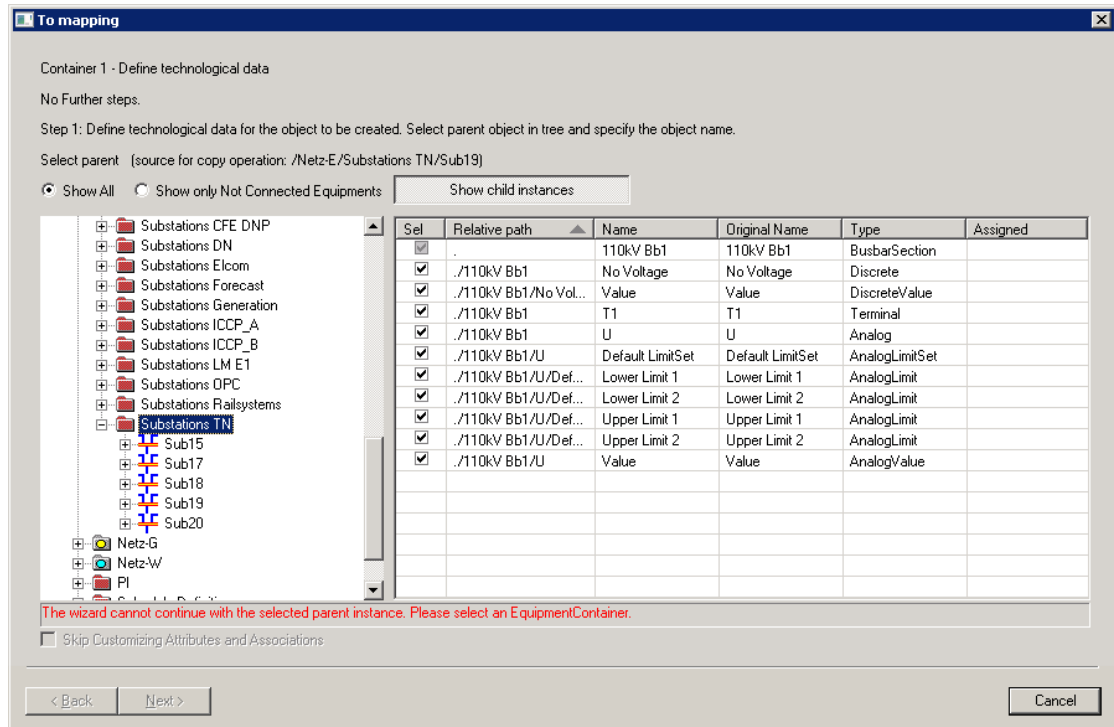
You can skip the customizing step by checking the hook for **Skip Customizing Attributes and Associations**. The hook remains set as long as you do not uncheck it explicitly within a later operation.

Button Area

The button area consists of the buttons **< Back**, **Next >**, **Finish**, and **Cancel**. The visibility of these buttons depends on the current status of the performed workflow.

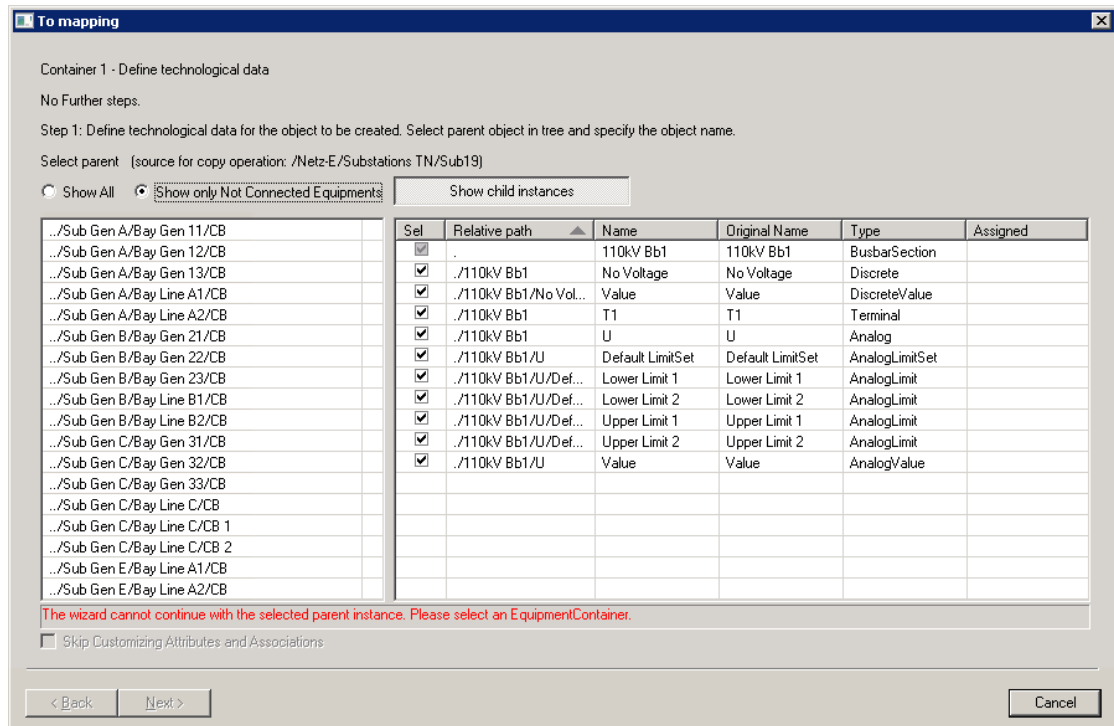
5.2.7 Technological Object Mapping Window

The mapping of graphical objects to existing domain data based on existing graphics or templates linked to domain data is performed using the **TO mapping** window.



[imgge024-150313-01.tif, 1, en_US]

Figure 5-30 TO Mapping Window - Show All



[imgge025-150313-01.tif, 1, en_US]

Figure 5-31 TO Mapping Window – Show only Not Connected Equipment

Title Area

In the title area the intention what needs to be done is mentioned. Follow the instructions to finish the current step. If more than one step has to be performed this is also indicated.

Select Parent

You can choose between **Show All** and **Show only Not Connected Equipment**. The parent for the data is selected by clicking the desired entry in the area below.

- **Show All** shows the already existing domain data in the form of a directory structure. Directories and subdirectories can be opened and closed by clicking the + or – icons.
- **Show only Not Connected Equipment** shows not fully connected equipments within the already existing domain data. The equipment needs to have the **NAFlag** set to **True** in the domain data properties to get listed. Not fully connected means that two-terminal equipment is only connected at one terminal. A use case for that option is when ACLines get added to a substation display that cross two separate substation displays.

Domain Data List

The left area shows the associated domain data object within the currently performed workflow. It is structured within a tabular form. Click the column header of the desired column to sort the content of the tabular form. The columns are dependent of the current step.

- Column **Sel** indicates the domain data objects that are included within the operation. You can exclude domain data objects by means of the hook. If you deselect domain data objects that include subhierarchical domain data objects, all subhierarchical domain data objects get automatically deselected. You cannot deselect the greyed out selection boxes. Those domain data objects are linked with the already created graphical objects.

Deselected objects are no longer considered within the workflow and will not be created.

- Column **Relative Path** indicates the hierarchical position of the domain data object based on the selected parent.
- Column **Name** specifies the domain data names that are to be mapped. The names are editable
- Column **Original Name** includes the domain data names that are stored in the library template. Within the copy and paste workflow, the names of the copied domain data are visible.
- Column **Type** identifies the CIM type of the domain data object.
- Column **Assigned** shows what will happen with the domain data instance.

Yes means that all domain data including all child instances contained within the step will be mapped to existing domain data.

Partial means that not all domain data contained within the step can be mapped to existing domain data. This happens if at least one child instance has been manually deselected or is not available in the existing domain data.

No means that the domain data has been unchecked in the column **Sel**.

Show child instances

In the mapping workflow, the **Show child instances** button is pressed by default and displays all associated child instance domain data objects in the Domain Data List according to the currently performed workflow.

Error Area

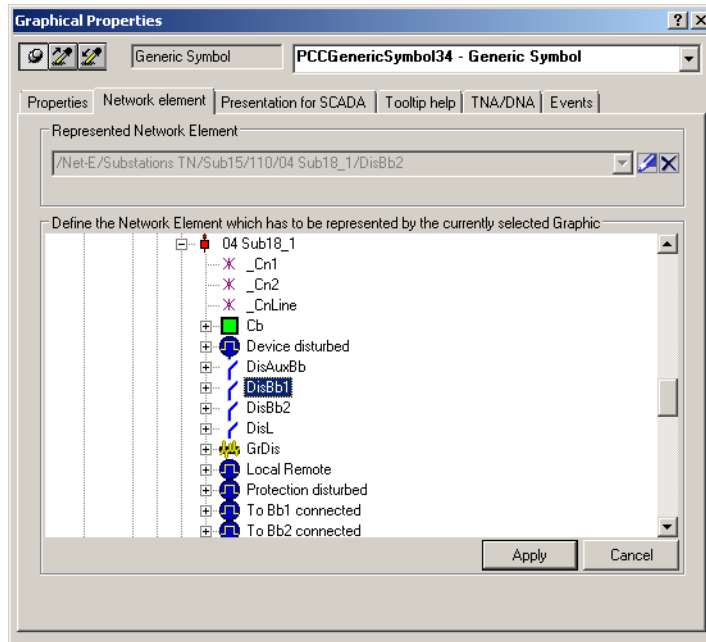
The error area gives you the reasons if something cannot be performed with the actual selection for the current step. As long as errors are displayed, the domain data list column **Assigned** is empty.

Button Area

The button area consists of the buttons **< Back**, **Next >**, **Finish**, and **Cancel**. The visibility of these buttons depends on the current status of the performed workflow.

5.2.8 Network Element Tab of Graphical Properties Window

The **Network Element** tab is used to create, modify or delete the link between the selected graphical object and its domain data instance.



[immrg015-180711-03.tif, 1, en_US]

Figure 5-32 Graphical Properties – Network Element

The functionality is provided for all **Standard Objects**, except the **Picture Section** within the **Object Palette**. The domain data object types that can be linked depend on the selected graphical object:

- Graphical object **Bus** can only link to domain data instance BusbarSection
- Graphical object **Connector** can only link to domain data instance ConnectivityNode
- Other graphical objects can link to any domain data instance except Terminals

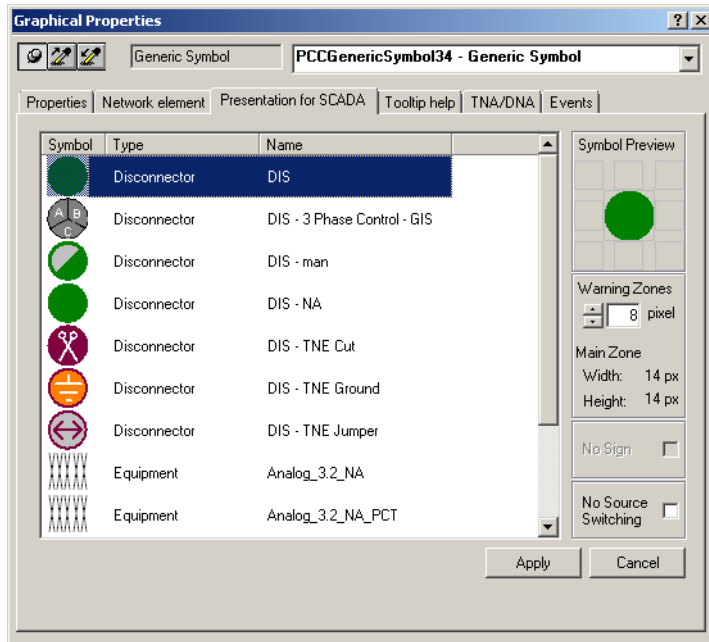


NOTE

In general, the graphical object **Generic Symbol** is used if various dynamic states are to be visualized at runtime by means of presentation logic assignments. The remaining other graphical objects can be used to link container domain data instance objects like Substation, VoltageLevel, Bay and so on.

5.2.9 Presentation Tab of Graphical Properties Window

The **Presentation Tab** is used to assign the presentation logic for the selected graphical object **Generic Symbol** and its domain data instance.



[rgimm015-180711-01.tif, 1, en_US]

Figure 5-33 Graphical Properties – Presentation

No Source Switching

If the checkbox is enabled, source switching between SCADA and NA at runtime is prevented for this graphical object.

Presentation Type List

The available presentation types (preview of the default presentation calculation rule) together with the path of the presentation logics within the IMM domain data hierarchy is visible. Select the desired presentation logic. Click the column header of the desired column to sort the content of the tabular form.

Symbol Preview

A preview of the selected presentation logic, including the warning zones, is shown on the top right side. The preview is determined based on the configured default presentation rule for the used presentation logic in IMM.

Warning Zone

The warning zone dimension of the graphical object can be specified.

Main Zone

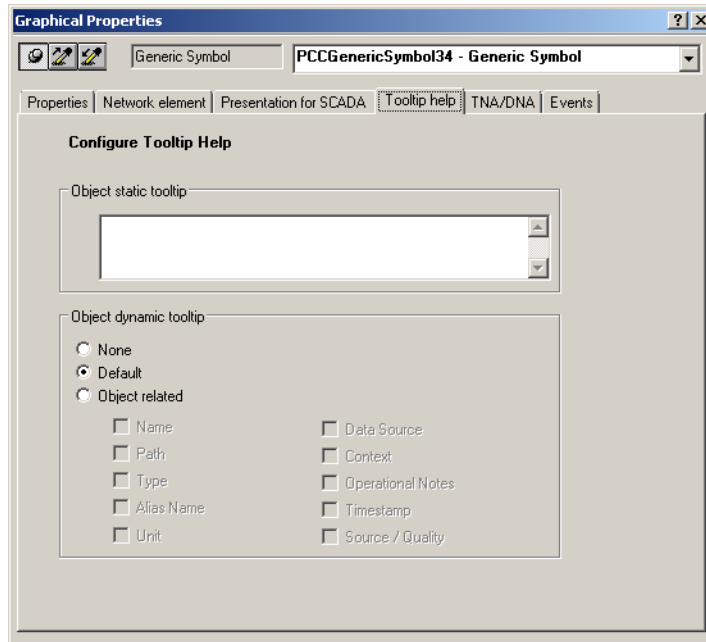
The main zone height and width of the graphical object is shown.

No Sign

If the checkbox is enabled, analog value representations are always without the sign.

5.2.10 Tooltip Help Tab of Graphical Properties Window

The **Tooltip help** tab is used to configure a tool-tip for the selected graphical object Generic Symbol (or topological) object and its domain data instance that is to be shown at runtime.



[immge003-180711-01.tif, 1, en_US]

Figure 5-34 Graphical Properties – Tooltip help

Object Static Tooltip

The entry field defines the static tooltip text that is displayed at runtime for this object. If empty, static tooltip is not shown at runtime.

Object Dynamic Tooltip

- **None**
Defines that only the text entered in the object static tooltip is displayed at runtime for this object.
- **Default**
Defines that the default object tooltip that is configured in **IMM** AC data set together with the object static tooltip, if defined, is displayed at runtime for this object.
- **Object related**
Allows to select a subset of information together with the object static tooltip, if defined, is displayed at runtime for this object.

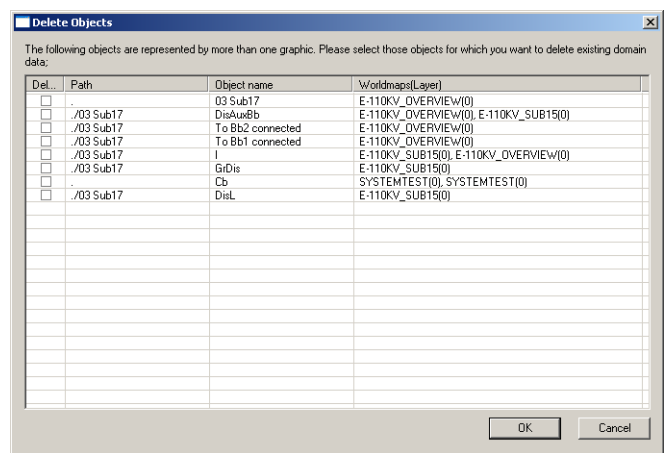


NOTE

If the configured data is not available for a particular object at runtime, it is not shown.
The **Default** dynamic tooltip does not support online activation if changed in **IMM** AC data set.

5.2.11 Technological Delete Objects Window

During deletion of domain data by deleting linked graphical objects that are graphically represented in more than one worldmap or within different layers of the active worldmap, the **Delete Objects** window is shown.



[immg017-100210-01.tif, 1, en_US]

Figure 5-35 Delete Objects Window

The domain data objects to be deleted are shown within a tabular form. You can perform the domain data deletion by means of the hook in the row **Delete**. The rows **Path** and **Object name** identify the domain data object(s). The row **Worldmaps(Layer)** shows in which worldmaps and layers the domain data objects have also representation with graphical objects. If you delete the domain data of the selected object with this operation you have to delete the graphical object also in the mentioned worldmaps and layers as those objects are now not longer linked to domain data any more.

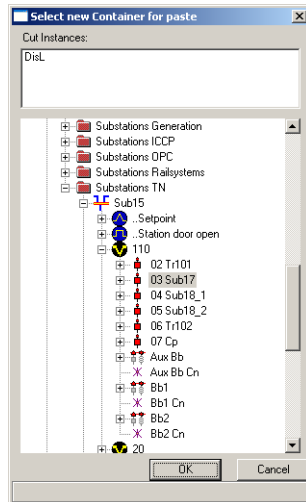


NOTE

Deletion of objects with **<SHIFT + DEL>** deletes always linked domain data too. When the graphical object is only presented in the active worldmap and layer where the deletion was initialized, deletion of linked domain data happens without explicit information to the user. To prevent deletion of linked domain data in this case use **Delete Graphic Only** within the context menu of the selected object with **< DEL>** or make sure that the technological editing mode is not active.

5.2.12 Technological Cut and Paste Window

During cut of graphical objects that are linked to domain data and paste it afterwards at a different location or worldmap, the **Select new Container for paste** window is shown when the technological editing mode is active.



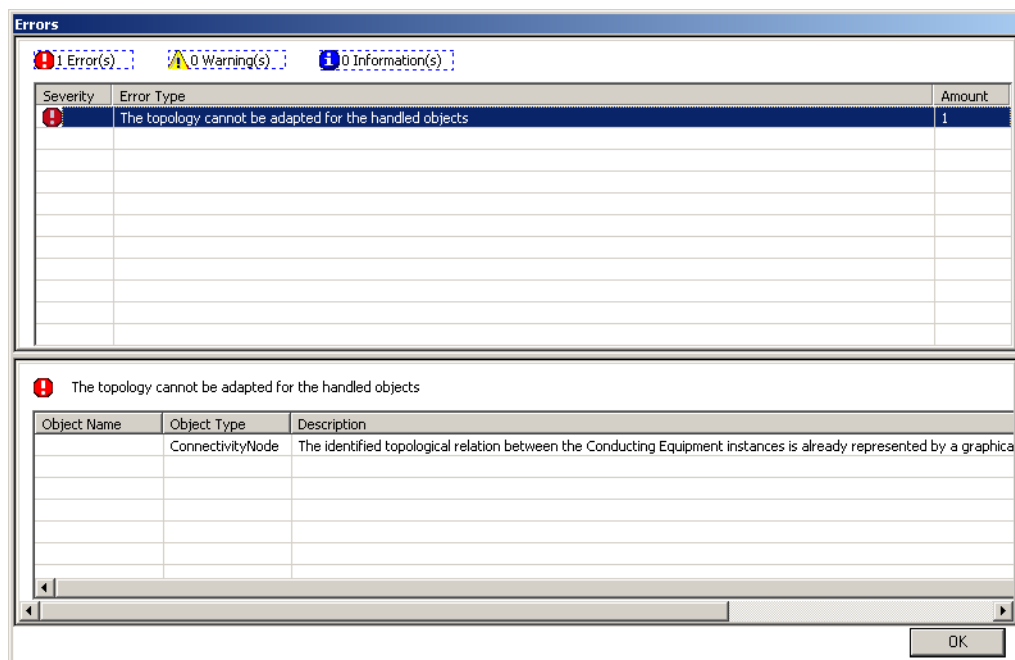
[immg018-100210-18.tif, 1, en_US]

Figure 5-36 Select New Container for Paste

The cut instances are shown in the upper area of the window. The lower area shows the already existing domain data in the form of a directory structure. Directories and subdirectories can be opened and closed by clicking the + or – icons. The cut instances will be pasted to the selected container within this operation.

5.2.13 Error Dialog for Graphical and Technological Editing

If the user has to be notified about problems and errors during graphical and technological editing workflows, the **Errors** dialog opens automatically.



[immg019-180711-01.tif, 1, en_US]

Figure 5-37 Errors Window

Error Type

An icon displays the type of error and the amount of each type displayed in the error overview section. You can use the 3 buttons to filter according the type.

Error Overview

The error overview section contains the filtered errors. The severity, the error type and how often this error type occurs is shown in a tabular form. Selecting an error displays the details in the error detail section.

Error Details

The error details section contains the details for the selected error type.

- **Object Name**
Contains the name of the domain data object that is affected by the error.
- **Object Type**
Contains the object type of the domain data object that is affected by the error.
- **Error Description**
Contains a detailed description of the error.

Button OK

OK closes the window.

Erroneous objects are discarded from the operation. You have to correct the errors manually afterwards.

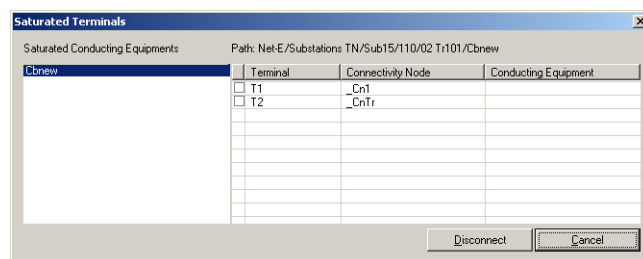
5.2.14 Creating and Manipulating the Topology - Saturated Terminals Window

The difficulty for the creation of the topological connection is the location of the **ConnectivityNode** instance within the domain data. The following rules are executed to determine the **ConnectivityNode** location (ordered by priority):

- Each **BusbarSection** has per definition one corresponding **ConnectivityNode** instance. This instance is normally placed in the same container as the **BusbarSection** (sibling) ignoring the location of the other connected **ConductingEquipment** instances.
- Each **ConductingEquipment** instance has a parent instance. If both connected TO have the same parent and its type is an **EquipmentContainer**, the **ConnectivityNode** instance will be located there too.
- In case the connected TO has no common parent except the system category, the **ConnectivityNode** will be placed in the more specialized **EquipmentContainer** instance (**Bay** prior **VoltageLevel** prior **Substation**). If at least 2 connected TO are placed in equal container types the graphical first connected TO dictates the location.
- CIM regulates the location of a **ConnectivityNode** within a **EquipmentContainer** instance (so at least a **Substation**) but propose as preferable location a **VoltageLevel** container.

If a new **ConnectivityNode** instance has to be created, its location will be determined by one of the above rules. The last rule is the fallback alternative if the others do not match.

If the technological connection is not possible, the **Saturated Terminals** window opens. One example is that all available **Terminals** for a **Breaker** are already connected topologically within the domain data.



[immg021-100210-03.tif, 1, en_US]

Figure 5-38 Errors Window

Saturated Conducting Equipments

The **ConductingEquipment** that is already saturated is mentioned in the field below. Its path within the domain data hierarchy is also displayed.

The list on the right side of the window displays the current domain data topology for the selected saturated **ConductingEquipment** without considering the current operation.

By selecting the hook, the corresponding Terminal is considered within the **Disconnect** operation.

Buttons Disconnect and Cancel

Cancel executes a complete rollback of the performed operation. Newly created graphical objects get deleted and the domain data topology stays as it was before in this case.

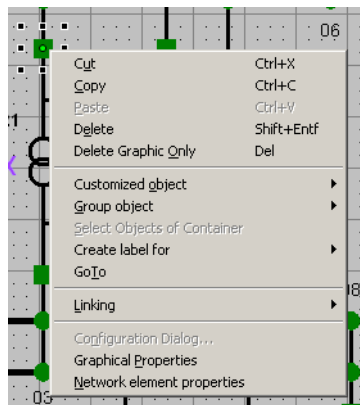
Disconnect executes the performed operation.

The hooked **Terminals** are disconnected within the existing domain data topology and are newly connected regarding your graphical operation.

If you do not hook **Terminals**, the existing domain topology remains as it was before. The **Connector** is graphically drawn, but not connected with the **Conducting Equipment**. You have to correct the errors manually afterwards.

5.2.15 The Context Menu of a Selected Object

The context menu of a selected object gives access to operations regarding this object.



[rgimm016-180711-01.tif, 1, en_US]

Figure 5-39 Context Menu

Cut

Cuts selected graphical objects together with linked IMM domain data and copies them onto the clipboard.

Copy

Copies selected graphical objects together with linked IMM domain data onto the clipboard.

Paste

Pastes copied or cut objects under the current mouse position. Linked IMM domain data is also included in the operation.

Delete

Deletes graphical objects together with linked IMM domain data.

Delete Only Graphic

Deletes only the graphical objects and keeps linked IMM domain data as they are.

Customized object

For creating, deleting, or editing customized objects.
For additional information, refer to Working with Group or Customized Objects.

Group object

For grouping objects or ungrouping a group.
For additional information, refer to Working with Group or Customized Objects.

Select Objects of Container

If the selected object is linked to an IMM domain data container, all graphical objects inked to IMM domain data instances within the selected container get also selected.

Create Label for

If the selected object is linked to IMM domain data, a text object is added containing the IMM domain data name.

GoTo

If the selected object is linked to an IMM domain data and is represented also in an additional worldmap, the additional worldmap can be opened.

Linking

For linking tag connections and texts.
For additional information, refer to Working with Objects.

Configuration Dialog

Some native WinCC objects provide a configuration dialog for specific object settings. Opens the specific configuration dialogs.

Graphical Properties

Opens the **Graphical Properties** window for the selected objects.

Network Element Properties

Opens the **IMM Single Instance Editor** window for the selected object. It is done if the graphical object is linked to IMM domain data.

6 Working with Groups or Customized Objects

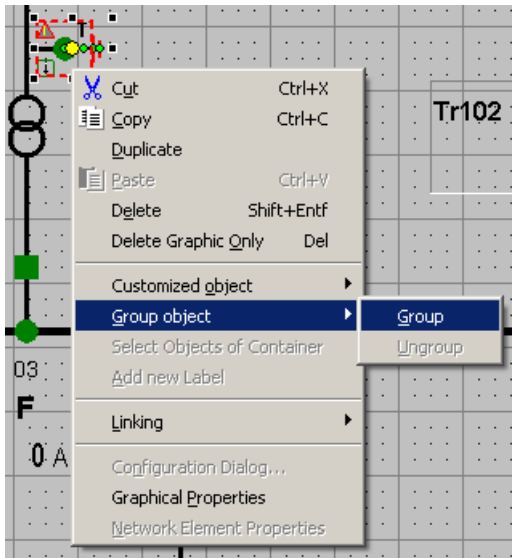
6.1	Operational Steps	228
6.2	User Interface Description	242

6.1 Operational Steps

6.1.1 Grouping Objects

The goal of this task is to group multiple objects.

- Select at least 2 objects.
- Open the context menu of the multiple selection or the **Edit** menu.
- Select the entry **Group object > Group**.



[Image008-280510-01.tif, 1, en_US]

Figure 6-1 Context Menu - Group Object

The selected objects are combined into a group. The selection frame of the multiple selection becomes the rectangle around the group. Selection marks as well as the reference point are shown for the group only.



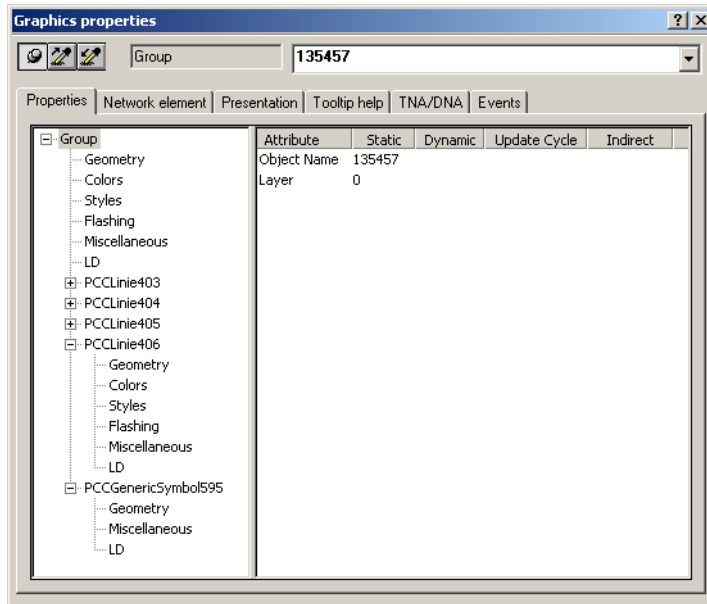
NOTE

Single groups can be combined into larger groups with the **Group** command. This allows a group hierarchy to be built up. Groups can be ungrouped in the reverse order of grouping.

6.1.2 Changing the Properties of a Group

The goal of this task is to modify properties of an object within a group or all objects within the group by assigning new value to accompanying attribute.

- Select a grouped object
- Open the **Graphical Properties** window.



[image009-280510-01.tif, 1, en_US]

Figure 6-2 Graphical Properties Window of a Group Object

- Select the property group of the desired individual object within the group or the common property groups in the **Properties** tab that contains the attribute you want to change.

Alternative Operation

You can also change the properties of a group (for example the geometric properties) with the mouse. The complete group is edited in the same way as single objects.



NOTE

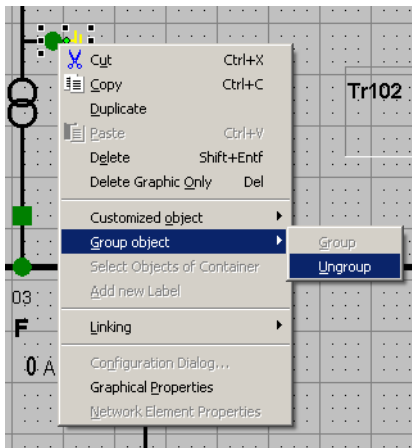
The objects included in a group can also be edited individually by changing their attributes. The objects are displayed in the **Graphical Properties** window with their property groups as a component of the Group object type.

6.1.3 Ungrouping a Group

The goal of this task is to ungroup a group.

- Select a grouped object
- Open the context menu or the **Edit** menu.

- Select **Group Object > Ungroup**.



[image010-280510-01.tif, 1, en_US]

Figure 6-3 Context Menu - Ungroup Group Object

The group selection marks disappear. The objects of the group are shown as a multiple selection.



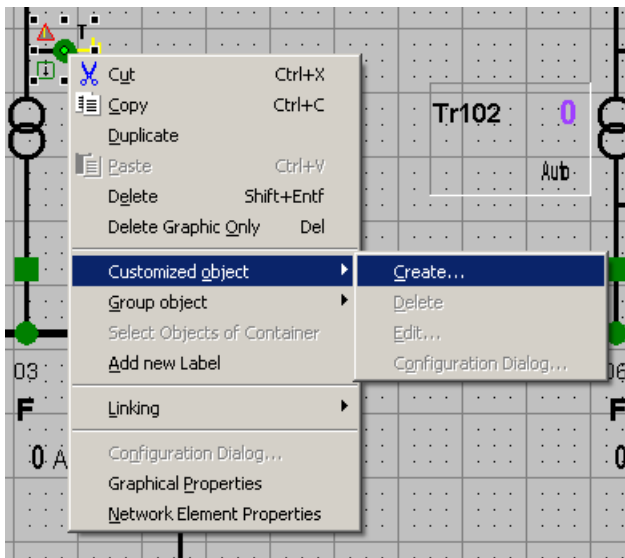
NOTE

The **Group Object** command can be used to group single objects again and thus to build up a hierarchy of groups. Groups can be ungrouped in the reverse order of grouping.

6.1.4 Creating a Customized Object

The goal of this task is creating a customized object.

- Select an object.
- Open the context menu or the **Edit** menu.
- Select the entry **Customized object > Create...**



[image011-280510-01.tif, 1, en_US]

Figure 6-4 Context Menu - Customized Object

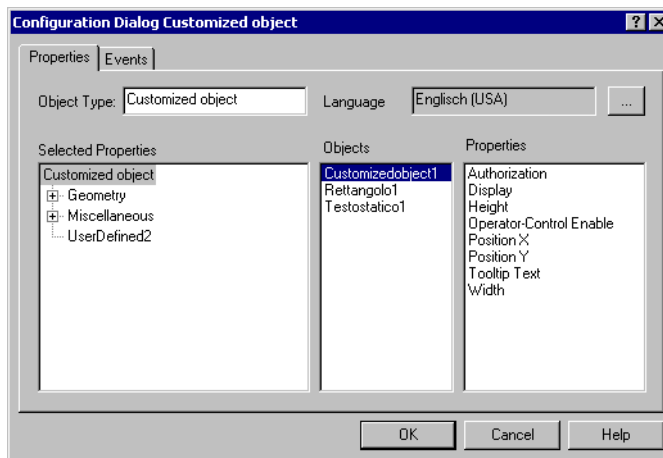
The **Configuration Dialog Customized Object** opens.

- Configure the available properties and events of the customized object.
- Confirm your configuration with **OK**.
The selected objects are combined into a customized object. The selection frame of the multiple selection becomes the rectangle around the customized object together with the common reference point.

6.1.5 Configuring a Customized Object

The goal of this task is to configure a customized object using the **Configuration Dialog Customized Object**. **Configuration Dialog Customized Object** appears after creating a customized object or can be opened using the **Edit** menu or the context menu of the customized object.

- Select the entry **Customized Object > Configuration Dialog....**
The **Configuration Dialog Customized Object** opens.



[immg030-101208-01.tif, 1, en_US]

Figure 6-5 Configuration of Customized Object

- Select the desired attribute and click to select one of the languages installed in WinCC.
Property groups and attributes can be inserted or removed from the **Selected Properties** area as required.
- Click the + or - symbols to show or hide subordinate entries.
- Select an object within the objects area for which you want to show the available attributes in the **Properties** area.
- Double-click the desired attribute of the properties area to insert it into the assigned position in the **Selected Properties** area.



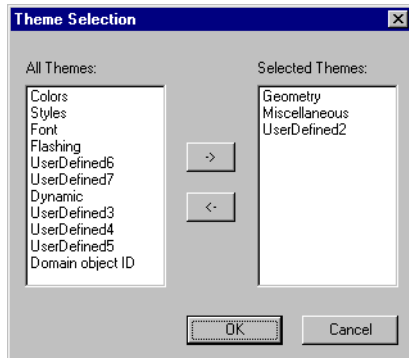
NOTE

An object can only receive an operator authorization as a whole. A customized object can have an operator authorization but its subordinate objects cannot.

6.1.6 Configuring the Property Groups of the Customized Object

The goal of this task is to configure the property groups of a customized object.

- Select the **Theme Selection...** entry in the context menu within of the **Selected Properties** area of the **Configuration Dialog Customized Object**.
The **Theme Selection** dialog opens.



[immg032-101208-01.tif, 1, en_US]

Figure 6-6 Theme Selection Dialog

- Select the desired property groups and move them with the arrow buttons.
- Confirm your configuration with **OK**.

User-defined property groups can also be used for a customized object. By default they contain the name **user-defined** and a sequential number. The name of the user-defined property groups can be changed.

- Select **Rename** in the context menu of the property group within the **Selected Properties** area of the **Configuration Dialog Customized Object**.
The name of the property group can now be edited.
- Enter the new name for the property group.
- Confirm your entry with **<ENTER>**.

6.1.7 Adding and Deleting Attributes of a Customized Object

The goal of this task is to add and delete attributes in the **Selected Properties** area of the **Configuration Dialog Customized Object**.

Adding Attributes

- Select an object in the **Objects** area from which you want to provide an attribute for the customized object.
The available attributes of the selected object are shown in the **Properties** area.
- **Double-click** the desired attribute in the **Properties** area.
The selected attribute is added at the relevant position in the **Selected Properties** area. The attribute name is extended by the object name of the object.
- Repeat the previous steps until all desired attributes are in the **Selected Properties** area.

Alternatively you can drag attributes to the desired position in the **Selected Properties** area with the mouse.



NOTE

The attributes of the customized object are automatically assigned to the associated property group. The default assignments cannot be changed.

Deleting Attributes

- Select the attributes of the customized object that you want to delete in the **Selected Properties** area.
- Select **Delete** in the context menu of the selected attribute. The selected attribute is removed from the selection.

6.1.8 Configuring the Properties of Attributes of the Customized Object

The goal of this task is to configure the properties of attributes of a customized object.

- Select the attribute of the customized object whose properties you want to configure in the **Selected Properties** area of the **Configuration Dialog Customized Object**.
- Select **Properties...** in the context menu of the selected attribute. The **Properties** dialog opens.
- Change the settings and confirm your input with **OK**.

6.1.9 Linking a Customized Object with a Structure Tag

The goal of this task is to link a customized object with a tag.

A structure type and at least one structure tag must be configured. For example, in the customized object, you have added 2 objects **Static Text** with the property **Text** in selected properties to **Userdefined2**. In **Graphics Editor**, the **Tag** dialog must be open. Using the menu **View>Toolbars...** activate the **Tag** dialog.

- In **Configuration Dialog Customized Object** in the area **Selected Properties**, select one of the static texts for the **Customized Object**.
- In the **Properties for Attribute Name** dialog, enter the name **StructureType**.
- In **Configuration Dialog Customized Object** in the area **Selected Properties**, select the second static text for the **Customized Object**.
- In the **Properties for Attribute Name** dialog, enter the name.
- Confirm your settings with **OK**.
- In the **Graphical Properties** dialog of the **Customized Object**, enter under attribute **StructureType** the name of the structure type as text.
- Dynamisize the desired properties with an element of a valid structure tag.
- If you now select the **Customized Object**, the open **Tag** dialog displays the available tags of the assigned structure type.
- Using drag and drop, you connect the desired structure tag from the **Tag** dialog with the customized object. The structure tag is entered into the object property tag name

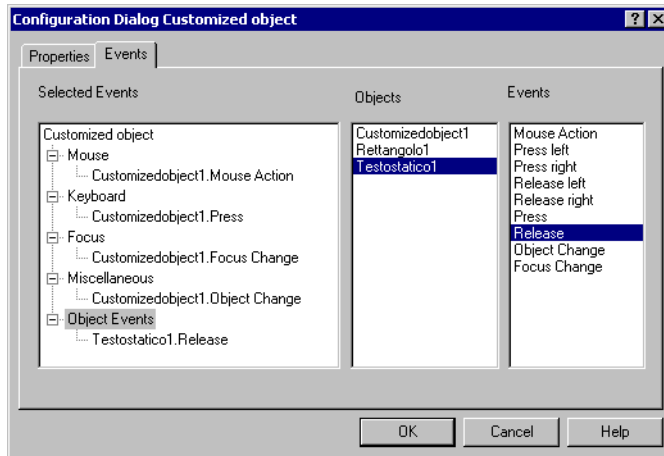
Duplicate customized object is assigned a tag of the same structure type using drag and drop.

6.1.10 Configuring the Events of a Customized Object

The goal of this task is to configure the events of a customized object using the **Configuration Dialog Customized Object**.

The **Configuration Dialog Customized Object** appears after creating a customized object or can be opened using the **Edit** menu or the context menu of the customized object.

- Select the entry **Customized Object > Configuration Dialog...**.
The **Configuration Dialog Customized Object** opens.
- Select the **Events** tab.



[immrg037-101208-01.tif, 1, en_US]

Figure 6-7 Configuration Dialog - Events Tab

Adding Events

- In the **Objects** area, select an object for which you want to configure an event in the customized object.
The events available for the selected object are displayed in the **Events** area.
- Double-click the desired event in the area **Events**.
The selected event is inserted into the respective event group in the **Selected Events** area. The event name is extended by the object name of the object.
- Repeat the previous steps until all desired events are in the **Selected Events** area.
- Confirm your configuration with **OK**.

Alternatively you can also add events by double-clicking the desired event. The event is added at the relevant position in the **Selected Events** area.



NOTE

The events configured for the customized object are automatically assigned to the associated event group. The default assignments cannot be changed.

Deleting Events

- Select the event of the customized object that you want to delete in the **Selected Events** area.
- Select **Delete** in the context menu of the selected event. The selected event is removed from the selection.

6.1.11 Editing a Customized Object

The goal of this task is to edit a customized object.

Activating Edit Mode

- Select a customized object.
- Open the context menu or the **Edit** menu.

- Select **Customized object > Edit...**
The edit mode is activated. Like multiple selection of objects, selection marks for the single objects contained in the customized object are shown. The selection marks are shown as shade squares with a white border in edit mode.

Exit Edit Mode

- Select a customized object currently in **Edit Mode**.
- Open the context menu or the **Edit** menu.
- Select **Customized object > Finish Editing**.
Edit mode is deactivated. When edit mode is exited, only the selected objects are imported into the customized object. If no object is selected, the customized object is discarded.

6.1.12 Deleting a Customized Object

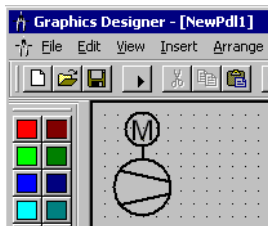
The goal of this task is to delete a customized object.

- Select a customized object.
- Open the context menu or the **Edit** menu.
- Select **Customized Object > Delete...** The selection marks of the customized object disappear. The individual objects of the customized object are shown as a multiple selection.

6.1.13 Example: Creating a Compressor as a Customized Object

The goal of this task is to understand how to create a compressor as a customized object.

- Create a graphic to show the compressor.

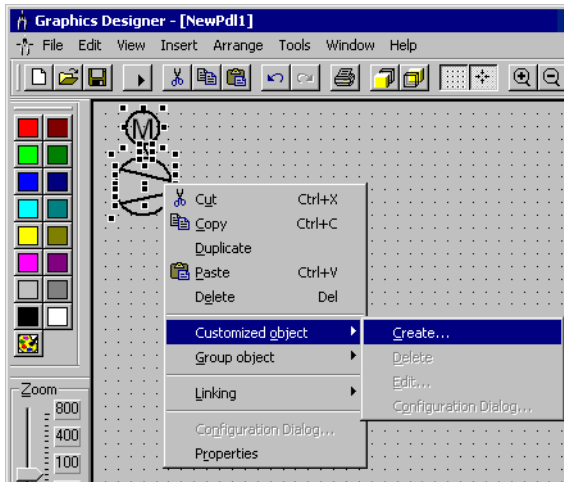


[immg028-161208-01.tif, 1, en_US]

Figure 6-8 Creating a Graphic

The compressor is created from the following standard objects: 2 circles, 3 lines, and a static text.

- Select all objects from which the customized object must be created. Select **Customized Object > Create...** in the multiple selection context menu.

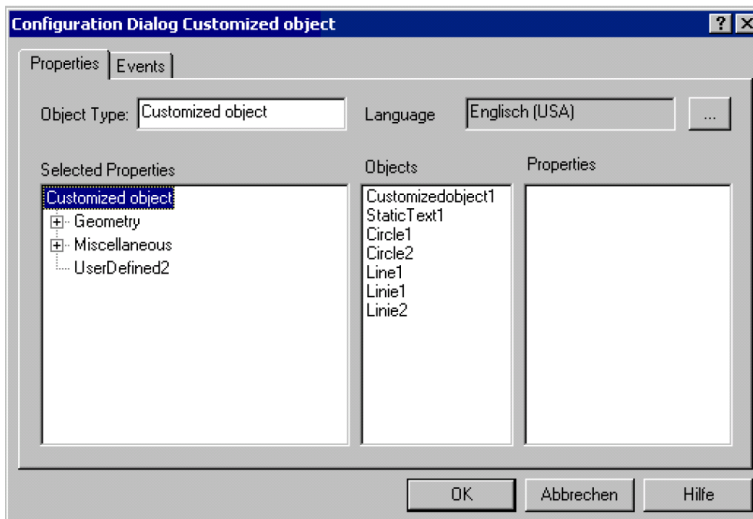


[immrg029-161208-01.tif, 1, en_US]

Figure 6-9 Multiselection Context Menu

The **Customized Object Configuration** dialog opens. The **Configuration Dialog Customized Object** shows all the single objects contained in the customized object with their properties and events.

- Select the **Properties** tab.

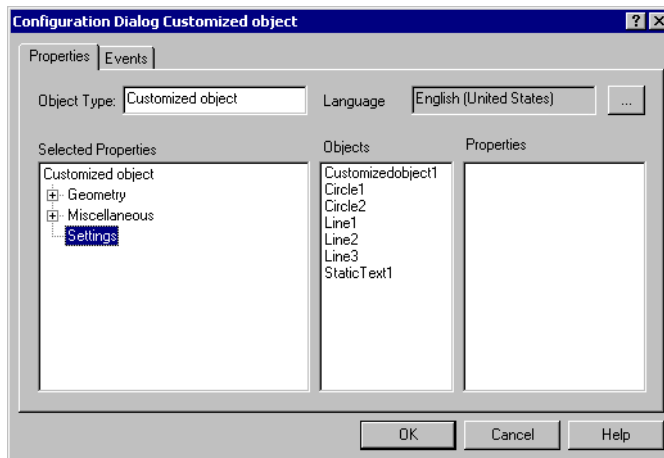


[rgimm013-180210-01.tif, 1, en_US]

Figure 6-10 Configuration Dialog Customized Object - Properties Tab

In this example the **UserDefined2** property group is used to define the desired properties and events of the customized object.

- Select **Rename** in the context menu of the UserDefined2 property group.

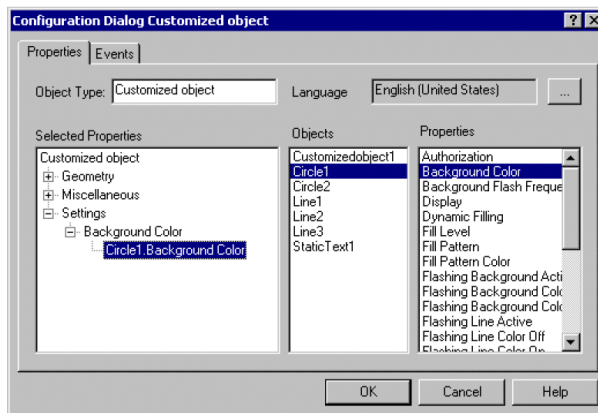


[immg031-161208-01.tif, 1, en_US]

Figure 6-11 Configuration Dialog

In this example the term settings is entered as the new name of the **UserDefined2** property group. The **On and Off** states must be visualized by changing the background color of **Circle1**.

- Select the object **Circle1**.
- Drag the background color property by drag and drop to the **Settings Property** group.

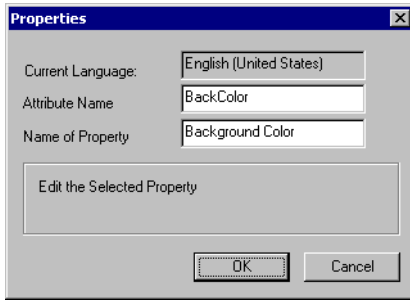


[rgimm014-180210-01.tif, 1, en_US]

Figure 6-12 Configuration Dialog - Objects

The customized object now has the **Circle1**. Background color attribute in the **Settings / Background Color** property group.

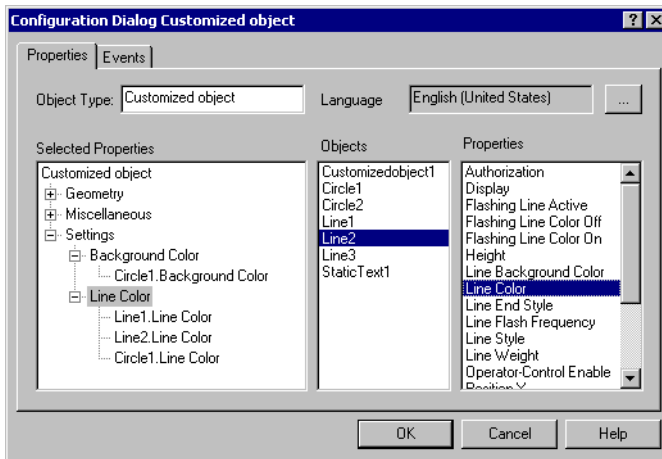
- Select **Properties...** in the context menu of the attribute **Circle1**.
Background color. Opens the **Properties** dialog.



[immrg033-161208-01.tif, 1, en_US]

Figure 6-13 Properties Dialog

- Enter a name for the dynamics of the attribute by C or VBS actions in the **Attribute Name** field. The **Name of Property** field contains the exact name of the attribute for the display in the **Object Properties** window and can also be changed. The **Approach** and **Retract** states must be displayed by flashing lines: The line colors of all objects, which are flashing and configured for, can be combined into an attribute of the customized object.
- Select the desired objects in sequence and drag the property **Line Color** using drag and drop into the property group settings.

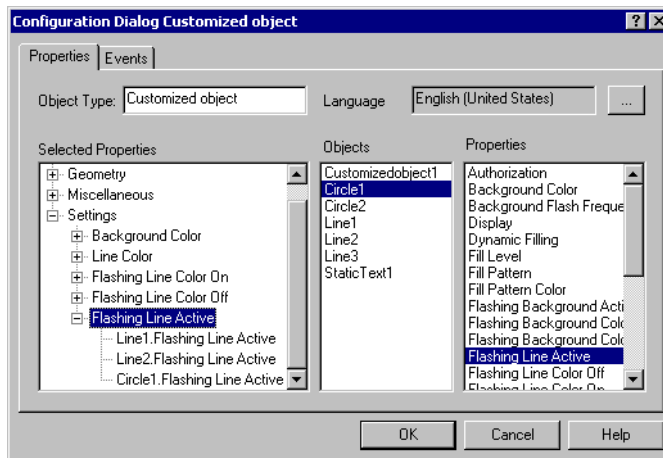


[immrg035-161208-01.tif, 1, en_US]

Figure 6-14 Configuration Dialog Customized Object - Line Color

The customized object now has the **Object. Line Color** attributes in the **Settings Property** group.

- Repeat this process for all attributes that the customized object requires.

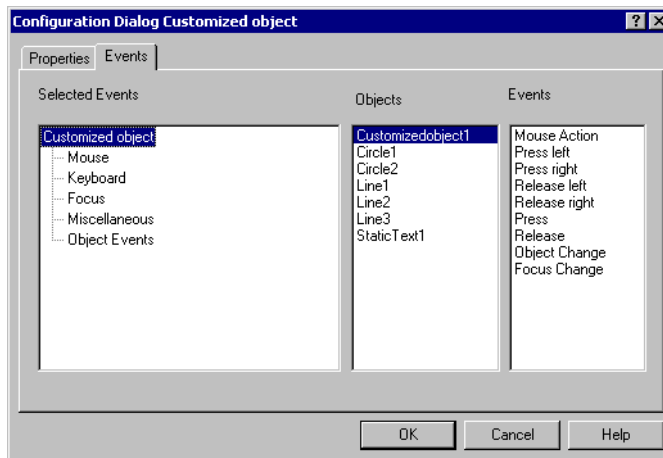


[rgimm015-180210-01.tif, 1, en_US]

Figure 6-15 Configuration Dialog Customized Object - Flashing Line Color

In this example the attributes **Flashing Line Color On**, **Flashing Line Color Off** and **Flashing Line Active** are added to various single objects.

- Select the **Events** tab to configure the required events for the customized object.



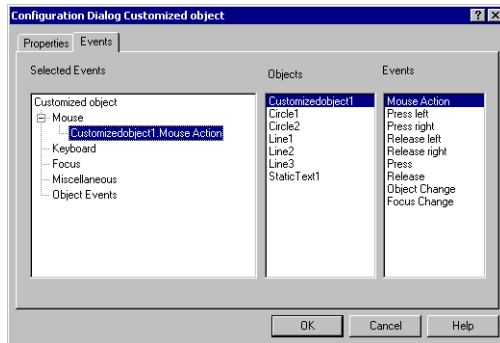
[rgimm016-180210-01.tif, 1, en_US]

Figure 6-16 Configuration Dialog Customized Object - CustomizedObject

In this example only the mouse click event is configured.

- Select the event **Mouse**.
Select the **CustomizedObject** object that is to respond to the desired event. If the event is configured for a contained single object, select the single object only.

- Double-click the event mouse click. The customized object now has the **CustomizedObject1**. Mouse Action event in the **Mouse** event group. Add more events in the same manner.

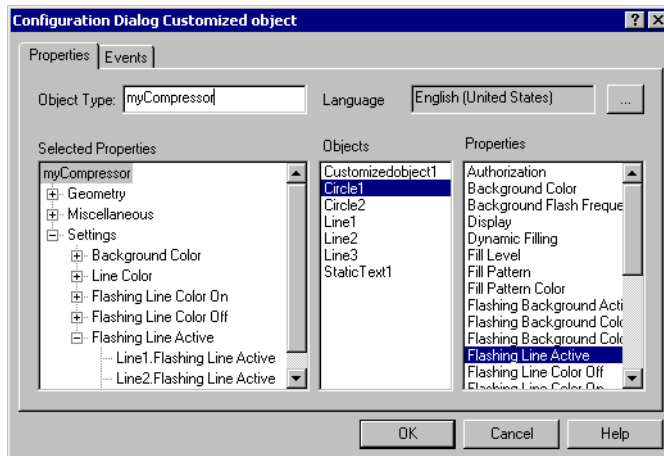


[rgimm017-180210-01.tif, 1, en_US]

Figure 6-17 Configuration Dialog Customized Object - CustomizedObject1

The customized object has the name **myCompressor**.

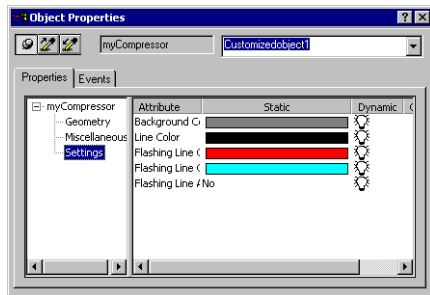
- Select the **Properties** tab.
- Enter the name **myCompressor** in the field object type.
- Click the entry **Customized Object** in the **Selected Properties** area to accept the name.
- Confirm the configuration with **OK** to finish creating the **myCompressor** customized object. The **Configuration Dialog Customized Object** is closed.



[rgimm018-180210-01.tif, 1, en_US]

Figure 6-18 Configuration Dialog Customized Object - myCompressor

- Select the new customized object in your worldmap.
Open the window **Object Properties** to check the configuration of the customized object.
The **Settings Property** group only shows the properties and events that you defined in the **Configuration Dialog Customized Object**.



[immg127-290109-01.tif, 1, en_US]

Figure 6-19 Configuration Dialog Customized Object - Settings Property

6.2 User Interface Description

6.2.1 Introduction

The objects within a worldmap can be combined by forming a group or a customized object from a multiple selection. In addition, a group or a combined object can be inserted into the **Library**. This makes the group or the combined object available as a library object for reuse in other worldmaps or projects.



NOTE

A group is only considered within the **Graphics Editor** and does not have any meaning in the runtime environment of the Spectrum Power Control Center.



NOTE

A group is internally treated as additional object on top of the individual objects of the group. Extensive usage of groups within a worldmap therefore increases the total number of graphical objects and the performance impact for large worldmaps. To decrease the total number of graphical objects for performance reasons, ungroup the groups within a worldmap.

6.2.2 Working with Groups

A multiple selection of objects can be combined into a group. A group of objects can be edited like a single object in the **Graphics Editor**. It is also possible to edit objects in the group individually.

Unlike multiple selection, the selection marks of the individual objects in a group are not shown. The selection marks cover the entire group. The selection frame of the multiple selection becomes the rectangle around all the objects of the group. The group has one common reference point.

Object Group

The objects of a multiple selection can be combined into a group with the **Group object > Group** command.

Properties of Objects Within a Group

The objects included in a group can be edited individually by changing their attributes. The objects are displayed in the **Graphics Properties** window with their property groups as a component of the Group object type.

Properties of a Group

A group can be edited in the **Graphics Editor** like all other objects.

The following elements are shown as components of the Group object type in the **Graphics Properties** window:

- Property groups and attributes that have at least one of the selected objects.
- All objects in the group with their own property groups and attributes.

Changing a common group attribute affects all single objects that have this attribute. However, for some attributes this effect has special rules that only apply to the group.

Examples

- If the origin of a group is moved, the values of the individual objects are configured for the new coordinates. The attributes **Position X** and **Position Y** of the individual objects do not change relative to the origin of the group.
- If the size of a group is changed, the attributes **Height** and **Width** of the individual objects are adjusted by percentage to match the size.

- If a rotation has been specified for single objects of a group in runtime, these settings are matched to the size of the group. The reference points for the rotation refer to the origin of the group, not the origin of the single object. In runtime, the objects for which no rotation was specified remain unchanged.
- Linear objects that are in a group with surface objects always take the color that was last selected for the background color or the line color in the properties of the group.
- If a group is to be operated with a keyboard shortcut, it must only be assigned to the single button object in the group. A keyboard shortcut assigned to the entire group does not work in runtime.

Ungroup a Group

Use the **Group object > Ungroup** command to split a group into its individual objects. The individual objects are then shown as a multiple selection.



NOTE

The **Group** command can be used to group single objects again and thus to build up a hierarchy of groups. Groups can be ungrouped in the reverse order of grouping.

6.2.3 Customized Objects

A customized object allows the properties and events that are displayed and changed in the **Graphics Properties** window to be configured individually.

A customized object can be edited in the **Graphics Editor** like a single object of the **Object Palette**. As in a group, the selection marks of the individual objects are no longer shown after creation of a customized object. The selection marks surround the entire customized object. The selection frame of the multiple selection becomes the rectangle around all the objects of the customized object. The customized object has one common reference point.

Note the following features for working with customized objects:

- The **Undo** function is not available for editing a customized object.
- Configured events of the single objects are deleted when the customized object is created.
- If editing mode is exited without an object having been selected, the customized object is deleted. When edit mode is exited, only the selected objects are imported into the customized object.
- Assigning keyboard shortcuts and specifying a tab sequence for objects within the customized object is not supported.
- If a group display is embedded in a customized object, the attributes group value, group relevant, and bit pattern group display are available for the customized object to capture and acknowledge states.
- It is not possible to create an attribute name twice. These attributes are preassigned by the system (standard attributes). If the attributes of a customized object are deleted, these attribute names cannot be used in a user-specific attribute.
- User-specific attribute names of the customized object must not contain special characters.
- Use the **Configuration Dialog Customized Object** to display the properties for the customized object. If a standard property is deleted there, this property is not displayed on the **Graphics Editor** user interface. Access to the property is still possible using VBS. This relates to the following properties: Object name, Layer, Left, Top, Width, Height, Visible, Enabled, and Password Level.
- Properties with the prefix @ are not electronically numbered (enum) by VBA. Therefore, these properties cannot be edited by VBA either.
- Properties with the prefix @ cannot be accessed by VBS.

- When addressing the user-specific properties of a customized object, the spelling used in the VB Scripts must correspond to the spelling in the **Configuration** dialog. Particular attention must be paid to upper or lower case.
- The property name is dependent on the language set. In the case of customized object with multilanguage user-specific properties, the language must be set in the **Configuration Dialog Customized Object** before editing the customized object. The required language can then be selected.



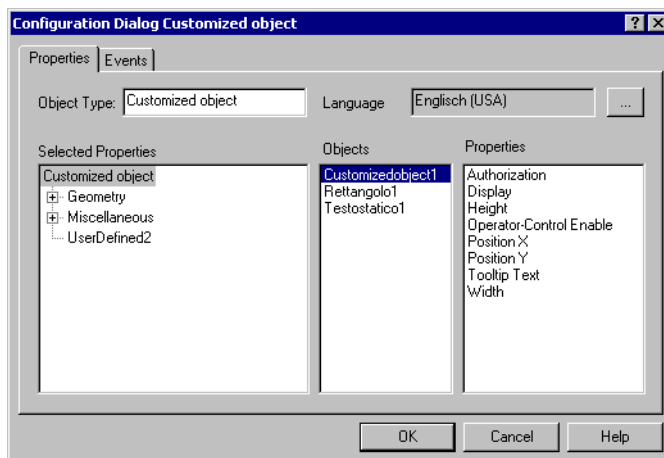
NOTE

The text properties of the text object must be dynamic to allow changes from external sources so the language change can function.

The objects of a multiple selection can be combined into a customized object with the command **Customized Object > Create...** in the context menu or the **Edit** menu. A customized object can be created from one single object. The object types Application Window, Picture Window, OLE element, Group, or another customized object cannot be imported into a customized object.

6.2.4 Configuration on a Customized Object

The **Configuration Dialog Customized Object** can be used to specify which properties and events can be configured for the customized object. The **Graphics Properties** window shows only the properties and events defined in the configuration dialog for the customized object. The settings are configured in the **Properties** and **Events** tab.



[immg030-101208-01.tif, 1, en_US]

Figure 6-20 Configuration of Customized Object

The **Configuration Dialog Customized Object** can be opened from the context menu of a selected customized object.

Properties Tab

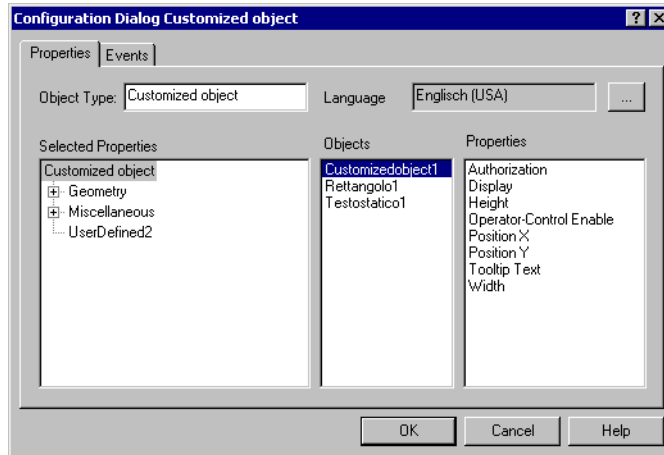
A selection of the available property groups and attributes can be defined in the **Properties** tab. The name and configured language of an available attribute can be changed. Every attribute of the customized object can also be linked to attributes of the included single objects.

Event Tab

The events available for the customized object are configured with the **Events** tab. It is also possible to make object events available for the included single objects.

6.2.5 Properties of a Customized Object

A selection of the available property groups and attributes is defined in the **Properties** tab. The name and configured language of an available attribute can be changed. Every attribute of the customized object can also be linked to attributes of the included single objects.



[immg031-101208-01.tif, 1, en_US]

Object Type

The name of the selected customized object is shown and can be changed. Enter the desired name.

Language

The customized object and the attributes of the selected properties can be configured for different languages.

Selected Properties

This area contains the property groups and attributes that are displayed as properties of the customized object in the **Graphics Properties** window. Property groups and attributes can be inserted or removed from the selection as required.

Objects

This area contains all objects included in the customized object. Select an object for which you want to show the available attributes in the **Properties** area.

Properties

This area contains all available attributes for the object selected in the objects area. Double-click the desired attribute to insert it into the assigned position in the **Selected Properties** area.



NOTE

An object can only receive an operator authorization as a whole. A customized object can have an operator authorization but its subordinate objects cannot.

Inherit Status Attribute

The statuses of the Display and Use release properties of the user object are inherited by the individual objects of the user object.

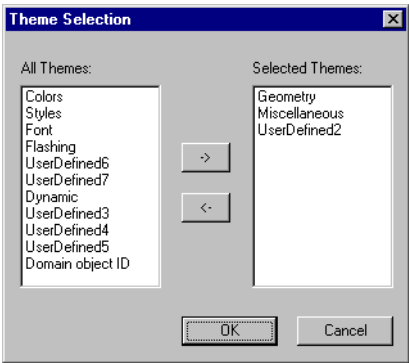
Example:

You have configured the Inherit status properties for the user object with Yes and display as Visible.
An object of the user object has the property invisible.

If the user object is made invisible and then again visible, then all objects of the user object are shown as visible. It occurs even if 1 object of the user object has the property invisible.

6.2.6 Property Groups of the Customized Object

Only the property groups that are listed in the **Selected Properties** area of the **Configuration Dialog Customized Object** are shown for the customized object in the **Graphics Properties** window. The selection of the property groups is configured with the **Theme Selection** dialog. The predefined name of the user-defined property groups can be changed.



[immg032-101208-01.tif, 1, en_US]
Figure 6-21 Theme Selection Dialog

All Themes

All property groups that can be made available for the customized object are listed in the left section of the dialog.

Selected Themes

All property groups that have been selected for the customized object are listed in the right section of the dialog.

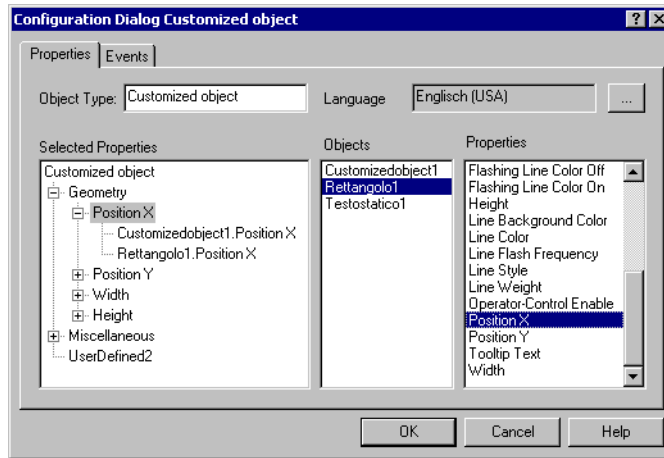
Arrow Buttons

Table 6-1 Use the arrow buttons to move property groups in the direction of the arrow:

	Moves the selected property groups from the left to the right section.
	Moves the selected property groups from the right to the left section.

6.2.7 Attributes of a Customized Object

Attributes of the customized object that are to be shown in the **Graphics Properties** window. It can be added or deleted from the selection in the **Selected Properties** area of the **Configuration Dialog Customized Object**.

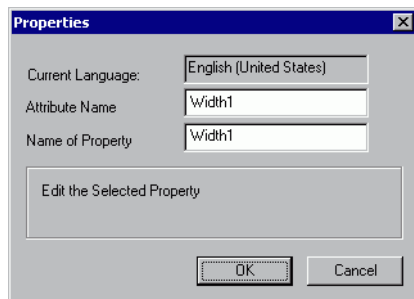


[immrg035-101208-01.tif, 1, en_US]

Figure 6-22 Configuration Dialog - Selected Properties Area

6.2.8 Properties of Attributes of the Customized Object

The properties of the attributes available in the customized object can be changed for display in the **Graphics Properties** window. The properties of the attributes can be configured with the **Properties** dialog.



[immrg036-101208-01.tif, 1, en_US]

Figure 6-23 Properties Dialog

Current Language

Displays the language for which the attribute was designed.

Attribute Name

Displays the attribute names under which the attribute is registered in WinCC. The **OLE Automation Name** must be unique. A change of the attribute name affects all languages configured for the attribute.

Property Name

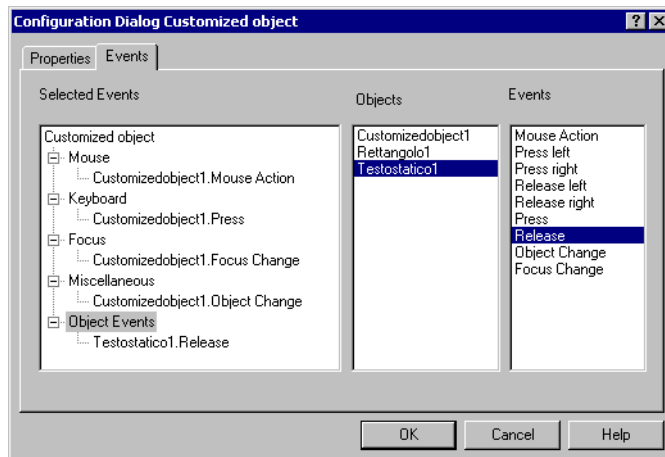
Property names with which the attribute is displayed in the **Graphics Properties** window.

6.2.9 Linking a Customized Object with a Structure Tag

A customized object is linked with a structure tag with an open **Tag** dialog in **Graphics Editor** simply by using drag and drop. Using the object properties of 2 text boxes, structure type and unique identification of the object are characterized using the structure tag.

6.2.10 Configuring the Events of a Customized Object

The events available for the customized object are configured with the **Events** tab of the **Configuration Dialog Customized Object**. It is also possible to make object events available for the included single objects.



[immg037-101208-01.tif, 1, en_US]

Figure 6-24 Configuration Dialog - Events Tab

Selected Events

This area contains the event groups:

- Mouse
- Keyboard
- Focus
- Miscellaneous
- Object events

The events designed for the customized object are assigned to these event groups. All events that are designed for the single objects in the customized object are assigned to the **Object Events** event group.

Objects

This area contains all objects included in the customized object. Select an object for which you want to show the available events in the **Properties** area.

Event

This area contains all available events for the object selected in the objects area. Double-click the desired event to insert it into the assigned position in the **Selected Events** area.

6.2.11 Editing a Customized Object

The edit mode for the customized object is activated with the **Customized object > Edit...** function. In edit mode, the properties of all included single objects can be separately changed.

Single objects can be added or deleted. A single object contained in a customized object is edited as described in Working with objects. The selection marks of the single objects are shown as dim squares with a white border in edit mode. When edit mode is exited, only the selected objects are imported into the customized object.

6.2.12 Deleting a Customized Object

A customized object can be split into its component objects with the command **Customized Object > Delete** in the context menu or the **Edit** menu. The individual objects are then shown as a multiple selection.

7 Working with Controls

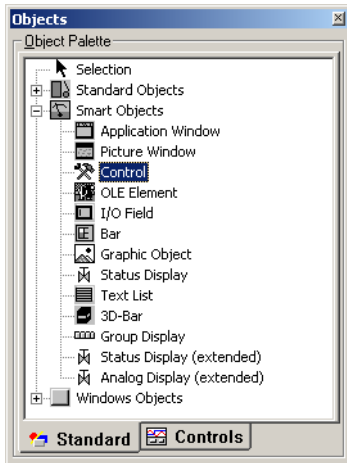
7.1	Operational Steps	252
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7.1 Operational Steps

7.1.1 Inserting a Control as Smart Object

The goal of this task is to insert a control from the **Standard** tab of the object palette into a worldmap.

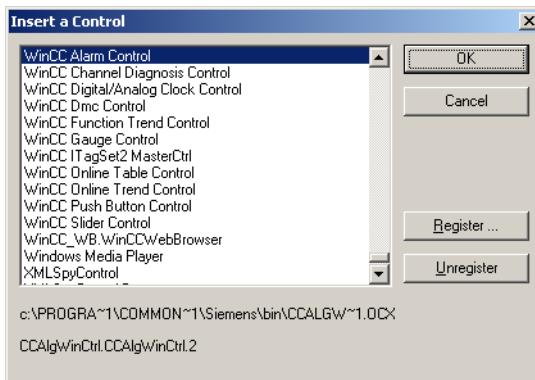
- In the **Object Palette**, click the smart object **Control**.



[image012-280510-01.tif, 1, en_US]

Figure 7-1 Object Palette

- Place the mouse pointer at the position in the worldmap to insert the control.
The mouse pointer changes into a crosshair with a control object symbol attached.
- Drag the control to the desired size.
The **Insert a Control** dialog opens.



[immg139-140409-01.tif, 1, en_US]

Figure 7-2 Insert a Control Dialog

- Select one of the controls that are registered in the operating system.
Click **OK** to apply the selection.
The associated configuration dialog opens for multiple controls.
- Adapt the settings of the configuration dialog as needed.
- Click **OK**.
The insertion process for the control object is finished.

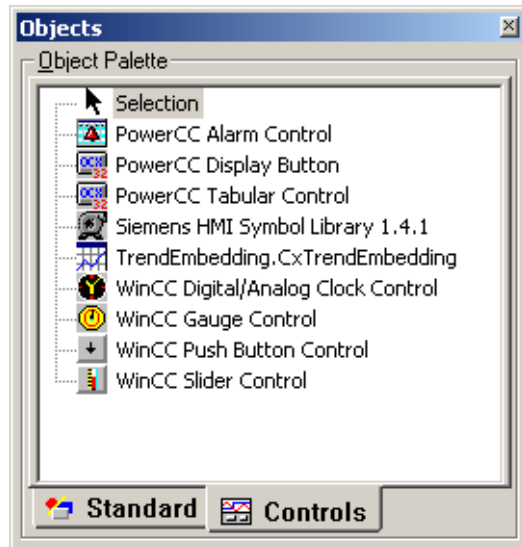
Alternative Operation

- Double-click in the **Object Palette** on the smart object **Control**, to insert it with the default properties into the worldmap.

7.1.2 Inserting a Control from the Controls Tab

The goal of this task is to insert a control from the **Controls** tab of the object palette into a worldmap.

- In the **Object Palette**, click the **Controls** tab. Select the control.



[rgimmg20-080210-01.tif, 1, en_US]

Figure 7-3 Object Palette

- Place the mouse pointer at the position in the worldmap to insert the control.
The mouse pointer changes into a crosshair with a control object symbol attached.
- Drag the control to the desired size.
The associated configuration dialog opens for multiple controls.
- Adapt the settings of the configuration dialog as needed.
- Click **OK**.
The insertion process for the selected control is finished.

Alternative Operation

- Double-click on the control in the **Object Palette**, to insert it with the default properties into the worldmap.



NOTE

WinCC Controls are native WinCC functionality and can only be used in a limited way within the Spectrum Power 5 environment. Linked WinCC and PowerCC Tags can only be read during Runtime. Writing data or setting values is not possible within the Spectrum Power 5 environment.

7.1.3 Resizing a Control

The goal of this task is to resize an existing control already placed within a worldmap.

- Position the mouse pointer on a handle of the selected control.
The mouse pointer changes to a double arrow. The alignment of the double arrow indicates the directions in which the handle is moveable:
 - Horizontal double arrow
The handles on the horizontal center axis is used to change the width of the control.
 - Vertical double arrow
The handles on the vertical center axis is used to change the height of the control.
 - Diagonal double arrow
The handles on the corners of the control is used to change the height and width of the control.
- Use the mouse to drag the handle to the position.
If the **Snap to Grid** function is activated the control is scaled with the mouse according to the settings for the grid.

Alternative Operation

- Changing the values of the **Width** and **Height** attributes in the **Graphics properties** window.

7.1.4 Positioning a Control

The goal of this task is to reposition an existing control already placed within a worldmap.

- Position the mouse pointer on the selected control.
The mouse pointer changes to a crosshair with arrowheads.
- Click the control and hold down the left mouse button.
The mouse pointer changes into an arrow with an outline icon (small rectangle with broken line).
- Move the mouse pointer.
The rectangle surrounding the control is displayed and shows a possible new position for the control. The control initially maintains its original position.
If the **Snap to Grid** function is activated the control is positioned with the mouse according to the settings for the grid.
- Release the mouse button.
The control is moved to the position previously displayed. The mouse pointer again changes into a crosshair with arrow ends.

Alternative Operation

- Moving the selected control with the aid of the arrow keys for mouse pointer control.
- Changing the **Position X** and **Position Y** attributes in the **Graphics properties** window.

7.1.5 Changing the Properties of a Control

The goal of this task is to adopt the properties of an existing control already placed within a worldmap according to your needs.

Changing the attributes in the **Graphics properties** dialog:

- In the context menu of a control select the entry Graphical properties to open the **Graphics properties** dialog.
The change of an attribute is executed immediately.



NOTE

For additional information of the **Graphics properties** dialog, refer to Graphics Properties Window.

Changing the settings in the control specific configuration dialogs:

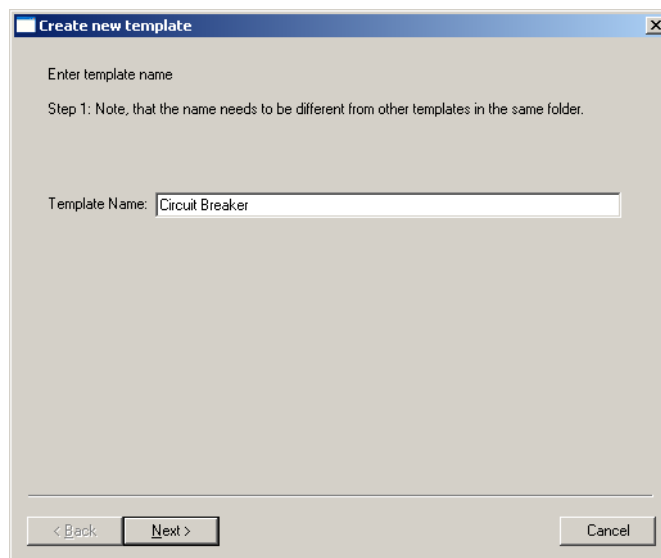
- Double-click a control to open the related configuration dialog.
- Make the desired changes.
- Click **Apply** to assign the new settings to the control.
- Click **OK**.
The configuration is stored in the control.

7.1.6 Storing Controls into the Library as Templates

The goal of this task is to insert controls and OLE objects into the library of the **Graphics Editor**. The library controls can be used later as templates for the worldmap editing.

- Select or create a subdirectory of the library in the library structure area. The library objects in this subdirectory are shown in the library objects details area.
- In the toolbar of the details area of the library click the icon  to insert an ActiveX control to the selected library folder.
- Alternatively, in the toolbar of the details area of the library click the icon  to insert an OLE object to the selected library folder.

The **Create new template** window to define the name of the template opens.

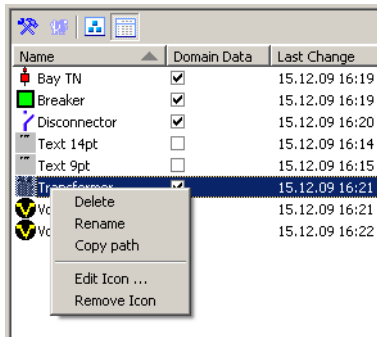


[immg128-140409-01.tif, 1, en_US]

Figure 7-4 Create New Template - Name of Template

- Define an unambiguous name for the new template within the library subdirectory and click **Next >**.
- Later additional configuration is required for that new template in this case. The **Insert a Control** window for further configuration issues opens automatically for an ActiveX control.
Alternatively, the **Insert Object** window for further configuration issues opens automatically for an OLE object.
For additional information on Controls, refer to [7.1.1 Inserting a Control as Smart Object](#).
For additional information on OLE objects, refer to Working with Objects.
- The library object gets created.

- If you want to assign a user-defined icon for the new library object, select **Edit Icon...** in the context menu of the new library object.



[rgimmg21-080210-01.tif, 1, en_US]

Figure 7-5 Library Element Context Menu

The **OPEN** menu to browse for a new icon appears.

- Browse for the icon, select the desired icon and confirm with **Open**.

Alternative Operation

- Select or create a subdirectory of the library in the library structure area. The library objects in this subdirectory are shown in the library objects details area.
- Select the graphical object that you want to add to the selected folder in the active worldmap.
- Hold down the mouse button and drag the selected object into the library details area. When you release the mouse button, the **Create new template** window to define the name of the template is opened.

7.1.7 Using and Configuring the Spectrum Power 5 Alarm Control

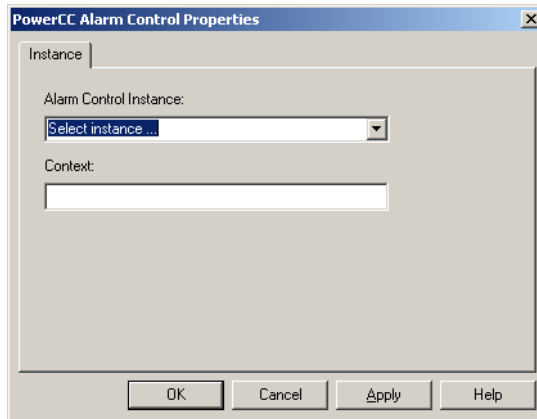
A **Spectrum Power 5 Alarm Control** is used to display any summary in a worldmap. The goal of this task is to insert a **Spectrum Power 5 Alarm Control** in a worldmap and configure it to display a summary.

- Insert the **Power 5 Alarm Control** from the **Object Palette** into a worldmap.

For additional information, refer to [7.1.1 Inserting a Control as Smart Object](#) or [7.1.2 Inserting a Control from the Controls Tab](#).

Defining the Summary

- After the **Spectrum Power 5 Alarm Control** is inserted into the worldmap, double-click the control to open the **Power 5 Alarm Control Properties** dialog.



[jimmrg141-140409-01.tif, 1, en_US]

Figure 7-6 Alarm Control Properties

- Select the desired summary within the list box **Alarm Control Instance**.
- Click **OK** or **Apply**.
The configuration is stored in the control.



NOTE

The entry field **Context** is reserved for future enhancements and not considered in this release.

7.1.8 Using and Configuring the Spectrum Power 5 Display Button

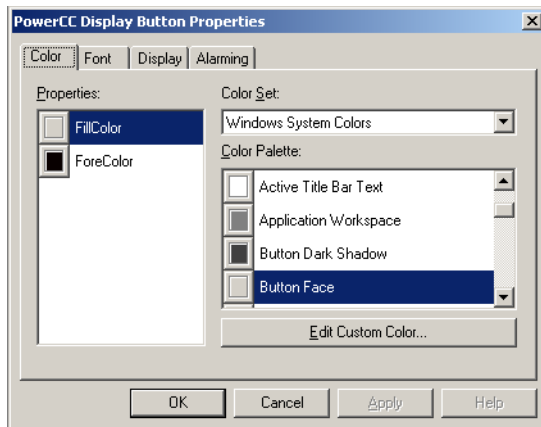
A **Spectrum Power 5 Display Button Control** is inserted into a worldmap to allow opening another worldmap or worldmap view at runtime. Alarming conditions displayed within the control at runtime can be configured. The goal of this task is to insert a **Spectrum Power 5 Display Button Control** in a worldmap and configure it to allow opening another worldmap view and to display alarming conditions.

- Insert the control, **Power 5 Display Button** from the **Object Palette** into a worldmap.

For additional information, refer to [7.1.1 Inserting a Control as Smart Object](#) or [7.1.2 Inserting a Control from the Controls Tab](#).

Defining the Appearance and Display Callup Target

- After the **Spectrum Power 5 Display Button Control** is inserted into the worldmap, double-click the control to open the **Power 5 Display Button Properties** dialog.



[immg142-140409-01.tif, 1, en_US]

Figure 7-7 Display Button Properties

- Select the desired static button colors within the **Color** tab.
- Select the desired font of the button caption within the **Font** tab.
- Define the desired button caption text, display callup target or user task within the **Display** tab.
- Define the desired alarming behavior within the **Alarming** tab.
- Click **OK** or **Apply**.

The configuration is stored in the control.

7.1.9 Using and Configuring the Spectrum Power 5 Tabular Control

7.1.9.1 Overview

With the **Spectrum Power 5 Tabular Control**, integrate a tabular display into a worldmap. In Runtime, the configured tabular is displayed. There are multiple options to configure the Look and Feel of the tabular. The goal of this overall task is to insert a **Spectrum Power 5 Tabular Control** in a worldmap and configure one or more time series data.

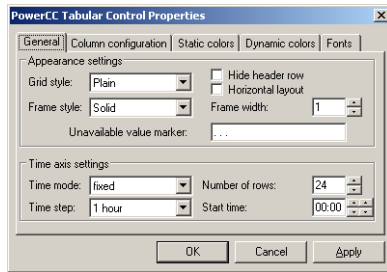
- Insert the control, **Power 5 Tabular Control** from the **Object Palette** into a worldmap.

For additional information, refer to [7.1.1 Inserting a Control as Smart Object](#) or [7.1.2 Inserting a Control from the Controls Tab](#).

7.1.9.2 Defining the Appearance of the Grid

The goal of this task is to adapt the general grid appearance of the **Spectrum Power 5 Tabular Control** together with the desired time range and the auto-scrolling behavior.

- Double-click the control to open the **Power 5 Tabular Control Properties** dialog.
- Click the **General** tab.



[rgimm002-190210-01.tif, 1, en_US]

Figure 7-8 Tabular Control Properties

- Configure the **Appearance settings** of the Grid as desired.
- Configure the **Time axis settings** of the Grid as desired.



NOTE

The layout of the **Time axis settings** area within the **Power 5 Tabular Control Properties** dialog depends on the selected **Time mode**.

- Click the **Column configuration** tab.
- Click **Add column** to add a new column to the grid.
- Select an already configured column and click **Edit column** to modify the configuration for the column of the grid.
- Select an already configured column and click **Delete column** to delete the column from the grid. The column entry in the overview area disappears.
- Select an already configured column and click the arrow up and down buttons to change the column order of the grid. The column is increased or decreased in the overview list.
- Click the **Static colors** tab.
- Configure the various settings for static coloring of the grid, rows, columns, and cells.
- Click the **Dynamic colors** tab.
- Define the cell text coloring for some measurement quality codes if desired.
- Define the cell background colors for maximum and minimum value highlighting.
- Define the text colors for limit highlighting.
- Click the **Fonts** tab.
- Define the desired font and effects for the column headers and rows.
- Click **OK** or **Apply**. The configuration is stored in the control.

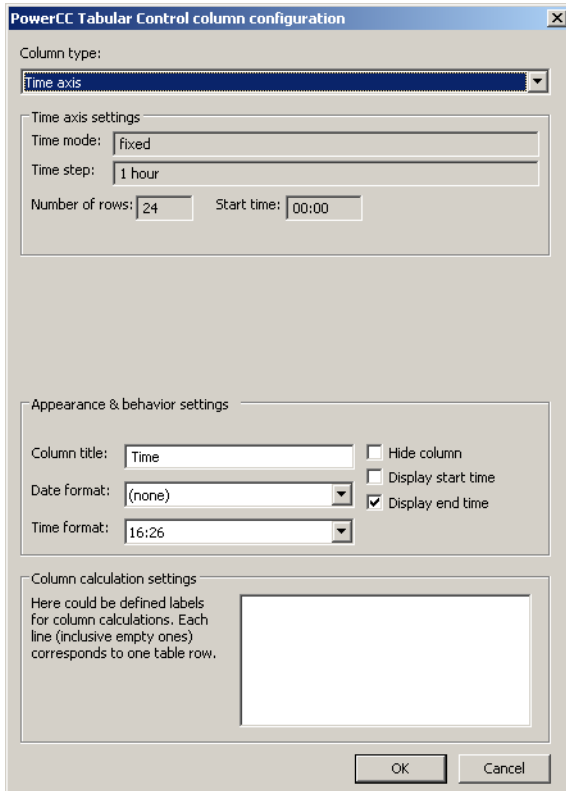
7.1.9.3 Adding or Modifying Time Axis Column Types

The goal of this task is to add new or modify existing columns (Type: Time axis) within the **Spectrum Power 5 Tabular Control**.

- Click **Add column** within the **Column configuration** tab of the **Power 5 Tabular Control Properties** dialog to add a new column to the grid.
- Alternately, select a column entry of **Type Time** in the overview area of the **Column configuration** tab of the **Power 5 Tabular Control Properties** dialog and click **Edit column** to modify the configuration for the selected column entry.

The **Power 5 Tabular Control column configuration** dialog opens.

- Choose **Time axis** from the list box of **Column type**.



[rgimmg23-080210-01.tif, 1, en_US]

Figure 7-9 Column Configuration – Time Axis

- Define the various settings within this dialog.
- Click **OK**.

After the **Power 5 Tabular Control column configuration** dialog is closed, the corresponding configured grid column is displayed in the overview area of the **Column configuration** tab of the **Power 5 Tabular Control Properties** dialog.



NOTE

The layout of the **Time axis settings** area within the **Power 5 Tabular Control column configuration** dialog depends on the selected **Time mode** configured within the **General** tab.

7.1.9.4 Adding or Modifying Time Series Data Column Types

The goal of this task is to add new or modify existing columns (Type: Time series data) within the **Spectrum Power 5 Tabular Control**.

- Click **Add column** within the **Column configuration** tab of the **Power 5 Tabular Control Properties** dialog to add a new column to the grid.
- Alternatively, select a column entry of **Type TS-API** in the overview area of the **Column configuration** tab of the **Power 5 Tabular Control Properties** dialog and click **Edit column** to modify the configuration for the selected column entry.

The **Power 5 Tabular Control column configuration** dialog opens.

- Choose **Time series data** from the list box of **Column type**.

[rgimm003-190210-01.tif, 1, en_US]

Figure 7-10 Column Configuration – Time Series Data

- Click **Signal settings**.

The **Signal Settings** dialog opens.

[rgimmg25-080210-01.tif, 1, en_US]

Figure 7-11 Signal Settings Dialog

- Define the various settings within this dialog.
- Click **OK**.

After the **Signal Settings** dialog is closed, the corresponding configured measurement is displayed in various fields of the former opened **Column configuration** dialog.



NOTE

The layout of the right section of the **Signal Settings** dialog depends on the selected data **category**.

- Define the various additional settings within this **Column configuration** dialog.
- Click **OK**

After the **Power 5 Tabular Control column configuration** dialog is closed, the corresponding configured grid column is displayed in the overview area of the **Column configuration** tab of the **Power 5 Tabular Control Properties** dialog.

7.1.9.5 Adding or Modifying Local Calculation Column Types

The goal of this task is to add new or modify existing columns (Type: Local calculation) within the **Spectrum Power 5 Tabular Control**.

- Click **Add column** within the **Column configuration** tab of the **Power 5 Tabular Control Properties** dialog to add a new column to the grid.
- Alternatively, select a column entry of **Type Calc.** in the overview area of the **Column configuration** tab of the **Power 5 Tabular Control Properties** dialog and click **Edit column** to modify the configuration for the selected column entry.

The **Power 5 Tabular Control column configuration** dialog opens.

- Choose **Local calculation** from the list box of **Column type**.

[rgimmg26-080210-01.tif, 1, en_US]

Figure 7-12 Column Configuration - Local Calculation

- Define the various settings within this dialog.
- Click **OK**.

After the **Power 5 Tabular Control column configuration** dialog is closed, the corresponding configured grid column is displayed in the overview area of the **Column configuration** tab of the **Power 5 Tabular Control Properties** dialog.

7.1.9.6 Adjust Column Width

The goal of this task is to adapt the width of columns within the **Spectrum Power 5 Tabular Control**.

- CTRL + Double-click the control to allow changing the width of certain columns.
- Click the frame in the header row and change the column width by dragging to the desired position. For **vertical time axis** layout, it is possible to set column width individually for each column. For **horizontal time axis** layout, all columns except the header column have the same size. Change of any column size affects all columns in the tabular.
- Click anywhere outside the **Spectrum Power 5 Tabular Control** to terminate the column width adjustment mode. The relative width of the columns is maintained when the **Spectrum Power 5 Tabular Control** is resized due to bounding rectangle change.



NOTE

For **vertical time axis** layout, when overall height of the **Spectrum Power 5 Tabular Control** is not enough to accommodate all rows with height based on configured font, they are distributed regularly to fill the available area. Otherwise all rows except header row have the same height calculated to accommodate 1 line of text while using the specified font. Any superfluous space is added to the header row.

For **horizontal time axis** layout, all rows are distributed regularly to fill the available area.

7.1.10 Using and Configuring the Siemens HMI Symbol Library

The **Siemens HMI Symbol Library Control** contains a comprehensive collection of prepared symbols. These symbols are used for the display of systems and system components in worldmaps.

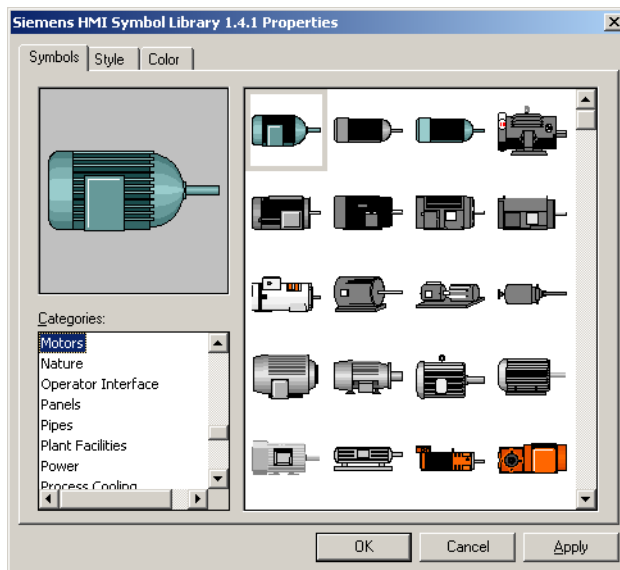
The goal of this task is to insert a particular symbol of the **Siemens HMI Symbol Library** in a worldmap and configure it as desired.

- Insert the control, **Siemens HMI Symbol Library** from the **Object Palette** into a worldmap.

For additional information, refer to [7.1.1 Inserting a Control as Smart Object](#) or [7.1.2 Inserting a Control from the Controls Tab](#).

Defining the Symbol and Appearance

- After the **Siemens HMI Symbol Library Control** is inserted into the worldmap, double-click the control to open the **Siemens HMI Symbol Library Properties** dialog.



[immrg146-140409-01.tif, 1, en_US]

Figure 7-13 Siemens HMI Symbol Library Properties

- Select the desired symbol within the **Symbols** tab.
- Define the symbol style and alignment for the selected symbol within the **Style** tab.
- Define the desired coloring for the selected symbol within the **Color** tab.
- Click **OK** or **Apply**.

The configuration is stored in the control.

7.1.11 Using and Configuring the Spectrum Power 5 Trend Control

7.1.11.1 Overview

With the **Spectrum Power 5 Trend Control**, integrate a trend display into a worldmap. In Runtime, the configured curves are displayed. There are multiple options to configure the look and feel of the corresponding trend and curves.

The goal of this overall task is to insert a **Spectrum Power 5 Trend Control** in a worldmap and configure one or more curves to be trended.

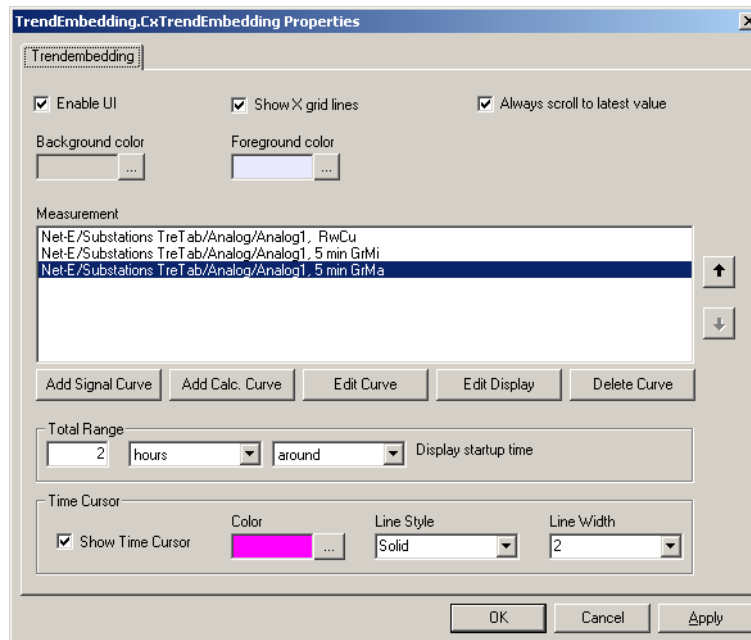
- Insert the control, **TrendEmbedding.CxTrendEmbedding** from the **Object Palette** into a worldmap.

For additional information, refer to [7.1.1 Inserting a Control as Smart Object](#) or [7.1.2 Inserting a Control from the Controls Tab](#).

7.1.11.2 Defining the Appearance

The goal of this task is to adapt the general appearance of the **Spectrum Power 5 Trend Control** together with the desired total time range visible and the auto-scrolling behavior.

- Double-click the control to open the **TrendEmbedding.CxTrendEmbedding Properties** dialog.



[image013-280510-01.tif, 1, en_US]

Figure 7-14 TrendEmbedding Control Properties

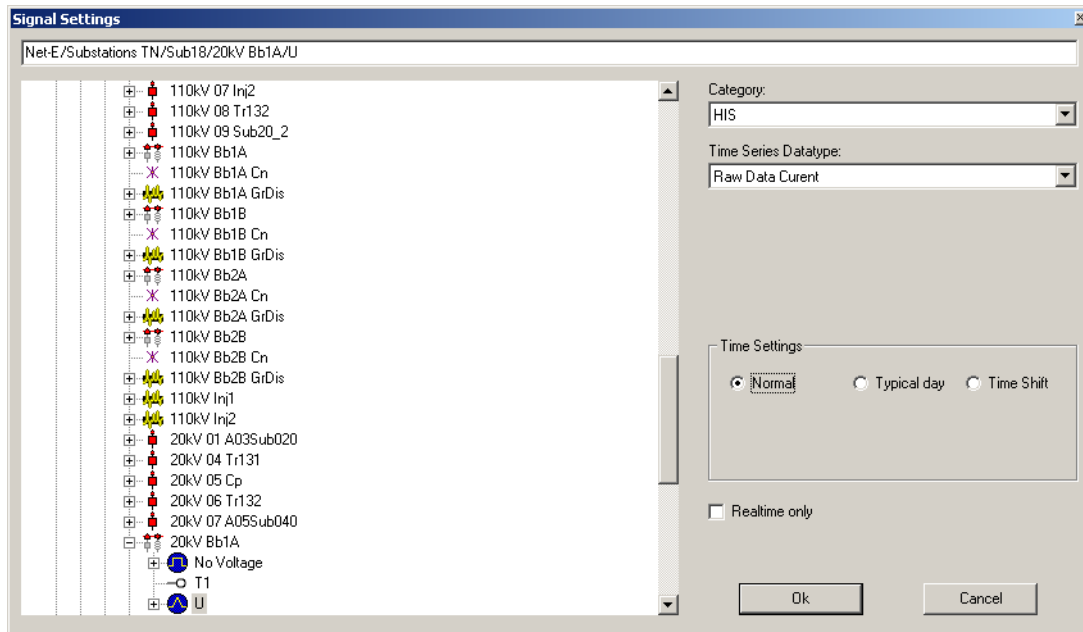
- Select the desired view (**Enable UI**, **Show X grid lines**, **Background color**, and **Foreground color**), scrolling behavior (**Always scroll to latest value**), and total time range (**Total Range**) to be presented within the control at runtime.
- Select the presentation (**Color**, **Line Style** and **Line Width**) of the time cursor, if desired.
- Click **Add Signal Curve** to add a new curve to the chart area.
- Click **Add Calc. Curve** to add a new calculated curve to the chart area and to configure the calculation using already existing curves.
- Select an already configured measurement to be trended and click **Edit Curve** to modify the configuration for the curve.
- Select an already configured measurement to be trended and click **Delete Curve** to delete the curve from the chart area.
The measurement entry in the **Measurement** area disappears.
- Select an already configured measurement to be trended and click **Edit Display** to configure the graphical layout of the curve and the chart area.
- Select an already configured measurement and click the arrow up and down buttons to change the z-order of the curves for runtime.
The curve is increased or decreased in the measurement list.
- Click **OK** or **Apply**.
The configuration is stored in the control.

7.1.11.3 Adding or Modifying Curves

The goal of this task is to add new or modify existing measurements to be trended within the **Spectrum Power 5 Trend Control**. In addition, parameters like signal source, data type, typical day, and time shift is defined.

- Click **Add Signal Curve** within the **TrendEmbedding.CxTrendEmbedding Properties** dialog to add a new measurement to be trended.
- Alternatively, select a measurement entry in the measurement area of the **TrendEmbedding.CxTrendEmbedding Properties** dialog and click **Edit Curve** to modify the signal settings for the selected measurement entry.

The **Signal Settings** dialog opens.



[rgimmg28-080210-01.tif, 1, en_US]

Figure 7-15 Signal Settings Dialog

- Define the various settings within this dialog.
- Click **OK**.
After the **Signal Settings** dialog is closed, the corresponding configured measurement is displayed in the **Measurement** area of the **TrendEmbedding.CxTrendEmbedding Properties** dialog.



NOTE

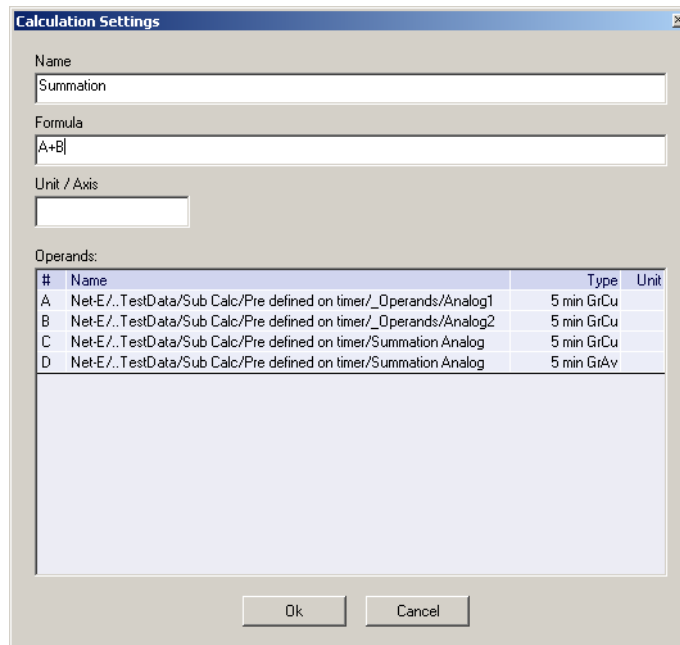
The layout of the right section of the **Signal Settings** dialog depends on the selected data category and time settings.

7.1.11.4 Adding or Modifying Calculated Curves

The goal of this task is to add new or modify existing calculated curves within the **Spectrum Power 5 Trend Control**.

- Click **Add Calc. Curve** within the **TrendEmbedding.CxTrendEmbedding Properties** dialog to add a new calculated curve.
- Alternatively, select a calculated curve entry in the measurement area of the **TrendEmbedding.CxTrendEmbedding Properties** dialog and click **Edit Curve** to modify the settings for the selected calculated curve.

The **Calculation Settings** dialog opens.



[immg014-280510-01.tif, 1, en_US]

Figure 7-16 Calculation Settings Dialog

- Define the various settings within this dialog.
- Click **OK**.

After the **Calculation Settings** dialog is closed, the corresponding configured calculated curve is displayed in the **Measurement** area of the **TrendEmbedding.CxTrendEmbedding Properties** dialog.



NOTE

Local calculations of curves can only be done if the curves used as operands do have the same cycles (same cycle unit and same cycle multiplier).

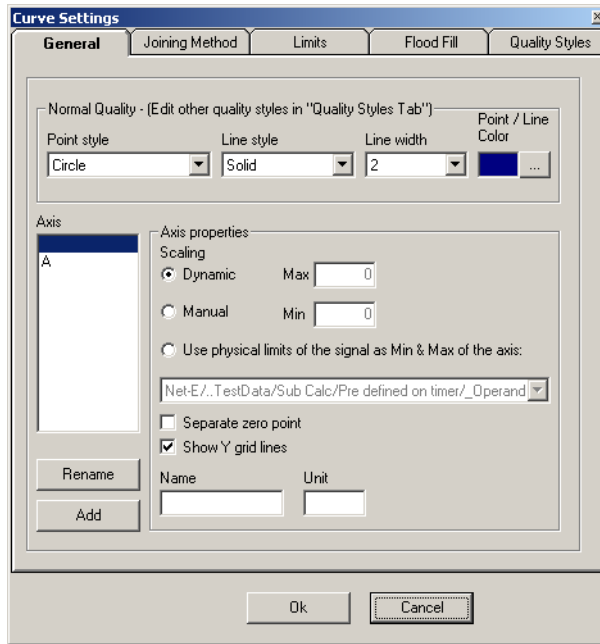
Local calculations using curves as operands with raw data is not possible at all.

7.1.11.5 Editing the Curves Presentation Properties

The goal of this task is to define the layout for configured curves of the **Spectrum Power 5 Trend Control**.

- Select a measurement entry in the measurement area of the **TrendEmbedding.CxTrendEmbedding Properties** dialog.
- Alternatively, click **Edit Display** to define the curve presentation properties for the selected measurement.

The **Curve Settings** dialog opens.



[image015-280510-01.tif, 1, en_US]

Figure 7-17 Curve Setting Dialog

- Define the various curve settings, grouped by tabs, within this dialog.
- Click **OK** to confirm the entries.



NOTE

You can approve and double-check your settings and the presentation at runtime if you use the **Curves and Trend UI**. This is carried out within the Spectrum Power 5 runtime environment already before activating the changes made in the worldmap.

The **Signal Settings**, **Calculation Settings**, and the **Curve Settings** dialog are the same for the **Curves and Trend UI** and the **Spectrum Power 5 Trend Control**.

For additional information, refer to Reference Guide User Interface Operator Support.

7.1.12 Using and Configuring the WinCC Digital/Analog Clock Control

A **WinCC Digital/Analog Clock Control** is inserted into a worldmap to display the current system time of the operating system at runtime. The time is displayed as analog or digital. In addition, the digital display contains the current date.

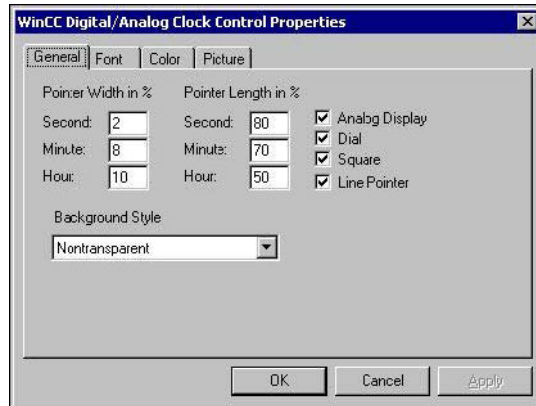
The goal of this task is to insert a **WinCC Digital/Analog Clock Control** in a worldmap and configure it as desired.

- Insert the control, **WinCC Digital/Analog Control** from the **Object Palette** into a worldmap.

For additional information, refer to [7.1.1 Inserting a Control as Smart Object](#) or [7.1.2 Inserting a Control from the Controls Tab](#).

Defining the Appearance

- After the **WinCC Digital/Analog Clock Control** is inserted into a worldmap, double-click the control to open the **WinCC Digital/Analog Control Properties** dialog.



[immrg048-111208-01.tif, 1, en_US]

Figure 7-18 WinCC Digital/Analog Clock Control Properties - General Tab

- Select the desired clock type, the general appearing and geometry of the pointers within the **General** tab.
- Define the font of a digital clock display within the **Font** tab. These settings do not affect an analog clock display.
- Define the desired coloring of the clock within the **Color** tab.
- Assign a background picture to the clock within the **Picture** tab.
- Click **OK** or **Apply**.
The configuration is stored in the control.

7.1.13 Using and Configuring the WinCC Gauge Control

The **WinCC Gauge Control** is used to display the monitored measured values in the form of an analog measurement clock. Warning and danger areas as well as the extreme values of the pointer movement is marked in colors.

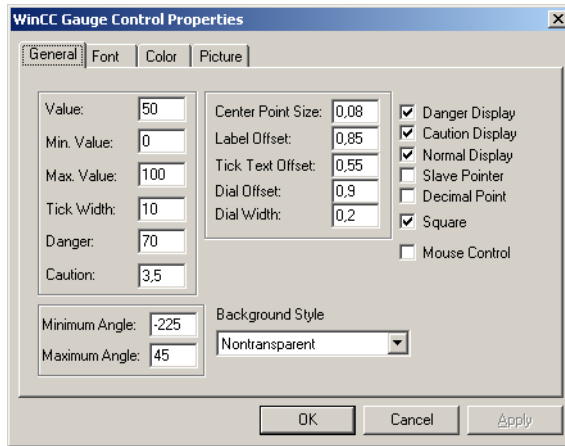
The goal of this task is to insert a **WinCC Gauge Control** in a worldmap and configure it as desired.

- Insert the control, **WinCC Gauge Control** from the **Object Palette** into a worldmap.

For additional information, refer to [7.1.1 Inserting a Control as Smart Object](#) or [7.1.2 Inserting a Control from the Controls Tab](#).

Defining the Appearance

- After the **WinCC Gauge Control** is inserted into a worldmap, double-click the control to open the **WinCC Gauge Control Properties** dialog.



[immrg155-140409-01.tif, 1, en_US]

Figure 7-19 WinCC Gauge Control Properties - General Tab

- Select the general appearing and measuring range scaling of the gauge control within the **General** tab.
- Define the font for the different labeling elements of the gauge within the **Font** tab.
- Define the desired coloring of the gauge within the **Color** tab.
- Assign a background and frame picture to the gauge within the **Picture** tab.
- Click **OK** or **Apply**.

The configuration is stored in the control.

7.1.14 Using and Configuring the WinCC Push Button Control

The **WinCC Push Button Control** is used to configure a button, which is connected to the execution of a command. In runtime, the button adopts the states **Pressed** and **Not pressed**. Both statuses are assigned with a different image, which shows the current state of the button.

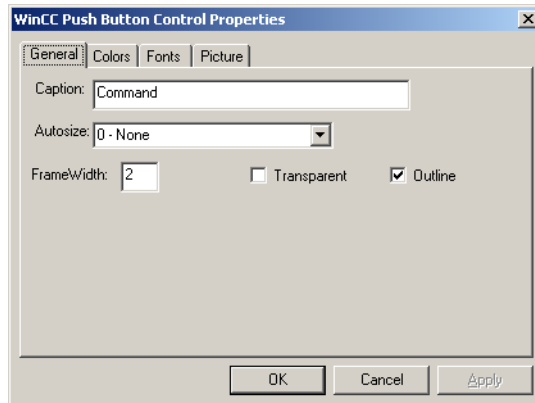
The goal of this task is to insert a **WinCC Push Button Control** in a worldmap and configure it as desired.

- Insert the control, **WinCC Push Button Control** from the **Object Palette** into a worldmap.

For additional information, refer to [7.1.1 Inserting a Control as Smart Object](#) or [7.1.2 Inserting a Control from the Controls Tab](#).

Defining the Appearance

- After the **WinCC Push Button Control** is inserted into a worldmap, double-click the inserted control to open the **WinCC Push Button Control Properties** dialog.



[immrg160-140409-01.tif, 1, en_US]

Figure 7-20 WinCC Push Button Control Properties – General Tab

- Select the general appearing of the **Push Button Control** within the **General** tab.
- Define the desired coloring of the push button within the **Color** tab.
- Define the font for the push button label within the **Font** tab.
- Assign pictures for the statuses pressed and not pressed to the push button within the **Picture** tab.
- Click **OK** or **Apply**.
The configuration is stored in the control.

7.1.15 Using and Configuring the WinCC Slider Control

The **WinCC Slider Control** is used to display the monitored measured values in the form of a slider control. The current value is displayed under the slider and the controlled measurement area is displayed as scale labeling.

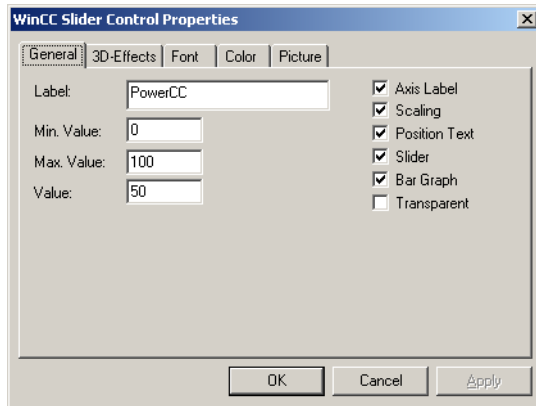
The goal of this task is to insert a **WinCC Slider Control** in a worldmap and configure it as desired.

- Insert the control, **WinCC Slider Control** from the **Object Palette** into a worldmap.

For additional information, refer to [7.1.1 Inserting a Control as Smart Object](#) or [7.1.2 Inserting a Control from the Controls Tab](#).

Defining the Appearance

- After the **WinCC Slider Control** is inserted into a worldmap, double-click the inserted control to open the **WinCC Slider Control Properties** dialog.



[immrg164-140409-01.tif, 1, en_US]

Figure 7-21 WinCC Slider Control Properties – General Tab

- Select the general appearing of the **WinCC Slider Control** within the **General** tab.
- Define the style and width for the 3D display of borders and scales within the **3D-Effects** tab.
- Define the font for the different labeling elements of the slider within the **Font** tab.
- Define the desired coloring of the slider within the **Color** tab.
- Assign pictures for the background and the slider within the **Picture** tab.
- Click **OK** or **Apply**.
The configuration is stored in the control.

7.2 User Interface Description

7.2.1 Introduction

ActiveX controls offer the option of integrating elements for controlling and monitoring into a worldmap. All ActiveX controls that are registered in the operating system are available for use in Spectrum Power 5. It also applies to controls from other manufacturers. However, in this chapter only the use of controls provided by WinCC and Spectrum Power 5 is explained. The controls are changed as required and embedded dynamically in the worldmap. The size and properties which a control accepts in runtime are defined in the Graphics Editor.

7.2.2 Description of the WinCC Controls and Spectrum Power 5 Controls

ActiveX controls serve for the monitoring and visualization of measured values and system parameters. By appropriate dynamics they are employed as control elements of the process control. The WinCC Controls and the Spectrum Power 5 Controls are installed during system installation.

Overview

Table 7-1 Control Description

Control	Abbreviation	Function
Power 5 Alarm Control	Spectrum Power 5 Alarm Control	Used to display summaries in a worldmap.
Power 5 Display Button	Spectrum Power 5 Display Button	A worldmap contains buttons for opening another view of a worldmap. Spectrum Power 5 Display Button determines the view that is opened. It also shows additional information like display title and toolbar and whether the view is opened in a new window. Parameters is set for the button to signal alarming conditions.
Power 5 Tabular Control	Spectrum Power 5 Tabular Control	The Tabular Control is used to display a tabular form of one or more time series data points in a worldmap.
Siemens HMI Symbol Library	Symbol Library	The Symbol Library contains a comprehensive collection of ready symbols for the display of systems and system components in worldmaps.
TrendEmbedding. CxTrendEmbedding	Curves and Trend Control	Used to display time series data of one or more data sources in a worldmap.
WinCC Digital/Analog Clock Control	Clock Control	The Clock Control is used to integrate a time display into a worldmap.
WinCC Gauge Control	Gauge Control	The Gauge Control is used to display the monitored measured values in the form of an analog measurement clock.
WinCC Push Button Control	Push Button	The Push Button is used to configure a button, which is connected to the execution of a specific command.
WinCC Slider Control	Slider control	The Slider Control is used to display the monitored measured values in the form of a slider control.

7.2.3 Using Control Objects

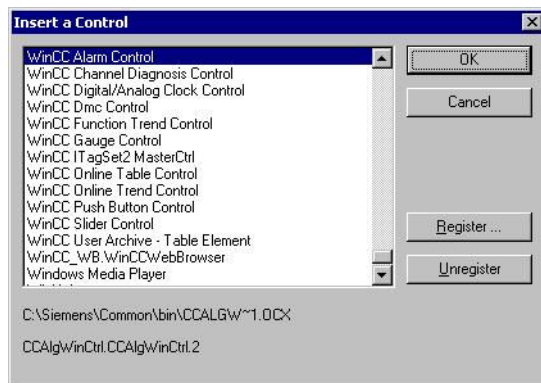
An ActiveX control is inserted from the **Object Palette** into a worldmap. A control is entered as **Smart Object > Control** from the **Standard** tab or from the **Controls** tab.

At the end of the insertion process, for some controls a configuration dialog is automatically opened, which the properties of the control are adapted to. This dialog is opened later by double-clicking the control. Detailed description of the configuration dialog is found in the descriptions of the related controls.

When a control is inserted, the object name assigned as standard is the description of the control with a consecutive number. This name is changed with the **Object Name** attribute in the **Graphics properties** window.

Smart Object >Control

If the **Smart Object > Control** is inserted, the selection of the desired control is done during the insertion process from the dialog **Insert a Control**. See all ActiveX controls that are registered in the operating system.



[immg001-101208-01.tif, 1, en_US]

Figure 7-22 Insert a Control Dialog

Register and Unregister

ActiveX controls must be configured in the operating system to be available for applications. Normally the registration is a component of the installation procedure. For troubleshooting or upgrading of individual controls, a registration or unregistration can be done manually. For additional information, refer to Changing the Registration of a Control.



NOTE

The use of ActiveX controls from third-party providers can lead to errors as well as drops in performance or system blocks. You are responsible for problems caused by the deployment of external ActiveX controls. It is recommended to check for safe functioning before use.



NOTE

WinCC Controls are native WinCC functionality and can only be used in a limited way within the Spectrum Power 5 environment. Linked WinCC and PowerCC Tags can only be read during Runtime. Writing data or setting values is not possible within the Spectrum Power 5 environment.

Controls from Tab Controls

If a control is inserted from the tab **Controls**, the selection of the selected control is done at the beginning of the insertion process. The tab **Controls** shows a default selection of WinCC and Spectrum Power 5 Controls, which is inserted directly into a worldmap. This control selection is configurable if necessary.



[rgimm004-190210-02.tif, 1, en_US]

Figure 7-23 Object Palette - Controls Tab

7.2.4 Control Manipulations

For control objects, also the principles regarding the rectangle surrounding the object applies.
For additional information, refer to Working with Objects.

Resize

Controls within worldmaps are resizable. The size of a control is defined by the geometry of the rectangle surrounding the object. The rectangle surrounding the control is symbolized on selecting a control by showing the handles.

Position

Controls within worldmaps are repositionable. The coordinates of the control origin define the position of the control. The left upper corner of the rectangle surrounding the control is called the object origin.

Properties

The properties of a control are adopted according to the requirements by:

- Changing the attributes in the **Graphics properties** window.
- Changing the settings in the various configuration dialogs.

7.2.5 Control Selection of the Object Palette

The **Controls** tab in the **Object** palette of the **Graphics Editor** contains a selection of controls. Use the **Select OCX Controls** dialog to change this selection as required. Individual controls are removeable from the selection. However, the list can be supplemented by any ActiveX controls that are registered in the operating system.

For additional information, refer to Customizing the Working Environment.



NOTE

The use of ActiveX controls from third-party providers can lead to errors as well as drops in performance or system blocks. You are responsible for problems caused by the deployment of external ActiveX controls. It is recommended to thorough test before implementing it.

7.2.6 Spectrum Power 5 Alarm Control

With the **Spectrum Power 5 Alarm Control**, summaries are configured for use during runtime.

Time Stamp	Path 1	Path 2	Path 3	Path 4	Path 5	Status	Value	Ind	Lii	Un	User	Console	Comment
02/13/09 15:20:20.000	Sub15	110/05 Sub18_2	Cb	Status		Ack	Close				SPCAdmin	DEB215TC	
02/13/09 15:20:20.000	Sub19	20kV 02 A07Sub075	DisBb1 truck	Status		Ack	Close				SPCAdmin	DEB215TC	
02/13/09 15:20:20.000	A05Sub052	A05Sub053	GrDis	Status		Ack	Open				SPCAdmin	DEB215TC	
02/13/09 15:20:20.000	A05Sub051	UEBERG	Circuit Breaker	Status		Ack	Close				SPCAdmin	DEB215TC	
02/13/09 15:20:20.000	Sub19	110kV 05 Sub18	DisBb2	Status		Ack	Open				SPCAdmin	DEB215TC	
02/13/09 15:20:20.000	Sub20	110kV/BRCST1	Dis 2	Status		Ack	Close	Spont			SPCAdmin	DEB215TC	
02/13/09 15:20:20.000	Sub17	110/00 Inj1	DisL	Status		Ack	Close				SPCAdmin	DEB215TC	
02/13/09 15:20:20.000	A03Sub023	A03Sub024	LoDis	Status		Ack	Close				SPCAdmin	DEB215TC	
02/13/09 15:20:20.000	A04Sub033	Transformer 1	LoDis	Status		Ack	Close				SPCAdmin	DEB215TC	
02/13/09 15:20:20.000	Sub20	110kV/BRCST1	Dis2 CB	Status		Ack	Open	Spont			SPCAdmin	DEB215TC	
02/13/09 15:20:20.000	Sub17	110/00 Inj1	DisBb2	Status		Ack	Open				SPCAdmin	DEB215TC	
02/13/09 15:20:20.000	Sub15	20/02 A01Sub001	DisRBb1	Status		Ack	Open				SPCAdmin	DEB215TC	
02/13/09 15:20:20.000	Sub17	110/05 Cp	DisBb2	Status		Ack	Open				SPCAdmin	DEB215TC	
02/13/09 15:20:20.000	Sub19	110kV 07 Sub20_1	DisBb2	Status		Ack	Open				SPCAdmin	DEB215TC	
02/13/09 15:20:20.000	Sub20	110kV/BRCST1	CB 1	Status		Ack	Close	Spont			SPCAdmin	DEB215TC	
02/13/09 15:20:20.000	Sub15	110/02 Tr101	Cb	Status		Ack	Close				SPCAdmin	DEB215TC	
02/13/09 15:20:20.000	Sub15	20/01 A02Sub010	DisRBb1	Status		Ack	Close				SPCAdmin	DEB215TC	
02/13/09 15:20:20.000	A05Sub050	Transformer 2	Circuit Breaker	Status		Ack	Close				SPCAdmin	DEB215TC	
02/13/09 15:20:54.000	Sub18	110kV 06 Sub15_2	I	Marker - Note		App	Set	Marker			SPCAdmin	DEB20Q6C	
02/13/09 15:21:00.000	Counter	Counter3		Main Raw Value		App	2673						Value missec
02/13/09 15:21:00.000	Counter	Counter5		Main Raw Value		App	2673						Value missec
02/13/09 15:21:00.000	Counter	Counter4		Main Raw Value		App	2673						Value missec
02/13/09 15:21:00.000	Counter	Counter2		Main Raw Value		App	2673						Value missec
02/13/09 15:21:00.000	Counter	Counter1		Main Raw Value		App	2673						Value missec

[uiosa013-240209-01.tif, 1, en_US]

Figure 7-24 Summary Layout

The **Power 5 Alarm Control Properties** dialog is used to define the parameters to display summaries within the **Spectrum Power 5 Trend Control**.

PowerCC Alarm Control Properties

Instance

Alarm Control Instance:

Select instance...

Context:

OK Cancel Apply Help

[immg141-140409-01.tif, 1, en_US]

Figure 7-25 Alarm Control Properties

Alarm Control Instance

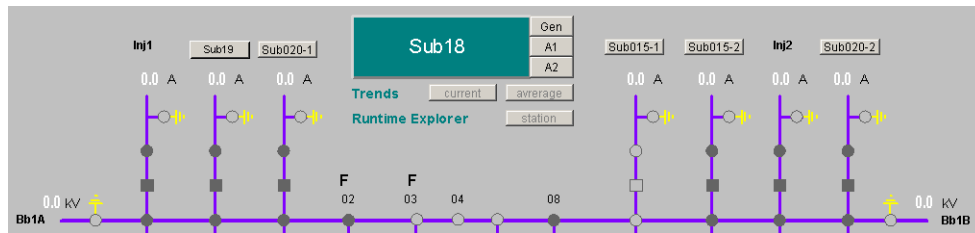
Open the list box and select the desired summary.

Context

Reserved for future usage. Currently not used and supported.

7.2.7 Spectrum Power 5 Display Button

With the **Spectrum Power 5 Display Button**, you can integrate display callup buttons and alarming condition visualizations into a worldmap. Multiple options to configure the presentation and functionality at runtime are provided.



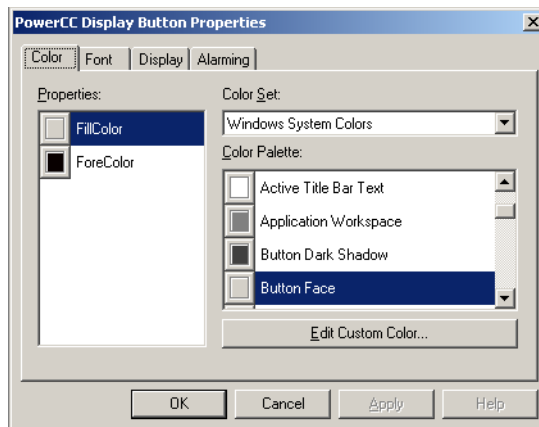
[immrg002-220709-02.tif, 1, en_US]

Figure 7-26 Example Usages of Spectrum Power 5 Display Buttons at Runtime

The **Power 5 Display Button Properties** dialog is used to define the parameters and functionality within the **Spectrum Power 5 Display Button**.

It consists of the following tabs:

- Color
- Font
- Display
- Alarming



[immrg142-140409-01.tif, 1, en_US]

Figure 7-27 Display Button Properties – Color

Properties

Select the color attribute you wish to change:

- The property **FillColor** displays the currently configured background color.
- The property **ForeColor** displays the currently configured color for the button caption.

Color Set

Select one of the color groups in the list box.

The related colors are displayed in the **Color Palette** section:

- The **Standard Colors** color group contains the 16 standard colors of the operating system
- The **Windows System Colors** color group contains the colors, which are set in the operating system currently as the display options at system start.

Color Palette

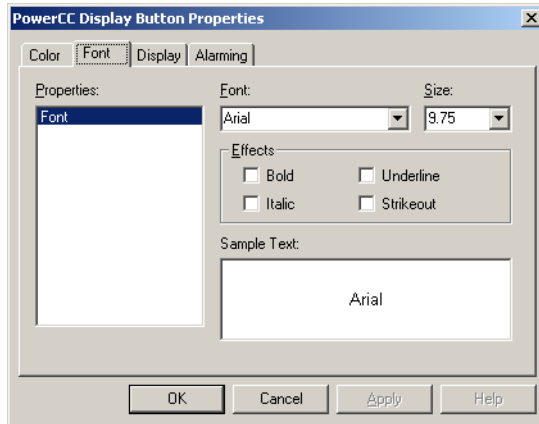
Select the desired color.

Click **Apply** to assign the new color to the selected color attribute.

Edit Custom Color

Click this button to access the **Color** dialog.

Define the desired color freely based on the graphic settings of the operating system.



[immg143-140409-01.tif, 1, en_US]

Figure 7-28 Display Button Properties – Font

Properties

The currently selected property is shown.

Font, Size

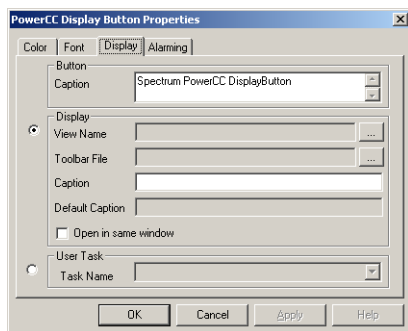
Select the desired font and size (specified in points) for the button caption from the list boxes. Use any of the fonts registered in the operating system.

Effects

Select one or more font effects.

Sample Text

The selected settings are displayed in a preview.



[rgimm005-190210-01.tif, 1, en_US]

Figure 7-29 Display Button Properties – Display

Button

Define the button caption here.

Display

The display parameters allow opening the defined worldmap view at runtime.

The settings are used and enabled if the option button on the left side of the **Display** area is checked.

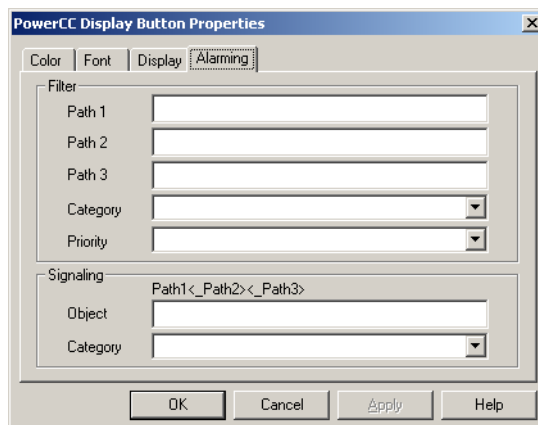
- **View Name**
Choose the desired view from the display tree view dialog after pressing the button on the right.
The view name and the default caption name within the display title bar at runtime are shown in the fields **View Name** and **Default Caption**.
- **Toolbar File**
Choose the desired toolbar from the find file dialog after pressing the button on the right. The toolbar file name used for the display toolbar at runtime is shown in the field **Toolbar File**.
- **Caption**
Define the name of the display that is shown in the display title bar at runtime if it is different from the one shown in the **Default Caption** field.
- **Default Caption**
Refer to View Name description.
- **Open in same window**
Defines whether the new worldmap view opens in a separate window or in the same window as the worldmap from which is selected at runtime.

User Task

The user task parameter allows performing a task at runtime.

The settings are used and enabled if the option button on the left side of the **User Task** area is checked.

- **Task Name**
Choose the desired user task to perform from the list box.



[immg145-140409-01.tif, 1, en_US]

Figure 7-30 Display Button Properties - Alarming

Filter

The filter parameters allow filtering of summaries (used for display callup) of the defined worldmap view at runtime.

- **Path 1, Path 2, and Path 3**
Define the desired path filter for the summary if a filter applies. It is possible to use wildcards within the path names. Examples are: * , A* , *A, A*B, *A*.
- **Category**
Define a desired category filter from the list box for the summary if a filter applies.
- **Priority**
Define a desired priority filter from the list box for the summary if a filter applies.

Signaling

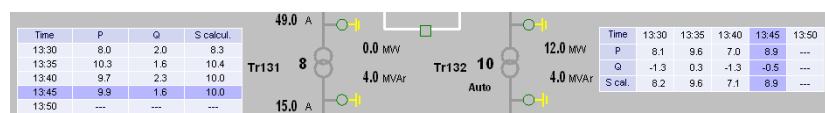
The signaling parameters allow flashing and dynamic coloring of the button if an alarm defined by the **Object** filter is active or if there are unacknowledged alarms at runtime.

- **Object**
Define the desired path filter for alarming consideration according the syntax above the entry field.
- **Category**
Define a desired category filter from the list box for alarming condition presentation.

7.2.8 Spectrum Power 5 Tabular Control

7.2.8.1 Overview

With the **Spectrum Power 5 Tabular Control**, integrate a tabular display into a worldmap. In runtime, the configured tabular is displayed. The **Spectrum Power 5 Tabular Control** displays historical data and realtime data as well as forecast data and schedules. Simple local calculation facilities allow performing basic operations over retrieved data at runtime, such as difference between planned and actual value. The content of the control is configurable in the Graphics Editor and partially modifiable at runtime by drag & drop operations. Multiple options to configure the Look and Feel of the corresponding tabular are provided.



[rgimmg36-080210-01.tif, 1, en_US]

Figure 7-31 Example for Tabular Layouts Within Runtime

7.2.8.2 Properties

The **Power 5 Tabular Control Properties** dialog is used to define the Look and Feel and the content of the grid at runtime.

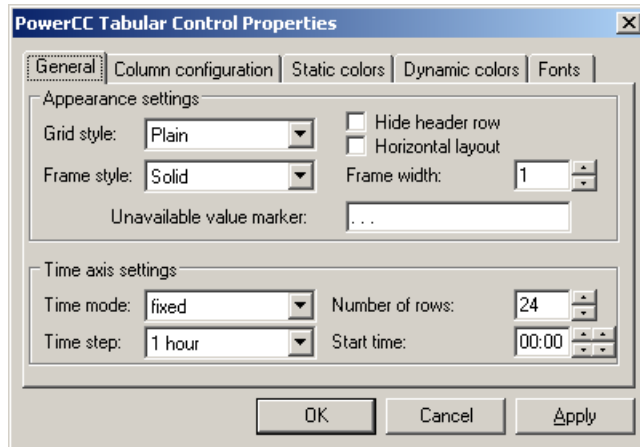
It consists of the following tabs:

- General
- Column configuration
- Static colors
- Dynamic colors
- Fonts



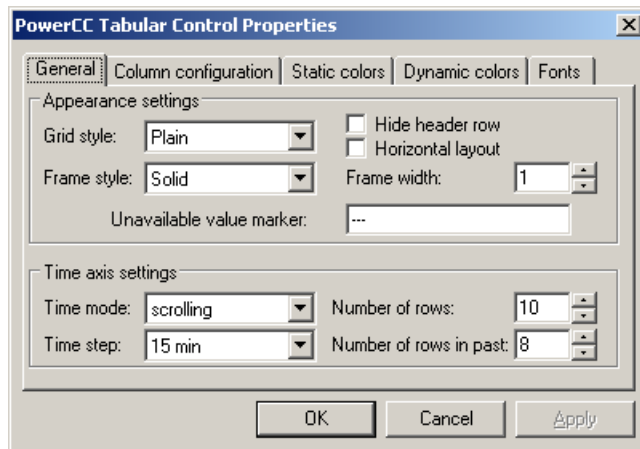
NOTE

The layout of the **General** tab of the dialog within the **Time axis settings** section depends on the selected time mode.



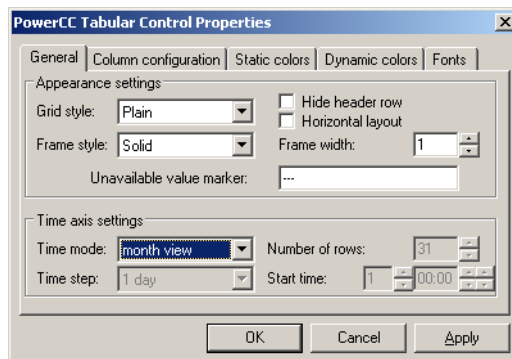
[rgimmg37-080210-01.tif, 1, en_US]

Figure 7-32 Tabular Control Properties – General, for Chosen Fixed Time Axis Mode



[rgimmg39-080210-01.tif, 1, en_US]

Figure 7-33 Tabular Control Properties – General, for Chosen Scrolling Time Axis Mode



[immge016-280510-01.tif, 1, en_US]

Figure 7-34 Tabular Control Properties – General, for Chosen Month View Time Axis Mode

Appearance Settings

Defines the general layout for the tabular.

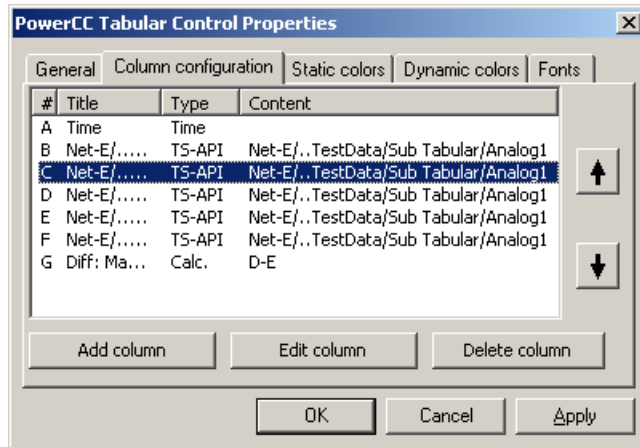
- **Grid style**
Choose the shape of the grid (excluding the header row).
If **Plain** is used, the configured **Cell background #2** within **Static colors** tab have no effect.

- **Frame style and Frame width**
Choose the style and the width of the grid lines.
- **Hide header row**
Column headers are shown or not within the tabular.
- **Horizontal layout**
Flips the layout of the tabular in the way, that the time axis of the tabular is horizontal instead of vertical.
- **Unavailable value marker**
Define the text that is shown within the grid cell if no value for the configured measurement is available at runtime.

Time Axis Settings

Defines the general layout for the tabular regarding the time range and scrolling behavior.

- **Time mode**
Choose the desired tabular scrolling behavior at runtime.
Fixed defines that the time column at runtime shows always the configured time range.
Fixed (day) is basically the same as **Fixed**. The difference is that rows that are in the future of the current time are forced to be empty (marked with the **Unavailable value marker**) and are never containing values from yesterday. This is of special importance if the grid starts with time frames later than the current time.
Month view defines that the tabular contains always 31 rows for each day of a month starting at the 1st day. Months less than 31 days will have empty entries for not available days.
Scrolling defines that the time column at runtime is updated according to the current time. The complete grid gets scrolled if the current time interval is changed at runtime.
- **Time step**
Choose the time step between the configured rows in the grid. By nature the time step has to fit to the query cycle of the signal sources configured within the columns.
- **Number of rows**
Define the number of rows that are visible for the grid. For time mode **Month view** it is fixed to 31.
- **Start time** (not for time mode: scrolling):
Define the start time for the first row within the configured complete number of rows. The layout and number of the entry fields depend on the chosen time step. For time mode **Month view** it is fixed to the 1st day of a month. For **Time step** 1 day an additional entry field to define the start day is displayed.
- **Number of rows in past** (only for time mode: scrolling): Define the number of rows pointing into the past within the configured complete number of rows. The current time interval therefore has always the same position within a scrolling grid configuration.



[rgimmg40-080210-01.tif, 1, en_US]

Figure 7-35 Tabular Control Properties – Column Configuration

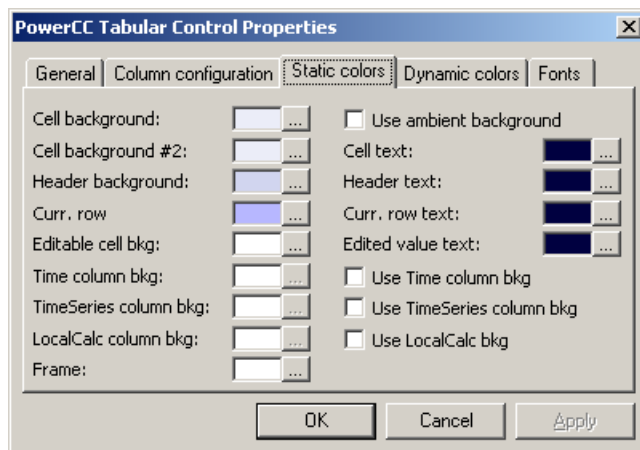
Grid Column Overview

The grid column overview area shows already configured grid columns, together with the header title, the data type and the associated measurement.

The up and down arrows on the right side change the column order of the grid.

Add Column, Edit Column, and Delete Column

- **Add column** opens the **Power 5 Tabular Control column configuration** dialog to add a new column to the grid.
- **Edit column** opens the **Power 5 Tabular Control column configuration** dialog to modify the column settings for the selected grid column.
- **Delete column** deletes the selected column from the grid.



[rgimmg41-080210-01.tif, 1, en_US]

Figure 7-36 Tabular Control Properties – Static Colors

Cell Background and Cell Background #2

Choose the background color of the grid cells from the color palettes. The cell background color #2 settings are only visible if **Plain** is not configured as grid style within the **General** tab.

Header Background and Current Row

Choose the background color of the grid header and the actual grid row at runtime from the color palettes.

Editable Cell Bkg

Choose the background color for editing cells (example for manual value update) at runtime from the color palette.

Time Column Bkg and Use Time Column Bkg

Check **Use Time column bkg** if an independent coloring for the cell background of time columns is used.

Choose the time column background color from the color palette.

Time series Column Bkg and Use TimeSeries Column Bkg

Check **Use TimeSeries column bkg** if an independent coloring for the cell background of time series columns is used.

Choose the time series column background color from the color palette.

LocalCalc Column Bkg and Use LocalCalc Bkg

Check **Use LocalCalc bkg** if an independent coloring for the cell background of calculation columns is used.

Choose the local calculation column background color from the color palette

Frame

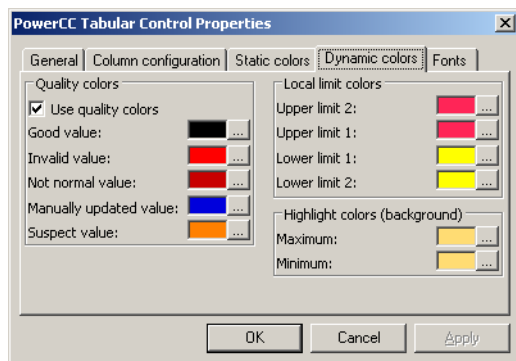
Choose the grid lines color from the color palette.

Use Ambient Background

If checked, the **Cell background** color is set to automatically consider the configured background color of the worldmap.

Cell Text, Header Text, Curr. Row Text, and Edited Value Text

Choose the static text color for the different subjects from the color palettes.



[image017-280510-01.tif, 1, en_US]

Figure 7-37 Tabular Control Properties – Dynamic Colors

Quality Colors

If checked, the text color within the cells is set according to the quality of the value. The static text color settings are overruled with the quality definitions.

Good Value, Invalid Value, Not Normal Value, Manually Updated, and Suspect Value

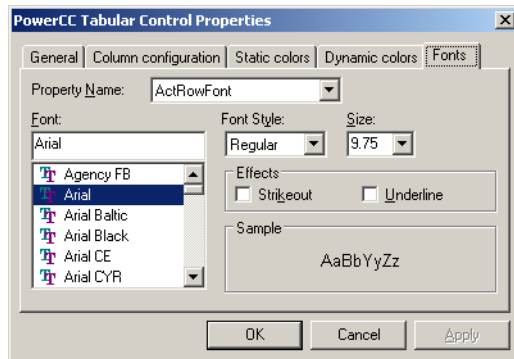
Choose the dynamic text color for the different subjects from the color palettes.

Highlight Colors (background)

Choose the dynamic background color of the maximal and minimal value that is to be highlighted in the grid from the color palettes. Whether the values are to be highlighted or not is set in the **Power 5 Tabular Control column configuration** dialog for each column separately.

Local Limit Colors

Choose the dynamic text color for the different limit values from the color palettes. Whether the values are to be highlighted or not is set in the **Power 5 Tabular Control column configuration** dialog for each column separately. The value of the limits is also set in the **Power 5 Tabular Control column configuration** dialog for each column separately.



[immrg006-220709-01.tif, 1, en_US]

Figure 7-38 Tabular Control Properties – Fonts

Property Name

Enables to define the font for different areas within the tabular:

- Use **ActRowFont** to define the font properties for the actual row at runtime.
- Use **DefaultFont** to define the font properties for the grid cells except the actual row at runtime.
- Use **HeaderFont** to define the font properties for the header row.

Font

If required, the currently selected font is shown and changed.

Use any of the fonts registered in the operating system.

Font Style, Size

Select the desired font style and size (specified in points) from the list boxes.

Effects

Select one or more font effects.

Sample

The selected settings are displayed in a preview.

7.2.8.3 Column Configuration for Time Axis Columns

The **Power 5 Tabular Control column configuration** dialog is used to define the column types, the content of the column and some additional functionality at runtime.

The layout of the dialog depends on the chosen **Column type**.

PowerCC Tabular Control column configuration

Column type:
Time axis

Time axis settings

Time mode: Fixed
Time step: 1 hour
Number of rows: 24 Start time: 00:00

Appearance & behavior settings

Column title: Time Hide column
Date format: (none) Display start time
Time format: 16:26 Display end time

Column calculation settings

Here could be defined labels for column calculations. Each line (inclusive empty ones) corresponds to one table row.

OK Cancel

[rgimmg43-080210-01.tif, 1, en_US]

Figure 7-39 Tabular Control Column Configuration – Time Axis

Time Axis Settings

Shows the configuration made in the **General** tab of the **Power 5 Tabular Control Properties** dialog.



NOTE

The layout of the **Time axis settings** area depends on the selected time mode.
For additional information, refer to Properties of the Power 5 Tabular Control.

Appearance and Behavior Settings

- **Column title**
Define the column title text to be shown within the tabular.
- **Date format and Time format**
Select the desired date and time format from the list boxes.
- **Hide column**
Check **Hide column** if the time axis column is not to be displayed within the tabular at runtime.
- **Display start time, Display end time**
Check if the rows start time or the end time or both is visible within the column axis column at runtime.

- **Column Calculation Settings**

You can define column calculations for time series data columns and local calculation columns within the **Power 5 Tabular Control column configuration** dialogs for these types of columns. The column calculations are inserted at the bottom or on the right side (in case of horizontal layout) of the grid and expand the number of configured columns for the grid. Multiple column calculations are possible for one column. Within the entry field you can define the label for the column calculations. The label is displayed within the time axis column. Each line represents 1 additional grid row for 1 column calculation. You have to seriously check the number, text and order of the labels with the configured column calculations for the time series data columns and local calculation columns.

7.2.8.4 Column Configuration for Time Series Data Columns

The **Power 5 Tabular Control column configuration** dialog is used to define the column types, the content of the column and some additional functionality at runtime.

The layout of the dialog depends on the chosen **Column type**.

PowerCC Tabular Control column configuration

Column type: **Time series data**

Time series settings

Data point: **Net-E/Substations TreTab/Analog/Analog1**

Data type: **Grid Data Current**

Query cycle: **1 minute**

Time shift: **0** **hour** **Signal settings**

Limit settings

☒ Perform limit check

Lower limit 1: **70.0** Upper limit 1: **330.0**

Lower limit 2: **30.0** Upper limit 2: **370.0**

Appearance & behavior settings

☒ Value ☐ TimeStamp ☐ Quality

Column title: **Net-E/Substations TreTab/Analog/1** ☐ Hide column

Num. of int. places: **4** Unit: **A** ☐ Drag & Drop lock

Dec. places: **1** ☐ Append unit to value

☐ Use default dec. places

Column calculation settings

☒ Highlight column min. ☒ Highlight column max.

☐ Highlight column min. till curr. row ☐ Highlight column max. till curr. row

Separator, Max, Min, Avg, Sum

Sep.	Use all values:	Sum	Avg	Min	Max
Clear	Use values till curr. row:	Sum CR	Avg CR	Min CR	Max CR

OK **Cancel**

[rgimmg44-080210-01.tif, 1, en_US]

Figure 7-40 Tabular Control Column Configuration – Time Series Data

Time Series Settings

- **Data point, Data type, Query cycle**

Shows the configuration made in the **Signal Settings** dialog for this column.

- **Time shift**

Select the desired time shift of the configured time series data from the list boxes. Positive time shifts for configured signal source categories HIS within the **Signal Settings** dialog does not display values as they are pointing into the future.

- **Signal Settings**

Opens the **Signal Settings** dialog to define a measurement for this column or modify existing signal settings.

Limit Settings

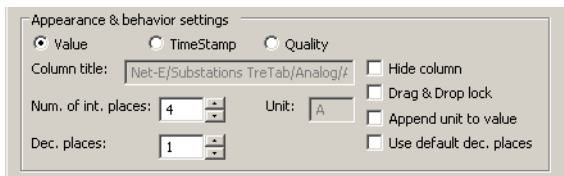
- **Perform limit check**

Check if limit violations are to highlight according the color definitions made in the **Tabular Control Properties – Dynamic colors** dialog. If checked you can edit the limits valid for this time series column in the 4 entry fields.

Appearance and Behavior Settings

- **Value, TimeStamp and Quality**

Select the desired type of information for the measurement, assigned in the **Signal Settings** dialog, that shall be presented in the time series column cells. Value presents the value for the given time cycle. TimeStamp presents the effective time stamp of the value and Quality presents the quality of the value for the given time cycle.



[imgge018-280510-01.tif, 1, en_US]

Figure 7-41 Appearance and Behavior Settings – Value

- **Column title**

Shows the text of the column title.

The text is fix if the column allows exchange of the measurement per drag and drop at runtime. In this case the path of the measurement is used.

The text is freely editable if the column does not allow exchange of the measurement per drag and drop at runtime.

- **Num. of int. places and Dec. places**

Select the desired value format from the list boxes.

- **Unit**

Shows the configured unit of the associated measurement.

- **Hide column**

Check **Hide column** if the time series data column is not to be displayed within the tabular at runtime.

- **Drag & Drop lock**

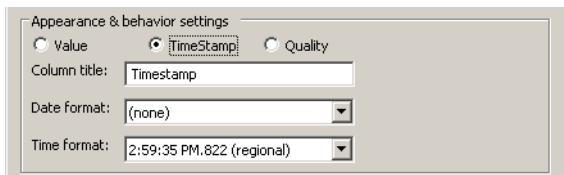
Check if the column does not allow to be a drag and drop target at runtime. If checked, the **Column title** is editable.

- **Append unit to value**

Check if the column values is included in the **Unit** within the cells at runtime.

- **Use default dec. places**

Uncheck if the decimal places within the columns is different from the configured default numbers. If unchecked the **Dec. places** are changeable.



[imgge019-280510-01.tif, 1, en_US]

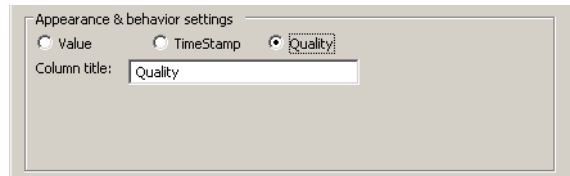
Figure 7-42 Appearance and Behavior Settings – TimeStamp

- **Column title**

Define the text of the column title.

- **Date format and Time format**

Select the desired date and time format from the list boxes.



[image020-280510-01.tif, 1, en_US]

Figure 7-43 Appearance and Behavior Settings – Quality

- **Column title**

Define the text of the column title.

Column Calculation Settings

- **Highlight column min. and Highlight column max.**

Check if the background color for the column cells including the minimum and/or the maximum value of the visible grid column are to be highlighted according to the color definitions made in the **Tabular Control Properties – Dynamic colors** dialog.

- **Highlight column min. till curr. row and Highlight column max. till curr. row**

Check if the background color for the column cells including the minimum and/or the maximum value of the visible grid column are to be highlighted according to the color definitions made in the **Tabular Control Properties – Dynamic colors** dialog. The difference to the former option is that the maximum and minimum calculation is only considering the visible grid column till the current row.

- **Buttons: Sum, Avg, Min, Max, Sum CR, Avg CR, Min CR, Max CR**

Every click on the buttons introduces a column calculation. The column calculations are inserted at the bottom or on the right side (in case of horizontal layout) of the grid and expand the number of configured columns for the grid. The sequence of the pressed buttons is visible in the output field above the buttons. The number of pressed buttons equals to the additional numbers of rows to the grid.

Sum calculates the total value of the visible column cells.

Avg calculates the average value of the visible column cells.

Min calculates the minimum value of the visible column cells.

Max calculates the maximum value of the visible column cells.

Sum CR calculates the total value of the visible column cells till the current row.

Avg CR calculates the average value of the visible column cells till the current row.

Min CR calculates the minimum value of the visible column cells till the current row.

Max CR calculates the maximum value of the visible column cells till the current row.

- **Button: Sep**

Every click on the button introduces an empty column. The empty columns are inserted at the bottom or on the right side (in case of horizontal layout) of the grid and expand the number of configured columns for the grid. The sequence of the pressed buttons is visible in the output field above the buttons. The number of pressed buttons equals to the additional numbers of rows to the grid.

- **Button: Clear**

A click on the button deletes all definitions done for the column calculations. The output field above the buttons is empty afterwards.



NOTE

The content of the additional rows introduced with the **Column Calculation Settings** buttons are separately configurable for each grid column of type time series data columns and local calculation columns.

The labels for these additional rows are configurable in the grid column of type time axis.

You have to check carefully that your definitions are consistent for the complete tabular.

7.2.8.5 Column Configuration for Local Calculations Columns

The **Power 5 Tabular Control column configuration** dialog is used to define the column types, the content of the column and some additional functionality at runtime.

The layout of the dialog depends on the chosen **Column type**.

The screenshot shows the 'PowerCC Tabular Control column configuration' dialog box. The 'Column type' dropdown is set to 'Local calculation'. The 'Local calculation settings' section has an 'Expression' field containing 'B+C'. The 'Limit settings' section includes a 'Perform limit check' checkbox (unchecked) and four input fields for 'Lower limit 1', 'Upper limit 1', 'Lower limit 2', and 'Upper limit 2', all set to '0'. The 'Appearance & behavior settings' section includes a 'Column title' field with 'Add', a 'Hide column' checkbox (unchecked), 'Num. of int. places' set to '4', 'Unit' field, and 'Dec. places' set to '0'. The 'Column calculation settings' section has four checkboxes: 'Highlight column min.' (unchecked), 'Highlight column max.' (unchecked), 'Highlight column min. till curr. row' (unchecked), and 'Highlight column max. till curr. row' (unchecked). At the bottom, there are buttons for 'Sep.', 'Clear', 'Use all values:', 'Use values till curr. row:', and a grid of calculation functions: Sum, Avg, Min, Max, Sum CR, Avg CR, Min CR, and Max CR. 'OK' and 'Cancel' buttons are at the bottom right.

[rgimmg45-080210-01.tif, 1, en_US]

Figure 7-44 Tabular Control Column Configuration – Local Calculation

Local Calculation Settings

Define the calculation within the **Expression** entry field. B and C in the screenshot are the column identifiers shown within the grid column overview area of the **Column configuration** tab of the **Power 5 Tabular Control Properties** dialog.

The following operands are supported:

- Floating point constants
- Operators: (,), +, -, *, /, sqrt()

Expressions are defined similar as in MS Excel.

For example, $(A+B)/2$ where **A** and **B** are implicit identifiers for 1st and 2nd column in the tabular.

Limit Settings

- **Perform limit check**
Check if limit violations are to highlight according the color definitions made in the **Tabular Control Properties – Dynamic colors** dialog. If checked you can edit the limits valid for this local calculation column in the 4 entry fields.

Appearance and Behavior Settings

- **Column title**
Define the text of the column title.

- **Num. of int. places and Dec. places**
Select the desired value format from the list boxes.
- **Unit**
Define the unit of the calculated result. If there is an entry in the field, the unit text is displayed together with the column values at runtime.
- **Hide column**
Check **Hide column** if the local calculation column is not displayed within the tabular at runtime.

Column Calculation Settings

- **Highlight column min. and Highlight column max.**
Check if the background color for the column cells including the minimum and/or the maximum value of the visible grid column are to be highlighted according to the color definitions made in the **Tabular Control Properties – Dynamic colors** dialog.
- **Highlight column min. till curr. row and Highlight column max. till curr. row**
Check if the background color for the column cells including the minimum and/or the maximum value of the visible grid column are to be highlighted according to the color definitions made in the **Tabular Control Properties – Dynamic colors** dialog. The difference to the former option is that the maximum and minimum calculation is only considering the visible grid column till the current row.
- **Buttons: Sum, Avg, Min, Max, Sum CR, Avg CR, Min CR, Max CR**
Every click on the buttons introduces a column calculation. The column calculations are inserted at the bottom or on the right side (in case of horizontal layout) of the grid and expand the number of configured columns for the grid. The sequence of the pressed buttons is visible in the output field above the buttons. The number of pressed buttons equals to the additional numbers of rows to the grid.
Sum calculates the total value of the visible column cells.
Avg calculates the average value of the visible column cells.
Min calculates the minimum value of the visible column cells.
Max calculates the maximum value of the visible column cells.
Sum CR calculates the total value of the visible column cells till the current row.
Avg CR calculates the average value of the visible column cells till the current row.
Min CR calculates the minimum value of the visible column cells till the current row.
Max CR calculates the maximum value of the visible column cells till the current row.
- **Button: Sep.**
Every click on the button introduces an empty column. The empty columns are inserted at the bottom or on the right side (in case of horizontal layout) of the grid and expand the number of configured columns for the grid. The sequence of the pressed buttons is visible in the output field above the buttons. The number of pressed buttons equals to the additional numbers of rows to the grid.
- **Button: Clear**
A click on the button deletes all definitions done for the column calculations. The output field above the buttons is empty afterwards.



NOTE

The content of the additional rows introduced with the **Column Calculation Settings** buttons are separately configurable for each grid column of type time series data columns and local calculation columns. The labels for these additional rows are configurable in the grid column of type time axis. You have to check carefully that your definitions are consistent for the complete tabular.

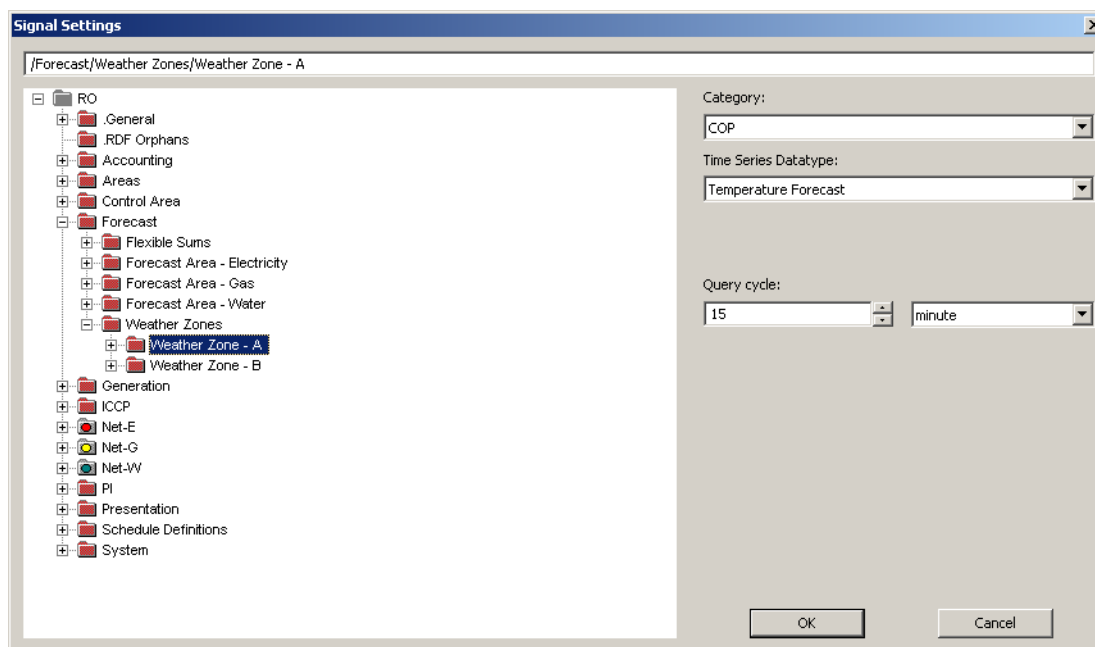
7.2.8.6 Signal Settings

The **Signal Settings** dialog is used to define parameters for the signal source of a particular time series data column within the **Spectrum Power 5 Tabular Control**.



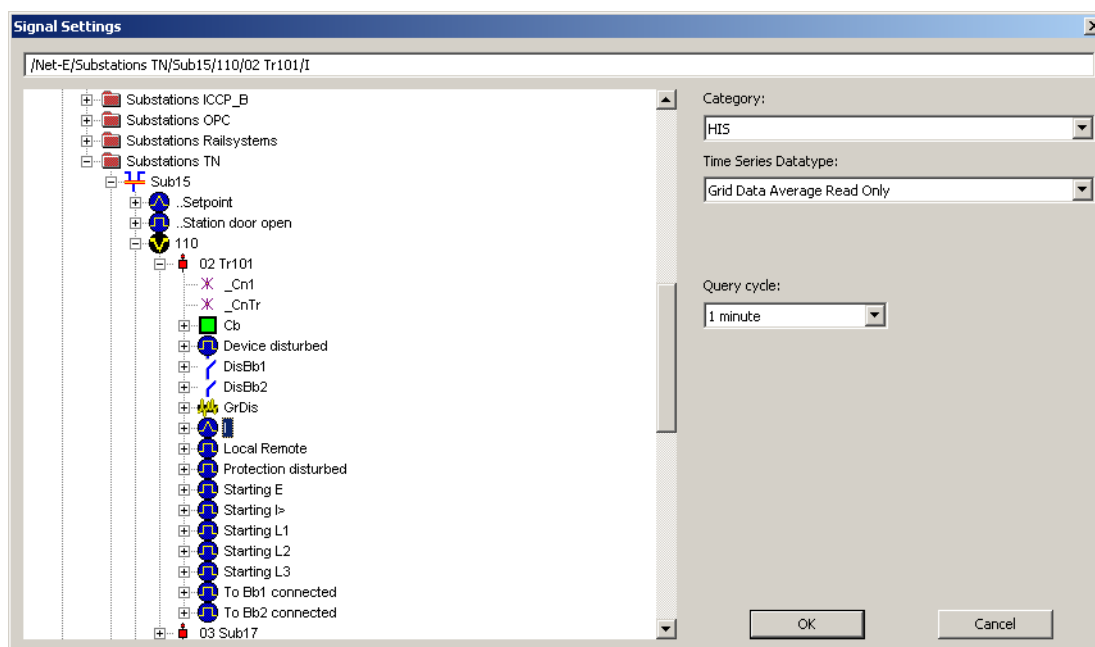
NOTE

The layout of the right section of the **Signal Settings** dialog depends on the selected data category:



[immg007-220709-01.tif, 1, en_US]

Figure 7-45 Signal Settings Dialog for Chosen Category COP



[rgimg48-080210-01.tif, 1, en_US]

Figure 7-46 Signal Settings Dialog for Chosen Category HIS

Category

Defines the signal source category:

- **COP** displays schedule data for the column.
- **HIS** displays historical and current data for the column.

Time Series Datatype

Defines the time series datatype for the column. The available selections are configured for the project.

Query Cycle (for Category COP)

Defines the number of cycles and the unit of the query for the column. Available units are second, minute, hour, day, and week.

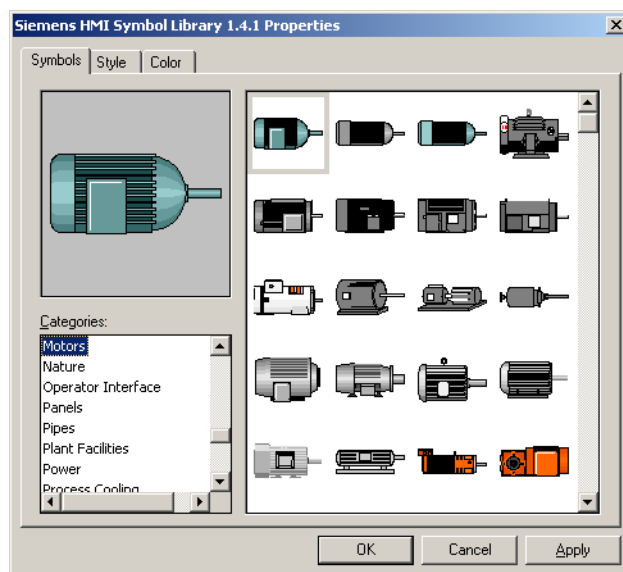
Query Cycle (for Category HIS)

Defines the cycle of the query for the column.

7.2.9 Siemens HMI Symbol Library

The **Siemens HMI Symbol Library** control contains a comprehensive collection of prepared symbols. These symbols are used for the display of systems and system components in worldmaps.

The **Siemens HMI Symbol Library Properties** dialog is used to select a symbol and define the Look and Feel for the selected symbol.



[immrg146-140409-01.tif, 1, en_US]

Figure 7-47 Properties of Siemens HMI Symbol Library - Symbols

Preview

In the area at the top left, the symbol selected in the selection is displayed. All settings of the **Siemens HMI Symbol Library** dialog are displayed in the preview.

Categories

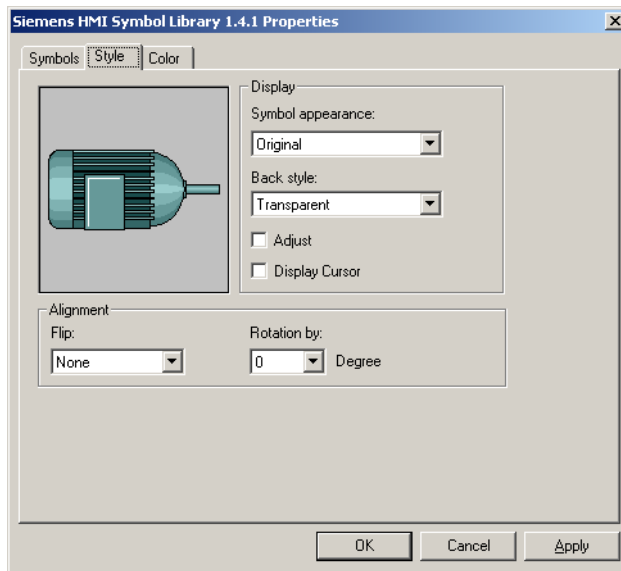
In the area at the bottom left all categories containing symbols of the **Symbol Library** are displayed.

Select the category that contains the desired symbol. All symbols of the selected category are displayed in the selection.

Selection

The right area displays all the symbols available for the selected category. Position the mouse pointer on a symbol, to show a short instruction.

Click the desired symbol to select it. The selected symbol is displayed in the preview.



[immrg147-140409-01.tif, 1, en_US]

Figure 7-48 Properties of Siemens HMI Symbol Library Dialog - Style

Alignment

In the bottom left area the alignment of the symbol is changeable. The symbol is mirrored and rotated.

Mirroring

Open the list box and select the desired mirroring type.

The symbol is mirrored on the horizontal or vertical central axis of the symbol respectively. The symbols can be mirrored both horizontally and vertically.

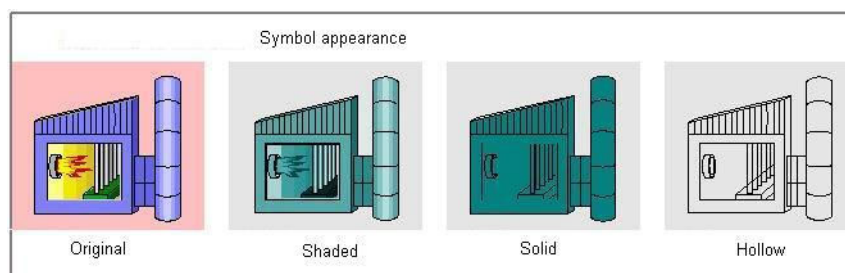
Rotation

Open the list box and select the desired angle for the rotation.

The rotation is done around the central axis of the symbol. The rotation is done counter clockwise in steps of 90, 180, and 270 degrees.

Symbol Appearance

Defines the appearance form of the symbol.



[immrg043-111208-01.tif, 1, en_US]

Original

The appearance form corresponds to the multi-colored display in the selection of the **Symbols** tab.

Shaded

Lines of the color black are maintained as contour lines. Elements of the symbols in other colors are displayed as light grades of the current foreground color.

Solid

Solid lines of the color black are maintained as contour lines.

All the elements of the symbol in other colors are assigned the color value of the current foreground color.

Hollow

Outline lines of the color black are maintained as contour lines. All the elements of the symbol in other colors are assigned the color value of the current background color.

Back Style

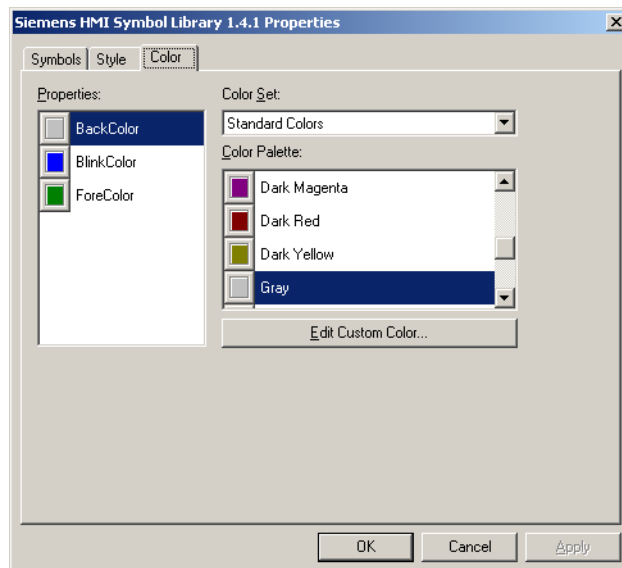
The background is displayed either as **Nontransparent** or **Transparent**.

Adjust

When selecting the **Adjust** option, the picture contents are adapted to the object size of the selected symbol.

Display Cursor

When selecting the **Display Cursor** option, the cursor is not hidden in runtime.



[immg148-140409-01.tif, 1, en_US]

Figure 7-49 Changing the Colors

Properties

Select the color attribute to change.

- The **BlinkColor** acquires the symbol in the flashing status. The flashing is only visible in runtime.
- The **BackColor** defines the background color. Defines the background of the symbol for the background type Nontransparent.
- In the appearance forms of **Shadow** and **Solid**, the symbol is shown in the **ForeColor**.

Color Set

- Select one of the color groups in the list box.
- The related colors are displayed in the **Color Palette** section.
- The **Standard Colors** group contains the 16 standard colors of the operating system.
 - The **Windows System Colors** group contains the colors, which are set in the operating system currently as the display options at system start.

Color Palette

- Select the desired color.
- Click **Apply** to assign the new color to the selected color attribute.

Editing Custom Color...

Click this button to access the **Color** dialog.

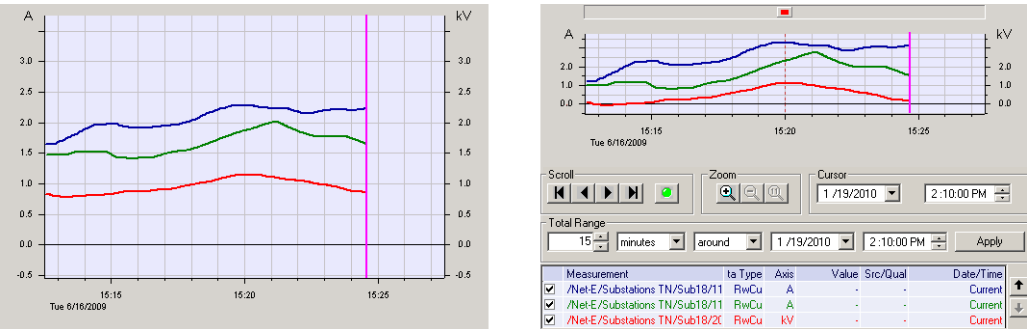
Define the desired color freely based on the graphic settings of the operating system.

7.2.10 Spectrum Power 5 Trend Control

7.2.10.1 Overview

With the **Spectrum Power 5 Trend Control**, you integrate a trend display into a worldmap. In runtime, the configured curves are displayed. The **Spectrum Power 5 Trend Control** displays historical data and realtime data as well as forecast data and schedules. Simple local calculation facilities allow performing basic operations over retrieved data at runtime, such as difference between planned and actual value curves.

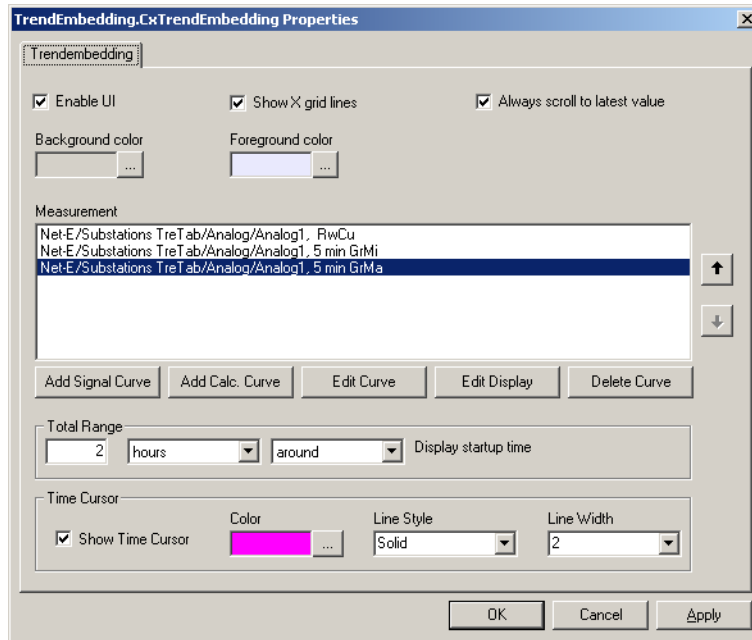
Multiple options to configure the Look and Feel of the corresponding trend and curves are provided. Basically there are 2 different runtime layouts available.



[rgimmg50-080210-01.tif, 1, en_US]
Figure 7-50 Available Runtime Layouts of a Spectrum Power 5 Trend Control

7.2.10.2 Properties

The **TrendEmbedding.CxTrendEmbedding Properties** dialog is the root dialog for configuring the **Spectrum Power 5 Trend Control**. Basically, the Look and Feel of the control at runtime is defined here.



[image013-280510-01.tif, 1, en_US]

Figure 7-51 TrendEmbedding Control Properties

Enable UI

Either the complete Trend UI (chart area, graphic presentation control and measurement list) or only the chart area is shown within the worldmap. For configurations with enabled UI, almost all settings described later are changeable during runtime.

Show X grid lines

X-grid lines or no x-grid lines for the chart area are shown at runtime.

Always Scroll to Latest Value

Defines either auto-scrolling of the time axis or fixed time axis while the worldmap is opened at runtime.

Background Color and Foreground Color

Select the background and foreground color of the embedded Trend UI from the color palettes.

Measurement

The measurements area shows already configured measurements to be trended, together with the data type and special date or time configurations. Calculated curves are also included in this area.

The up and down arrows on the right side change the z-order of the trended curves on the chart area.

Add Signal Curve, Add Calc. Curve, Edit Curve, Edit Display, and Delete Curve

- **Add Signal Curve** opens the **Signal Settings** dialog to add a new measurement to be trended.
- **Add Calc. Curve** opens the **Calculation Settings** dialog to add a new calculated curve based on existing curves as operands.
- **Edit Curve** opens the **Signal Settings** dialog to modify the signal settings for the selected measurement.
- **Edit Display** opens the **Curve Settings** dialog to modify the layout for the selected measurement.
- **Delete Curve** deletes the selected measurement from the chart.

Total Range

Defines the total time range within the worldmap.

- Use **after** for beginning the time axis with the current time at worldmap call-up at runtime (pointing to future).
- Use **before** for ending the time axis with the current time at worldmap call-up at runtime (pointing to past).
- Use **around** for defining that the time axis has the current time at worldmap call-up at runtime in the middle of the chart area.

Time Cursor

Either the current time cursor is displayed in the chart area or not. For a desired time cursor select the color from the color palette and define the line style and line width (specified in points) with the list boxes.

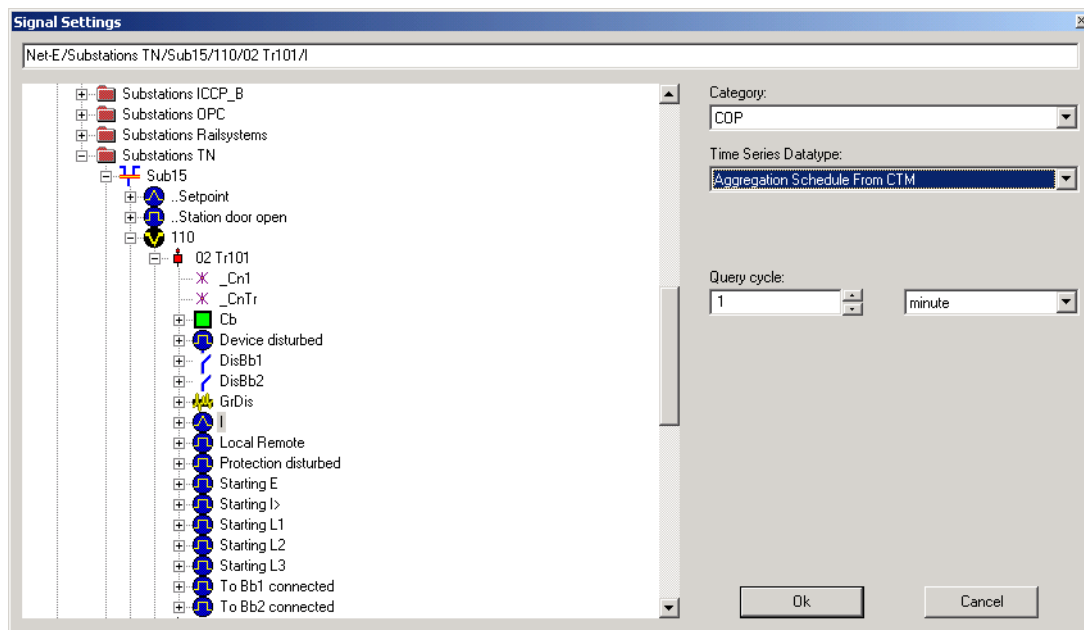
7.2.10.3 Signal Settings

The **Signal Settings** dialog is used to define parameters for the signal source of a particular curve within the **Spectrum Power 5 Trend Control**.



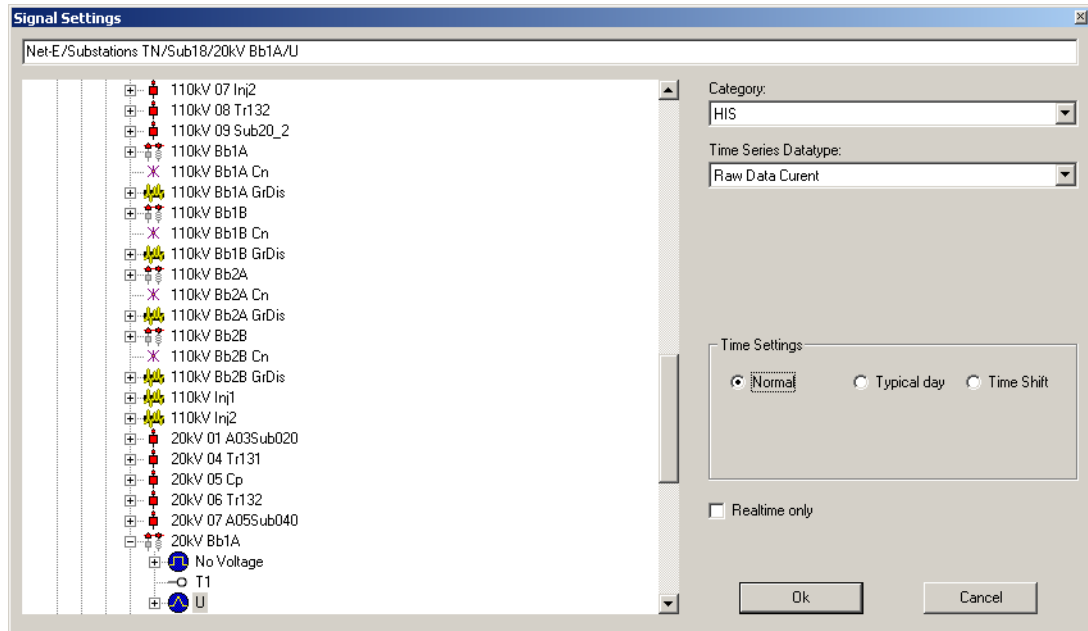
NOTE

The layout of the right section of the **Signal Settings** dialog depends on the selected data category.



[immrg022-230709-01.tif, 1, en_US]

Figure 7-52 Signal Settings Dialog for Chosen Category COP



[rgimmg53-090210-01.tif, 1, en_US]

Figure 7-53 Signal Settings Dialog for Chosen Category HIS



NOTE

The layout of the right section of the **Signal Settings** dialog of the chosen category HIS depends on the selected time series data type and time settings.

Category

Defines the signal source category:

- **COP** displays schedule data for the curve.
- **HIS** displays historical and current data for the curve.

Time Series Datatype

Defines the time series datatype for the curve. The available selections are configured for the project.

Query Cycle (for Category COP)

Defines the number of cycles and the unit of the query for the curve. Available units are second, minute, hour, day, and week.

Query Cycle (for Category HIS)

Defines the cycle of the query for the curve. Possible query cycles are dependent from the selected time series datatype.

The **Query cycle** is not available for time series datatype **Raw Data Current**.

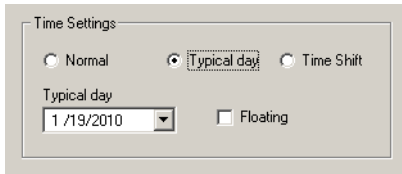
Time Settings (for Category HIS)

Defines the time settings for the curve trended.

- **Normal**
No special time settings, the current time is used.

- **Typical day**

Defined by the absolute date in the past with a drop-down calendar field after the selection. The typical day is repeated for every day, except **Floating** is checked. Then the typical days are changed according to time history when scrolling to another day.

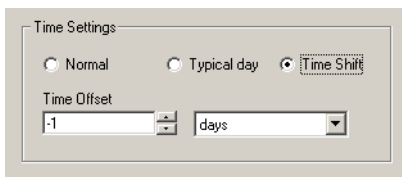


[image021-280510-01.tif, 1, en_US]

Figure 7-54 Time Settings – Typical Day

- **Time Shift**

Define the time offset within the 2 entry fields after the selection. The curve drawn according to the specified time offset to current time at runtime.



[image022-280510-01.tif, 1, en_US]

Figure 7-55 Time Settings – Time Shift

Realtime only (for Category HIS)

The **Realtime only** selection box is only shown if time series datatype **Raw Data Current** and **Time Settings Normal** has been selected. If realtime only is selected, the curve is a pure realtime trend and does not display values stored in the **Historical Information System (HIS)**.

7.2.10.4 Calculation Settings

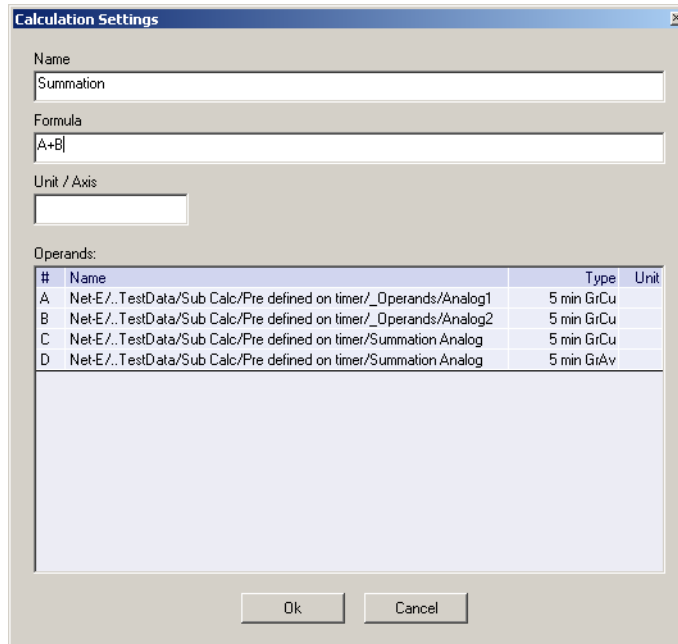
The **Calculation Settings** dialog is used to define curves that are the result of local calculations based on existing curves as operands within the **Spectrum Power 5 Trend Control**.



NOTE

Local calculations of curves can only be done if the curves used as operands do have the same cycles (same cycle unit and same cycle multiplier).

Local calculations using as operands curves with raw data is not possible at all.



[immg014-280510-01.tif, 1, en_US]

Figure 7-56 Calculation Settings Dialog

Name

Define the name of the calculated curve. The name will appear in the measurement list of the **TrendEmbedding.CxTrendEmbedding Properties** dialog.

Formula

Define the calculation within the **Formula** entry field. A and B in the screenshot are the curve identifiers shown within the operands overview area of this dialog.

The following operands are supported:

- Floating point constants
- Operators: (,), +, -, *, /, sqrt()

Expressions are defined similar as in MS Excel.

For example, **A+B** where **A** and **B** are implicit identifiers for 1st and 2nd curve in the chart.

Unit / Axis

Define the unit (y-axis) of the calculated curve.

Operands

The read-only operands area shows already configured measurements to be trended, together with the data type, unit and an identifier to be used in the formula.



NOTE

The unit of the local calculation allows the assignment of the local calculation to an axis which might be new or already existing because shared with another curve in the chart.

The formula gets validated with the other entries of the dialog when **Ok** gets pressed. If the validations fails, you get a message box displaying an error message to allow correcting the entries.

Following validation rules are checked:

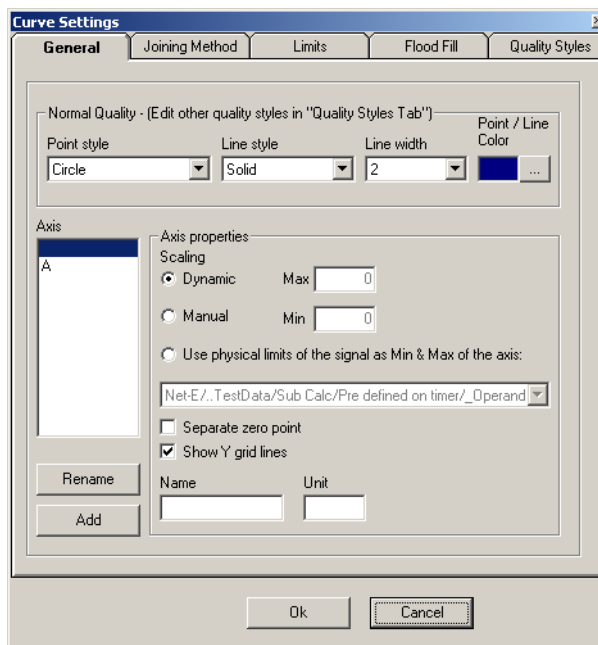
- Curves with raw data type cannot be used as operands
- All curves must have the same cycles
- Time-shifted curves can only get used if the time-shift matches with the cycle time. For example, a curve with 1-Day-Average which gets shifted with 1 hour cannot get combined in a calculation with a normal curve with 1-Day-Average
- The formula must be a valid algebraic expression.
- Cyclic dependencies of calculations are not allowed.

Local calculations can get used as operands of derived calculations as long as the validation rules above are respected.

The local calculation is performed value-wise according to the formula. The joining method of the local calculation can be set freely. But this means that the resulting curve of the local calculation will not be necessarily the exact mathematical calculation 2 functions.

7.2.10.5 Curve Settings

The **Curve Settings** dialog is used to define the layout for the configured curves within the **Spectrum Power 5 Trend Control**.



[image015-280510-01.tif, 1, en_US]

Figure 7-57 Curve Settings Dialog – General Tab

Point Style, Line Style, Line Width, and Point/Line Color

Defines the point style, line style, and line width (specified in points) of the curve for the quality **Normal** with the list boxes. Select the color from the color palette.

Axis

The corresponding y-axis of the curve is selected out of the available y-axis within the chart configuration.

- **Rename**
Enables renaming of the available y-axis within the chart.
The entry fields **Name** and **Unit** is enabled to allow the changes.
- **Add**
Adds an additional y-axis and the unit of the y-axis within the chart and assigns this axis to the selected curve.
The entry fields **Name** and **Unit** is enabled to allow the changes.

Axis Properties

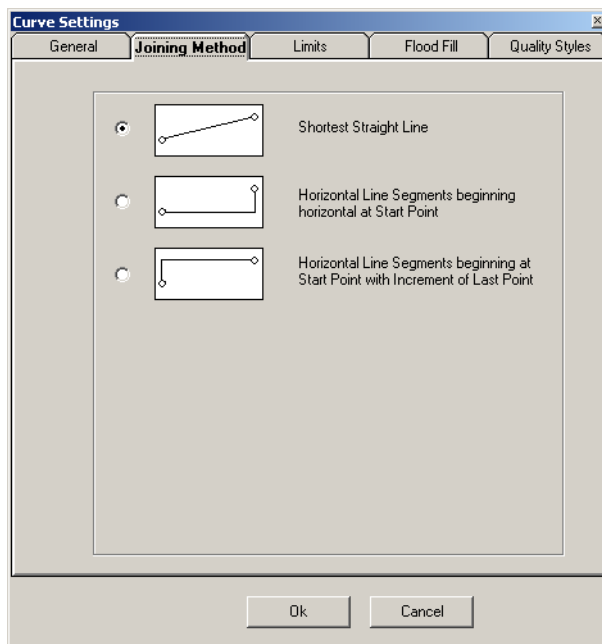
The corresponding y-axis properties within the chart configuration is selected for the curve.

- **Dynamic**
Allows dynamic scaling of the y-axis of the corresponding curve according the current values at runtime.
- **Manual**
Forces a fix scale of the y-axis of the corresponding curve at runtime.
Within the **Min** and **Max** fields the values of the manually scaled y-axis is configured. Both fields are enabled if **Manual** has been selected.
- **Use physical limits of the signal as Min & Max of the axis**
Sets the **Min** and **Max** values to the physical limits of the related signal source of the corresponding curve. The list box below is enabled if more than 1 curves share the y-axis. You can than choose the signal source for the physical limits.
- **Separate zero point**
If **Separate zero point** is unchecked, multiple y-axis use a common zero point. Otherwise the zero points of the different y-axis are separated.
- **Show Y grid lines**
Y-grid lines or no y-grid lines for the selected y-axis are shown at runtime.
- **Name and Unit**
Refer to Axis description.



NOTE

Multiple curves may use the same y-axis. The changes you are configuring for **Axis** and **Axis Properties** apply to all curves using the same y-axis.



[image023-280510-01.tif, 1, en_US]

Figure 7-58 Curve Settings Dialog – Joining Method Tab

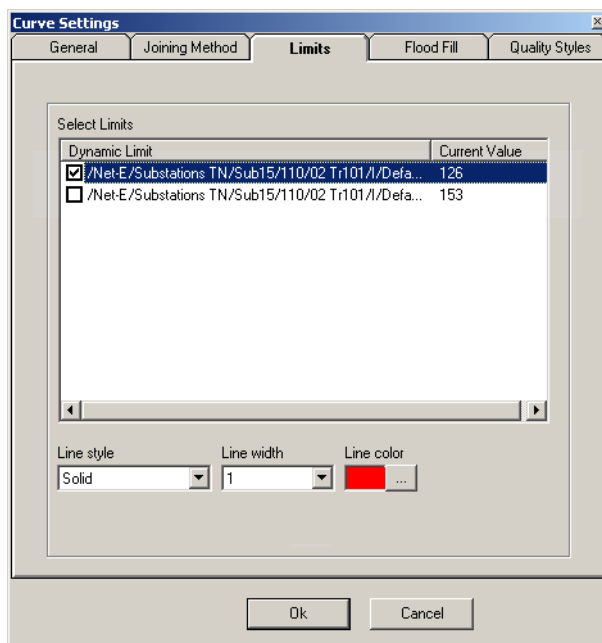
Option Buttons

Defines the type of connection between 2 curve points for the selected curve.



NOTE

By nature, trending current values always use the display type **Horizontal Line Segment beginning horizontal at Start Point**.



[image024-280510-01.tif, 1, en_US]

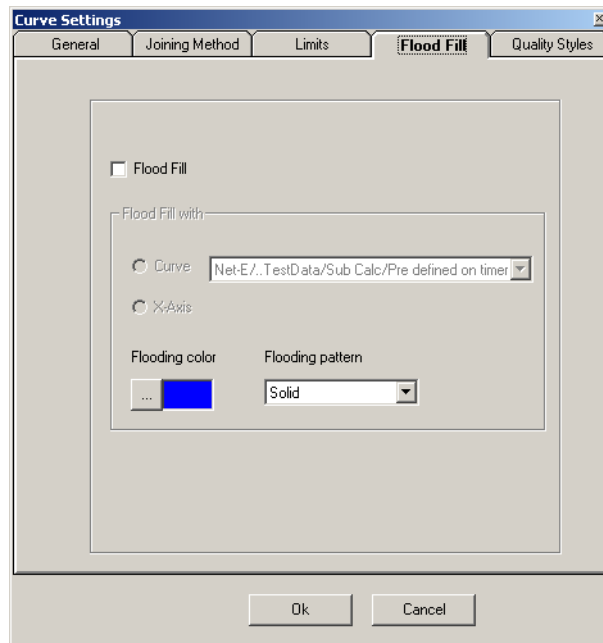
Figure 7-59 Curve Settings Dialog – Limits Tab

Select Limits

Shows the defined limits for the selected measurements to be trended, together with the current limit value.
Select the limits to display as a line within the chart with the checkboxes on the left side.

Line Width, Line Style, and Line Color

Defines the limit line style and limit line width (specified in points) of the selected dynamic limit with the list boxes.
Choose the color from the color palette.
Unchecked limits are not shown within the chart.



[image025-280510-01.tif, 1, en_US]

Figure 7-60 Curve Settings Dialog – Flood Fill Tab

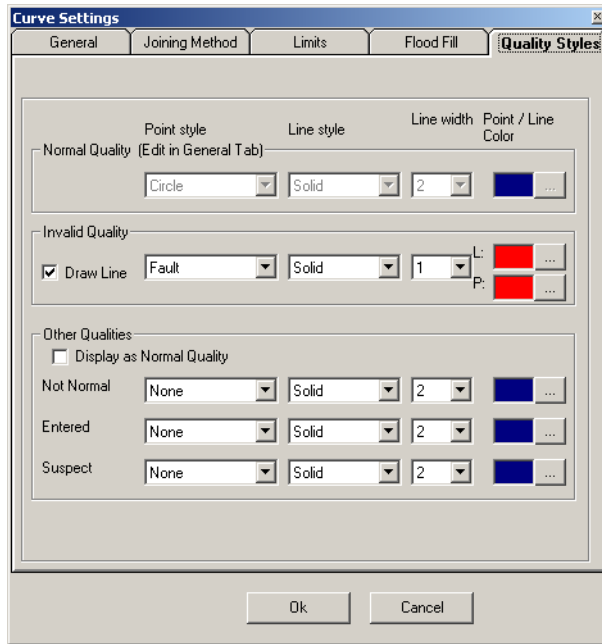
Flood Fill

The check box enables filling the space between the selected curve and another one or the x-axis with a selectable color and pattern. The settings are used if **Flood Fill** is enabled.

Flood Fill with

Defines the corresponding flooding properties of the selected curve.

- **Curve and X-Axis**
Defines flooding of the selected curve with the x-axis or with another curve configured for the chart (from the list box on the right).
- **Flooding color**
Choose the color from the color palette.
- **Flooding pattern**
Choose the pattern for flooding the area between the selection.



[image026-280510-01.tif, 1, en_US]

Figure 7-61 Curve Settings Dialog – Quality Styles Tab

Normal Quality

- **Point Style, Line Style, Line Width, and Point/Line Color**

Shows the point style, line style, and line width (specified in points) as well as the point/line color of the curve for the quality **Normal** that has been defined in the **General** tab of this dialog.

Invalid Quality

- **Draw Line**
Choose between displaying still a curve in case of invalid values or not. If no line shall be drawn, the settings for line style, and line width (specified in points) as well as the line color are disabled.
- **Point Style, Line Style, Line Width, Point Color, and Line Color**
Define the point style, line style, and line width (specified in points) of the curve for the quality **Invalid** with the list boxes. Select the color for Line and Point from the color palettes. If no line shall be drawn, only the point style and point color properties are enabled.

Other Qualities

- **Display as Normal Quality**
Choose between displaying the qualities Not Normal, Entered and Suspect of a curve according the settings or not. If checked, the settings for point style, line style, and line width (specified in points) as well as the point/line color are disabled.
- **Point Style, Line Style, Line Width, Point/Line Color**
Define the point style, line style, and line width (specified in points) of the curve for the qualities Not Normal, Entered and Suspect with the list boxes. Select the colors for Line/Point from the color palette.



NOTE

Quality coloring of the curves is only applicable for HIS as datasources. COP schedules do not have any quality codes, thus COP schedule curves are displayed with **Normal** quality.



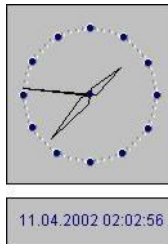
NOTE

The free configuration and combination of these style properties allow flexibility to cover several use cases:

- Many curves in the same chart
Dynamic style and coloring is achieved through line style and widths rather than with colors because each curve already has its own color
- 1 or 2 curves in the same chart
Dynamic style and coloring can get achieved through coloring instead of line style and width. The clarity of the chart is still guaranteed

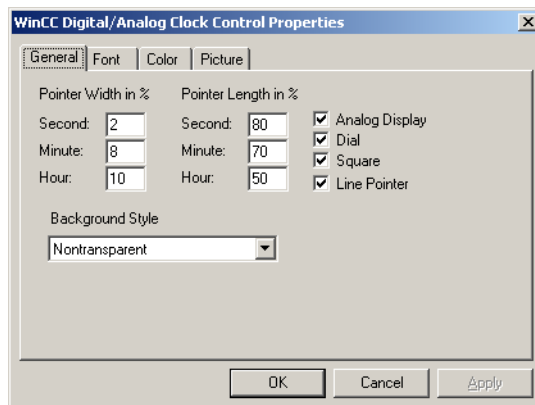
7.2.11 WinCC Digital/Analog Clock Control

With the **WinCC Digital/Analog Clock Control**, you integrate a time display into a worldmap. In runtime, the current system time of the operating system is displayed. The time is displayed as analog or digital.



[immg046-111208-01.tif, 1, en_US]

The **WinCC Digital or Analog Clock Control Properties** dialog is used to define the Look and Feel of the clock.



[immg150-140409-01.tif, 1, en_US]

Figure 7-62 WinCC Digital Analog Clock Control Properties - General Tab

Pointer Width in %, Pointer Length in %

For the analog display, set the width and the length of the second, minute and hour pointer. The values are understood as percent share of the pointer length respective of the radius of the dial.

Analog Display

Choose between analog and digital display of the clock.

Dial

Hide or unhide the hour marks for the analog display.

Square

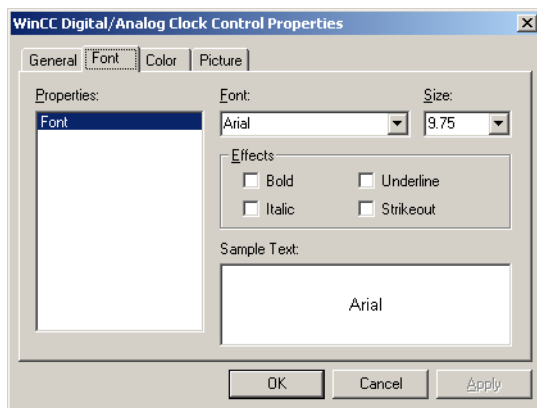
Select whether the analog clock is displayed in square shape.

Line Pointer

Specify whether for the analog display only the border or also the filling color of the pointer is displayed.

Background Style

- **Nontransparent**
Displays the analog clock with a frame.
- **Transparent border**
Displays the colored dial without a border.
- **Transparent**
Displays the analog clock without background.



[immg151-140409-01.tif, 1, en_US]

Figure 7-63 WinCC Digital Analog Clock Control Properties - Font Tab

Properties

The currently selected property is shown.

Font, Size

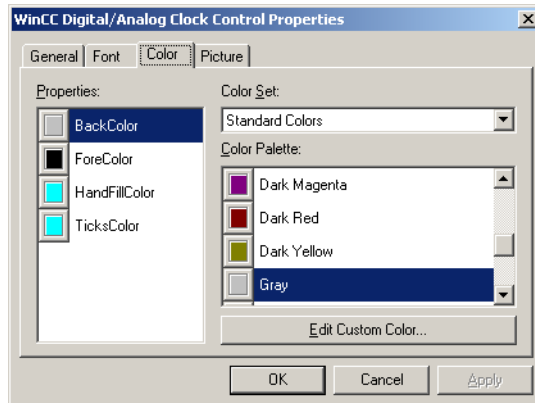
Select the desired font and size (specified in points) for the digital display of the clock from the list boxes.
Use any of the fonts registered in the operating system.

Effects

Select one or more font effects.

Sample Text

The selected settings are displayed in a preview.



[immg152-140409-01.tif, 1, en_US]

Figure 7-64 WinCC Digital Analog Clock Control Properties - Color Tab

Properties

Select the color attribute you wish to change.

Color Set

Select one of the color groups in the list box.

The related colors are displayed in the **Color Palette** section:

- The **Standard Colors** color group contains the 16 standard colors of the operating system.
- The **Windows System Colors** color group contains the colors, which are set in the operating system currently as the display options at system start.

Color Palette

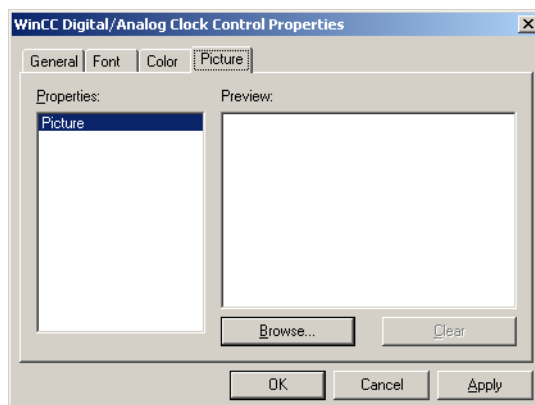
Select the desired color.

- Click **Apply** to assign the new color to the selected color attribute.

Edit Custom Color...

Click this button to access the **Color** dialog.

Depending on the graphic settings of the operating system, define the desired color freely.



[immg153-140409-01.tif, 1, en_US]

Figure 7-65 WinCC Digital Analog Clock Control Properties - Picture Tab

Browsing

- Click **Browse...** to access the **Browse Picture** dialog.
- Select the graphic file, to which you want to assign the selected property. Pictures in the following formats are inserted: BMP, DIB, ICO, CUR, EMF, WMF, GIF, and JPG.

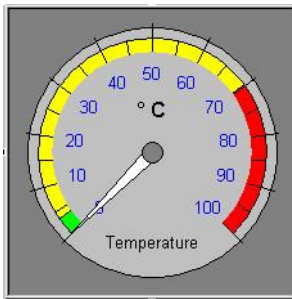
Deleting

Click **Clear** to remove the assigned picture from the selected property.

7.2.12 WinCC Gauge Control

7.2.12.1 Overview

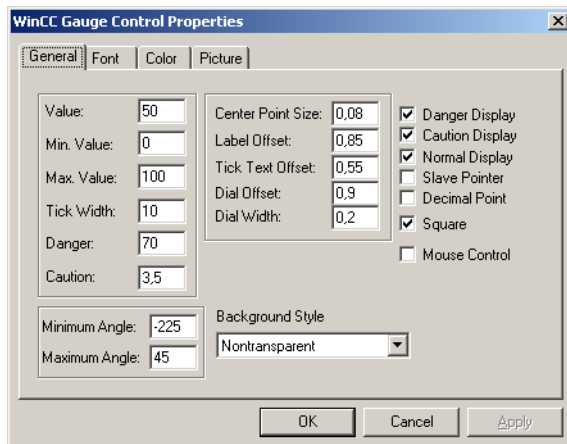
The **WinCC Gauge Control** is used to display the monitored measured values in the form of an analog measurement clock. Warning and danger areas as well as the extreme values of the pointer movement is marked in colors.



[immrg053-111208-01.tif, 1, en_US]

7.2.12.2 Properties

The **WinCC Gauge Control Properties** dialog is used to define the Look and Feel of the gauge.



[immrg154-140409-01.tif, 1, en_US]

Figure 7-66 WinCC Gauge Control Properties - General Tab

Value

Specify a start value for the pointer of the gauge. If process driver connections are missing, this value is shown in runtime.

Min. Value

Enter the lower limiting value of the measuring range, this value is shown as the min. scale value.

Max. Value

Enter the upper limiting value of the measuring range, this value is shown as the max. scale value.

Tick Width

Enter the distance of the main tick marks as the difference of 2 neighboring measured values.
The area between 2 main tick marks is divided by default by a wider tick mark of half the length.

Danger

Set the limiting value for the start of the danger area.

Caution

Set the limiting value for the start of the warning area.

Minimum Angle

Enter the angle for the display at the scale start. This value is entered in degrees.

Maximum Angle

Enter the angle for the display at the scale end. This value is entered in degrees.

Center Point Size

Specify the radius for the display of the center point, this value is given in decimal format as percent share of the diameter of the gauge.

Label Offset

Specify the position for the labeling of the gauge, this value is given in decimal format.
From the topmost border line of the rectangle encompassing the object, the value is the percent share of the object height.

Tick Text Offset

Specify the inner radius of the scale labeling. This value is given in decimal format as percent share of the radius of the gauge.

Dial Offset

Specify the outer radius of the scale. The value is given in decimal format as percent share of the radius of the gauge.

Dial Width

Specify the length of the main tick marks and thus the width of the scale. This value is given in decimal format as percent share of the radius of the gauge.

Danger Display

Display the scale area starting at the limiting value **Danger** in the **Danger color** (default color: red).

Caution Display

Display the scale area starting at the limiting value **Warning** in the **Warning color** (default color: yellow).

Normal Display

Display the scale area ending at the limiting value **Warning** in the **Normal color** (default color: green).

Slave Pointer

Activate a marker function for the maximum and the minimum pointer movement.
It marks the real measurement area in runtime with colored lines:

- A green line marks the minimum pointer movement.
- A red line marks the maximum pointer movement.

Decimal Point

Display the values of the scale labeling as whole numbers or as decimal numbers with one decimal place.

Square

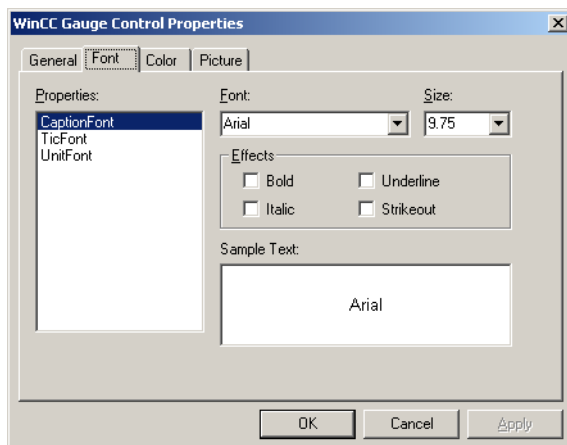
Select whether the gauge is displayed in square shape.

Mouse Control

The mouse pointer is shown in runtime, if it is placed on the symbol.

Background Style

- **Nontransparent**
Displays the gauge clock with a frame.
- **Transparent border**
Displays the gauge clock without a border.
- **Transparent**
Displays only the scale of the gauge clock.



[immg157-140409-01.tif, 1, en_US]

Figure 7-67 Changing the Font of the Gauge Control - Font Tab

Properties

Enables to define the font for different properties within the gauge:

- Use **CaptionFont** to change the font for the name of the gauge.
- Use **TicFont** to change the scale labeling of the gauge.
- Use **UnitFont** to change the font for the display of the physical unit.

Font

The current font for the selected property is shown and changed.
Use any of the fonts registered in the operating system.

Size

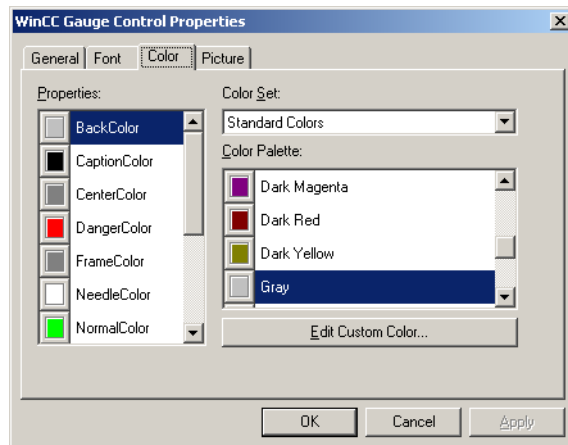
Select the desired font size (specified in points) from the list box.

Effects

Select one or more font effects.

Sample Text

The selected settings are displayed in a preview.



[immg158-140409-01.tif, 1, en_US]

Figure 7-68 Changing the Color of Gauge Control - Color Tab

Properties

Select the color attribute you wish to change.

Color Set

Select one of the color groups in the list box.

The related colors are displayed in the **Color Palette** section:

- The **Standard Colors** color group contains the 16 standard colors of the operating system.
- The **Windows System Colors** group contains the colors, which are set in the operating system currently as the display options at system start.

Color Palette

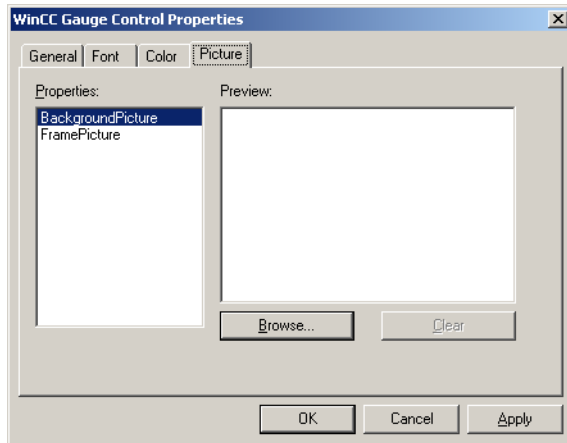
Select the desired color.

- Click **Apply** to assign the new color to the selected color attribute.

Edit Custom Color...

Click this button to access the **Color** dialog.

Depending on the graphic settings of the operating system, define the desired color freely.



[immrg159-140409-01.tif, 1, en_US]

Figure 7-69 Changing the Pictures of the Gauge Control – Picture Tab

Properties

Select the property for which you want to change the picture assignment.

The related pictures are displayed in the **Preview** section:

- The **BackgroundPicture** property defines an image to the circular background of the gauge.
- The **FramePicture** property defines an image to the rectangular background of the control.

Browsing

- Click **Browse...** to access the **Browse Picture** dialog.
- Select the graphic file, to which you want to assign the selected property. Pictures in the following formats are inserted: BMP, DIB, ICO, CUR, EMF, WMF, GIF, and JPG.

Deleting

Click **Clear** to remove the assigned picture from the selected property.

7.2.12.3 Additional Control Properties

The properties of the control are changed in the **WinCC Gauge Control Properties** dialog or in the window **Graphics properties**.

The following attributes are changed in the window **Graphics properties** for the selected control.

Caption

Enter a text to display within the gauge (for example the type of the controlled value).

By default, the label **SIMATIC** is configured.

Unit Text

Enter a text to display within the gauge (for example, the physical unit of the displayed value).

By default, this field is blank.

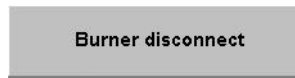
Unit Offset

Specify the position for the display of the physical unit on the gauge.

The value is given in decimal format. The value is the percent share of the object height, starting from the topmost border line of the rectangle encompassing the object.

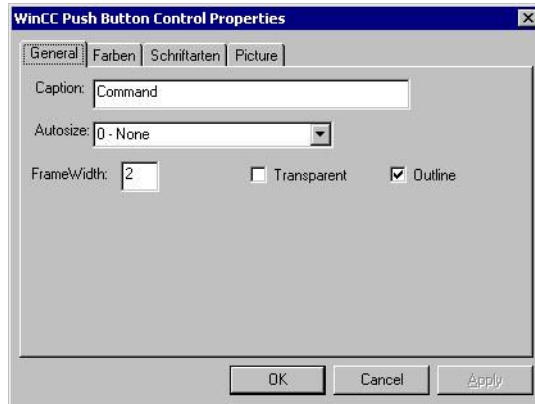
7.2.13 WinCC Push Button Control

The **WinCC Push Button Control** is used to configure a button, which is connected to the execution of a command. In runtime, the button adopts the states pressed and not pressed. Both statuses are assigned with a different image, which shows the current state of the button.



[immrg061-111208-01.tif, 1, en_US]

The **WinCC Push Button Control Properties** dialog is used to define the Look and Feel of the push button.



[immrg062-111208-01.tif, 1, en_US]

Figure 7-70 WinCC Push Button Control Properties – General Tab

Caption

Enter a text for the labeling of the button. This label, for instance, contain a function description.

Autosize

For the geometry of the button, configure automatic size adaptation:

- Select **0 - None**, to disable the automatic size adaptation.
- Select **1 - Adjust Picture Size To Button**, to adapt the size of the image to the geometry of the button.
- Select **2 - Adjust Button Size To Picture**, to adapt the geometry of the button to the size of an assigned picture.

Frame Width

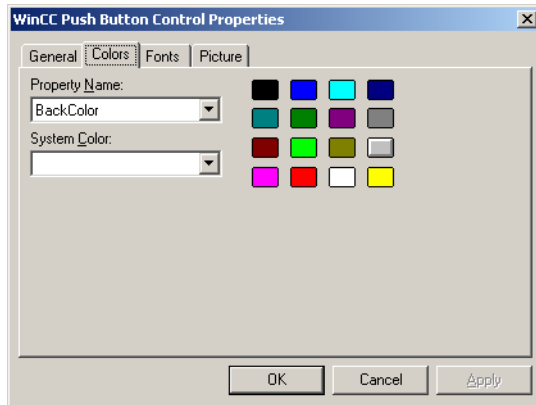
Enter a value for the 3D border width of the button in pixel.

Transparent

The background of the button is displayed transparent.
In this case, the configured background color is not shown.

Outline

The 3D border of the button is surrounded with an additional outer border line.



[immrg161-140409-01.tif, 1, en_US]

Figure 7-71 WinCC Push Button Control Properties – Colors Tab

Property Name

Select the color attribute that you wish to change from the list box.

System Color

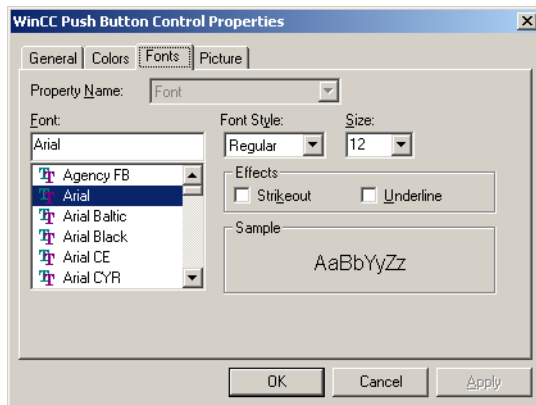
Assign a system color.

The list box contains all the picture elements, the display options of which is adapted in the operating control panel.

Basic Color Palette

Select one of the 16 standard colors of the operating system in the right-hand area.

Either **System Color** or colors from the **Basic Color Palette** is chosen.



[immrg162-140409-01.tif, 1, en_US]

Figure 7-72 WinCC Push Button Control Properties – Fonts Tab

Property Name

The currently selected property is shown.

Font

The currently selected font for the control label is shown and is changeable.

Use any of the fonts registered in the operating system.

Font Style, Size

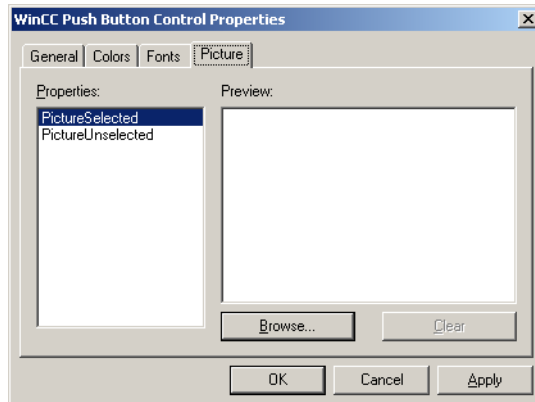
Select the desired font style and size (specified in points) from the list boxes.

Effects

Select one or more font effects.

Sample

The selected settings are displayed in a preview.



[immg163-140409-01.tif, 1, en_US]

Figure 7-73 WinCC Push Button Control Properties – Pictures Tab

Properties

Select the property for which you want to change the picture assignment.

The related pictures are displayed in the **Preview** section:

- The **PictureSelected** property defines an image that is displayed only if the button is clicked at runtime (status pressed).
- The **PictureUnselected** property defines an image that is displayed as long as the button is not pressed at runtime.

Browsing

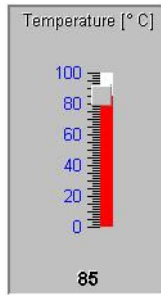
- Click **Browse...** to access the **Browse Picture** dialog.
- Select the graphic file, to which you want to assign the selected property.
Pictures in the following formats are inserted: BMP, DIB, ICO, CUR, EMF, WMF, GIF, and JPG.

Deleting

Click **Clear** to remove the assigned picture from the selected property.

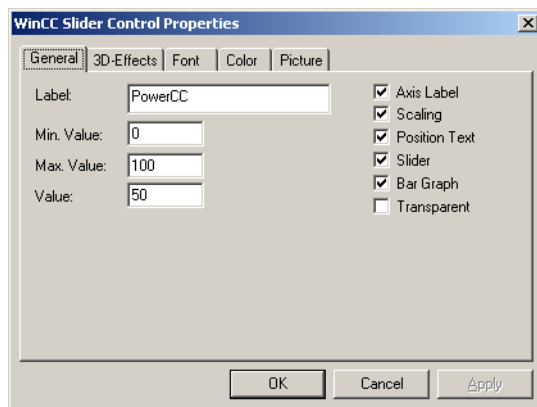
7.2.14 WinCC Slider Control

The **WinCC Slider Control** is used to display the monitored measured values in the form of a slider control. The current value is displayed under the slider and the controlled measurement area is displayed as scale labeling.



[immrg005-121208-01.tif, 1, en_US]

The **WinCC Slider Control Properties** dialog is used to define the Look and Feel of the slider.



[immrg164-140409-01.tif, 1, en_US]

Figure 7-74 WinCC Slider Control Properties – General Tab

Label

Enter a text to display, for example the type of the controlled value at the slider.

Min. Value

Enter the lower limiting value of the measuring range. This value is shown as the minimum scale value of the slider control.

Max. Value

Enter the upper limiting value of the measuring range. This value is shown as the maximum scale value of the slider control.

Value

Specify a start value for the position of the slider. If the process driver connections are missing, the slider is displayed at this position in runtime.

Axis Label

The tick marks of the scale is labeled with measured values. The step size is set depending on the specified measurement area and the size of the control automatically.

Scaling

On the left side of the slider, tick marks are shown for the display of a scale.

Position Text

The value of the current position of the slider is displayed under the slider.

Slider

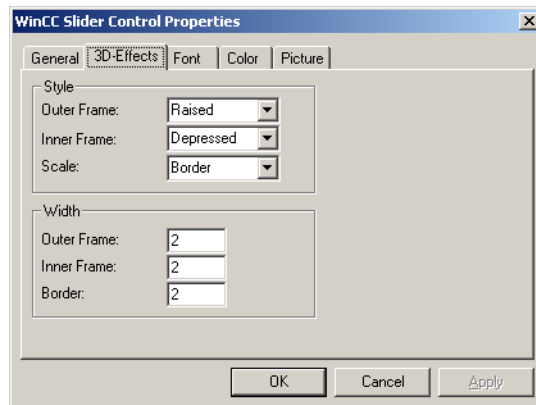
The slider is hidden.

Bar Graph

The display of the sliding areas above and below the slider is hidden.

Transparent

The background of the control is displayed transparent. The configured background color is hidden.



[immrg165-140409-01.tif, 1, en_US]

Figure 7-75 WinCC Slider Control Properties - 3D-Effects Tab

Style

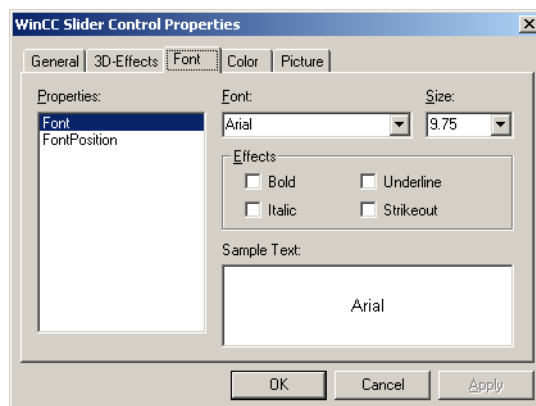
Select a style for the display of the **Inner** and **Outer Frame** as well as the **Scale**.

The border of the **WinCC Slider Control** consists of the 3 elements outer frame, border, and inner frame. A 3D effect is created by the two-colored display of the outer and inner frames as well as the tick marks of the scaling.

The styles depressed and raised are distinguished by the changed alignment of the 2 color attributes **Upper Frame Color** and **Bottom Frame Color**. The **Border** style for the scale affects that the elements are displayed single-colored.

Width

Enter a value for the width of the border elements in pixels.



[immrg166-140409-01.tif, 1, en_US]

Figure 7-76 WinCC Slider Control Properties – Font Tab

Properties

Enables to define the font for different properties within the slider.

- Use **Font** to change the labeling and the scale font of the slider.
- Use **FontPosition** to change the font of the value, which is displayed by the current position of the slider.

Font

The current font for the selected property is shown and changed.
Use any of the fonts registered in the operating system.

Size

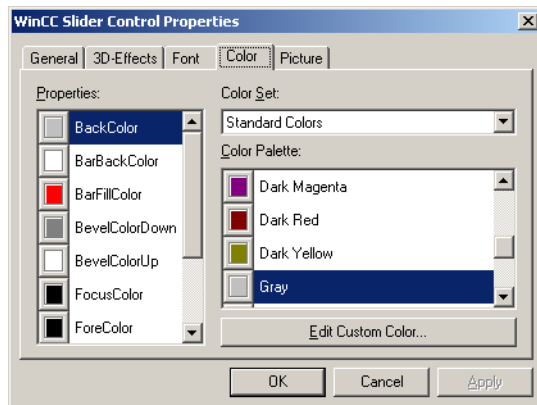
Select the desired font size (specified in points) from the list box.

Effects

Select one or more font effects.

Sample Text

The selected settings are displayed in a preview.



[immg167-140409-01.tif, 1, en_US]

Figure 7-77 WinCC Slider Control Properties – Color Tab

Properties

Select the color attribute you wish to change.

Color Set

Select one of the color groups in the list box.

The related colors are displayed in the **Color Palette** section:

- The **Standard Colors** color group contains the 16 standard colors of the operating system.
- The **Windows System Colors** group contains the colors, which are set in the operating system currently as the display options at system start.

Color Palette

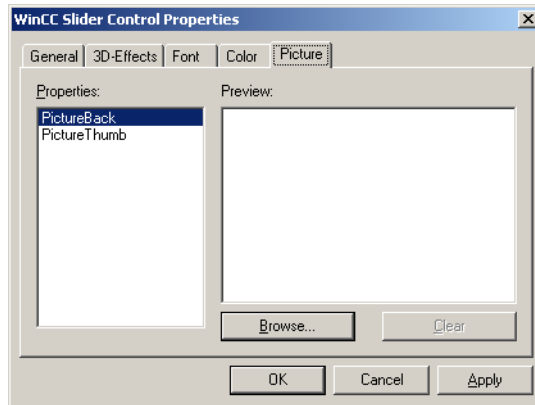
Select the desired color.

- Click **Apply** to assign the new color to the selected color attribute.

Edit Custom Color...

Click this button to access the **Color** dialog.

Depending on the graphic settings of the operating system, define the desired color freely.



[immrg168-140409-01.tif, 1, en_US]

Figure 7-78 WinCC Slider Control Properties – Picture Tab

Properties

Select the property for which you want to change the picture assignment.

The related pictures are displayed in the **Preview** section:

- The **PictureBack** property defines an image to the background of the slider.
- The **PictureThump** property defines an image on the slider of the control.

Browsing

- Click **Browse...** to access the **Browse Picture** dialog.
- Select the graphic file, to which you want to assign the selected property.
Pictures in the following formats are inserted: BMP, DIB, ICO, CUR, EMF, WMF, GIF, and JPG.

Deleting

Click **Clear** to remove the assigned picture from the selected property.

